

A SELF-STUDY BOOK FOR THE ROUNDS AFTER THE CODING TEST

The Interview Rounds

HR · Behavioural · Projects · Puzzles

Structures you fill with your own true stories — never scripts you memorise

THE ONE RULE OF THIS BOOK

Every answer in this book is shown as a **structure** plus a **filled example**. The filled example is there so you can see the shape. It is not your answer. You substitute your own real project, your own real team, your own real numbers — every time. Nobody in this book is coached to lie, and no weak answer is fixed by inventing a better one.

TIER 1 Standard HR screen — TCS, Accenture, Infosys, Wipro, Capgemini, Cognizant

TIER 2 Cross-border and regional — Grab, Sea/Shopee, GIC, DBS, Agoda, SCB

TIER 3 Structured behavioural, deep probing — Google, Amazon, Microsoft, Goldman Sachs

8 chapters · weak answer vs strong answer, side by side · labelled STAR · 24 puzzles · 7 guesstimates · one-page revision cards

A NOTE ON WHERE THIS MATERIAL COMES FROM

Every question, answer, story, puzzle and example in this book is **original**. Each one was written for this book.

Nothing here is copied, quoted or paraphrased from any published book, course, question bank, coaching handout or website.

No company's real interview is reproduced here. No transcript, no question paper, no candidate's recollection. Real hiring questions and real interview conversations are private; they are not public documents, and none were used.

When something is labelled, for example, "TCS · pattern" or "Amazon · pattern", that label means one thing only: it was written to match the **publicly documented shape** of that kind of round — its usual topics, its usual depth of follow-up, and its usual time pressure, as described in openly published careers pages, job posts and the companies' own public descriptions of how they interview. The label is never a claim that this question, or anything like it, was actually asked at that company.

Company names appear only to describe that shape. This book is not affiliated with, endorsed by, or connected to any company named in it. All trademarks belong to their owners.

Every filled answer, project and story in this book is **invented for teaching**. The people in them do not exist. They show you the structure, the length and the level of detail. Your own answer must come from your own life — at Tier 3 the follow-up to the follow-up finds an invented story, and a background check finds the rest.

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How to use this book

Two pages. Read them once, then start Chapter 1.

1 • What the three tiers mean in this book

In the coding books a tier means how hard the problem is. Here it means **how deep the interviewer digs into the same answer**. One honest story, told at three depths.

Tier	Who runs the round	What it feels like	Follow-ups
0 Warm-up	you, with a timer	One slot of a structure at a time: write the NOW sentence, or say the AC-KNOWLEDGE line and hold three seconds of silence. Mechanics only.	0
1 Standard HR	TCS, Accenture, Infosys, Wipro, Capgemini, Cognizant	The famous ten — yourself, strengths, weakness, why us, five years, relocation, bond, notice period. Asked once, taken at face value.	0–1
2 Cross-border	Grab, Sea/Shopee, GIC, DBS, Agoda, SCB	The same questions plus the regional layer: work pass and relocation reality, why South East Asia, communication style, CTC vs in-hand in two currencies.	1–2
3 Structured behavioural	Google, Amazon, Microsoft, Goldman Sachs	Principle-driven. One story, then “why that choice?”, “what would you do differently?”, “what did the other person say?”. The follow-ups are the interview.	3–5

Work Tier 1 first even for a Tier 3 company. A Tier 3 answer is not a different answer — it is the same Tier 1 answer that survived five layers of “why”.

2 • How to use the book

Weak vs strong, side by side	Each big question shows a WEAK and a STRONG answer to the same question, then a table naming what changed. The change table is the lesson — not the sample answer.
Structure, then filled example	The structure (STAR, NOW–PATH–PROOF–POINT, A–F–E–F, PAUSE, SPEH) is yours to keep. Every filled example is invented for teaching. Never carry one into a real room.
Out loud, on a timer	An answer that reads well at your desk and runs three minutes out loud has already failed. Record yourself weekly and play it back. And build the story bank (Chapter 4) before you write any answer — the other order produces invented stories.

TRAP

The four failures that cost the most offers: **no number in the answer**; **“we” where it should be “I”**; **a weakness that is secretly a boast**; **a project you cannot trace one request through**.

3 • The six-week plan

About 90 minutes a day, six days a week. Day 7 is the checkpoint — spoken, on a timer, from memory.

Week	Chapters	Daily target	Checkpoint (day 7)
1	1, 2	Ch 1 on days 1–2: the funnel, and what each round scores. Ch 2 on days 3–6: rewrite five resume bullets a day as VERB + WHAT + HOW + NUMBER.	A one-page resume where every bullet holds a number you can defend.
2	3	One HR question a day, out loud, on a timer: warm-up slot, then Tier 1, then the Tier 2/3 version of the same question.	Questions 1–6 back to back, recorded, none over 90 seconds.
3	4	Days 1–2: build the story bank of 12 real events. Days 3–6: two stories a day in labelled STAR, each with a measured result.	Someone picks three of your stories and probes each three levels deep.
4	5	One Project Defence Sheet per project, then a day each on: what did YOU do, the 60-second architecture, the code you must explain, the extension question.	Trace one request end to end on a blank page, then defend two trade-offs.
5	6, 7	Days 1–3: salary, bond, notice period, backlogs and gaps in A–F–E–F, 45 seconds each. Days 4–6: four puzzles a day using PAUSE, then two guesstimates.	Five awkward questions cold, then one puzzle and one guesstimate, thinking aloud.
6	8, revision	Days 1–4: GD entries with SPEH, the follow-up email, questions to ask, CTC vs in-hand. Days 5–6: all eight revision cards and the quick reference.	Full mock: 10 HR questions, 2 STAR probes, 1 project defence, 1 puzzle.

4 • The 10-day crash plan

Only if your interview is inside two weeks. It cuts practice volume and Tier 3 depth — not the structures.

Day	Chapters	Finish that day
1	1, 2	The funnel and the score sheets. Fix your top five resume bullets: every one gets a number.
2–3	3	“Tell me about yourself” in NOW–PATH–PROOF–POINT, strengths, weakness; then why us (20 minutes of real research), five years, and your two questions to ask.
4–5	4	Story bank of 12 real events, then all 12 written in labelled STAR. Practise a 30-second and a 2-minute version of each.
6	5	One Project Defence Sheet per resume project. Trace one request end to end, out loud.
7–8	5, 6	The “I don’t know” protocol. Then salary, bond, notice period, backlogs, gaps and CGPA — 45 seconds each, flat voice, then silence.
9	7	PAUSE on eight puzzles, then three guesstimates. Memorise only the anchor numbers table.
10	8, appendix	GD with SPEH, the follow-up email, CTC vs in-hand. Then the quick reference at the back.

TRAP

The crash plan fails in exactly one way: reading it silently. Every line says **out loud** or **recorded** for a reason — the room tests speech, not recognition.

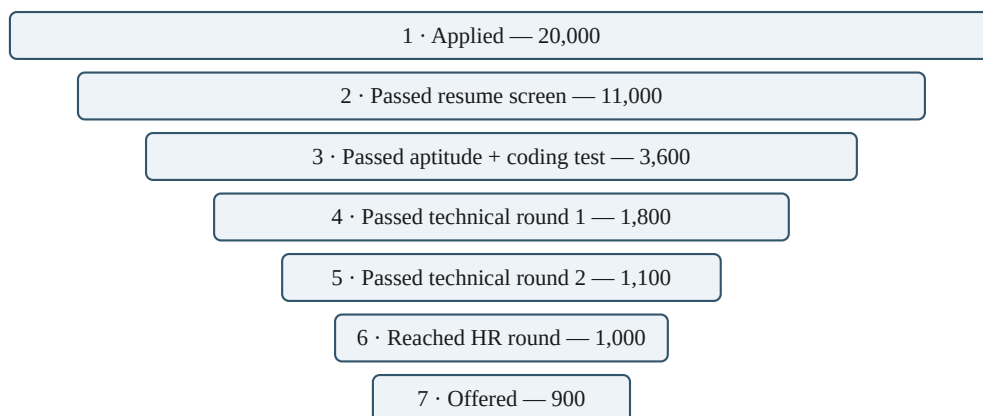
How Hiring Actually Works

Learn the machine before you argue with it

Pattern in one page

You are not being judged by one person who likes you or does not like you. You are moving through a **machine**. The machine has stages. Each stage has a different owner, a different score sheet, and a different reason to say no. Most students prepare for one stage — the coding test — and then lose at a stage they never studied. This chapter shows you every stage, who runs it, what it scores, and what a “no” at that stage actually means.

The funnel, drawn



Each bar is a stage. Width is how many people are left.

Look at the last two bars. They are almost the same width. **If you reach the HR round, you are very likely to get the offer.** That is why an HR round feels friendly. It is not a formality though — the 10% who fail it fail for reasons you can control completely.

THE SEVEN STAGES, AND WHO OWNS EACH

Stage 1 — Apply. You send a resume. Owner: a form or a portal. Cost of a no: zero information. You are usually not told.

Stage 2 — Screen. Software matches keywords; then a human recruiter looks for about seven seconds. Owner: recruiter. Scores: does this person look like the job post?

Stage 3 — Test. Aptitude, or coding, or both. Owner: software. Scores: a number. This is the biggest single cut in the whole funnel.

Stage 4 — Technical round. One or two engineers. Scores: can you write code, explain it, and be corrected without breaking?

Stage 5 — Project / deep-dive round. Scores: is the thing on your resume real, and do you understand the choices inside it?

Stage 6 — HR / behavioural / managerial. Scores: will you accept, will you stay, will you be a problem, can you work with people.

Stage 7 — Offer and paperwork. Salary, bond or service agreement, joining date, notice period, background check.

The one rule that explains everything: each stage exists to make the next stage cheaper. A test costs the company nothing. An engineer's hour costs a lot. So the cheap stages cut hard and the expensive stages cut gently.

The arithmetic, run

These proportions are a typical **shape** for a large campus drive. They are not any one company's real figures. Run the program and read the last column.

funnel.js — survival at each stage

JS

```
// How many applicants survive each stage of a mass campus drive.
// Numbers are a TYPICAL SHAPE, not any one company's real data.

const stages = [
  ["Applied", 1.00],
  ["Resume / ATS pass", 0.55],
  ["Aptitude + coding", 0.18],
  ["Technical round 1", 0.09],
  ["Technical round 2", 0.055],
  ["HR round", 0.050],
  ["Offer", 0.045],
];

const applied = 20000;
let prev = applied;
console.log("stage reaching % of applied survived this stage");
for (const [name, keep] of stages) {
  const here = Math.round(applied * keep);
  const pass = prev === here ? "-" : (100 * here / prev).toFixed(0) + "%";
  console.log(
    name.padEnd(20) + String(here).padStart(9) +
    (100 * keep).toFixed(1).padStart(13) + "%" + pass.padStart(20));
  prev = here;
}
```

Output

TEXT

stage	reaching	% of applied	survived this stage
Applied	20000	100.0%	-
Resume / ATS pass	11000	55.0%	55%
Aptitude + coding	3600	18.0%	33%
Technical round 1	1800	9.0%	50%
Technical round 2	1100	5.5%	61%
HR round	1000	5.0%	91%
Offer	900	4.5%	90%

Read the right-hand column as “if I get here, what is my chance of moving on?”

- Resume screen: 55%. Almost half of all effort dies here, before any human talks to you.
- Test: 33%. Two out of three people who sit the test are removed by it.
- HR round: 91%. The friendliest number in the table.

SHORTCUT

Where should a week of preparation go? Put it where the survival rate is worst **and** your control is highest. That is the resume screen (Chapter 2) and the test. But do not put **zero** time on the HR and behavioural rounds — a 9% failure rate at stage 6 means roughly 1 student in 11 who had already won the job talked their way out of it.

What each round is actually scoring

An interviewer does not write “nice guy” on a form. In most structured companies they fill a **scorecard** with fixed lines. Your job is to hand them the words for each line.

Round	Who runs it	What the form asks	What a “no” means
Resume screen	Recruiter + keyword software	Does this person match the must-have list? Is the resume readable in 7 seconds?	You did not use the words the job post used.
Aptitude test	Software	Score above the cut-off, section-wise.	Speed or accuracy. Nothing personal.
Coding test	Software	Test cases passed. Sometimes time and code quality.	You could not finish under pressure.
Technical 1	Engineer, 1–3 years ahead of you	Can they code? Do they explain while coding? Do they take a hint?	Silence, or refusing the hint.
Technical 2 / project	Senior engineer	Is the project real? Can they defend a design choice? Do they know the limits of what they built?	The project sounded copied.
Managerial	The hiring manager	Will this person be useful in 6 months? Will they need babysitting?	No evidence of finishing anything alone.
HR	HR / talent team	Will they accept? Will they stay? Any red flags in the story?	Gaps you would not explain, or a story that changed.

A scorecard, filled in

This is the shape of a structured behavioural scorecard. Learn it, because it tells you what to say. Ratings are usually on a fixed scale and the interviewer must write **evidence**, not opinion.

Line on the form	Rating	Evidence the interviewer wrote
Ownership	Strong	Found the slow query themselves, nobody asked them to.
Dealing with ambiguity	Mixed	Waited two days for the mentor before trying anything.
Communication	Strong	Explained the index choice in plain words, no jargon hiding.
Technical depth	Mixed	Knew what an index does, could not say what it costs on writes.
Learns from failure	Strong	Named their own bug and what they changed after it.
Recommendation	Hire	Depth is normal for a fresher; ownership is above bar.

Notice every “evidence” cell is a **fact with a subject and a verb**. “Good attitude” would be rejected by a reviewer. So an answer that gives no facts gives the interviewer nothing to write, and an empty form is a no.

The resume screen, in detail

Two filters sit here, in this order.

Filter 1 — keyword matching software (often called an ATS, applicant tracking system). It does not understand your resume. It matches strings.

Filter 2 — a human, for about seven seconds. They look at: current/last title, company or college, dates, and whether the page is readable.

Here is a small model of filter 1. It is not any company's real software. It shows the **shape** of the decision so you stop guessing.

ats.js — a keyword screen, modelled

JS

```
// A small model of how a keyword screen scores a resume against a job post.
// This is NOT the real software any company runs. It shows the SHAPE of the
// decision.

const normalise = (s) =>
  s.toLowerCase()
    .replace(/[^\a-z0-9+\. ]/g, " ") // keep + # . so c++, c#, node.js
    survive
    .split(/\s+/)
    .map((w) => w.replace(/^\.|\.+$/g, "")) // drop sentence dots, keep inner ones
    .filter(Boolean);

function score(resumeText, mustHave, niceToHave) {
  const words = new Set(normalise(resumeText));
  const hit = (k) => k.split(" ").every((w) => words.has(w));
  const must = mustHave.filter(hit);
  const nice = niceToHave.filter(hit);
  const missing = mustHave.filter((k) => !hit(k));
  const pct = Math.round((100 * (2 * must.length + nice.length)) /
    (2 * mustHave.length + niceToHave.length));
  return { pct, must: must.length, nice: nice.length, missing };
}

const MUST = ["javascript", "node", "rest api", "sql", "git"];
const NICE = ["docker", "react", "aws", "testing", "mongodb"];

const weak = `
Hardworking and dedicated fresher seeking a challenging position in a reputed
organization where I can utilize my skills and grow along with the company.
Skills: C, programming, web technologies, database, good communication.
Project: Final year project on a web based system.
`;

const strong = `
B.Tech CSE 2026. Skills: JavaScript, Node.js, Express, REST API, SQL, MongoDB,
Git, Docker, unit testing.
Project: Bus seat booking service - Node.js + Express REST API, MySQL schema,
JWT auth, 38 Jest tests, shipped in Docker.
`;

const fixed = strong.replace("Node.js, Express", "Node.js (Node), Express");

for (const [name, text] of [["WEAK", weak], ["STRONG", strong], ["FIXED", fixed]]) {
  const r = score(text, MUST, NICE);
  console.log(`${name.padEnd(7)} score=${String(r.pct).padStart(3)}% \ ` +
    `must ${r.must}/${MUST.length} nice ${r.nice}/${NICE.length} \ ` +
```

```
}  
    `missing: ${r.missing.join(", ") || "none"}`);
```

Output					TEXT
WEAK	score= 0%	must 0/5	nice 0/5	missing: javascript, node, rest api, sql, git	
STRONG	score= 73%	must 4/5	nice 3/5	missing: node	
FIXED	score= 87%	must 5/5	nice 3/5	missing: none	

Three lessons, and the third one is the one nobody tells you.

1. The weak resume scores **zero**. Every word in it is about feelings. It contains no noun that a job post would contain. “Hardworking” is not a skill. “Web technologies” is not a technology.
2. The strong resume scores 73% with the same person, same project, better nouns.
3. The strong resume still **missed one must-have**: the post said node, the resume said Node.js. A string matcher does not know they are the same thing. Writing Node.js (Node) once fixed it and moved the score to 87%.

TRAP

The abbreviation trap. A screen looks for the exact token. Write both forms, once, the first time you use it: Node.js (Node) · PostgreSQL (Postgres) · Amazon Web Services (AWS) · Continuous Integration (CI) · Object Oriented Programming (OOP)

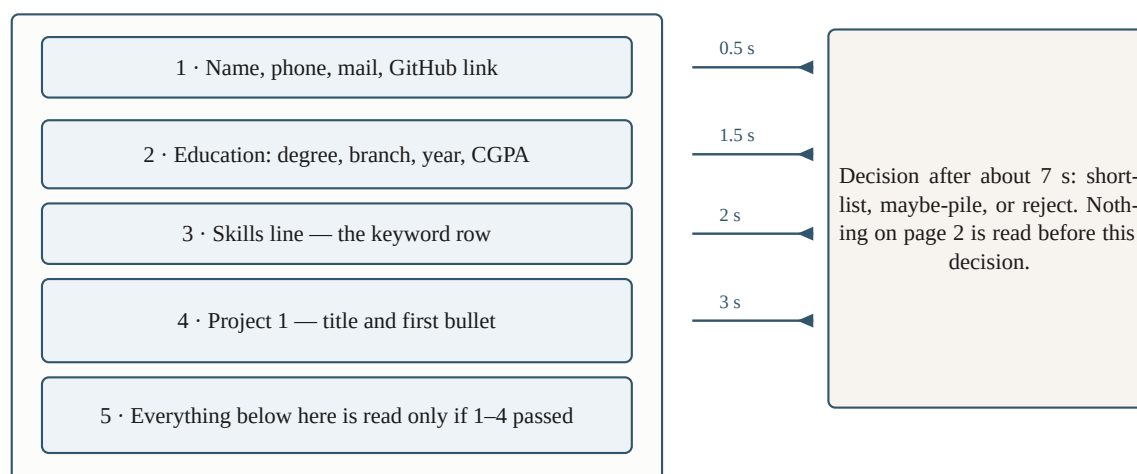
Do not spam the word twenty times. One honest pair of forms is enough, and a human reading the same line must not feel you are gaming it.

TRAP

Things that break the software filter, every time. Skills in a text box inside an image. Skills in the page header or footer. Two-column layouts where the parser reads across the columns and glues unrelated words together. Fancy fonts where the letters do not extract. A .png or a scanned .pdf instead of a text .pdf. Send a text-selectable PDF. Test it: open the PDF, press select-all, copy, paste into a plain text editor. If what you paste is scrambled or empty, the software sees the same scrambled or empty text.

The seven-second human scan

If the resume survives the software, a human opens it. Eye-tracking studies of recruiters have been done for decades and they all find the same thing: the first pass is a **Z** across the top third, then a stop wherever a number or a familiar word appears.



Where the recruiter's eyes actually go on page one.

So the top third of page one carries the whole decision. Three consequences:

1. **Your best project goes first**, not in date order. Nobody is checking your chronology.
2. **The skills line goes above the projects**, because the recruiter is ticking a checklist.
3. **Never put anything important on page two**. For a fresher there should be no page two.

Seconds	What they are checking	What kills you here
0–1	Name, contact, links	A dead GitHub link. A mail id like coolboy_143.
1–3	Degree, branch, year of passing	Year of passing missing, so they cannot tell if you are eligible.
3–5	Skills row against the must-have list	Skills written as “web technologies”, “programming”.
5–7	First project title and first bullet	A title like “Final Year Project”. Say what it does .

TRAP

The coolboy_143 problem is real. Your mail id appears next to your name at the top of every application and on every calendar invite. Make one that is `firstname.lastname` plus digits if needed. It costs five minutes and it removes a bad first impression you will never be told about.

TRAP

Dead links are worse than no links. If your resume says GitHub and the profile has three empty repositories from 2023, the recruiter now has evidence **against** you. Either fix the profile before you send the resume, or remove the link. Same for a portfolio site that does not load.

Three hiring machines, not one

The same word “interview” means three different machines. This book uses three tiers, and in **this** book the tiers mean the following.

Tier	What the round is	Typical companies	What wins
1	Standard HR screen. Tell me about yourself, strengths, why us, relocation, bond.	TCS, Accenture, Infosys, Wipro, Capgemini, Cognizant	Clarity, calm, a clear yes on mobility and joining.

2	Cross-cultural and regional. Work pass reality, communication style, why this region.	Grab, Sea/Shopee, GIC, DBS, Agoda, SCB, LINE MAN, Razer	Specific, practical answers about moving and working across cultures.
3	Structured behavioural. Deep STAR probing, follow-ups, written score-cards.	Google, Amazon, Microsoft, Goldman Sachs, D. E. Shaw, Adobe, Uber	One real story, told in detail, that survives five follow-up questions.

What changes between the tiers

	Tier 1	Tier 2	Tier 3
Length of HR round	10–20 min	25–40 min	45–60 min
How many stories needed	1–2	3–4	6–10
Follow-up depth	rarely	2 levels	4–6 levels
Notes taken	checklist	notes	written evidence + rating
Decided by	the interviewer	panel	panel + an independent reviewer
Biggest risk to you	sounding rehearsed	being vague about relocation	a story that falls apart under probing

SHORTCUT

The climb is real. A Tier-1 answer is a Tier-3 answer with the follow-ups removed. If you build **one** honest story properly — with the numbers, the names, the thing that went wrong — you can shorten it for Tier 1 and expand it for Tier 3. You do not need three sets of material. You need one set, told at three depths.

STAR — the four labels you will use all book

Every behavioural answer in this book is written in four labelled parts. Learn the labels now; Chapter 4 goes deep.

STAR

S — Situation. Where, when, who, and what was going on. Two sentences. No more.

T — Task. What **you personally** had to do. Not “we”. The word “I” must appear here.

A — Action. What you actually did, step by step. This is the longest part — about half the whole answer. Verbs, decisions, and why you chose each one.

R — Result. What changed. A number if one exists. Then what you learned or changed about how you work.

Rough shape: S 15%, T 10%, A 55%, R 20%.

TRAP

The number one STAR failure is the Action part collapsing into one sentence. “So I fixed it and it worked.” You just deleted the only part the interviewer was scoring.

Worked examples

Every example below shows a **weak** answer and a **strong** answer for the same student, and names exactly what changed. The strong answers here are **filled examples of a structure**.

READ THIS BEFORE EVERY EXAMPLE IN THIS BOOK

You must substitute your own real experience. Do not memorise these words. If you say a story that is not yours, the first follow-up question will end the interview, because the follow-up asks for a detail only the real person would know. Use the **structure**. Put your own project, your own numbers, your own mistake into it.

This book will never teach you to invent experience. Where you have a real weakness — a backlog, a gap year, a low CGPA, a rejection — you will be taught to state it plainly and then move to evidence of improvement. That works. Lying does not, and it is also how offers get withdrawn after a background check.

TIER 1

TCS NQT · Accenture · Infosys · Wipro · Capgemini

Example 1 [TCS NQT · pattern]

“Walk me through what you understand about our hiring process, and where you think you are strongest.”

SOLUTION

This looks like small talk. It is a test of whether you researched anything. Structure: **(a)** name the stages you have been through, **(b)** name the stage you are in, **(c)** name your strongest stage with one piece of evidence, **(d)** name the stage you are still growing in, honestly.

Weak version first.

TRAP

WEAK: “Sir, I have applied through the portal and given the test. I think I am strong in everything. I am a quick learner and hardworking, so any process is fine for me.”

SOLUTION

STRONG: “I have cleared the online test — the aptitude and the coding section — and this is my HR round. I think my strongest part is the project round: I built the booking service myself, so I can explain any line in it. The part I am still building is speed on data-structure questions under a timer. I have been doing two timed problems a day since January; I have gone from finishing about one problem in 45 minutes to two.”

What changed, exactly:

- “Strong in everything” → one named strength with one named piece of evidence.
- “Quick learner” → a measurable rate of learning: 1 problem per 45 min → 2.
- Added an honest weak area. This raises trust instead of lowering it.
- Named the actual stages, which proves you read something about the company.

Example 2 [Infosys · pattern]

“We hire for many locations. Are you okay with relocation, and with a shift that may be at night?”

SOLUTION

The company is not curious. It is protecting itself from an offer that gets rejected in month two. It wants a **specific, informed yes** — or an honest constraint stated early.

Structure: (1) direct answer in the first sentence, (2) evidence that you have thought about it practically, (3) any genuine constraint, stated once and without drama.

TRAP

WEAK: “Yes sir, anything is fine, I am very flexible, wherever you send me I will go.”

Why this is weak: it is the same sentence every candidate says, so it carries no information. Worse, recruiters have learned that maximum flexibility stated instantly often becomes a decline letter later. It reads as “I have not thought about this.”

SOLUTION

STRONG: “Yes to relocation. I have already looked at what it means: I would need about three weeks after the offer letter to arrange a place and close things at home, and I have family support for that. On night shifts — yes, I can do rotational nights. I did a six-week stretch of 10 p.m. to 2 a.m. work during my internship because the deploys were at night, so I know what it does to my sleep and how to manage it. The one thing I would ask for is the rotation calendar in advance so I can plan.”

What changed, exactly:

- Vague yes → yes plus a concrete lead time (three weeks).
- Claim of flexibility → evidence of having actually done nights before.
- Added one reasonable ask. Asking for a calendar is not a red flag. It reads as an adult.

If your honest answer is **no** — say a medical reason, or a family responsibility — say it in the first sentence, say what you **can** do, and stop. “I cannot do permanent night shifts because of a health condition I manage. Rotational days and evenings are fine, and I am fully open on location.” A clear constraint costs you some roles. A hidden constraint costs you the job in month three, with a bad exit on your record.

Example 3 [Wipro · pattern]

“Why did you apply to us? You have applied to six companies today.”

SOLUTION

Structure: (1) one specific true thing about the company, (2) the link to something you actually did, (3) what you want from the first two years. Never say “it is a reputed company”, because it is true of every company on the list and therefore says nothing.

TRAP

WEAK: “It is a very reputed company with a good work culture and a global presence. It will be a good platform for my career growth and I can learn a lot here.”

Why this is weak: swap in any company name and the sentence is unchanged. A sentence that survives a name-swap is a sentence that carries no information.

SOLUTION

STRONG: “Two reasons. First, the role is a Node and SQL backend role, and that is exactly the stack I built my booking project in — I am not switching stacks to take this job. Second, you train freshers in a structured programme with a defined stream allocation, and I want the first year to be structured rather than being dropped on a client alone. What I want from two years here is to move from writing endpoints to owning a service, including its on-call.”

What changed, exactly:

- “Reputed” → a reason that would be false for a company with a different stack.
- Added the link to their own work (the booking project).
- Added a concrete two-year goal, which answers the unasked question “will you leave in eight months?”

Example 4 [Capgemini · pattern]

“Our offer includes a service agreement. Are you comfortable with that?”

This is the first hard-money question in the funnel. Chapter 6 goes deep. Here is the honest version in short.

SOLUTION

What a service agreement (often called a bond) usually is. You agree to stay for a period — commonly 12 to 24 months. If you leave earlier, you owe a stated amount, often framed as recovery of training cost. Sometimes original certificates are asked for. Sometimes an amount is held back from salary and returned at the end.

What you must find out before you sign, in writing:

1. The exact lock-in period, in months, and the date it starts — offer date, joining date, or end of training?
2. The exact amount, and whether it reduces month by month or stays flat until the last day.
3. Whether original degree certificates are retained. Ask for a written receipt if they are.
4. What counts as an exit — does termination by the company, or a medical exit, also trigger it?

5. Whether the amount is payable by you or by a future employer, and whether the company will give a relieving letter on payment.

Structure for answering in the room: (1) a calm yes or a conditional yes, (2) two factual questions, (3) no negotiation attempt at the HR-round stage.

TRAP

WEAK (aggressive): “Bonds are not legally valid, everyone says so. I will sign but I can leave anytime.”

WEAK (scared): “Yes yes, no problem, whatever you say, I will sign anything.”

The first ends the conversation and sometimes the offer. The second means you sign a two-year lock-in without knowing the amount, and find out in month nine.

SOLUTION

STRONG: “Yes, I am comfortable with a service agreement in principle — I am looking for a first job I stay in, not one I leave in six months. Two things I would want in the offer letter so I can plan: the exact duration and from which date it counts, and the exact recovery amount. If the company holds original certificates, I would want a written acknowledgement. Once I see those, I do not expect any problem.”

What changed, exactly:

- Argument or panic → a yes with two factual, non-emotional questions.
- Moved the whole issue to **the written offer**, which is where it belongs.
- Said the thing HR wants to hear — “I intend to stay” — and it is true, so it is safe.

Real strategy, plainly: the time to negotiate a bond is **after** you hold the written offer and **before** you sign it, with the recruiter, over email. Not in the HR round. In the HR round your only job is to stay in the process and get the terms on paper.

Example 5 [Cognizant · pattern]

“That is all from my side. Do you have any questions for us?”

This is the last question in almost every round, and most candidates throw it away.

SOLUTION

It is scored. Not heavily, but it is the last thing written on the form, and “asked no questions” reads as “not really interested” or “did not prepare”.

Structure: ask two questions, in this order — one about **the work**, one about **how you will be judged**. Then thank them. Do not ask more than three; the interviewer has a next candidate.

TRAP

WEAK: “No ma’am, you have explained everything very well. I have no questions.”

ALSO WEAK: “What is the salary?” and “How many leaves do I get?” as the **first** question in a technical or HR round. These are fair questions — ask them of the recruiter, in the offer conversation, not of the engineer who just interviewed you.

SOLUTION

STRONG: “Two questions. First, for someone joining this team as a fresher, what does the first ninety days look like — is there a training track, and do I get a defined area to own after it? Second, six months in, what would make you say this hire went well? I would like to know what I am being measured on before I start, not after.”

What changed, exactly:

- Silence → two questions that show you are thinking about **doing the job**.
- The second question is the strong one. It asks for the success criteria, which is exactly the thinking a manager wants to see, and it gives you real information.
- Neither question can be answered from the company website, so it proves you are listening rather than reciting.

Three more you can keep in your pocket:

- “What is the biggest thing that slows this team down right now?”
- “How does work reach me — a ticket, a mentor, a standup?”
- “What is the one thing a fresher usually gets wrong here in the first month?”

Example 6 [Accenture · pattern]

“If another company gives you an offer with a higher package, will you leave us?”

SOLUTION

This is a **risk** question wearing a **loyalty** costume. The company is trying to predict a decline. Answering with a promise of undying loyalty is not believed and is not needed.

Structure: (1) do not pretend money does not matter, (2) name the thing that actually decides it for you, (3) give a concrete reason this role satisfies that thing, (4) stop.

TRAP

WEAK (the over-promise): “No sir, never. I will stay in your company for my whole life. Money is not at all important to me, only learning matters.”

Nobody believes this, and it is easy to disprove: if money truly did not matter, you would not have asked about the package. An answer the interviewer does not believe is worse than a careful one.

TRAP

WEAK (the honest-but-blunt): “Obviously I will take whichever offer pays more.”

True for many people. Still a decline risk on a form, stated by you, in your own words, for free.

SOLUTION

STRONG: “Money matters to me — I would not pretend otherwise. But at this stage the thing that decides it for me is what I will be doing in the first year, because that is what sets up the next five. This role is backend on Node and SQL, which is where I want to get deep, and there is a defined training track. A higher number for a role where I do support tickets for a year would not be a better deal for me. If I accept here, I am accepting to stay and learn, and I do not treat an acceptance as something to shop around afterwards.”

What changed, exactly:

- Denial of a real motive → acknowledging it in one clause and moving on.
- “I will never leave” → a **decision rule** they can believe: first-year work beats a short-term number.
- Added the specific thing about this role that satisfies the rule.
- Ended with the sentence HR actually needed: an acceptance here is not a bargaining chip. Only say this if it is true for you.

If you genuinely are holding another offer, do not hide it and do not brandish it. “I do have another offer in hand with a deadline of the 14th. I would rather join here, and I am telling you the date so we can be realistic about timelines.” That is a fact, given early, that helps both sides. Chapter 6 covers the full playbook for competing offers.

TIER 2

Singapore & Thailand · Grab · Shopee · GIC · DBS · Agoda · SCB

Example 7 [Grab · pattern]

“You are applying from India for a role in Singapore. Talk me through how you have thought about the move.”

SOLUTION

This is a **risk** question, not a **passion** question. A cross-border hire that falls through costs the company months. Your answer must reduce three risks: the pass risk, the money risk, and the will-they-actually-come risk.

Structure: (1) I know it needs an employment pass and that the company sponsors it, (2) here is my realistic timeline, (3) here is what I have already done, (4) here is what I would need from you.

TRAP

WEAK: “Singapore is a beautiful country and it is a hub for technology. I have always dreamed of working abroad and I am very excited about this opportunity. I am ready to join immediately.”

Why this is weak: “always dreamed” is about you, not about the risk they carry. “Join immediately” is not possible for a cross-border hire and shows you have not checked.

SOLUTION

STRONG: “I understand the role needs an employment pass and that the company files it, so my part is documents and timing. Realistically I am looking at about two weeks to get my degree and transcript attestations in order, then the filing time on your side, then two to four weeks for flights and housing. So eight to ten weeks from an offer, and I would rather give you a date I can keep than say immediately. I have kept my passport valid until 2032 and my transcripts are already digitised. What I would want from you is which month you need me to start, so I work backwards from that, and whether there is relocation support for the first month of housing.”

What changed, exactly:

- Emotion about the country → a timeline with named steps.
- “Immediately” → a defensible 8–10 week estimate with the reasoning shown.
- Added proof of preparation: passport validity, digitised transcripts.
- Ended by asking for their constraint, which turns it into a planning conversation.

Be honest about what you do not know. Pass rules, salary thresholds and quotas change, and they differ by country and by year. Never state a rule as fact in an interview. Say “my understanding is X — is that still current?” You get the right answer and you look careful instead of wrong.

Example 8 [Shopee · pattern]

“Our team is spread across four countries and three time zones. How do you work with people you never meet?”

SOLUTION

Structure: (1) name the specific problem of distance — it is not “communication”, it is **delay**, (2) one real example of how you handled delay, (3) the habit you built from it.

TRAP

WEAK: “I have very good communication skills and I am a good team player. I can adjust with anyone and I am always available on call.”

“Always available” across three time zones means “I have not thought about this”. It also promises something that burns people out in two months.

SOLUTION

STRONG: “The real problem is not language, it is that a question costs a whole day if the other person is asleep when I ask it. In my final-year project my two teammates were on a different schedule from me — they worked nights. We lost about three days in the first two weeks to back-and-forth. So we changed two things: every question had to be written with what I had already tried and what I would do if there was no reply by my evening, and we kept one shared document with decisions and dates instead of deciding in chat. After that we did not lose a full day to a blocked question again. The habit I kept is writing the fall-back into the question itself.”

What changed, exactly:

- “Good communicator” → a named failure mode (a blocked question costs a day).
- Added a real cost: three days lost.
- Added two specific mechanisms, not adjectives.
- Ended with a portable habit, which is what the interviewer will write on the form.

Example 9 [DBS · pattern]

“In your culture, is it acceptable to tell a senior person they are wrong? How would you do it here?”

SOLUTION

This is genuinely asked, and it is not a trap about your country. The interviewer wants to know whether a bad decision in a meeting will be **corrected** or **silently followed**.

Structure: (1) answer the direct question honestly, (2) separate **what** you would say from **how and where**, (3) one real example, (4) what you would do if overruled.

TRAP

WEAK: “No sir, in our culture we always respect seniors, so I would accept whatever they say and do it.”

This is a straight fail. Read the scorecard line again: will a bad decision be corrected?

SOLUTION

STRONG: “I would say it, and I would be careful about where. In college I worked on a project where the team lead wanted to store seat availability as a count only. I thought it would double-book under two simultaneous bookings. I did not argue in the group meeting — I messaged him with a four-line example showing two requests interleaving, and asked if I had misunderstood. He had not seen that case, and we moved to a row lock per seat. Two things I try to keep: disagree with an example rather than an opinion, and do it in the smallest room possible first. If I am overruled after that and it is not a safety or data-loss issue, I write down my concern once, then commit fully and help make it work.”

What changed, exactly:

- “I would accept whatever they say” → “I would say it, and here is how.”
- Added a concrete technical example, which proves the story is real.
- Added the **disagree-then-commit** ending, which is exactly what the form is scoring.
- Separated the question of **whether** to speak from **where** to speak.

Example 10 [Agoda · pattern]

“Our working language is English, and the team includes people whose first language is not English either. How comfortable are you?”

SOLUTION

This is not a grammar test. The interviewer is checking whether you will **say you did not understand** instead of nodding and getting it wrong.

Structure: (1) an honest statement of your level, (2) the habit you use when you do not understand, (3) the habit you use when **they** may not have understood you.

TRAP

WEAK: “My English is excellent, I have no problem at all, I scored very well in English in school.”

School marks are not the measure. Also, a candidate who claims zero difficulty is exactly the candidate who will nod through a misunderstood requirement.

SOLUTION

STRONG: “I am comfortable in English for technical work — I read documentation in it and I wrote my project report in it. Two things I am honest about. When I lose a sentence, usually because of speed or an accent I am new to, I say so and repeat back what I understood in my own words rather than nodding. In my internship my mentor was from Kerala and spoke fast, and for the first week I got two tasks slightly wrong before I started doing that. And when I explain something, I write the key decision in a message afterwards, so there is a text version that does not depend on anyone catching every word.”

What changed, exactly:

- A claim about fluency → a claim about **behaviour when communication fails**, which is the actual risk.
- Added a real cost (two tasks slightly wrong) and the correction.
- Added the written-summary habit, which is the single most valuable cross-border habit and the one the form will record.

Example 11 [Sea/Shopee · pattern]

“Most people who move for a job go home after two or three years. Will you?”

SOLUTION

An expensive question for them: relocation, a pass, and onboarding cost real money, and they lose it if you leave at month 30.

Structure: (1) do not promise forever, (2) give the **conditions** under which you would stay, (3) show you have thought about the practical side of staying, (4) be truthful about family obligations rather than hiding them.

TRAP

WEAK: “No no, I will never go back, I want to settle there permanently.”

Said by almost everyone, believed by almost nobody, and it also hides a real constraint that will surface later.

SOLUTION

STRONG: “I will not promise forever, because I do not think you would believe it and I do not know what year ten looks like. What I can tell you is what would make me stay, and it is the work: if I am still learning something I could not learn elsewhere, I stay. Practically, I have looked at what a longer stay involves — the pass renewal cycle, and that my parents would visit rather than move. My mother has a health condition that is stable, and my brother is in the same city as them, so I am not the person who has to be physically present. I am planning on a first commitment of at least three to four years, and I would rather say that number and keep it.”

What changed, exactly:

- An unbelievable “forever” → a stated, keepable number with the conditions attached.
- Added evidence of practical thinking (renewal cycle, who covers family responsibility).
- Disclosed a real family constraint calmly, with the mitigation, instead of hiding it.
- Gave them the decision rule — “if I am still learning, I stay” — which is something a manager can actually act on.

TIER 3

Google · Amazon · Microsoft · Goldman Sachs · D. E. Shaw · Adobe

Example 12 [Amazon · pattern]

“Tell me about a time you took ownership of something that was not your job.”

Then expect these follow-ups: “**What exactly did you do first?**” → “**Why that and not the obvious thing?**” → “**What did it cost?**” → “**What did your teammate think?**” → “**What would you do differently?**”

SOLUTION

This is the tier where your answer is written down as evidence. Structure is STAR, and then the follow-ups eat the **Action** part, so the Action part must be deep enough to survive five questions.

Weak first, so you can see the difference in size.

TRAP

WEAK: “In our final year project the website was very slow. So I took ownership and fixed the database. After my changes the website became fast and my team was very happy. I learnt that ownership is very important.”

Count what an interviewer can write on the form from this: nothing. No number, no decision, no cost, no alternative considered. It also dies on follow-up one — “what exactly did you do first?” has no answer inside it.

SOLUTION

STRONG, with the four parts labelled:

Situation. “In my final year we built a bus seat booking service for the college transport office — three of us, about ten weeks. Two weeks before the demo, the transport clerk told us search felt slow, but nobody owned performance; we had split the work by feature, not by concern.”

Task. “It was not assigned to me — I was on the payments screen. I decided to own it because I was the only one who had touched the database schema, and if it was still slow at the demo the whole project would read as broken.”

Action. “First I refused to guess. I wrote a small script that hit the search endpoint 200 times with the real route data and printed the median and the 95th percentile, because the clerk’s complaint was about the bad cases, not the average. The median was 0.4 seconds and the p95 was 1.9 seconds, so it really was the tail.

Then I turned on the slow query log and found a single query joining bookings to seats and filtering by route and date. It was scanning the whole bookings table — about 90,000 rows — because the only index was on the primary key.

I considered three fixes. Caching the results: fastest to write, but the data changes on every booking, so a stale cache would show a seat that was already sold — I ruled it out on correctness. Denormalising a seats-available count onto the route row: fast, but it introduces a write-conflict problem under simultaneous bookings, and we had no time to get locking right. Adding a composite index on route and travel date: cheap, reversible, no change to application code.

I took the index. I measured again with the same 200-request script: p95 went from 1.9 seconds to 0.31 seconds. I also measured the write side, because an index is not free — booking insert went from about 6 ms to about 8 ms, which was fine for our volume, and I said so in the report rather than hiding it.

Then I did the part that was actually uncomfortable: I told my teammate who wrote the search feature, before I told the group. I showed him the log line, not my opinion. He added the migration himself so it stayed his code.”

Result. “p95 dropped from 1.9 s to 0.31 s, about 6 times faster on the bad cases. The demo ran without anyone mentioning speed, and the transport office used it for two terms.

What I changed permanently about how I work: I now measure the 95th percentile, not the average, because the average hid the exact thing the user was complaining about. And when I find someone else’s problem I take it to them first with data, not to the group.”

What changed between weak and strong, exactly:

- “Very slow” → measured p95 of 1.9 s. A complaint became a number.
- “Fixed the database” → found the query, named the table size, named the missing index.
- Added **rejected alternatives with reasons**. This is the single biggest upgrade. It turns “I did a thing” into “I made a decision.”
- Added the **cost** of the fix (writes 6 ms → 8 ms). Admitting a trade-off raises your score; pretending there is none lowers it.
- Added the human action — telling the teammate privately, letting him own the commit.
- “I learnt ownership is important” → two specific, portable habits.

Surviving the follow-ups

The strong answer above was built so that every follow-up already has an answer inside it. Here is the probe, and the sentence in the answer that survives it.

Follow-up probe	What you say, drawn from the story you already told
What did you do first?	Measured it. 200 requests, median and p95, before touching anything.
Why p95 and not the average?	The complaint was about bad cases. The average hid them: 0.4 s looked fine.
Why not caching?	Data changes on every booking; a stale cache can sell a sold seat. Correctness beat speed.
What did the fix cost?	Writes went 6 ms → 8 ms, and one more index to maintain in migrations.
What did your teammate think?	I showed him the slow query log privately and he wrote the migration himself.

What would you do differently?	Measure in week two, not week eight. We had no performance owner at all.
What if the index had not worked?	Next step was the denormalised count with a row lock per seat; I had the design but not the time.

SHORTCUT

Build one story, then attack it yourself. Write your story, then ask yourself the seven probes above. Any probe you cannot answer is a hole. Fill it with the truth — go and check the real number if you still can. Do not fill it with an invention: at Tier 3 the interviewer will ask a follow-up to your follow-up, and inventions do not survive two levels.

Example 13 [Google · pattern]

“Tell me about a time you failed.”

SOLUTION

Structure: (1) a real failure with a real cost, (2) your own contribution to it, in the first person, (3) what you did **immediately after**, (4) the change that stuck, (5) evidence that the change stuck.

The scoring line is **“learns from failure”**, and the evidence they need is step 5. Most candidates give steps 1–4 and stop.

TRAP

WEAK (the fake failure): “My biggest failure is that I am a perfectionist, so I spend too much time on details. Once I worked so hard on a project that I did not sleep.”

This is the oldest non-answer there is, and it reads as “I will not tell you anything true”. The scorecard line gets a “no evidence”, which is scored the same as a bad answer.

TRAP

WEAK (the blame version): “We failed because my teammate did not do his part and the faculty guide kept changing the requirements.”

Everything might be true. It still fails, because the form asks what **you** learnt, and the answer contains no action by you.

SOLUTION

STRONG, four parts labelled:

Situation. “In my third year I ran the technical event for our department fest. 180 students registered online.”

Task. “I owned registration and the day-of check-in, alone.”

Action. “I built the registration form myself and stored entries in a spreadsheet. I never tested it above about 20 entries. On the morning of the event, check-in used the same spreadsheet over the college wifi, and with four volunteers editing at once it started overwriting rows. About 25 students’ entries were lost. I did not have a backup and I did not have a paper list.

What I did in the moment: I stopped using the sheet entirely, put two volunteers on paper registers split alphabetically, and re-entered everything that evening. The event started 35 minutes late.

What I did after: I found the 25 affected students from the payment records and mailed each of them the same day with what happened and their confirmed slot. Nobody dropped out.”

Result. “The failure was mine: I tested a system at 20 rows and ran it at 180, and I had no fallback for a tool I did not control.

What changed and stuck: for the next event, six months later, I ran a dry run at full size — I generated 200 fake entries and had four people hit check-in at once, which broke it again in testing, where breaking is free. We

moved to a simple database-backed form and kept a printed list as a paper fallback. 240 students checked in with no data loss and we started on time. I still do the two things I learned there: test at the size you will actually run at, and always have one fallback that does not need the network.“

What changed, exactly:

- Fake weakness → a real failure with a real, countable cost (25 records, 35 minutes).
- No blame. The words are “I never tested”, “I did not have a backup”.
- Added what was done **in the moment**, which shows behaviour under stress.
- Added the **repair** — mailing the 25 students. Interviewers score this heavily.
- Added step 5, the evidence it stuck: the second event, at 240 people, with the measured result. Without step 5 this is a story. With step 5 it is a pattern.

Example 14 [Microsoft · pattern]

“Give me an example of a decision you made with incomplete information.”

SOLUTION

Structure: (1) what was missing and why you could not just go and get it, (2) the deadline that forced a decision, (3) how you **reduced** the uncertainty cheaply, (4) the decision and the reversibility of it, (5) what you would have done if it went wrong.

Point 4 is the one that separates the top answers: **did you choose a decision you could undo?**

TRAP

WEAK: “I did not have enough information but I trusted my gut and took the decision, and luckily it worked out well.”

“Luckily” hands the interviewer a rating for you, and it is not a good one.

SOLUTION

STRONG, four parts labelled:

Situation. “During my internship I was asked to add a CSV export to an internal report page, three days before a monthly review.”

Task. “Nobody could tell me how big the export would get. The person who owned the data had left, and the report team was on leave until after the review.”

Action. “I could not get the real answer, so I bought information cheaply instead. I ran a count on the table for the last three months: about 40,000 rows a month, growing roughly 8% a month. That gave me a range rather than a fact: somewhere between 40,000 and maybe 70,000 rows within a year.

Then I chose based on the range, not on the middle. Building the CSV fully in memory would be 30 lines and would probably hold at 70,000 rows, but would fall over silently at some unknown point beyond it. Streaming the rows out was about 60 lines and had no size limit.

I took the streaming version, because the failure mode of the in-memory one was a crash in production on a day nobody was watching, and the extra cost was half a day of my time. I also added a hard cap with a clear error message at 500,000 rows, so that if my growth guess was badly wrong the system would say so instead of dying quietly.“

Result. “The export shipped in time for the review. It was still running a year later; the table had crossed 90,000 rows a month by then, which the in-memory version would not have survived.

What I took from it: when I cannot get the number, I get the **range** and design for the bad end of it, and I prefer the decision that fails loudly over the one that fails silently.“

What changed, exactly:

- “Trusted my gut” → bought cheap information (a row count and a growth rate).
- A single guess → a range, with the decision made against the bad end.
- Added the **failure mode** comparison. Crash-in-production versus half a day of work is a real engineering trade-off and it is what a senior interviewer is listening for.
- Added the safety valve (the 500,000-row error), which shows you planned for being wrong.

- “Luckily it worked out” → evidence a year later.

Example 15 [Goldman Sachs · pattern]

“You have been rejected by us before. Why should this time be different?”

This is asked, it is uncomfortable, and it has a good answer.

SOLUTION

Structure: (1) accept it in one sentence, no defensiveness, (2) name the actual reason, as specifically as you know it, (3) the work you did, with dates and volume, (4) evidence that is **external** to your own opinion.

Never say “the interviewer did not understand my answer”. Even when that is what happened.

TRAP

WEAK: “Last time I was unlucky, the questions were from a topic that was out of syllabus, and the interviewer was in a hurry. This time I am more confident.”

Confidence is not evidence. “Unlucky” tells the panel that a second rejection would also be luck, so there is no reason to expect a different result.

SOLUTION

STRONG: “Yes — I was rejected in the second technical round in August last year. The feedback I got was that I could not finish the coding question in time; I got a working brute-force solution and did not reach the optimal one before the clock ran out. That was accurate, and it was a speed problem, not a concept problem.

Since then I have done timed practice four days a week — 210 problems logged with the time taken for each. In August my median time on a medium problem was about 41 minutes. Over the last month it is 22. I also changed how I start: I now spend the first three minutes writing the brute force in comments and its complexity before writing any code, which is what stops me from rewriting halfway through.

The external evidence is that I cleared the online round this time with all test cases passing in 38 of the 60 minutes, which I did not do last year.”

What changed, exactly:

- “Unlucky” → the real reason, stated in the company’s own words from the feedback.
- Vague effort → a logged volume (210 problems) and a measured before/after (41 → 22 minutes).
- Added a **method** change, not just more hours. Anyone can say they practised more.
- Ended on evidence from outside the candidate’s own head: this year’s test result.

If you were given no feedback, say so and then say what you diagnosed yourself: “I was not given specific feedback. My own read was that I was weak on X, because that is where I froze. So here is what I did about X.”

Example 16 [Uber · pattern]

“Tell me about a time you disagreed with someone and you turned out to be wrong.”

Note the twist. Most candidates prepare the version where they were right.

SOLUTION

The scorecard line is usually something like “**seeks the truth over being right**”. The evidence they want is: did you change your mind when the data changed, and how fast?

Structure: (1) the disagreement, stated fairly from both sides, (2) what evidence you demanded, (3) the moment you were shown to be wrong, (4) what you did in the next hour, (5) whether you changed how you form opinions.

TRAP

WEAK: “I disagreed about which framework to use. Later I realised my teammate was right and I accepted it gracefully. I learnt that we should always listen to others.”

No evidence, no cost, no moment. “Always listen to others” is a proverb, not a behaviour.

SOLUTION

STRONG, four parts labelled:

Situation. “In a four-person hackathon team we had to store about 300 MB of sensor readings and query them by time range. I wanted a relational database. A teammate wanted flat CSV files read with a script.”

Task. “I was the loudest voice for the database, so if we went that way and it was wrong, the lost time was on me.”

Action. “I argued from principle: indexes, types, no parsing cost. He argued from the clock — we had 26 hours. Rather than keep arguing I asked for a test we could both accept: load one day of data both ways and time a range query. That took 40 minutes and it settled it.

The result was not what I expected. His version answered a one-day range query in about 0.9 seconds. Mine took 6 minutes to import before it answered anything, and we had to re-import every time the sensor format changed, which it did twice that night. For a 26-hour project where the data was written once and read a few times, his solution was correct and mine was theory.

I said so out loud, in front of the team, in one sentence: ‘The measurement says you are right, we are doing it your way.’ Then I spent the next hour writing the query script with him rather than sulking about it. I did add one thing from my side that he agreed with — a check that rejected malformed rows on load, because the format kept changing.”

Result. “We finished with four hours to spare and the parsing check caught 1,100 bad rows that would have been silent wrong answers.

What changed in how I argue: I now try to convert a disagreement into the cheapest experiment that would settle it, and I try to say what evidence would change my mind **before** I see it. If I cannot name that evidence, I am not arguing, I am just preferring.”

What changed, exactly:

- “I realised he was right” → a named experiment with a number that proved it.
- Added the cost of the experiment (40 minutes) — it shows judgement about how much to spend settling an argument.
- Added the **public** concession. Interviewers score changing your mind visibly.
- Added what you contributed **after** losing the argument, which shows you did not disengage.
- “Always listen” → a concrete, portable rule: name the evidence that would change your mind, in advance.

Example 17 [Adobe · pattern]

“Tell me about a time you had to say no, or push back on something you were asked to do.”

SOLUTION

The line being scored is usually about **judgement and backbone** — will this person take on an impossible amount of work and then quietly miss everything?

Structure: (1) what was asked and by whom, (2) why yes would have been the wrong answer, in numbers, (3) what you offered **instead** — a no is only strong when it comes with an alternative, (4) the outcome, (5) how the relationship survived.

TRAP

WEAK (the pushover): “I never say no. If my senior asks for something I always find a way to do it, even if I have to work all night.”

This reads as: cannot estimate, cannot prioritise, will silently fail and tell nobody.

TRAP

WEAK (the refuser): “I told him it was not my responsibility and that he should ask someone else.”

Correct boundary, zero judgement, no alternative offered. This scores badly too.

SOLUTION

STRONG, four parts labelled:

Situation. “Two days before our project review, our guide asked us to also add a mobile version of the dashboard.”

Task. “I was doing the dashboard, so the estimate and the answer were mine to give.”

Action. “I did not answer immediately in the meeting; I asked for an hour. I listed what a usable mobile version actually needed — the chart library we used did not resize, so it was a replacement, not a tweak: roughly 9 to 11 hours of work plus re-testing. Against that I had 14 hours left, and 6 of them were already committed to the parts of the review that were definitely being graded.

So I went back with a no and two alternatives instead of just a no. Option A: I make the existing dashboard readable on a phone — larger fonts, single column, no chart resizing — in about 2 hours, so it does not look broken if someone opens it on a phone in the review. Option B: I do the full mobile version and drop the export feature, which he had asked for the previous week. I said plainly that I would not attempt both and hit the deadline, and that if I tried, the risk was not that mobile would be rough, it was that **nothing** would be finished.

He took option A.“

Result. “The review went ahead with everything finished. He opened it on his phone during the review, and it was readable, which was the actual thing he cared about — he had not been asking for a full rebuild, he had been asking not to be embarrassed on a phone screen.

What I took from it: ask what the request is **for** before estimating it, and never deliver a bare no. A no plus two costed options is a decision handed to the person who should be making it, not a refusal.“

What changed, exactly:

- “I always say yes” or “not my job” → a costed estimate (9–11 hours against 14 available).
- Added the **named risk** of saying yes: not “it will be rough” but “nothing will be finished”.
- Added two alternatives with prices. This is the whole answer, really.
- Added the discovery that the real requirement was smaller than the stated one, which is a senior-level observation and very well scored.

After the last round: offer, money, paper

Chapter 6 covers these in full. Here is the honest map so you know what is coming, and so a recruiter’s phone call does not catch you unprepared.

The timeline

Step	Typical wait	What you should do in this window
Last round ends	–	Send a short thank-you mail the same day. Four lines.
Verbal offer / call	2 days – 3 weeks	Get the number, the location, the bond and the joining month on this call . Write them down.
Written offer letter	2 days – 2 weeks after the call	Read every clause. This is the only document that matters.
Your acceptance	they will push for same-day	It is normal to ask for 2–3 working days. Ask in writing.
Background check	1–4 weeks	Make sure every date and title on your resume matches your documents exactly .
Joining	1–6 months	Keep in touch monthly. Offers do get deferred; a silent candidate is easier to defer.

Salary: the real strategy for a fresher

WHAT IS ACTUALLY NEGOTIABLE, AND WHEN

Mass campus hiring (Tier 1). The number is fixed by the drive, printed in the notice, and the recruiter genuinely cannot change it. Negotiating here wastes goodwill. What **can** sometimes move: the location, the stream or technology allocation, and the joining date. Ask for those instead.

Off-campus and Tier 2 / Tier 3. There is usually a band. The band moves for exactly two reasons: (a) a competing written offer, (b) proven skills the role needs that other candidates do not have. It does not move for need, effort, or marks.

The single most useful habit: do not say a number first if you can avoid it, and never say a number you have not worked out.

When they ask “what is your expected salary?”

TRAP

WEAK (the giveaway): “Sir, as per company standards, anything you give is fine. I just want to learn.”
You have now capped yourself at the bottom of the band and told them you did not research.

TRAP

WEAK (the wild number): “35 lakhs per annum.”

For a fresher role advertised at a fraction of that, this ends the conversation and marks you as someone who has not checked reality. Note this is **not** about ambition. It is about having no data.

SOLUTION

STRONG (structure: deflect once, then give a researched range with a reason):

“For a fresher backend role in Bengaluru, from what I have seen from seniors who joined last year and from what is advertised, the range is roughly A to B. I would be looking in that range, and I am more interested in the total package — including the bond terms and whether there is a location allowance — than in one number. If you can share the band for this role I can tell you straight away whether it works.”

Why this works:

- Gives a range, not a point. A range is negotiable; a point is a target to push down.
- The range is **justified** by a source, so it does not sound invented.
- Moves the conversation to total package, which is where a fresher genuinely has room.
- Asks them for the band, which is the question you actually wanted answered.

Getting A and B: ask three seniors from your own college who joined the same kind of company last year. This is the highest-quality data available to you and it costs three messages. Adjust for city and for the year. Do not use figures from a website comment section.

Notice period, if you already have a job

THE THREE NUMBERS YOU NEED

1. **Your contractual notice period.** Read your own appointment letter. 30, 60 and 90 days are all common.
2. **Whether buyout is allowed,** and who pays. Some employers permit you to pay for the unserved days. Some refuse regardless of payment.
3. **Your last working day, calculated from the date you would resign** — not a vague “two months”.

What to say: “My notice period is 60 days from the date I resign. If I resign on the 1st, my last working day is the 30th of the following month, so the earliest realistic joining date is the 1st after that. Buyout of up to 30 days is allowed in my contract at my own cost, so if you need me earlier I can try for that, but I cannot promise it because it needs my manager’s approval.”

Never promise a joining date that requires your current employer to be generous. Promise the date you can achieve without anyone’s permission, and offer the earlier date as a possibility.

Reading a job post like an interviewer

Everything above is easier if you read the post properly. A job post has four parts and they are not equally important.

Part of the post	What it really is	Use it for
Title + team	The keyword the screen matches on	Your resume headline
Requirements / must-have	The checklist the recruiter ticks	Your skills line, in their words
Nice to have	Tie-breakers between shortlisted people	Your project bullets
Responsibilities	What you will be asked about in rounds 4–6	Choosing which stories to prepare
Culture / values section	The actual behavioural scorecard, published	Mapping one story per value

SHORTCUT

The values section of a company’s careers page is not marketing filler — at Tier 3 it is often literally the list of lines on the scorecard. Print it. Write one honest story of your own beside each line. If a line has no story, that is your gap, and you have weeks to go and create a real experience that fills it: fix a bug in an open-source project, run an event, take over a piece of a group project nobody wants. Creating the experience is legitimate. Inventing it is not.

A six-week plan that matches the funnel

Put effort where the survival rate is worst. This is one honest allocation.

Week	Main work	Time split
1	Rewrite the resume against three real job posts (Chapter 2). Fix the PDF text test.	70% resume, 30% coding
2	Aptitude drills, timed. Coding: two problems a day, timed.	80% test prep
3	Same, plus write your six core stories in STAR (Chapter 4).	60% test, 40% stories
4	Project deep-dive: be able to defend every choice you made (Chapter 5).	50% project, 50% test
5	Hard questions: salary, bond, gaps, backlogs, rejections (Chapter 6).	40% hard Qs, 60% mixed
6	Mock rounds out loud, with a timer and a recording. Fix the worst one.	100% mocks

Out loud is not optional. An answer that is perfect in your head is often 90 seconds too long, or full of “basically” and “actually”, when it leaves your mouth. Record one answer on your phone and listen to it once. It is unpleasant and it is the fastest improvement available to you.

Online rounds: the part nobody prepares

Most first rounds now happen over a video call or a proctored browser tab. Losing here is avoidable and painful.

THE 20-MINUTE CHECKLIST, DONE THE DAY BEFORE

1. **Test the actual platform**, not “my laptop works”. Join a test call on the same site. Some proctored tools refuse a browser you have never used.
2. **A second device with mobile data**, charged, with the meeting link already open. Home wifi failing mid-round is the single most common technical disaster.
3. **Power cable plugged in**. Not “the battery was at 60%”.
4. **Camera at eye level**. A laptop on a table points up your nose. Put it on some books.
5. **Light in front of you, not behind**. A window behind your head turns you into a silhouette.
6. **A plain wall or a tidy corner**. Not a virtual background — they glitch and eat your hands when you gesture.
7. **Headphones with a mic**. Laptop speakers cause echo, and echo makes an interviewer tired, and a tired interviewer writes shorter notes.

8. **Tell the house.** A door opening mid-answer costs you the thread.
9. **A pen and blank paper** in frame. Allowed almost everywhere, and thinking on paper is faster than thinking in a text box.
10. **Your resume printed or on a second screen**, because you will be asked “walk me through point three” and scrolling in the shared tab looks terrible.

TRAP

Reading from a script on a second monitor is visible. Your eyes track left-right in a pattern that does not match conversation, and your pauses fall in the wrong places. It is noticed more often than candidates think, and it is usually scored as dishonesty rather than nervousness. Keep **bullet keywords** on paper — five words per story, not sentences.

When something goes wrong mid-round

What happens	What to say and do, in order
Your internet drops	Rejoin from the phone immediately, then say: “Sorry, my connection dropped, I am back on mobile data. I was at the part where...” Do not apologise three times.
You did not hear the question	Ask once, plainly: “Could you repeat the last part?” Never answer a question you did not hear.
You do not know the answer	Say so, then say what you would do: “I have not used that. My guess would be X because of Y — is that the right direction?”
You gave a wrong answer 5 minutes ago	Go back to it yourself: “Can I correct something I said earlier?” This raises your score. Interviewers notice self-correction.
You go blank on your own project	Say: “Give me ten seconds, I want to get the number right.” Silence for ten seconds is fine. Waffling for ninety is not.
Someone walks in	Mute, deal with it in five seconds, unmute, one short apology, continue.

What a rejection actually means

A rejection is a **stage number**, not a verdict on you. Knowing which stage you died at tells you exactly what to fix. Ask the recruiter — politely, once, by mail — and many will tell you.

Rejected at	The most likely real reason	What to fix, specifically
No reply at all	The screen. Your resume did not match the post’s words.	Chapter 2. Rewrite against three real posts.
Aptitude test	A sectional cut-off, usually the one you find boring.	Find which section. Drill that section only.
Coding test	Time, not concept. You had a working brute force at the buzzer.	Timed practice. Log the minutes per problem.
Technical round 1	You went silent, or you refused a hint, or you could not explain your own code.	Practise narrating out loud while you code.
Project round	The project sounded borrowed. You could not say why you made a choice.	Chapter 5. Know one project to the bottom.
Managerial / HR	No evidence of finishing something alone; or a gap or constraint that surfaced late.	Chapters 4 and 6. Prepare six real stories and the hard answers.
After the offer	Background check mismatch, or you went silent for two months.	Match every date to your documents. Reply to every mail.

Asking for feedback without sounding bitter

TRAP

WEAK: “Sir, I gave my best and still I was rejected. Please tell me what was the reason, because I feel the evaluation was not fair. I request you to reconsider.”

Two problems. It asks them to defend a decision, so the safe reply is silence. And “reconsider” turns a feedback request into an appeal, which recruiters are trained to close.

SOLUTION

STRONG (a four-line mail):

“Thank you for letting me know, and for the time your team spent.

If it is possible, I would find it genuinely useful to know which round was weakest, even in one line — I am preparing for the next cycle and I would rather work on the right thing.

Either way, I would like to apply again when there is a suitable opening.

Thanks again.”

What changed, exactly:

- No appeal. Nothing to defend, so replying is cheap for them.
- Asks for **one line about a round**, which is a question they can actually answer, instead of “why did you reject me”, which they cannot.
- States the intent to reapply, which is the sentence that gets you into the “keep warm” list.

Reapplying is normal. Many companies have a waiting period — commonly six months — before you may apply again. Ask what it is. Then reapply **with something new on the resume**, and say in one line what changed: “Since my last application I have shipped X and my timed-round result improved from A to B.” Reapplying with an identical resume gets an identical result.

Eight things students believe that are not true

The belief	What is actually true
A referral guarantees an interview.	A referral gets your resume looked at by a human . It skips filter one, not filter two. A weak resume with a referral is still a weak resume.
More pages means more experience.	For a fresher, page two is usually not read at all. One page, dense, is stronger.
You must never say “I do not know”.	Saying it, then giving your reasoning and a way to find out, scores better than a confident wrong answer, which scores worse than silence.
HR rounds are just a formality.	About 1 in 11 candidates who reach it are removed there, and it is the stage where you have the most control.
You should always agree with the interviewer.	At Tier 2 and 3 you are explicitly scored on whether you will correct a wrong decision. Agreeing with everything is a fail on that line.
A low CGPA cannot be recovered.	Many cut-offs are firm and some are not. Where a cut-off is firm, the route is off-campus roles that screen on skills. Where it is not, evidence of an upward trend plus shipped work is a real answer. See Chapter 6.
Interviewers want to trick you.	Interviewers want to stop interviewing and go back to their work. Make it easy to say yes: give facts they can write down.
Talking more shows confidence.	A 4-minute answer to a 40-second question is the most common cause of running out of time before your best story.

Top traps in this chapter

TRAP

1. Preparing only for the coding test, then losing at the resume screen you never looked at.
2. Writing a resume in words you like instead of the words the job post used.
3. Sending a scanned or image-based PDF that the parser reads as empty.
4. Saying “I am flexible about everything” — it is the one answer that carries no information.
5. Giving a fake weakness (“I am a perfectionist”). It scores the same as no answer.
6. Telling a story with no number in it, and no cost, and no rejected alternative.
7. Blaming a teammate, a guide, or an interviewer for a past rejection.
8. Arguing about the legality of a bond in the interview room instead of reading the clause afterwards.
9. Naming a salary figure you have not worked out, in either direction.
10. Promising a joining date that depends on your current manager being kind.
11. Memorising an answer from a book — including this one — instead of substituting your own real experience.

Practice

PRACTICE · TIER 1 target: 25 min, spoken out loud

1. Write the seven stages of the funnel from memory. Beside each one, write the single question that stage is really asking.
2. Take one job post you have actually seen. List its must-have keywords. Now check your current resume against that list, word by word, and count the misses.
3. Answer out loud, in under 60 seconds: “Are you comfortable with relocation?” Record it. Listen. Did you give a lead time and evidence, or an adjective?
4. Answer out loud: “Why our company?” Now swap in a competitor’s name. If the answer still makes sense, rewrite it.
5. Write down, honestly, what you would ask about a service agreement before signing. Aim for five questions.

ANSWER KEY

1. Apply, screen, test, technical 1, technical 2 / project, HR / managerial, offer. The questions: **does the form accept me · do I match the post · can I score above the cut-off · can I code and be corrected · is my project real · will I accept and stay · will I sign.**
2. The count of misses is the only number that matters here. Every miss is a coin flip you lose at the screen.
3. A good answer has three parts: yes/no in sentence one, a lead time in sentence two, evidence in sentence three.
4. The name-swap test is the fastest quality check there is for a “why us” answer.
5. Duration and start date · exact amount and whether it reduces · certificates retained (get a receipt) · what events trigger it · relieving letter on payment.

PRACTICE · TIER 2 target: 30 min

1. You are asked: “Why do you want to work in Singapore rather than in India?” Write a four-sentence answer containing (a) one thing about the **role**, (b) one thing about the **market or the product**, (c) your realistic timeline, (d) one question back to them. Nothing about weather or beauty.
2. Write a three-sentence answer to “how do you handle a teammate in another time zone?” that contains one number and one mechanism. Not one adjective.
3. A manager makes a decision you believe is wrong, in a meeting of eight people. Write exactly what you would do, in order, in four steps. Include what you do if you are overruled.

4. Estimate your own realistic weeks-from-offer-to-joining for a cross-border role, and list the three steps that make up most of it.

ANSWER KEY

1. Good structure: role fit (the stack or the domain matches what I have built) → market reason (scale or type of product I cannot get here) → “eight to ten weeks from offer because of attestations and housing” → “which month do you need me to start?”
2. Example shape: “We lost three days in two weeks to blocked questions. We fixed it by writing the fall-back plan into the question itself and keeping decisions in one dated document.”
3. (1) Do not argue in the meeting of eight. (2) Check I have understood, in writing, with a concrete example. (3) Raise it in the smallest room possible. (4) If overruled and it is not a data-loss or safety issue, record the concern once and commit fully.
4. Attestation and documents, the filing window, and flights plus housing. Give the sum as a range and offer to work backwards from their date.

PRACTICE · TIER 3 target: 45 min, written then spoken

1. Write one full STAR story from your own life, with the four parts labelled, where the **Action** part is at least half the words. It must contain: one measured number, one alternative you rejected and why, and one cost of your chosen fix.
2. Now attack it. Answer all seven follow-up probes from the table in this chapter. Mark every probe you cannot answer.
3. For each hole you marked, go and find the real answer if you still can. Write down what you will say for the ones where the information is genuinely gone. (“I did not measure the write cost at the time. If I did it again I would, and here is roughly what I would expect and why.”)
4. Write a failure story that includes step 5 — the evidence the change stuck. If you have no step 5 yet, that is a signal: the change has not been tested, and you have time to test it.
5. Take the values or culture list from one real company’s careers page. Write one honest story of your own beside each line. Circle the lines with no story.

ANSWER KEY

1. Self-check: if you deleted your Situation and Task, would the Action still show a decision? If not, you described a task, not ownership.
2. A story with three or more unanswerable probes is not ready. It is not a bad story — it is an unfinished one.
3. “I do not know, and here is how I would find out” is an acceptable Tier-3 answer. “I think it was around...” when you are guessing is not — say you are estimating.
4. No step 5 is the most common reason a good failure story scores as average.
5. A circled line is your next six weeks. Go and do the thing, then you will have the story honestly.

ONE-PAGE REVISION CARD

The funnel. Apply → screen → test → technical 1 → technical 2 / project → HR / managerial → offer. Cheap stages cut hard, expensive stages cut gently.

Survival, typical shape. Screen 55% · test 33% · tech 1 50% · tech 2 61% · HR 91% · offer 90%. The test is the biggest cut. The HR round is the friendliest — and still removes about 1 in 11.

What each round scores. Screen: do you match the post. Test: a number. Technical: can you code **and** be corrected. Project: is it real. HR: will you accept and stay.

The screen is two filters. String-matching software, then a human for 7 seconds. Write both forms of every abbreviation once: Node . js (Node) . Send a text-selectable PDF and test it with select-all, copy, paste.

STAR. Situation 15% · Task 10% · **Action 55%** · Result 20%. The Action part is what is scored. Label the four parts in your own head every time.

Three upgrades that turn any weak story strong.

1. Put a measured number where an adjective was.
2. Name an alternative you rejected, and why.
3. Name the cost of the fix you chose.

Tier depth. Tier 1 = the story, short. Tier 2 = the story plus practical detail about moving and working across distance. Tier 3 = the story plus five levels of follow-up. One set of honest material, told at three depths.

The three failure answers that score zero. The fake weakness. The blamed teammate. The story with no number.

Money and paper, in one line each.

- Campus salary is fixed; negotiate location, stream, joining date instead.
- Off-campus: give a **researched range** with a source, never a single number, never “anything is fine”.
- Bond: get duration, start date, amount, certificate handling and exit triggers **in writing**, and negotiate after the written offer, not in the room.
- Notice period: promise only the date you can hit without anyone’s permission.

The rule for every answer in this book. Use the structure. Substitute your own real experience. Never invent one — at Tier 3 the follow-up to the follow-up will find it, and a background check finds the rest.

Resume & Profile

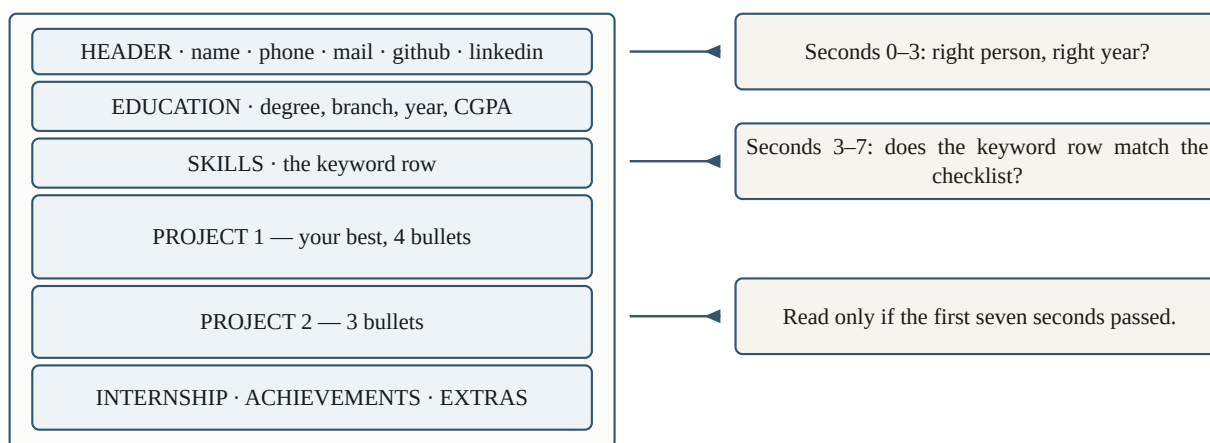
One page whose only job is to earn the next ten minutes

Pattern in one page

Your resume has exactly one job: **get you to the next stage**. It is not a biography. It is not a certificate collection. It is not an honest full account of your life. It is an argument, one page long, that says “talking to me for thirty minutes is worth your time.”

In Chapter 1 you saw the two filters it must pass: string-matching software, then a human for about seven seconds. This chapter builds a page that passes both, bullet by bullet.

The page, drawn



One page. The decision happens in the top third.

THE SEVEN BLOCKS, IN THIS ORDER

1. **Header** — name, one phone, one mail, GitHub, LinkedIn. Three lines, no more.
2. **Education** — degree, branch, institute, year of passing, CGPA. One line each.
3. **Skills** — the keyword row, grouped, written in the job post’s words.
4. **Projects** — your two or three best, **best first**, each with 3–4 bullets.
5. **Experience / Internships** — same bullet rules as projects.
6. **Achievements** — only things with a number or a name attached.
7. **Extras** — positions of responsibility, open source, certifications that actually mean something. Cut this block first when you run out of room.

The order is not negotiable for a fresher. Skills above projects, because the recruiter is ticking a checklist. Projects above experience if your project is better than your internship — and for most students it is.

What is NOT on the page: an objective paragraph, a photo, date of birth, marital status, father’s name, full address, a declaration line with your signature, “References available on request”, or a skills bar chart showing “JavaScript 80%”.

Will it fit? Count lines, not feelings

Students write four pages and then shrink the font to 8pt. That fails both filters: a human will not read it, and it looks desperate. Count instead.

```
// Will your resume fit on ONE page? Count lines, not feelings.
// A4, 10.5pt, normal margins -> about 46 usable lines. 95 characters fit on a line.

const BUDGET = 46, CPL = 95;

const blocks = [
  ["Header (name, phone, mail, links)", 3, 0],
  ["Education", 4, 0],
  ["Skills", 3, 0],
  ["Project 1 title + bullets", 1, 4],
  ["Project 2 title + bullets", 1, 3],
  ["Internship title + bullets", 1, 3],
  ["Achievements", 3, 0],
  ["Gaps between sections", 7, 0],
];

function fit(avgBulletChars) {
  const wrap = Math.ceil(avgBulletChars / CPL);
  let total = 0;
  for (const [, fixed, bullets] of blocks) total += fixed + bullets * wrap;
  const verdict = total <= BUDGET
    ? "fits on one page"
    : `OVER by ${total - BUDGET} lines -> drop ${Math.ceil((total - BUDGET) / wrap)}
bullets, ` +
    `or trim every bullet under ${CPL} characters`;
  console.log(`avg bullet ${String(avgBulletChars).padStart(3)} chars ` +
    `-> ${wrap} line(s) each -> ${String(total).padStart(2)} lines total :
${verdict}`);
}

[80, 130, 170, 220].forEach(fit);
```

Output

TEXT

```
avg bullet 80 chars -> 1 line(s) each -> 33 lines total : fits on one page
avg bullet 130 chars -> 2 line(s) each -> 43 lines total : fits on one page
avg bullet 170 chars -> 2 line(s) each -> 43 lines total : fits on one page
avg bullet 220 chars -> 3 line(s) each -> 53 lines total : OVER by 7 lines -> drop 3
bullets, or trim every bullet under 95 characters
```

Read the 130 and 170 rows together. Both cost two lines. So a 130-character bullet is **wasting nothing** by growing to 170 — the line is already spent. But crossing to 220 costs a third line on every bullet and pushes you off the page.

SHORTCUT

The line-wrap rule. Bullets are cheapest at just under a whole number of lines. Write the bullet, then look at the last line: if it holds only two or three words, delete three words elsewhere and pull it up. You get a tighter bullet and a free line.

The bullet formula

Every bullet on your resume is built from the same four parts. Miss one and the bullet goes weak.

VERB + WHAT + HOW + NUMBER

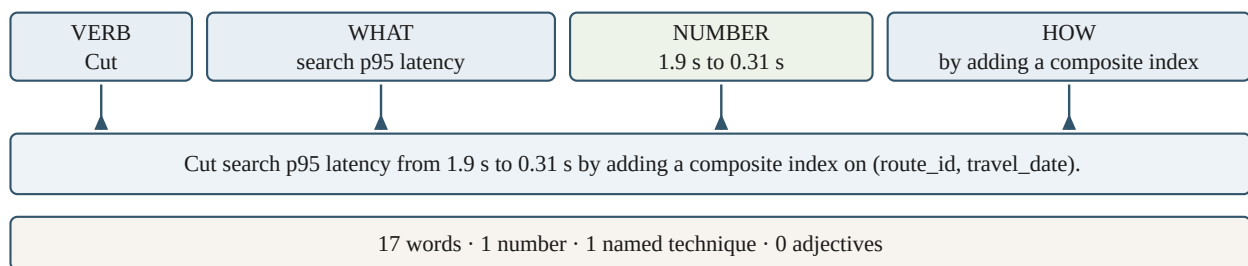
VERB — a past-tense action verb, first word, no “I”. Built, shipped, cut, raised, wrote, migrated, automated, indexed, measured, reduced, designed, tested, deployed, led.

WHAT — the thing you produced, named specifically. Not “a module”. “A seat-booking REST API.” “A CSV export.”

HOW — the technology or the method. This is where the keywords live. “with Node.js and Express”, “by adding a composite index”, “as a stream”.

NUMBER — the size, the speed, the count, the time saved, the users. Anything countable.

A bullet with no number is a claim. A bullet with a number is evidence.



The same sentence, taken apart.

Check your bullets with a script

Do not trust your own eyes on your own writing. Run this on your real bullets.

bullet.js — four rules per bullet

JS

```
// Checks one resume bullet against four rules. Run it on YOUR OWN bullets.

const VERBS = ["built","shipped","cut","raised","wrote","added","fixed","migrated",
  "automated","designed","measured","reduced","led","tested","deployed","indexed"];
const WEAK = ["responsible for","worked on","involved in","helped in","was part of",
  "good knowledge of","familiar with","hardworking","team player"];

function check(b) {
  const low = b.toLowerCase();
  const first = low.split(/\s+/)[0];
  const problems = [];
  if (!VERBS.includes(first)) problems.push("no strong verb in position 1");
  if (!/\d/.test(b)) problems.push("no number anywhere");
  for (const w of WEAK) if (low.includes(w)) problems.push(`filler phrase: "${w}"`);
  const words = b.trim().split(/\s+/).length;
  if (words > 22) problems.push(`too long (${words} words, aim <= 22)`);
  return problems;
}

const bullets = [
  "Responsible for the backend module of the college project and worked on database.",
  "Built a Node.js REST API for bus seat booking; cut average search time from 1.9 s to 0.3 s by adding a composite MySQL index.",
  "Cut API p95 latency 1.9 s -> 0.3 s by adding a composite index on (route_id,
```

```

travel_date).",
];

for (const b of bullets) {
  const p = check(b);
  console.log("-".repeat(60));
  console.log(b);
  console.log(p.length ? "  FAIL: " + p.join(" | ") : "  PASS");
}

```

Output

TEXT

```

-----
Responsible for the backend module of the college project and worked on database.
  FAIL: no strong verb in position 1 | no number anywhere | filler phrase:
"responsible for" | filler phrase: "worked on"
-----
Built a Node.js REST API for bus seat booking; cut average search time from 1.9 s to
0.3 s by adding a composite MySQL index.
  FAIL: too long (25 words, aim <= 22)
-----
Cut API p95 latency 1.9 s -> 0.3 s by adding a composite index on (route_id,
travel_date).
  PASS

```

Notice bullet two. It is a **good** bullet that fails on length only. The fix is not to delete information — it is to delete the words that carry none: “Built a”, “cut average search time from”, “by adding a composite MySQL index” all shrink without losing a single fact.

The same check in Python, if that is what you have on your machine:

bullet.py — the same four rules

PY

```

import re
VERBS = {"built", "shipped", "cut", "raised", "wrote", "added", "fixed", "migrated",
"automated", "designed", "measured", "reduced", "led", "tested", "deployed", "indexed"}
WEAK = ["responsible for", "worked on", "involved in", "helped in", "was part of",
"good knowledge of", "familiar with", "hardworking", "team player"]

def check(b):
    low, out = b.lower(), []
    if low.split()[0] not in VERBS: out.append("no strong verb in position 1")
    if not re.search(r"\d", b): out.append("no number anywhere")
    out += [f'filler phrase: "{w}"' for w in WEAK if w in low]
    n = len(b.split())
    if n > 22: out.append(f"too long ({n} words, aim <= 22)")
    return out

for b in ["Responsible for the backend module of the college project and worked on
database.",
"Cut API p95 latency 1.9 s -> 0.3 s by adding a composite index on
(route_id, travel_date)."]:
    print(b); print(" ", check(b) or "PASS")

```

Audit the whole page at once

```
// Audit a whole resume at once. Paste YOUR bullets into `mine` and run it.

const FILLER = ["responsible for","worked on","involved in","helped in","was part of",
  "good knowledge","familiar with","hardworking","team player","various","etc",
  "successfully","as per requirement","using latest technologies"];

function audit(bullets) {
  const withNum = bullets.filter((b) => /\d/.test(b)).length;
  const fillerHits = bullets.flatMap((b) =>
    FILLER.filter((f) => b.toLowerCase().includes(f)).map((f) => [b, f]));
  const lens = bullets.map((b) => b.split(/\s+/).length);
  const avg = (lens.reduce((a, c) => a + c, 0) / lens.length).toFixed(1);
  const longest = Math.max(...lens);
  const firstWords = new Set(bullets.map((b) => b.split(/\s+/)[0].toLowerCase()));

  console.log(`bullets           : ${bullets.length}`);
  console.log(`with a number       : ${withNum}/${bullets.length}` +
    ` (${Math.round(100 * withNum / bullets.length)}%   target >= 70%`);
  console.log(`filler phrases found  : ${fillerHits.length} target 0`);
  fillerHits.forEach(([b, f]) => console.log(` "${f}" in: ${b.slice(0, 48)}...`));
  console.log(`avg words per bullet   : ${avg} target 14-22`);
  console.log(`longest bullet        : ${longest} words`);
  console.log(`distinct opening verbs : ${firstWords.size}/${bullets.length}` +
    ` target: all different`);
}

const mine = [
  "Responsible for the frontend of the college event portal using latest technologies.",
  "Worked on the database design and various queries for the admin panel.",
  "Built a Node.js REST API for seat booking; 12 endpoints, 38 Jest tests.",
  "Cut search p95 latency 1.9 s to 0.31 s by adding a composite index on route and date.",
  "Built the CSV export as a stream so it holds at 500,000 rows instead of 70,000.",
];

audit(mine);
```

Output

TEXT

```
bullets           : 5
with a number       : 3/5 (60%)   target >= 70%
filler phrases found  : 4 target 0
  "responsible for" in: Responsible for the frontend of the college even...
  "using latest technologies" in: Responsible for the frontend of the college even...
  "worked on" in: Worked on the database design and various querie...
  "various" in: Worked on the database design and various querie...
avg words per bullet   : 14.2 target 14-22
longest bullet        : 18 words
distinct opening verbs : 4/5 target: all different
```

The last line is a check students never do. Two bullets in that list both start with “Built”. Four bullets that all start with “Developed” read as one grey block and the recruiter’s eye skips them. Vary the verb.

The verb bank

Pick the verb that matches what you actually did. Using a bigger verb than the work deserves is the most common way a resume starts to sound false.

What you did	Verbs that fit
Made a thing that did not exist	Built · Wrote · Designed · Created · Shipped · Implemented
Made an existing thing faster or smaller	Cut · Reduced · Halved · Sped up · Trimmed · Optimised (only with a number)
Made an existing thing bigger or better	Extended · Raised · Increased · Improved (only with a number)
Removed a defect	Fixed · Debugged · Traced · Eliminated · Patched · Corrected
Moved something from A to B	Migrated · Ported · Converted · Replaced
Removed human effort	Automated · Scripted · Scheduled · Batched
Made something measurable	Measured · Instrumented · Load-tested · Profiled · Benchmarked
Made something safer	Validated · Locked · Secured · Rate-limited · Sandboxed
Worked with people	Led · Coordinated · Reviewed · Mentored · Onboarded · Ran
Wrote things down	Documented · Specified · Wrote up · Diagrammed

TRAP

Verbs that quietly erase you. “Responsible for”, “Worked on”, “Involved in”, “Helped in”, “Assisted with”, “Was part of”, “Participated in”, “Gained exposure to”, “Learnt about”, “Utilised”, “Handled”, “Managed” (when you managed nobody).

Every one of these describes a **position** rather than an **output**. Read a bullet that starts with one of them and ask: what came out at the end? If nothing did, delete the bullet.

TRAP

Verbs that are too big for fresher work. “Architected”, “Spearheaded”, “Revolutionised”, “Pioneered”, “Owned the end-to-end vision”, “Engineered a paradigm shift”. These do not read as confidence, they read as a candidate who has not seen real architecture work yet. “Built” is a completely respectable word. Use it.

Where your numbers come from

“But my project has no numbers.” It does. You have not counted them yet. Every one of these is countable **today**, from your own repository or your own memory, with no invention.

Kind of number	How you get it, honestly
Size of the thing you built	Count endpoints, screens, tables, components. <code>git ls-files</code> and a line count.
Size of the data	<code>SELECT COUNT(*)</code> on your own table. “Tested on 90,000 rows.”
Users	How many people actually used it. 3 is a number. “Used by the 4 members of the transport office” is honest and specific.
Speed	Time it. Two lines of code around the call. Before and after.
Tests	Count them. “38 Jest tests, 81% line coverage” — the tool prints coverage for you.
Time saved	What the manual process took, times how often. “Replaced a 20-minute weekly export.”
Errors removed	Count the rows the validation rejected. “Caught 1,100 malformed rows.”

Team and your share	“3-person team; I owned the API and the schema.” Honest scoping.
Duration	“10 weeks.” It tells the reader the scale of the work.
Scale you tested to	“Load-tested at 200 concurrent requests.” Even if real traffic was 3.

TRAP

Do not invent the number. Two reasons, and the second is the one that matters. First, at the project round you will be asked how you measured it, and a made-up number has no measuring method behind it. Second, a made-up number is usually **too round and too big** — “improved performance by 90%” — and experienced interviewers read round-and-big as invented. A real number is odd-looking: 6.1 times, 0.31 seconds, 1,100 rows.

SHORTCUT

If you genuinely cannot measure it now, go and measure it this week. The project is still in your repository. Clone it, run it, time it, count the rows. An afternoon of measuring turns six weak bullets into six strong ones, and it also prepares you for Chapter 5, where you have to defend every one of them.

Worked examples

READ THIS BEFORE EVERY EXAMPLE

You must substitute your own real experience. These are filled examples of a **structure**. If you copy the words, you will be asked “what index, on which columns, and why that order?” in the project round, and the answer will not be there.

Nothing in this chapter asks you to lie. Where you have a backlog, a gap, a low CGPA or a weak year, you will be shown how to state it plainly on the page and put the evidence of improvement right next to it. That is a strategy that works. Hiding it is not, because almost every offer is followed by a document check.

TIER 1

TCS NQT · Accenture · Infosys · Wipro · Capgemini

Example 1 [TCS NQT · pattern]

Rewrite the top of a fresher resume. The current version starts with an objective paragraph.

TRAP

WEAK:

RAHUL S.
Age: 21 | DOB: 14-08-2004 | Male | Unmarried
Father's Name: Suresh S. | Address: 4/221, Second Cross Street,
Ambattur, Chennai - 600053 | rahul_cool143@mail.example

CAREER OBJECTIVE

To obtain a challenging position in a reputed organization where I can utilize my technical skills and knowledge for the growth of the organization as well as for my own career development, and to work in a dynamic environment which gives me scope to enhance my knowledge and skills.

What this costs, line by line: six lines and about 20% of the page, spent before a single fact about your ability. Nothing here helps the recruiter tick a box. The objective is true of every candidate alive, so it carries zero

(Note: my CGPA was affected in semester 4 due to my mother's serious illness and hospitalisation during that period, which caused me to have 2 backlogs, both of which I have now cleared. I request you to kindly consider my situation.)

Three lines of apology on a one-page document. It makes the 6.4 the loudest thing on the page, and “kindly consider my situation” moves you from candidate to petitioner.

SOLUTION

STRONG:

EDUCATION

B.Tech Computer Science · XYZ Institute of Technology · 2026 · CGPA 6.4/10

Sem 5-7 CGPA 7.6 · no active backlogs

Class XII (CBSE) 2022 – 84% · Class X (CBSE) 2020 – 88%

What changed, exactly:

- The number is stated plainly, with its scale (/10), so nobody has to guess.
- Year of passing is present, which is what the mechanical filter needs.
- **The trend is given its own sub-line.** 6.4 overall next to 7.6 in the recent semesters tells a story of recovery without one word of explanation. This is the whole technique.
- “No active backlogs” is stated as a fact, in three words. Many drives have a rule about active backlogs, and answering it on the page saves a rejection-by-uncertainty.
- Class X and XII on one line — they matter for Tier-1 cut-offs and nowhere else, so they get one line, not three.

The spoken version belongs in the interview, and it is short: “My CGPA is 6.4. It dropped in my fourth semester when my mother was hospitalised and I was at home. I had two backlogs from that period and cleared both. From the fifth semester my CGPA has been 7.6.” State it, give the number, stop. Chapter 6 works through the follow-ups.

TRAP

Never round up, never convert dishonestly, never hide an active backlog. Marks are verified from your transcripts before joining, and a mismatch is treated as misrepresentation rather than as a mistake — which means a withdrawn offer, not a conversation. If you have an active backlog, say so, with the date of your next attempt.

Example 3 [Wipro · pattern]

Rewrite a skills section.

TRAP

WEAK:

TECHNICAL SKILLS

Languages : C, C++, Java, Python, JavaScript, PHP, Kotlin, Go

Web : HTML, CSS, Bootstrap, web technologies

Databases : DBMS, SQL, Oracle

Tools : MS Office, Windows, Internet

Soft Skills : Hardworking, Quick learner, Team player, Leadership

Four separate problems.

1. Eight languages from a 21-year-old is not believed, and it invites a question in a language you last touched in semester three. Every item here is a **promise to be tested**.
2. “Web technologies”, “DBMS”, “Internet” are not skills. “MS Office” and “Windows” are not skills for a software role.
3. Soft skills as a list of adjectives is the single most ignored block on any resume. Nobody has ever been shortlisted for writing “Hardworking”.

4. Nothing is grouped by how well you know it, so the reader must assume the weakest.

SOLUTION

STRONG:

SKILLS

Strong : JavaScript, Node.js (Node), Express, SQL (MySQL), REST API design, Git
Working : Python, MongoDB, Docker, Jest, Linux shell
Familiar : React, AWS (EC2, S3), Redis
Coursework : Data Structures, DBMS, Operating Systems, Computer Networks

What changed, exactly:

- 20+ items → 17, but **ranked into three honest bands**. This is the key move. It tells the interviewer where to aim, and it makes the top band believable.
- Both forms of every abbreviation written once — Node.js (Node), SQL (MySQL) — so a string matcher finds either spelling (Chapter 1).
- Fake skills removed. Soft-skill adjectives removed entirely; they belong in your **bullets**, demonstrated, never claimed.
- Coursework kept as a separate line, so it does not pretend to be experience.

The honesty ladder, stated plainly.

- **Strong:** I have built something with it and I can be questioned on it for 20 minutes.
- **Working:** I have used it in a real project and can use it again with the docs open.
- **Familiar:** I have followed a tutorial or used it once. I will say so if asked.

Being asked about a **Familiar** item and answering “I have only used that once, on a tutorial project — here is what I did with it” costs you nothing. Being caught overclaiming a **Strong** item costs you the round.

Example 4 [Capgemini · pattern]

Turn a typical college project into a project block.

TRAP

WEAK:

PROJECTS

1. College Management System
 - It is a web based application for college management.
 - Front end is HTML CSS and back end is PHP MySQL.
 - Responsible for the admin module.
 - Successfully completed the project with my team members.
2. Final Year Project
 - Machine learning based system using Python.

What is wrong: the title says nothing about what it **does**. “It is a web based application” restates the title. No number anywhere. “Responsible for” hides what you actually did. “Successfully completed” is not an outcome. And project two is a placeholder, not a project.

SOLUTION

STRONG:

PROJECTS

Bus Seat Booking Service – Node.js, Express, MySQL github.com/.../seatbook
Booking API for the college transport office; 3-person team, I owned the API and schema.

- Built 12 REST endpoints with JWT auth; 38 Jest tests, 81% line coverage.
- Cut search p95 latency 1.9 s → 0.31 s by adding a composite index on (route_id, travel_date); measured over 200 requests on 90,000 rows.
- Prevented double-booking with a row lock per seat after a load test at 50

- concurrent requests exposed 7 duplicate bookings.
- In use by the transport office for 2 terms; ~400 bookings per week.

Attendance Defaulter Report – Python, pandas, PostgreSQL github.com/.../attreport

- Replaced a 20-minute manual weekly export with a 6-second script; used by 4 staff.
- Validated input on load, rejecting 1,100 malformed rows that had been silently corrupting the old spreadsheet totals.

What changed, exactly:

- Title now says **what it does** and **what it is built with**, and carries a link.
- One italic context line gives team size and **your** scope — “I owned the API and schema”. Honest scoping is a strength, not a weakness.
- Every bullet starts with a different verb: Built, Cut, Prevented, In use.
- Every bullet has a number, and the numbers are odd-looking because they are real.
- Bullet three is the best one on the page: it contains a **problem found by testing**, which is the thing interviewers most want to see and students most often delete.
- The dead second project is replaced by a small, finished, useful one. A small finished thing beats a large unfinished one on a resume, every time.

SHORTCUT

The link is worth a bullet. A `github.com/...` link beside the project title converts the recruiter’s doubt into a two-second check. It only helps if the repository has a README that explains what the project does in the first three lines. A link to an empty repo is worse than no link at all — see the GitHub section later in this chapter.

Example 5 [Cognizant · pattern]

Write the internship block for a student whose internship was mostly small tickets and watching.

SOLUTION

Most fresher internships really are small tickets. That is fine. The mistake is either inflating it into architecture work or dismissing it into nothing.

Structure: what the team did (one clause) → what **you** shipped → the number → one thing you learned that you can be questioned on.

TRAP

WEAK (dismissing):

INTERNSHIP

ABC Softwares, Chennai – Intern (Jun 2025 - Aug 2025)

- Learnt about the company's technology stack and software development life cycle.
- Attended daily standup meetings and worked on small tasks given by my mentor.
- Gained good exposure to a professional working environment.

Nothing here is a thing you produced. “Gained exposure” is what happened **to** you. After reading this the recruiter knows only that you were physically present for eight weeks.

TRAP

WEAK (inflating):

- Designed and architected scalable microservices for a high-traffic production system serving millions of users.

You did not architect it. The first follow-up — “what were the service boundaries and who decided them?” — ends this badly, and it makes your honest bullets look inflated too.

SOLUTION

STRONG:

INTERNSHIP

ABC Softwares, Chennai – Backend Intern (Jun - Aug 2025, 8 weeks)

- Shipped 14 tickets to production on an internal billing service (Java Spring, MySQL),
 - reviewed by 2 engineers; 11 merged without changes requested.
- Fixed a report that double-counted refunds for same-day cancellations; traced it to a date-boundary comparison, added 4 regression tests, corrected 3 months of reports.
- Cut a nightly job from 22 min to 9 min by batching 1-row-at-a-time inserts into batches of 500. My mentor suggested batching; I measured, sized the batch and shipped it.
- Took over the on-call runbook page for that job: rewrote it after I needed it at 2 a.m. and found it wrong.

What changed, exactly:

- “Small tasks” → **14 tickets, 11 merged clean**. Counting your own tickets is honest, easy and specific. The merge-clean number is a quality signal you can actually produce.
- A named bug with a named cause. This is the bullet an interviewer will pick, and you can talk about date boundaries for ten minutes because you lived it.
- **Credit given where it is due**: “My mentor suggested batching; I measured, sized the batch and shipped it.” Giving credit does not shrink your contribution — it makes the whole entry believable, and it is exactly the behaviour a team wants.
- The last bullet is small and unglamorous and interviewers love it: you found a document was wrong and you fixed it, unasked.

Example 6 [Accenture · pattern]

What belongs in Achievements, and what quietly hurts you?

SOLUTION

The rule: an achievement needs a number, a rank, or a name. If it has none of those, it is not an achievement, it is a participation record, and it is taking a line from a bullet that would have worked harder.

TRAP

WEAK:

ACHIEVEMENTS AND CERTIFICATIONS

- Participated in various technical events and workshops at college level.
- Certificate of participation in national level hackathon.
- Completed online course on Python.
- Member of the college coding club.
- Certificate for attending a 2-day seminar on Artificial Intelligence.

Why it hurts. A reader scans this and finds no rank, no number, no selection, no output. Worse, five lines of participation create an impression that you had nothing better to list — which may not be true, and which those five lines are now preventing you from disproving.

SOLUTION

STRONG:

ACHIEVEMENTS

- Top 40 of 1,200 teams, [Name] national hackathon 2025 – built a 2-day offline-first attendance app; code linked above.
- Solved 400+ problems on [platform]; peak rating 1,640 (top ~15% of the site).
- Coding club: ran a 6-week beginner track, 40 attendees, materials open-sourced.

What changed, exactly:

- “Participated” → **a rank out of a field size**. “Top 40 of 1,200” is an achievement. “Participated in a national hackathon” is a Tuesday.
- The online course disappears, and the **thing built while learning** takes its place. Nobody is hired for a certificate; people are hired for what the certificate was supposed to produce.
- “Member of the coding club” → what you **did** in the club, with two numbers.
- Five weak lines → three strong ones, freeing two lines for a project bullet.

Certifications: the honest ranking.

- **Worth a line:** a certification that is genuinely hard and externally invigilated — a cloud associate-level exam, a database vendor exam. It is a real filter and recruiters know it.
- **Worth a line only if you built something with it:** a course platform certificate. List the project, mention the course in four words, not the reverse.
- **Not worth a line:** attendance and participation certificates. Anyone who turned up has one.

TIER 2

Singapore & Thailand · Grab · Shopee · GIC · DBS · Agoda · SCB

Example 7 [Grab · pattern]

You are applying from India to a backend role in Singapore. What changes on the page?

SOLUTION

Three things change, and nothing else should. Do not rewrite your personality for another country. Do remove the ambiguity that makes a cross-border recruiter hesitate.

Change 1 — the header answers the work-authorisation question before it is asked. Change 2 — the numbers get units and context a foreign reader can size. Change 3 — the date format stops being ambiguous.

TRAP

WEAK header:

RAHUL S. · Ambattur, Chennai - 600053 · +9198xxxxxxx · rahul.s.dev@mail.example
Available from 05/06/2026

A recruiter in Singapore does not know Ambattur, cannot tell whether you need sponsorship, and cannot tell whether 05/06/2026 is 5 June or 6 May. Each of those is a small hesitation, and hesitation at the screen stage means the next resume.

SOLUTION

STRONG header:

RAHUL S. Chennai, India (IST, UTC+5:30) · willing to relocate to Singapore
+91 98xxx xxxxx (WhatsApp) · rahul.s.dev@mail.example
github.com/rahul-s-dev · linkedin.com/in/rahul-s-dev
Work authorisation: will require employment pass sponsorship · available from 5 June 2026

What changed, exactly:

- City plus **country**, plus the time zone. A distributed team reads the time zone immediately.
- The relocation intent is stated, not implied.
- The sponsorship need is stated **plainly and without apology**. Hiding it wastes everyone’s time and it always comes out. Companies that sponsor are not scared of the sentence; companies that do not sponsor were never going to hire you anyway.
- A messaging app noted beside the phone number, because an international recruiter often cannot make an ordinary call to you.
- 05/06/2026 → 5 June 2026. Write the month as a word. This single habit removes a whole class of confusion across every date on the page.

And in the bullets: give numbers a unit and a comparison a stranger can size. “400 bookings per week across 6 routes” is readable anywhere. “Handled the bookings of our college” is not.

Example 8 [Agoda · pattern]

You have a 14-month gap between graduation and now. How does the page handle it?

SOLUTION

The rule: a gap you name is a fact; a gap the reader discovers is a red flag. An unexplained hole between two dates makes a recruiter imagine the worst available explanation. Two lines remove it.

Structure on the page: give the gap its own dated entry, with a neutral one-word reason and **what you did in it**.

TRAP

WEAK (leaving the hole):

B.Tech CSE, 2024

Intern, ABC Softwares – June 2025 to Aug 2025

The reader sees a hole from mid-2024 to mid-2025 and fills it in themselves. You are no longer in control of the story.

TRAP

WEAK (the vague cover-up):

2024 - 2025 : Self study and skill enhancement

Every gap in the world can be described this way, so it reads as a phrase chosen to avoid saying something. It also has no evidence attached.

SOLUTION

STRONG:

Aug 2024 - Sep 2025 · Family care responsibility, with part-time study

- Cared for a family member after surgery; this was the reason I did not take a role.
- Completed 3 courses (backend with Node.js, SQL, system design basics) and shipped 2 projects in this period – both linked below, with commit history from that time.
- Took freelance work for a local clinic: a 4-screen appointment tool, live since Mar 2025, about 60 bookings a month.

What changed, exactly:

- The gap has **dates and a name**. It is now an entry, not a hole.
- The reason is stated in one clause, without medical detail and without apology.
- **Evidence that survives checking:** shipped projects with commit history in that window. This is the part that converts a gap from a risk into a demonstration of self-direction.
- One paid piece of work, however small, with a live date and a usage number.

Other honest gap reasons, written the same way: “Preparing for competitive examinations (cleared prelims, did not clear mains)”. “Health — recovered, no ongoing restrictions”. “Startup attempt — built X, 40 users, shut down in Jan 2025; here is what I learned.” Each is a sentence, then evidence. None of them needs an apology, and none of them should be stretched into a paragraph.

Do not stretch a job’s dates to cover a gap. Dates are verified in the background check from payslips, offer letters or your provident fund record. A stretched date is found, and it is treated as falsification rather than as rounding. A named gap has never cost anyone an offer the way a caught date change does.

Example 9 [DBS · pattern]

Write a LinkedIn headline and About section for a cross-border application.

SOLUTION

LinkedIn is a **search surface**. Recruiters type keywords into a search box and read headlines in a list. So the headline is a keyword line, not a slogan.

Headline structure: [What you are] · [2-4 technologies] · [year / status] · [location or mobility]. About 15 words. Recruiters see roughly the first 10 on mobile, so front-load.

TRAP

WEAK headline: “Aspiring Software Engineer | Passionate about Technology | Seeking Opportunities | Open to Work | Let’s Connect!”

Zero searchable nouns. “Aspiring” is a word that means “not yet”. The exclamation mark is doing no work. A recruiter searching for Node.js Singapore will never find this profile.

SOLUTION

STRONG headline: “Backend engineer (fresher) · Node.js, Express, SQL, Docker · B.Tech 2026 · open to Singapore, pass sponsorship required”

What changed, exactly: every noun is now something a recruiter would actually type into the search box; the year tells them if you are hireable now; the mobility and sponsorship line is stated once, up front.

TRAP

WEAK About: “I am a hardworking and dedicated individual with a passion for coding. I am a quick learner and a good team player, always eager to learn new technologies and take on new challenges. Looking for an opportunity to grow.”

Six adjectives, no facts. It could belong to anybody.

SOLUTION

STRONG About (four short paragraphs, nothing longer):

Backend-focused CS final-year student. I build small services end to end and keep them running: API design, the SQL schema underneath, tests, and deployment.

Most recent: a seat-booking service for my college transport office – Node.js, Express, MySQL – in use for two terms, ~400 bookings a week. I cut search p95 latency from 1.9 s to 0.31 s by fixing an index, and stopped double-booking with a row lock after a load test found 7 duplicates.

Currently learning: system design fundamentals and PostgreSQL internals. Writing up what I learn at github.com/rahul-s-dev.

Open to backend roles in Singapore from June 2026 (employment pass sponsorship needed).
Happy to talk – rahul.s.dev@mail.example.

What changed, exactly:

- Adjectives → one sentence saying what you **do**, then evidence.
- A named recent project with real numbers, so the profile is checkable.
- A “currently learning” line, which honestly signals direction without claiming mastery.
- A clear, unembarrassed availability-and-sponsorship line and a direct way to contact you.

Example 10 [Shopee · pattern]

You are from Mechanical / Civil / Electronics, not Computer Science, and you are applying for a software role. What does the page do?

SOLUTION

The rule: the branch is a fact, not an argument. Win the argument with the Skills and Projects blocks instead, and do it above the fold.

A non-CS student loses at the screen for one reason: the top third of the page says “Mechanical” and nothing says “software” until halfway down. Fix the **order and the positioning line**, not the truth.

TRAP

WEAK:

RAHUL S.

CAREER OBJECTIVE: Though I am from a Mechanical background, I am very passionate about software and I am willing to learn any technology. I request you to give me a chance to prove myself in the IT field.

EDUCATION

B.E. Mechanical Engineering, 2026, CGPA 7.9

...

SKILLS

Python, SQL, basic web development

Two mistakes. It **apologises** for the branch in the first sentence, which makes the branch the headline. And the software evidence is far below the decision zone.

SOLUTION

STRONG:

RAHUL S. Chennai, India · open to relocation · available Jun 2026
+91 98xxx xxxxx · rahul.s.dev@mail.example · github.com/rahul-s-dev

Backend engineer (fresher) · Python, SQL, REST APIs · B.E. Mechanical 2026
Two shipped services; 400+ commits over 18 months.

SKILLS

Strong : Python, FastAPI, SQL (PostgreSQL), REST API design, Git

Working : Docker, pytest, Linux shell, pandas

Familiar : JavaScript, React, AWS (EC2, S3)

PROJECTS

Machine Downtime Tracker – Python, FastAPI, PostgreSQL [github.com/.../downtime](https://github.com/rahul-s-dev/downtime)
· Built for my department's workshop: logs machine stoppages, 9 endpoints, 62 pytest tests. In use by 2 lab technicians for 11 months; 3,800 entries logged.
· Cut the weekly report from a 40-minute manual spreadsheet to a 5-second query.
· Added CSV import that validates on load; rejected 210 malformed legacy rows that had
 been corrupting monthly totals.

Vibration Data Pipeline – Python, pandas, PostgreSQL [github.com/.../vibe](https://github.com/rahul-s-dev/vibe)
· Processed 1.2 M sensor readings into 5-minute buckets; query time for a one-day range
 dropped from 41 s (raw scan) to 0.6 s after bucketing and a time index.

EDUCATION

B.E. Mechanical Engineering · XYZ Institute · 2026 · CGPA 7.9/10

Self-taught software: 400+ commits over 18 months, 2 services in real use.

What changed, exactly:

- The apology is deleted. Nobody needs to be told that Mechanical is not CSE; they can read.
- A **positioning line** names what you are in the first four lines, with the branch stated plainly in the same line — not hidden, not apologised for.

- **Skills and Projects moved above Education.** This is the one structural change that matters for a branch switcher, and it is legitimate: you are putting your strongest true evidence where the seven seconds are spent.
- The projects use the **domain you actually come from**. A machine-downtime tracker and a vibration pipeline are better than another to-do app, because they are real, they have real data, and nobody else has them.
- A volume-of-work line (“400+ commits over 18 months”) answers the real doubt — is this serious or is this a weekend hobby? — with a checkable fact.

In the interview, the question comes anyway, and the honest answer is short: “Mechanical was my entrance-exam result, not my choice. I started writing Python in my second year for my own lab work because the spreadsheet process was painful, and I have shipped two things that are still in use. I do have gaps — I have not done a formal compilers or OS course — and I have been closing those from [source]. I would rather be judged on the two services than on the branch.” Chapter 6 works through the follow-ups.

TIER 3

Google · Amazon · Microsoft · Goldman Sachs · D. E. Shaw · Adobe

Example 11 [Amazon · pattern]

Write a project block where every bullet survives five follow-up questions.

SOLUTION

At this tier the resume is **the question paper for your interview**. The interviewer picks a bullet and drills. So write bullets you **want** to be drilled on.

The test: for each bullet, can you answer — how did you measure it? what did you try first? what did it cost? what would you do differently? If not, either fix the bullet or fix your knowledge of your own project.

TRAP

WEAK (impressive-sounding, undefendable):

- Optimised the application and improved performance by 90%.
- Implemented a scalable microservices architecture using industry best practices.
- Utilised advanced machine learning algorithms to increase accuracy significantly.

Every one of these dies on the first follow-up. “90% of what, measured how?” “How many services, and why was one not enough for 400 bookings a week?” “Which algorithm, and what was your baseline?” These bullets are not a lie yet, but they are a trap you built for yourself, and they will be read as inflation.

SOLUTION

STRONG:

Bus Seat Booking Service – Node.js 20, Express, MySQL 8, Docker · 3-person team,
I owned the API, schema and deploy github.com/.../seatbook

- Cut search p95 from 1.9 s to 0.31 s (6x) by adding a composite index on (route_id, travel_date); measured with 200 requests against 90,000 rows. Write path cost 6 ms -> 8 ms, accepted for this write volume.
- Eliminated double-booking under concurrency: a load test at 50 parallel requests produced 7 duplicate seats; fixed with SELECT ... FOR UPDATE per seat row. Chose a row lock over a cached availability count because the count could go stale.
- Made the CSV export a stream after estimating 12 months of growth (40k -> ~90k rows per month); the in-memory version would have failed silently past ~70k.
- 38 Jest tests, 81% line coverage; CI runs them on every push.

What changed, exactly:

- “Improved by 90%” → a before, an after, a multiplier, **and the method of measurement**.
- Added **the cost** of the fix on the write path. Naming a trade-off you accepted is the single strongest signal on a fresher resume. Nobody expects a free win; senior people expect you to know the price.
- Added **the rejected alternative and why** — cache versus row lock. This is a design decision written in 12 words, and it invites exactly the follow-up you can answer.

- Added how the bug was **found** (a load test), not just that it was fixed.
- Version numbers on the stack, so the reader knows you actually ran it rather than read about it.

The five drills your bullets must survive

The drill	Where the answer already is, in the strong block above
How did you measure it?	200 requests against 90,000 rows; before and after.
What did you try first?	The obvious cache; rejected on staleness.
What did it cost?	Writes 6 ms → 8 ms; one more index in migrations.
How did you find the bug?	A load test at 50 parallel requests produced 7 duplicates.
What would you do differently?	Measure in week 2, not week 8 — there was no performance owner.

SHORTCUT

Write the bullet **after** you can answer the five drills, not before. If a drill has no answer, you have found a hole in your own understanding of your own project, and you still have time to go and close it. That is worth far more than a nicer sentence.

Example 12 [Google · pattern]

Your project was a 4-person team assignment. How do you write it without either stealing credit or disappearing?

SOLUTION

Structure: one context line names the team size and **your** scope. Then every bullet is about something **you** did. Where a result was collective, say so and name your contribution to it.

TRAP

WEAK (stealing):

- Built a full-stack food delivery platform with real-time order tracking, payment integration and an admin dashboard.

Written in the first person singular by implication, for four people's work. It collapses on the first question about the payment integration you never touched.

TRAP

WEAK (disappearing):

- Worked in a team of 4 on a food delivery platform. Helped in the backend part.

Now nothing is attributable to you at all. "Helped in" is the most self-erasing phrase on any resume. This scores lower than saying nothing.

SOLUTION

STRONG:

Food Delivery Platform – 4-person team; I owned order state and the driver-assignment service (Node.js, PostgreSQL, Redis) github.com/.../fooddrop

- Designed the order state machine (6 states, 9 transitions) and made every transition idempotent, which removed a class of duplicate-status bugs seen in week 3.
- Wrote the driver-assignment service: nearest-free-driver from a Redis geo set, median assignment 180 ms over 2,000 simulated orders.
- Team result: platform demoed with 60 test users, end to end. My part was orders and assignment; payments and the admin UI were teammates'.

What changed, exactly:

- Ownership scoped in the title line, so everything under it is fairly read as yours.
- Verbs are personal and specific: Designed, Wrote.
- The collective result is labelled “Team result” and your share is named **explicitly**. Far from weakening the entry, this is the line that makes the rest of it believable.
- Nothing is claimed that you cannot be drilled on. You can talk about the state machine for twenty minutes and you will never be embarrassed about the payment gateway.

Example 13 [Microsoft · pattern]

Your GitHub link is on the resume. What does the reviewer actually look at, and in what order?

SOLUTION

An engineer who opens your GitHub spends about a minute. They look at: your pinned repositories, the README of the top one, the commit history shape, and whether the code runs. They are not reading your code line by line. They are asking **is this person real?**

TRAP

WEAK profile: 27 repositories, 24 of which are forks of tutorials, all named things like project1, test, newfolder. No pinned repositories. Every repo’s last commit is a single commit saying “final code” on one day.

What the reviewer concludes: the code arrived from somewhere else on the last day. That may be unfair. It is also the only available reading of a one-commit history.

SOLUTION

STRONG profile, in four moves that take one evening:

1. **Pin three repositories** — your two best projects and one small, clean, useful thing. Pinning is one click and it controls everything the reviewer sees first.
2. **Write a README that answers four questions in the first screen:** what does it do, how do I run it in two commands, what is the architecture in five lines, what is not finished. That last one earns more trust than a feature list.
3. **Delete or archive the tutorial forks**, or at least un-pin them. A profile of 3 real things beats a profile of 27 things where 24 are noise.
4. **Let the commit history be honest.** Commits spread over the weeks you actually worked, with messages that say what changed, are themselves the evidence that the work is yours.

A README opening that works:

seatbook

Seat booking API for a college bus service. Node.js + Express + MySQL.
Used by our transport office for two terms (~400 bookings/week).

Run it: `docker compose up` then open `http://localhost:3000/docs`

How it works: 12 REST endpoints -> service layer -> MySQL. Seats are locked per row (SELECT ... FOR UPDATE) during booking, which is what stops double booking under concurrent requests. JWT for auth. 38 Jest tests.

Not done yet: no payment flow, no seat map UI, admin actions are API-only.

Why this works: the reviewer knows in ten seconds what it is, can run it in one command, is told the one interesting design decision, and is told the limits by you rather than finding them. “Not done yet” is the most persuasive section in the whole file.

Example 14 [Goldman Sachs · pattern]

A role you want asks for things you do not have. What do you put on the page?

SOLUTION

The rule: close the gap with real work, or state the nearest true thing. Never claim the skill.

- Take the post’s must-have list. For each item you are missing, you have exactly three honest moves.

Move	What it looks like on the page
Close it — build something small, this week	A two-day project using the missing technology, listed honestly as small: “Kafka — 2-day exercise: order events producer/consumer, 3 partitions, at-least-once handling.” A small labelled thing is credible; a big vague claim is not.
Map it — name the nearest thing you have done	Post asks for Kubernetes; you have Docker Compose. Write Docker , docker compose under Working and do not write Kubernetes at all. In the interview: “I have not run Kubernetes. I have containerised and orchestrated locally with Compose, so the concepts I know are images, networking and health checks; I have not done scheduling or rollouts.”
Drop it — say nothing	If you have neither the skill nor the nearest thing, leave it off entirely. A silent gap is normal. A claimed gap is a failed round.

TRAP

WEAK (the keyword stuffing move): adding Kubernetes, Kafka, Spark, TensorFlow to the skills line because the post mentioned them.

This does two bad things at once. It may get you **through** the screen, which means you now sit in a technical round being asked about Kafka partitions. And most interviewers can tell within two questions, at which point everything else on your resume becomes suspect too — including the parts that were true.

SOLUTION

STRONG (the same candidate, honestly):

SKILLS

Strong : JavaScript, Node.js (Node), Express, SQL (MySQL), REST API design, Git
 Working : Python, MongoDB, Docker & docker compose, Jest, Linux shell, CI (GitHub Actions)
 Familiar : React, AWS (EC2, S3), Redis, message queues (RabbitMQ – one 2-day exercise)

PROJECTS

Order Events Exercise – RabbitMQ, Node.js github.com/.../order-events

- 2-day exercise to learn message queues: producer/consumer for order events, at-least-once delivery, retry with a dead-letter queue after 3 failures.
- Wrote up what surprised me (duplicate deliveries are normal, so consumers must be idempotent) in the README.

What changed, exactly:

- The missing skill is neither claimed nor hidden — it is present at its real size, labelled “one 2-day exercise”, with a link.
- The write-up of **what surprised you** is the strongest part. It proves learning happened, and it gives the interviewer an easy, friendly question to ask you, which you can answer well.
- Everything else on the page stays trustworthy, which is the thing you are really protecting.

Example 15 [D. E. Shaw · pattern]

Your resume will **generate** the questions you are asked. Design that on purpose.

SOLUTION

At Tier 3, the interviewer reads your resume for two minutes before the call and picks the questions from it. That means you control the question paper — if you write deliberately.

The technique: every bullet should be a door you want opened. Put one **deliberate hook** in each project: an interesting trade-off, a bug with a real cause, a measurement. Remove anything you would rather not discuss.

If your bullet says...	...expect exactly this question
composite index on (route_id, travel_date)	Why that column order? What happens if the query filters on date only?
SELECT ... FOR UPDATE per seat row	What is the deadlock risk? What is your lock ordering?
38 Jest tests, 81% line coverage	What is not covered by that 19%? Is coverage a good target?
streamed the CSV export	What happens if the client disconnects halfway?
Redis geo set for nearest driver	What does Redis do if it restarts? Is your data durable?
JWT auth	How do you revoke a token before it expires?
microservices	Why? What did splitting cost you that a single service would not have?
machine learning model, 94% accuracy	What was the class balance? What is your baseline? Precision or recall?

TRAP

WEAK (undefended hooks): leaving microservices, 94% accuracy and blockchain on the page because they sound impressive, without being able to answer the question in the right-hand column.

Each of these is a door you opened and then could not walk through. An interviewer who asks about your own listed technology and gets a blank look will discount **everything** on the page, including the parts you did well.

SOLUTION

STRONG (hooks you want opened):

- Chose a composite index on (route_id, travel_date) in that order because every query filters route first; a date-only query still does a scan, which was acceptable here.
- Locked the seat row (SELECT ... FOR UPDATE) rather than caching an availability count, because the count could go stale between read and write.
- 38 Jest tests, 81% line coverage – the uncovered part is the admin refund path, which is not automated yet.

What changed, exactly:

- Each bullet now contains **the answer to its own follow-up**, in a clause. The interviewer still asks, and you elaborate — but the resume already proved you knew.
- Bullet three names **what is not covered**. Volunteering a limit is the fastest way to be trusted about everything else.
- The word “microservices” is simply gone, because the honest answer was “the course template had three services.” A word you cannot defend is a liability no matter how good it looks.

The final self-test before you send any resume to a Tier-3 company: read every line out loud and ask “what question does this invite, and can I talk about it for five minutes?” Any line where the answer is no gets rewritten or deleted. There is no third option.

Formatting so both filters can read it

THE RULES THAT MATTER, AND WHY

1. **One column.** Two-column layouts can be read across the columns by a parser, gluing your phone number to your CGPA. If you use two columns, run the copy-paste test below.

2. **No text inside images.** A skills graphic is invisible to the screen.
3. **Nothing important in the header or footer** of the page. Many parsers skip them.
4. **No tables for layout** if you can avoid it; simple tables are usually fine, nested ones are not.
5. **Standard section names.** “Education”, “Skills”, “Projects”, “Experience”. Not “My Journey” or “What I Bring”. The parser looks for the standard words.
6. **One common font**, 10–11pt body. No script fonts.
7. **Send PDF unless the post says otherwise.** If the post says .doc or .docx, send that. Follow the instruction — some portals genuinely only parse what they asked for.
8. **File name:** Rahul_S_Backend_Resume.pdf. Not resume_final_v3_updated.pdf, and never Untitled.pdf.
9. **Dates as Mon YYYY** — Jun 2026, not 06/2026.

The copy-paste test — do this every single time

60 SECONDS, AND IT CATCHES ALMOST EVERY PARSING DISASTER

1. Open your final PDF.
2. Select all. Copy.
3. Paste into a **plain** text editor — not a word processor.
4. Read what appeared.

Pass: your name is on its own line, the skills are in a readable row, the bullets are separate lines, and nothing is glued to anything else.

Fail: empty output (your PDF is an image), scrambled order, or words from two columns joined into nonsense like JavaScriptChennai Node.js600053.

Whatever you see in that text editor is close to what the screening software sees. If it is unreadable to you, it is unreadable to the filter, and no amount of good content fixes that.

SHORTCUT

On a machine with pdftotext installed you can do the same check from the terminal: `pdftotext -layout Rahul_S_Backend_Resume.pdf - | head -40`. Same idea, faster, and you can run it every time you export.

Three skeletons you can copy the shape of

Copy the **shape**, never the words. Substitute your own real projects, your own real numbers, your own real weak spots.

Skeleton A — no internship, projects only

NAME City, Country · open to relocation · available Mon YYYY
+CC phone · first.last@mail.example · github.com/handle · linkedin.com/in/handle

Backend engineer (fresher) · Node.js, SQL, REST · B.Tech CSE 2026

SKILLS

Strong : 5-6 items you can be drilled on for 20 minutes
Working : 4-5 items you have used in a real project
Familiar : 3-4 items you have tried once (say so if asked)
Coursework: DSA, DBMS, OS, Networks

PROJECTS

Project 1 – what it does, stack
1 context line: team size and YOUR scope

github link

- Built [thing]: [size number], [test number].
- Cut/Fixed [named problem] from [before] to [after] by [named method]; measured by [how].
- [The trade-off you accepted and its price.]
- In use by [who], [usage number].

Project 2 – what it does, stack

github link

- [Output bullet with a number.]
- [Bullet about a bug you found and how you found it.]

EDUCATION

B.Tech [Branch] • [Institute] • [Year] • CGPA x.x/10

[trend line if it helps you] • no active backlogs

Class XII [Board] YYYY – xx% Class X [Board] YYYY – xx%

ACHIEVEMENTS

- Rank out of field size, event, year – what you built.
- Volume with a rating or percentile.
- Something you ran, with an attendance number.

Skeleton B — with an internship

Same as A, with one change: the internship goes **above** projects only if it was genuinely better than your best project. Judge it by one question — which one can you talk about for twenty minutes?

EXPERIENCE

[Company], [City] – [Role] (Mon - Mon YYYY, N weeks)

- Shipped N tickets to production on [system] ([stack]); N merged without changes.
- Fixed [named bug]: traced to [named cause], added N regression tests.
- Cut [named job] from N to M by [method]. [Who suggested what – give credit.]
- [The unglamorous ownership bullet: a runbook, a doc, a test suite nobody wanted.]

PROJECTS

... as in Skeleton A, two projects ...

Skeleton C — a gap, a switch, or a weak year

NAME City, Country • open to relocation • available Mon YYYY
contacts • links

[What you are] • [3-4 technologies] • [degree and year]

[One evidence line: shipped things, commit volume, usage.]

SKILLS

... three honest bands ...

PROJECTS

... your two strongest, both with links and usage numbers ...

EXPERIENCE / OTHER ACTIVITY (dated, nothing left as a hole)

Mon YYYY - Mon YYYY • [Neutral name for the period]

- [One-clause reason, no apology, no medical detail.]
- [Evidence with dates INSIDE this window: shipped X, N users, commit history.]
- [Any paid work, however small, with a live date and a usage number.]

EDUCATION

[Degree] • [Institute] • [Year] • CGPA x.x/10

[trend line] • [backlog status stated as a fact]

Skeleton C is the one most students need and the one nobody shows them. Look at what it does: it never hides the weak part, it gives the weak part a name and a date, and it puts checkable evidence directly underneath. The page is not asking for sympathy anywhere. That is the whole technique, and it works because it is true.

Keep one master file, export many

Keep a **master** document with every bullet you have ever written — far longer than one page. Never send it. For each application, copy it, delete down to one page against that job post, rename, export, run the copy-paste test.

Two reasons this matters. You stop rewriting bullets from memory each time, which is where errors and inflation creep in. And when an interviewer asks about something from a version you sent four months ago, you still have that exact version. Keep the sent PDFs in a folder named by company and date: `2026-03-14_GrabSG_backend.pdf`.

SHORTCUT

Put the master file in a Git repository, even if it is just a plain text file. One commit per edit. Then “what did I send them?” is a one-command question, and you get a free record of how your resume improved — which is itself useful material for the “how have you grown?” question in Chapter 4.

Tailoring: 20 minutes per application

One resume for every company is the reason most applications vanish. You do not write a new resume each time. You change four things.

Minutes	What you change	Why
0–5	Copy the post’s must-have list into a scratch file. Underline every noun.	These are the exact strings both filters use.
5–10	Edit the skills row so every must-have you honestly have appears in the post’s own words.	This is where most of the screen score comes from.
10–15	Reorder projects so the most relevant one is first, and rewrite its first bullet to face the role.	The recruiter reads one bullet of one project.
15–18	Fix the one-line headline under your name to match the role title.	Title match is a heavy signal for both filters.
18–20	Rename the file, re-export, run the copy-paste test.	Catches the disaster before it costs you the application.

Tailoring is not lying. You are choosing which true things to put first. If a must-have is something you do not have, it does not go on the page — see Example 14. The four edits above change emphasis and vocabulary, never facts.

The mail you send with it

If you are mailing a resume to a person, the mail is read before the attachment.

TRAP

WEAK:

Subject: Job

“Respected Sir/Madam, I am writing this mail to apply for any suitable position in your esteemed organisation. I have attached my resume herewith for your kind perusal. I am a hardworking and dedicated fresher and I request you to kindly consider my application and give me an opportunity to prove myself. Awaiting your positive response. Thanking you.”

SOLUTION

STRONG:

Subject: Backend Engineer (fresher) – Rahul S. – Node.js/SQL – resume attached

Hello [name],

I am applying for the Backend Engineer (fresher) role, ref BE-2026-14.

Closest thing I have done to this job: I built and shipped a Node.js + MySQL booking API that my college transport office has used for two terms (~400 bookings a week). I fixed a search that took 1.9 s at p95 and got it to 0.31 s with a composite index, and stopped double-booking with a per-row lock after a load test found 7 duplicates.

Resume attached (1 page). Code: github.com/rahul-s-dev/seatbook

I am in Chennai, available from June 2026, open to relocation.

Thank you,

Rahul S. · +91 98xxx xxxxx

What changed, exactly:

- Subject line is now searchable and states the role and the reference number.
- “Any suitable position” → one named role. Applying to “anything” reads as applying to nothing.
- The middle paragraph is the **whole mail**: one piece of evidence, with numbers, that maps to the job. Nobody needs your adjectives.
- “Kindly consider... prove myself” removed. It lowers your standing and adds no information.
- Ends with the three facts a recruiter needs to act: location, availability, phone.
- Under 120 words. It is read on a phone.

Top traps in this chapter

TRAP

1. An objective paragraph. It costs a fifth of the page and says nothing.
2. A resume longer than one page as a fresher. Page two is not read before the decision.
3. Shrinking the font to 8pt to fit. Cut bullets instead.
4. “Responsible for”, “Worked on”, “Involved in”, “Helped in”. Every one erases you.
5. A bullet with no number. It is a claim, not evidence.
6. A number that is round and huge — “improved by 90%” — with no measurement behind it.
7. Listing eight programming languages. Each one is a promise to be tested.
8. Soft skills as adjectives. Demonstrate them in bullets or leave them out.
9. Claiming a technology because the job post mentioned it. It gets you into a room you cannot survive, and it poisons the true parts of your resume.
10. Hiding a CGPA behind “First Class”, or omitting the year of passing.
11. An unexplained date gap. Name it, date it, attach evidence.
12. Changing an employment date to cover a gap. This is found in the background check and treated as falsification.
13. A GitHub link to a profile of tutorial forks and one-commit repositories.
14. A photo, DOB, marital status or a signed declaration on the PDF you send.
15. Sending a scanned or image PDF. Run the copy-paste test every time.
16. Sending the same untailored resume to 40 companies and concluding the market is bad.

Practice

PRACTICE · TIER 1 target: 45 min, with your real resume open

1. Print or open your current resume. Count: how many lines before the first **fact about your ability**? If it is more than four, delete until it is four.
2. Paste all your bullets into `audit.js` and run it. Write down three numbers: percentage of bullets with a number, count of filler phrases, count of distinct opening verbs.
3. Rewrite your three worst bullets using VERB + WHAT + HOW + NUMBER. For each, write beside it **where the number came from** — the table, the log, the count.
4. Rewrite your skills section into the three honest bands: Strong, Working, Familiar. Be strict. Anything you could not defend for 20 minutes is not Strong.
5. Run the copy-paste test on your PDF. Write down exactly what came out wrong.

ANSWER KEY

1. Four lines maximum before Education. Name, contacts, links, one positioning line.
2. Targets: 70% or more of bullets carry a number · zero filler phrases · all opening verbs different.
3. If you cannot say where a number came from, you cannot use it — go and measure it, or drop it. Most numbers are one `COUNT(*)`, one `git log --oneline | wc -l`, or one timer away.
4. A common honest result is 4–6 Strong items, not 20. That is a good resume, not a thin one.
5. Typical failures: empty text (image PDF), columns glued together, icons turning into boxes, the header line missing entirely.

PRACTICE · TIER 2 target: 40 min

1. Take one real job post from a company outside your country. List its must-have nouns. Mark each one: **have it / nearest thing / do not have**. Do not write anything from the third group onto your page.
2. Rewrite your header for that application: country, time zone, mobility, work authorisation, availability date in 5 June 2026 form.
3. Rewrite two of your bullets so a reader who has never heard of your college can size the numbers. Add units and a comparison.
4. Write your LinkedIn headline using the formula. Count the words. Check the first ten words alone still say what you are.
5. If you have a gap, write the three-line dated entry for it: dates, one-clause reason, evidence with links.

ANSWER KEY

1. The “nearest thing” column is your interview script for that skill: name what you did do, name the boundary of it, and stop.
2. A header that answers work authorisation before it is asked removes the most common reason a cross-border resume is set aside.
3. “About 400 bookings a week across 6 routes” travels. “Handled our college’s bookings” does not.
4. If the first ten words are “Aspiring · Passionate · Seeking”, rewrite. They must be nouns a recruiter would type into a search box.
5. Evidence with a **date inside the gap window** is what turns the entry from a claim into a fact.

PRACTICE · TIER 3 target: 60 min, written

1. Take your best project. For every bullet, answer the five drills in writing: how measured, what tried first, what it cost, how the bug was found, what you would do differently. Mark every bullet with an unanswerable drill.
2. For one marked bullet, go into the repository and **get the real answer** — run it, time it, count the rows. Then rewrite the bullet with the real number.
3. Rewrite one bullet to include a trade-off you accepted and its price. (“Writes 6 ms → 8 ms, accepted at this volume.”)
4. Rewrite one team project entry so that ownership is scoped in the title line and every bullet is personally attributable, with one line labelled “Team result”.
5. Rewrite the README of your top repository to answer: what it does, how to run it in two commands, how it works in five lines, what is not finished.

6. Take a must-have from a real post that you do not have. Build the two-day version of it this week. Write the honest, size-labelled entry for it.

ANSWER KEY

1. A bullet whose drills you cannot answer is not a strong bullet; it is an invitation to a bad five minutes. Fix the knowledge or cut the bullet.
2. Real numbers look irregular. If your rewritten number is “50%” exactly, check it again.
3. A named trade-off is the clearest fresher-level signal that you understand engineering rather than features.
4. “Team result: ... My part was X; Y and Z were teammates’.” This sentence **increases** trust in everything above it.
5. The “Not done yet” section of a README is the most persuasive part of it.
6. A two-day exercise, labelled as a two-day exercise, is credible. The same work described as expertise is not.

ONE-PAGE REVISION CARD

What the page is for. One job: earn the next ten minutes. It passes a string matcher, then a human for seven seconds.

Order of blocks. Header → Education → Skills → Projects (best first) → Experience → Achievements → Extras. Skills above projects, because the recruiter is ticking a checklist.

Cut on sight. Objective paragraph · photo · DOB · marital status · father's name · full address · declaration and signature · “references available on request” · skill percentage bars. (Company application **forms** are different — fill those as asked.)

The bullet formula. VERB + WHAT + HOW + NUMBER. 14–22 words. A different opening verb every time. A bullet with no number is a claim; a bullet with a number is evidence.

Numbers you already have. endpoints · tables · rows · users · tests and coverage · before/after milliseconds · time saved per week · rows rejected · team size and your scope · weeks · load tested at N.

Real numbers look irregular. 6.1x, 0.31 s, 1,100 rows. Round and huge reads as invented.

The three honest skill bands. Strong = I can be drilled for 20 minutes. Working = I used it in a real project. Familiar = I tried it once and will say so.

Weakness on the page. State the fact, put the trend beside it, and stop.

- CGPA 6.4/10 with Sem 5-7 CGPA 7.6 · no active backlogs underneath.
- A gap gets dates, a one-clause reason, and evidence with commits inside that window.
- A missing skill gets closed (a labelled two-day exercise), mapped (name the nearest true thing), or dropped. Never claimed.
- Never change a date, never round up a CGPA. Both are verified, and both are treated as falsification, not error.

Team work. Scope ownership in the title line. Personal verbs in every bullet. One line labelled “Team result” with your share named.

Formatting. One column · no text in images · nothing in the page header/footer · standard section names · PDF unless told otherwise · Rahul_S_Backend_Resume.pdf · dates as Jun 2026.

The copy-paste test. Open the PDF, select all, copy, paste into a plain text editor. What you see is roughly what the filter sees. Run it on every export.

Tailoring, 20 minutes. Skills row into their words → best-matching project first → rewrite its first bullet → headline matches the role title → rename, re-export, retest.

The application mail. A searchable subject with the role and reference · one named role · one paragraph of evidence with numbers · resume and code link · location, availability, phone. Under 120 words.

And the rule that outranks every technique in this chapter: everything on the page must be true and must be defensible by you, out loud, for twenty minutes. Substitute your own real experience into every structure here. A resume you cannot defend is not an asset — it is the question paper for a round you will fail.

The Core HR Questions

Ten questions decide most HR rounds. Build an answer for each one, from your own life.

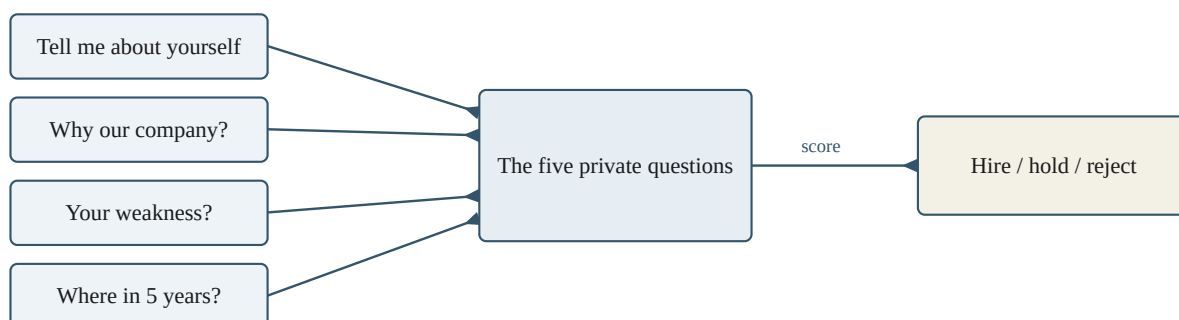
Pattern in one page

The HR round is short. It is often 15 to 25 minutes. In that time the interviewer is trying to answer five private questions. Every question they ask out loud is a way of getting at one of those five.

WHAT THE HR ROUND IS REALLY SCORING

1. **Can this person speak clearly?** Can they answer in 60 to 90 seconds without wandering?
 2. **Is the resume true?** If the resume says “built a REST API”, can the person describe it without help?
 3. **Will they actually join?** Companies lose money when an offer is rejected. They look for signs: did you research us, do you have a reason to be here, is the location workable?
 4. **Will they stay one or two years?** Not forever. Just long enough to be worth training.
 5. **Are they easy to work with?** Do they blame others? Do they listen? Do they interrupt?
- Nothing else is being scored. Not your vocabulary. Not big words. Not a fancy accent.

Read that list again before you prepare any answer. If an answer of yours does not move at least one of those five, it is decoration. Cut it.



The same five scores, hiding behind different questions

The rule that beats every script

You will find “sample answers” everywhere. Most students memorise one and repeat it. It fails, and it fails in a predictable way: the first 30 seconds sound polished, then the interviewer asks one follow-up and the whole thing collapses, because there was nothing underneath.

THE SUBSTITUTION RULE — THE MOST IMPORTANT LINE IN THIS CHAPTER

Every answer in this book is a **STRUCTURE plus a filled example**. The structure is yours to keep. The filled example is **not**. It is filled with a made-up student’s life so you can see the shape. **You must replace every fact in it with your own real experience.**

If you cannot fill a slot with something true, the answer is not ready. Go and do something real, however small, and then fill the slot. Do not invent it.

TRAP

Never claim work you did not do. Not an internship you did not have. Not a team size you did not lead. Not a number you did not measure. Interviewers ask follow-ups for a living. The moment a claim breaks under one “how did you measure that?”, the whole interview is over, and at many companies the rejection is recorded against your name for future openings too.

There is always a truthful version of your answer that is strong enough. This chapter shows you how to find it.

How long should an answer be?

Calm English speech is roughly 130 words per minute. That gives you a simple budget.

Check any answer against its time budget

JS

```
// Calm interview speech in English: about 130 words per minute.
const WPM = 130;

const speakSeconds = (text) => {
  const words = text.trim().split(/\s+/).filter(Boolean).length;
  return { words, seconds: Math.round((words / WPM) * 60) };
};

const check = (label, budgetSec, text) => {
  const { words, seconds } = speakSeconds(text);
  const verdict = seconds > budgetSec * 1.15 ? "T00 LONG - cut"
    : seconds < budgetSec * 0.5 ? "T00 SHORT - add proof"
    : "fits";
  console.log(` ${label.padEnd(14)} words=${String(words).padStart(3)} ` +
    ` ~${String(seconds).padStart(3)}s budget=${String(budgetSec)}s ->
    ${verdict} `);
};
```

Running it on three versions of the same introduction:

Measured output

TEXT

```
60s intro      words=105 ~ 48s budget=60s -> fits
too short      words= 19 ~  9s budget=60s -> T00 SHORT - add proof
rambling       words=210 ~ 97s budget=60s -> T00 LONG - cut
```

TIME BUDGETS TO MEMORISE

Question	Seconds	Words
Tell me about yourself	60–90	130–190
Strength / weakness	45–60	95–130
Why this company	45–60	95–130
Where in 5 years	40–50	85–110
A behavioural story (Chapter 4)	90–120	190–260
Do you have questions for us	ask 2–3	–

SHORTCUT

Practise with a clock, not with a feeling. Record yourself on your phone. Play it back once at normal speed. You will hear your own filler words in the first 20 seconds, and you will never un-hear them. That single recording is worth ten hours of reading.

Question 1 — “Tell me about yourself”

This is asked in almost every HR round. It is also the question students prepare worst, because it feels easy. It is not a biography request. It is a request for a **pitch**.

STRUCTURE: NOW - PATH - PROOF - POINT

NOW (1 sentence). Who you are today. Name, year, branch, college city.

PATH (1–2 sentences). The one line of your story that led here. Pick one thread only.

PROOF (2–3 sentences). The strongest concrete thing you built or did, with a number if you honestly have one. This is the part everyone skips and it is the part that gets you hired.

POINT (1 sentence). Why that path points at **this** job at **this** company.

Total: 5 to 7 sentences. 60 to 90 seconds. Then stop talking.

WARM-UP

build the reflex

Example 1 [warm-up]

Write the NOW sentence only. 20 words or fewer. Say your name, your year, your branch, your city, and one skill area.

SOLUTION

Filled example, to be replaced with your own facts:

“I am Meera Raghavan, a final-year Information Technology student in Pune, and my strongest area is backend work in Node.”

That is 22 words. Count yours. If it is over 25, drop the adjectives first.

Example 2 [warm-up]

Write the PROOF sentence only, for one thing you have actually built. It must contain a noun (what), a verb (what you did), and a measurement (how much) if you have one.

SOLUTION

Filled example:

“I built a canteen pre-order site that about 200 students used during our college fest, and I cut the order page from about four seconds to under one second.”

Notice the hedge words: “about 200”, “about four seconds”. Hedging is honest and it is **safe**. If you did not count exactly, say “about”. Never state a precise number you did not measure.

TIER 1

TCS NQT · Accenture · Infosys · Wipro · Capgemini

Example 3 [TCS NQT · pattern]

“Tell me something about yourself.”

TRAP

Weak answer.

“Good morning sir. Myself Meera Raghavan. I am from Pune. I did my schooling from a CBSE school and I got 89 percent in 12th. Then I took admission in IT branch. My father is a bank employee and my mother is a housewife. I have one younger brother who is studying in 10th standard. My hobbies are listening to music and reading books. I am a hard-working person, punctual and a quick learner. I want to work in a reputed company like yours where I can grow my career and utilise my skills for the growth of the organisation. Thank you.”

Now the same student, same facts, nothing invented:

SOLUTION

Strong answer.

“I am Meera Raghavan, a final-year IT student in Pune. **(NOW)**

I moved toward backend work in my second year, after I realised I enjoyed the database side of a project much more than the screens. **(PATH)**

The clearest example is a canteen pre-order site I built for our college fest. About 200 students used it over three days. The order page was taking about four seconds to load because I was fetching the full menu on every click. I cached the menu and added an index on the item table, and it came down to under a second. I also learned the hard way to handle two people ordering the last item at the same time. **(PROOF)**

I am applying here because your service teams do a lot of backend and data work for clients, and that is the part of a project I already choose on my own. **(POINT)**“

Name exactly what changed

What	Weak version	Strong version
Opening	Ritual words: “Myself...”, “reputed company”	Plain sentence: name, year, city, area
Family and school	Four sentences, zero signal	Removed. Nobody is scored on a brother in 10th
Adjectives	hard-working, punctual, quick learner – all un-proven	Replaced with one story that shows it
Evidence	None	200 users, 4s to under 1s, a concurrency bug
Closing	“grow my career, utilise my skills”	A named link between her work and their work
Follow-up safety	Nothing to ask about – interview stalls	Five easy follow-ups invited, all answerable

Look at the last row. A strong introduction **hands the interviewer their next question**. That is deliberate. You want them asking about the index, or the double-order bug, because you can talk about those for five minutes with real detail.

TRAP

“**Myself Meera Raghavan**” is not English. It is a very common habit in Indian interviews. Say “I am Meera Raghavan” or “My name is Meera Raghavan”. Similarly, drop “I am having 8.1 CGPA” (say “I have”), and drop “kindly do the needful” from your emails forever.

What if I have no internship and no big project?

Then you say so, plainly, and you show the small real thing instead. Small and true beats big and vague.

Example 4 [Wipro · pattern]

Same question, but the student has no internship, one college project, and a 7.1 CGPA.

SOLUTION

“I am Arun Kale, final-year Computer Engineering in Nagpur. **(NOW)**

I have not done an industry internship. What I have instead is a habit of finishing small things. **(PATH, honest)**

Last year our class had trouble tracking lab submissions, so I wrote a small attendance and submission tracker in Node with a SQLite file. It is not a big system. It is about 600 lines. But three faculty members used it for a full semester, and I had to fix two real bugs after release: one where a date was stored in local time and shifted by a day, and one where two teachers editing the same record overwrote each other. Fixing those taught me more than the building did. **(PROOF)**

My CGPA is 7.1. It was 6.4 after my second year, and it has gone up every semester since. **(honest weakness, with the direction of travel)**

I am applying for your development track because the work I have enjoyed most is exactly this: small tools that somebody actually uses. **(POINT)**

SHORTCUT

The direction of travel beats the absolute number. “7.1, and it was 6.4 two years ago” is a story about someone who fixes things. “7.1” alone is just a number. Use this move for CGPA, backlogs, and any weak patch. Chapter 6 goes deeper into it.

TIER 2

Singapore & Thailand · Grab · Shopee · GIC · DBS · Agoda · SCB

Example 5 [Shopee · pattern]

“Walk me through your background.” Role is a graduate backend engineer in Singapore, and the student is applying from India.

SOLUTION

Regional rounds add one thing to NOW-PATH-PROOF-POINT: a **location reason** that is not “salary” and not “abroad life”.

“I am Meera Raghavan, a final-year IT student in Pune. **(NOW)**

For the last two years my work has been on the backend and database side. **(PATH)**

The project I would point to is a canteen pre-order site used by about 200 students over three days. The interesting part was not building it, it was the two failures: the order page took about four seconds because I refetched the whole menu each click, and two people could buy the same last item. I fixed the first with caching and an index and got it under a second. For the second I moved the stock check into the same transaction as the order insert. **(PROOF)**

On why Singapore specifically: the products I use and read about that run at real regional scale are built here, across several countries and currencies at once. I want the problem where the same feature has to work in four markets, because that changes the design, not just the copy. I have also checked the practical side – I understand this role would be on an Employment Pass sponsored by the company, and I am ready for the timeline that involves. **(POINT, with location)**”

The last sentence does real work. Regional recruiters screen hard for candidates who have not thought about visas and will drop out halfway. One calm sentence showing you know the pass exists moves you ahead of most applicants. Do not overclaim: say what you know, not what you guess. Never state a pass or visa status you do not have.

TIER 3

Google · Amazon · Microsoft · Goldman Sachs · D. E. Shaw · Adobe

Example 6 [Amazon · pattern]

“Tell me about yourself.” Product company, 45-minute behavioural round, and the interviewer will drill into whatever you say.

SOLUTION

At Tier 3 the structure is the same, but the PROOF must be **drillable to three levels**. Before you speak a sentence, ask yourself: if they say “why?” three times, do I still have an answer?

“I am Meera Raghavan, final-year IT in Pune, and I work mostly on backends. **(NOW)**

The thread through the last two years is that I keep ending up as the person who measures things. **(PATH – a claim about character, which the PROOF must now earn)**

Concretely: a canteen pre-order site for our fest, about 200 users over three days. The order page took about four seconds. My first guess was that the server was slow. I put timers around three sections and found the server call itself was about 80 milliseconds, and the rest was the browser re-rendering a 120-item menu on every click. So I had been about to fix the wrong thing. I cached the menu client-side and rendered only the changed row, and it dropped to under a second. **(PROOF – and it contains a wrong first guess, on purpose)**

I am here because the thing I liked most in that project was being forced to measure before changing, and that is the default way of working I read about on your engineering blog. **(POINT)**“

Why the wrong guess is in there deliberately: it gives the interviewer a place to dig, and the digging makes you look better, not worse. “What made you check instead of just optimising the server?” is a question you **want**.

TRAP

At Tier 3 do not open with “I am passionate about technology”. It is the single most common opening line and it carries zero information. Passion is shown by what you did on a Sunday, not by the word “passionate”.

Question 2 — Strengths

STRUCTURE: NAME - PROOF - USE

NAME one strength. One. Not three.

PROOF: one short situation where it changed an outcome.

USE: one sentence on how it would help in this role.

Pick a strength that is **relevant to the job, provable**, and **not a personality adjective**. “Punctual” is not a strength for a developer. “I debug by measuring before changing” is.

Example 7 [Infosys · pattern]

“What are your strengths?”

TRAP

Weak answer. “My strengths are that I am a hard worker, a quick learner, I am punctual, I have good communication skills, and I am a good team player.”

Five adjectives, zero evidence, and every other candidate said the same five. The interviewer writes nothing down.

SOLUTION

Strong answer.

“I will pick one: I am stubborn about finding the real cause instead of patching the symptom. **(NAME)**

In my canteen project the order page was slow and I assumed the server was the problem. Before changing anything I put timers on three parts. The server was 80 milliseconds. The browser re-render was the rest. If I had trusted my guess I would have spent a week making a fast thing faster. **(PROOF)**

In a support or maintenance role that habit matters, because most of the cost of a bug is in finding it, not in fixing it. **(USE)**“

What	Weak	Strong
Count	Five strengths	One strength, defended
Type	Personality adjectives	A working habit
Evidence	None	A situation with numbers
Self-criticism	None	Admits a wrong first guess – reads as honest
Relevance	Generic	Tied to what this role actually does

SHORTCUT

Test any strength with this sentence: “Could a person who is bad at this job also claim it?” If yes, it is a weak choice. Anyone can claim “hard-working”. Almost nobody can claim “I timed it before I changed it” and back it up.

A menu of strengths that survive follow-ups

Choose from real behaviour, not from a list of adjectives. Each of these needs your own story.

Strength you can prove	The proof it needs
I measure before I change	A time your measurement contradicted your guess
I write things down so others can pick them up	A doc or README somebody else actually used
I ask early instead of being stuck silently	A time asking on day 1 saved a week
I finish the boring last 10 percent	A project you actually deployed, not just demoed
I test the ugly inputs	A bug you caught with an empty string or a duplicate
I can read somebody else’s code	A codebase you did not write and still changed safely

Question 3 — Weaknesses

This question is not a trap, but students turn it into one. The interviewer wants to know two things: can you see yourself honestly, and are you doing anything about it.

STRUCTURE: REAL - COST - FIX - EVIDENCE

REAL: a genuine weakness that is not fatal for this job.

COST: one concrete time it actually cost something. This is what makes it believable.

FIX: the specific mechanism you now use. A mechanism, not a wish.

EVIDENCE: one sign the mechanism is working.

The shape is: “Here is a true flaw. Here is the damage it did once. Here is the system I built so it does less damage. Here is proof the system runs.”

Example 8 [Capgemini · pattern]

“What is your biggest weakness?”

TRAP

Weak answer A (the fake weakness). “My weakness is that I am a perfectionist and I work too hard, sometimes I trust people too much.”

Every interviewer has heard this hundreds of times. It reads as “I will not tell you anything true”, which is worse than any real flaw.

TRAP

Weak answer B (the fatal one). “I get bored easily and I lose interest in a project after a few weeks.”

True, perhaps, but it answers the interviewer’s private question number 4 – will they stay – with a loud no. Honesty does not mean handing them a reason to reject you. Pick a **true** weakness that is **survivable**.

SOLUTION

Strong answer.

“I under-communicate when I am stuck. My instinct is to keep digging quietly. **(REAL)**

In our sixth-semester group project I lost about two days on a login bug before telling anyone. When I finally said it out loud in the group chat, a classmate recognised it in ten minutes – our session cookie was not being set because we were testing on two different ports. Those two days came out of the team’s buffer, not mine. **(COST)**

Since then I use a rule rather than good intentions: if I am stuck for 45 minutes, I write down what I tried and post it, even if it is embarrassing. I keep a timer visible. **(FIX)**

In my last two projects I have raised three things that way, and the longest one was solved in under an hour by somebody else. **(EVIDENCE)**“

What	Weak	Strong
Truth	A disguised brag	An actual flaw with a name
Damage	Claimed to have none	Two lost days, and whose buffer paid
Fix	Absent, or “I am working on it”	A 45-minute rule with a visible timer
Proof	None	Three real uses since
Risk	Answer B threatens retention	The flaw is real but survivable

Weaknesses that are safe to tell the truth about

Safe means: real, common, fixable with a mechanism, and not a direct threat to doing the job.

Safe and true	Do NOT use
I stay stuck too long before asking	I get bored and leave things
I over-explain and my updates are too long	I am bad with deadlines
I used to skip writing tests when rushed	I do not like being told what to do
Public speaking makes me nervous	I lose my temper with people
I say yes to too many tasks and then run late	I am not good at learning new things
I focus on code and forget to update the team	Anything about honesty or attendance

If your true weakness is on the right-hand list, that is a real problem to work on in your life – but an interview is not a confession booth. Choose a different true weakness from the left-hand list. Choosing which truth to tell is not lying. Inventing a truth is.

Question 4 — “Why do you want to join us?”

This is the retention question in disguise. A weak answer here is the most common silent reason for rejection, because it reads as “I applied everywhere”.

STRUCTURE: FACT - FIT - FUTURE

FACT: one specific, checkable thing about the company that you found yourself. Not “reputed”, not “market leader”, not “great work culture”.

FIT: the honest link between that fact and something you have actually done or want to do.

FUTURE: what you would want to be doing there in year one or two.

You need 20 minutes of research per company. That is the whole cost. Most candidates spend zero, so 20 minutes puts you near the top.

Where to find the FACT in 20 minutes

1. The company careers page: read the actual job description, twice. Write down three verbs from it (“maintain”, “migrate”, “support”).
2. The engineering blog, if they have one. One post, the most recent.
3. The last two quarters of company news. What did they launch, buy, or enter?
4. The tech stack listed in three of their open roles. Compare it to yours honestly.
5. One person’s public profile in the team you applied to, to learn what the work looks like.

Example 9 [TCS · pattern]

“Why do you want to join our company?”

TRAP

Weak answer. “Your company is a reputed multinational company with a very good work culture and a great brand value in the market. I want to work here so that I can enhance my skills and grow along with the organisation. Also it is one of the top IT companies in India.”

Nothing here is specific to them. Swap in any other company name and the sentence still works. That is the test, and it fails it.

SOLUTION

Strong answer.

“Two reasons, one about the work and one about me.

The job description for this role lists application support and migration work on existing client systems, and it mentions SQL tuning specifically. **(FACT – taken straight from their own JD)**

That matches what I have actually enjoyed. My project work was mostly reading and fixing code I did not write – our college attendance tool was handed to me half-finished, and the parts I liked were finding out why it broke and making the queries fast. Greenfield building was less interesting to me than that. **(FIT – honest, and it explains a preference most freshers cannot articulate)**

In the first year I would want to get properly good at one client’s system – good enough that people ask me before they change something. **(FUTURE)**“

What	Weak	Strong
Specificity	Works for any company	Quotes their own job description
Direction	What the company gives her	What she brings them , then what she learns
Honesty	Flattery	Admits she prefers maintenance to greenfield
Retention signal	None	A concrete year-one goal inside their work

SHORTCUT

The swap test. Read your “why us” answer out loud with a competitor’s name substituted. If it still makes sense, it is not an answer yet. Go find one checkable fact.

TIER 2

Singapore & Thailand · Grab · Shopee · GIC · DBS · Agoda · SCB

Example 10 [Grab · pattern]

“Why do you want to work here, and why in this region rather than in India?”

SOLUTION

Answer both halves. Students answer only the first and lose the round on the second.

“On the company: your business runs several different services on one account and one wallet, in countries with different rules and payment methods. That means almost every feature is a multi-country problem, not a single-market problem. **(FACT)**

On the work fit: the parts of my own project I found hardest were exactly the boring compatibility ones – currency rounding, time zones, one item being ordered twice at once. I would rather work on that class of problem than on one clean market. **(FIT)**

On the region: I want to be in the place where the product is actually designed and run, not in a team that receives specifications. I also want to work in a mixed team – most of my life I have worked with people who share my assumptions, and I think that has limits I cannot see yet. **(FUTURE, honest)**

On the practical side: I have looked at what this role involves for an Indian graduate – company sponsorship, and a salary floor the pass requires. I am not expecting the company to bend anything, and I would rather discuss it early than late.”

Notice what is **not** said: nothing about salary being higher, nothing about “abroad exposure”, nothing about settling permanently. Those may be true motivations, but as a stated reason they predict that you will leave when a better-paying country calls. Say the true reason that is also durable.

TIER 3

Google · Amazon · Microsoft · Goldman Sachs · D. E. Shaw · Adobe

Example 11 [Google · pattern]

“Why us? And be specific – what do you actually know about what this team does?”

SOLUTION

“I will answer the second part first, because I think it is the real question.

This team’s job listing says you own the internal tooling that other engineering teams use to ship. Two things stood out. The first is that your users are engineers, which means your users will tell you exactly what is wrong, loudly, and you cannot hide behind averages. The second is that the listing mentions reducing build feedback time, which is a latency problem with a very clear success metric. **(FACT – read carefully, and interpreted, not just repeated)**

Why that fits me: the work I have enjoyed most was internal. I built a submission tracker that three faculty members used, and the useful part was that they complained directly and I could fix it the same week. The feedback loop was one day, not one semester. **(FIT)**

Where I am honest about the gap: my experience is at 200 users, not at your scale. What I think transfers is the measuring habit, not the solutions. I expect most of what I believe about performance to be wrong at your scale, and I would want to find that out quickly rather than defend it. **(FUTURE, plus a calibrated admission)**“

SHORTCUT

Name your own gap before they find it. At Tier 3 the interviewer already knows a final-year student has not run anything at scale. Saying it yourself costs you nothing and buys a lot of credibility. Never do this with a **core** requirement of the job, only with a gap that is normal for your stage.

Question 5 — “Where do you see yourself in five years?”

The interviewer is asking: will you still be here, and do you have any direction at all.

STRUCTURE: DIRECTION - NEXT STEP - ANCHOR

DIRECTION: the kind of work you want to be doing, not a job title.

NEXT STEP: what you want to be trusted with in year one or two. Concrete and small.

ANCHOR: one line tying it to this company’s actual path.

Avoid two failures: the fantasy (“I want to be a manager in three years”) and the emptiness (“I want to grow with the organisation”).

Example 12 [Accenture · pattern]

“Where do you see yourself in five years?”

TRAP

Weak answer. “In five years I see myself in a managerial position in your esteemed organisation, leading a team and contributing to the growth of the company.”

Problems: a fresher who wants to manage in five years often means “I want to stop coding”, which is not what a delivery team is hiring for. It also names a promotion the company cannot promise.

TRAP

Weak answer B. “Sir, in five years I want to prepare for higher studies abroad and then maybe do my own startup.”

This may well be true. But you have just told the retention question its answer is “no”. If it **is** your firm plan, apply to companies where that is fine, and be honest – do not take an offer you know you will break. If it is one vague option among many, do not present it as a plan.

SOLUTION

Strong answer.

“I want to be the person a team relies on for one system end to end – the one who knows why it was built that way and what breaks it. (**DIRECTION – a kind of work, not a title**)

Getting there, in the first two years I want to move from ‘fixes tickets in a module’ to ‘owns that module’, including being the person on call for it. (**NEXT STEP**)

I know your delivery teams rotate people across client systems, so I would expect two or three domains in five years rather than one. That suits me – I would rather have depth in one and working knowledge of a few than a title. (**ANCHOR – and it shows she knows how they operate**)”

What	Weak	Strong
Unit of ambition	A title (“manager”)	A capability (“owns a system”)
Realism	Ignores how promotions work	Uses their real rotation model
Retention	Neutral or negative	Positive and specific
Checkability	Cannot be discussed further	Invites “which system would you want?”

Question 6 — “Why should we hire you?”

This one feels aggressive. It is not. It is a request for a summary, and it is your chance to repeat your best evidence one more time.

STRUCTURE: MATCH - EDGE - RISK

MATCH: the two or three requirements from their JD that you genuinely meet.

EDGE: the one thing about you that most other applicants at your level do not have.

RISK: what they would be taking on, and why it is manageable. Optional at Tier 1, strong at Tier 3.

Example 13 [Cognizant · pattern]

“Why should we hire you over the other candidates?”

TRAP

Weak answer. “Because I am a fresher, I am a fast learner, and I will work hard for your company. Whatever training you give me, I will learn it quickly and give my best.”

It promises effort, which is free. Everyone promises effort. It gives no reason to choose **this** person.

SOLUTION

Strong answer.

“Three things from your job description match what I have actually done: SQL work, supporting code somebody else wrote, and a rotating on-site shift. I have done the first two in college projects and I have already checked that the shift works for me. **(MATCH)**

The thing I would claim over most freshers is that I have maintained code I did not write. Our attendance tool was handed to me half-built, with no documentation and one comment in the whole file. I spent two weeks just reading it and writing down what each screen touched, before I changed anything. Most people my level have only built from scratch. **(EDGE)**

What you would be taking on: I have never worked to a client SLA and I do not know your process yet. The part of that I can control is speed of picking it up, and my evidence for that is the attendance tool. **(RISK)**“

What	Weak	Strong
Currency	Effort, which everyone offers	Evidence, which few have
Source	Self-description	Their own requirement list
Differentiation	None	One rare skill for a fresher, named exactly
Maturity	—	Names its own risk and limits it

Question 7 — The branch and choice questions

Common in Indian HR rounds, especially for students from non-CS branches.

THE THREE VARIANTS AND THE ONE STRUCTURE

“Why did you choose this branch?” · “You are from Mechanical – why IT?” · “Why not higher studies?”

Structure: TRUE START - TURNING POINT - PROOF OF COMMITMENT

The interviewer is checking that this is not a panic decision that will reverse in a year. So the answer must contain **costly evidence** – something you spent time on that a casual person would not have.

Example 14 [Infosys · pattern]

“You have a Mechanical Engineering degree. Why are you applying for a software role?”

TRAP

Weak answer. “Sir, actually there is no scope in core mechanical these days, there are no jobs in the market, and IT has more opportunities and better packages, so I shifted to software.”

Honest, but it says: the moment core jobs return, or another field pays more, I leave. It also says nothing about ability.

SOLUTION

Strong answer.

“I chose Mechanical at 17 because I liked machines and I did not really know what software work was. That part was not a mistake – I still use the habit of thinking about tolerance and failure. **(TRUE START – no apology, no fake passion)**

The turn came in my third semester. Our lab had to log vibration readings by hand into a register. I wrote a small script to read the sensor output file and chart it, mostly to avoid the boredom. It saved my lab partner and me about an hour a week, and I realised I had spent a whole weekend on it happily. **(TURNING POINT – a real, small, checkable moment)**

Since then: I have finished a full backend course and written about 4,000 lines of Node across three projects, and I sat the same coding test as the CS students in my college and cleared it. My CS fundamentals are self-taught, which means my theory has gaps – I have had to go back and learn operating systems properly on my own, and I am still working on databases. **(PROOF OF COMMITMENT, plus an honest limit)**

So it is not that mechanical had no jobs. It is that I found out which of the two I would keep doing on a Sunday.”

SHORTCUT

Costly evidence is the whole answer. Anyone can say they are interested. Only an interested person has a Sunday spent on it, a course finished, a repository with commits over months. Bring the cost, not the claim.

Example 15 [Wipro · pattern]

“Why are you not going for higher studies, like an MTech or a masters abroad?”

SOLUTION

“I considered it and I decided against it for now, for one reason. When I look at the things I learned fastest, all of them came from something being used and breaking – the attendance tool had three faculty using it and two real bugs, and I learned more from those two bugs than from the whole course that year.

I want a few years of that kind of learning before I decide whether a masters would add something. If I do go later, I would want to go knowing which specific area I want, rather than going because I did not know what else to do.

I am not ruling it out, and I would not want to say otherwise. What I can say is that I am not applying anywhere for it this year, and my plan for the next two years is work.”

This is the honest version. Do not promise “I will never do a masters” if that is untrue. Promise what you can actually keep: your plan for the next two years. Interviewers respect a bounded promise far more than an unbounded one they do not believe.

Question 8 — Family, hobbies, and the small talk

Indian HR rounds often open with family background or hobbies. Foreign interviewers rarely ask about family, and in some countries it is not allowed. Prepare a short, calm answer and move on.

RULES FOR SMALL TALK ANSWERS

Family: two sentences maximum. Who they are and one line about what you take from them. Never a list of every relative. Never income or caste details.

Hobbies: name hobbies you can actually discuss for two minutes. If you say “reading”, be ready to name the last book you finished and say one true thing about it. If you cannot, do not say “reading”.

Never claim a hobby to sound impressive. “Chess” invites “what opening do you play?” from an interviewer who plays.

Example 16 [TCS · pattern]

“Tell me about your family background.” Then: “What are your hobbies?”

TRAP

Weak. “My father is a bank employee, my mother is a housewife, I have one younger brother studying in 10th standard, we are originally from Nashik but we shifted to Pune in 2011, my uncle is also in the same field...” – and 90 seconds are gone with nothing scored.

Weak hobbies. “Reading books, listening to music, and playing cricket.” Then: “Which book did you read last?” and a long silence.

SOLUTION

Strong.

“My father works in a bank and my mother runs a small tailoring business from home. The thing I picked up from her is that she does her accounts every single evening, however small the day was – that is where my habit of writing things down comes from.” **(Two sentences, then stop.)**

Hobbies. “I play badminton twice a week at the college court, and I have been slowly reading through a book on how databases work – I am about halfway, at the chapter on indexes, which is what made me try adding one in my project. I am not a fast reader, so one technical book takes me a few months.”

That last answer is safe under follow-up, because every claim in it is small and true, and one of them links straight back to her project.

SHORTCUT

Link one hobby to one project. It makes the small talk earn its time and it gives the interviewer a clean bridge into technical questions on your terms.

Question 9 — Location, shift, travel and the “are you flexible” questions

These are pure retention questions. The company has been burned by people who said yes and then refused to move.

THE HONEST-FLEXIBILITY STRUCTURE

Answer the question directly first (yes / yes with a condition / no). Then give the reason. Then, if there is a real constraint, state it once, plainly.

Do not say yes to something you will refuse later. A withdrawn acceptance costs you the offer **and** your college’s relationship with that recruiter, which affects your juniors.

Example 17 [Accenture · pattern]

“Are you willing to relocate anywhere in India? And are you comfortable with rotational night shifts?”

TRAP

Weak answer A. “Yes sir, anything is fine, I am fully flexible, no problem at all.” – said by a student who has already decided he will ask for Pune later. This is the answer that ends with a rejected offer letter and a blacklisted college.

Weak answer B. “Actually I would prefer Pune because my family is here, and night shift is a little difficult for me because of health reasons, and also I cannot travel much...” – three conditions, no yes anywhere, and the interviewer stops listening at the second one.

SOLUTION

Strong answer (a genuine yes).

“Yes to relocation, anywhere in India. I have thought about it concretely – I have already checked what a shared flat costs in Chennai and Hyderabad, and I have lived away from home for two years in my hostel, so I know I can do it.

On night shifts: yes, for a rotation. I have done overnight work during our fest and I know I function on it, though I am slower the next morning, so I would want to know the rotation schedule in advance to plan around it. Is the rotation weekly or monthly here?”

SOLUTION

Strong answer (a genuine constraint – said properly).

“Relocation: yes, anywhere in India.

On night shifts I will be straight with you rather than say yes and renegotiate later. I have a medical condition that my doctor has asked me not to do rotating nights with. Fixed evening shifts are fine, and I have done those. If this role is night-rotation only, I would rather know now, and I would still be interested in other openings on your team.”

What	Weak	Strong
Order	Conditions first, answer buried	Direct answer first, then the reason
Count	Three vague constraints	One real constraint, stated once
Evidence	None	Hostel years, rent checked, a doctor’s instruction
Effect	Reads as “will renegotiate later”	Reads as “means what he says”
Extra	–	Ends with a question – turns it into a conversation

Saying no to one condition does not automatically end the process, especially if you say it early and calmly. Saying yes falsely almost always ends it later, and it ends it worse.

TIER 2

Singapore & Thailand · Grab · Shopee · GIC · DBS · Agoda · SCB

Example 18 [DBS · pattern]

“This role is based in Singapore. Do you have the right to work here, and what is your timeline?”

SOLUTION

Answer in three facts and nothing more. This is not a place for enthusiasm.

“I do not hold a work pass. I would need the company to sponsor an Employment Pass, and I understand that means the role has to meet the qualifying salary and that the application takes some weeks after an offer.

On timeline: my final semester ends in May and my degree certificate is usually issued in July. I can start any time after May with a provisional certificate if that is acceptable, or after July if the pass application needs the final one.

I have a valid passport with about eight years left, and no travel restrictions.”

TRAP

Never guess or overstate your immigration status. Do not say “I think I am eligible” about a pass you have not checked. Do not imply you hold a status you do not hold. This is one of the few places where a wrong statement can void an offer after you have joined, and in some countries it is a legal issue and not just an HR one. Say what you know. For anything else say: “I have not confirmed that – I would need to check the official source.”

What to actually know before a regional interview

You are not expected to be an immigration expert. You are expected to have looked once.

Know this	Why they ask
Whether the role sponsors, or requires existing status	Half of applicants never checked
Roughly how long sponsorship takes after an offer	It decides your start date
Your passport validity and any renewal needed	A common silent blocker
Your earliest honest start date, with reasons	They are staffing a specific quarter
Whether you need the final degree certificate or a provisional one works	Onboarding rule
That you will not ask them to bend a rule	They cannot, and asking ends it

Question 10 — “Do you have any questions for us?”

Saying “no sir, you have explained everything” is a small, real loss. Ask two or three. Write them down beforehand – it is fine to read from a notebook, and it looks prepared, not weak.

ASK ABOUT THE WORK, THE MEASUREMENT, AND THE FIRST YEAR

Good questions are ones the interviewer can answer from their own experience, and that a candidate who intends to do the job well would want to know.

Bad questions are ones you could have answered by reading their website, or that are really about leaving (leave policy, work from home on day one) rather than about working.

Hold salary, notice period and bond details for the HR or offer stage, not the first technical round. Chapter 6 covers exactly how and when to raise those.

Ask these	Why it lands
What does the first three months look like for someone in this role?	Shows you are planning to do the work
How is someone in this role judged at the end of year one?	Shows you want to be measured
What is the thing new joiners here find hardest?	Invites an honest answer and you learn a lot
Who would I be asking when I am stuck in week two?	Tests whether support exists
What did the last person in this role move on to do?	Reveals the real growth path
Is this role on an existing system or a new one?	Changes what you should prepare

TRAP

Do not ask, in a first round: “What is the salary?”, “How many leaves do I get?”, “Can I work from home?”, “Is there a bond?”, “How soon will I be promoted?”, “What does your company do?” The last one is fatal. The others are fair questions asked at the wrong time.

Example 19 [Microsoft · pattern]

“Any questions for me?” – last five minutes of a behavioural round.

SOLUTION

Two questions, and one follow-up on the answer. That last part is what separates a real question from a prepared one.

“Two things.

First: when something goes wrong in production on your team, what does the next day look like? I am asking because I want to know whether the focus is on the cause or on who did it.”

(They answer.)

“That is useful – you said you write a document afterwards. Who reads it? Is it only the team, or does it go wider?”

“Second: if I joined, what would you want me to be able to do on my own by month six that I cannot do now?”

SHORTCUT

Always follow up on their answer once. Anyone can read a prepared question off a page. Reacting to the answer proves you were listening, and listening is one of the five things being scored.

Question 11 — Money, bond and notice period: the short version

Chapter 6 covers these in full. But they can appear in a first HR round, so you need a safe, honest answer ready now. The rule is the same for all three: **do not improvise on money.**

SALARY: THE TWO-STAGE ANSWER

Stage 1 — deflect once, politely, and ask for their range. “I would like to understand the role a little more first. Do you have a band in mind for this position?” Most companies do, and most will say it. Then you are negotiating against a real number instead of a guess.

Stage 2 — if they push, give a RANGE with a REASON, not a single number. The bottom of your range must be a number you would actually accept, because that is the number you will get.

How to build the range. Two sources: (a) what your own college’s placement record shows for this company and role this year and last, (b) what public salary sites show for the same title and city, taking the middle, not the top. Say the source out loud – it turns a demand into a fact.

For a campus or mass-hiring offer the number is usually fixed. There is no negotiation on a standard fresher package, and pushing hard looks naive. Ask instead about what **is** variable: the location, the project, the training track, the joining date.

Example 20 [Accenture · pattern]

“What are your salary expectations?”

TRAP

Weak answer A. “Sir, as per company norms, whatever the company gives I am fine with it.” This hands away the entire negotiation. It also sounds like you have not valued your own work.

Weak answer B. “I am expecting 12 lakhs.” – said by a fresher with no reference point, for a role whose band is 4.5 to 6. The number is not the problem. The absence of a reason is. It reads as “he has no idea what this job is worth”, which is itself a data point about judgement.

SOLUTION

Strong answer (first round, no number yet).

“I do not want to give a number before I understand the role properly – the shift pattern and the location both change what makes sense. Do you have a band for this position? I am happy to tell you whether that works for me.”

SOLUTION

Strong answer (they insist).

“For a fresher backend role in Pune, from my college’s placement data for this role last year and this year, and from public listings for the same title, the range I have been working with is about 5 to 6.5 lakhs fixed. I would be comfortable inside that. If the role includes a night rotation I would look at the upper half of it, because that is what the listings I saw reflect.

If the number for this batch is fixed, that is completely fine – please tell me and I will not push. I would then want to talk about the location and the project instead.”

What	Weak	Strong
Anchor	Given away, or invented	A range with two named sources
Floor	Unstated – so it becomes the lowest offer	The bottom of the range is a real acceptance
Tone	Submissive or entitled	Factual, and it names a reason for the upper half
Exit	None	Explicitly accepts a fixed band without resentment

BOND AND SERVICE AGREEMENT: WHAT TO ACTUALLY ASK

Many Indian service companies attach a **service agreement** – you agree to stay a fixed period (often one to two years) or pay an amount if you leave early. It is a normal, legal commercial term. It is not a trick. But you must know four things **before you sign**, and it is entirely proper to ask them in the interview.

1. **How long** is the period, and when does it start – the joining date, or the end of training?
2. **How much** is the amount, and is it a flat figure or does it reduce month by month?
3. **What triggers it** – only resignation, or also being released, or failing training?
4. **Are original documents held?** A company asking to keep your original degree certificate is something to think very carefully about. A cheque or a signed undertaking is common; holding originals is a different matter.

Ask all four in one calm question. “Could you tell me about the service agreement – the duration, the amount, when the clock starts, and whether any original documents are held?”

Then get it in writing. Whatever you are told verbally, read the actual clause in the offer letter. If the letter and the conversation differ, the letter wins. Ask for a copy before you sign, and read the whole page. If you do not understand a clause, ask your placement officer – that is literally their job.

Signing a service agreement you intend to break is not a strategy, it is a debt. If the amount is more than you could pay, treat it as a real constraint on your choice, not as a formality. Decide before you sign, not after.

Example 21 [Infosys · pattern]

“Are you comfortable signing a two-year service agreement?”

TRAP

Weak. “Yes sir, no problem, I will sign anything.” – from someone who has not read it and who will panic in month seven when a better offer arrives.

SOLUTION

Strong.

“In principle yes, and I would want to confirm three things before signing: the exact duration, whether the clock starts at joining or after training, and the amount if it is broken.

I am asking because I would rather commit knowing the terms than sign and find out later. Two years matches what I said earlier about wanting to own one system properly, so the period itself is not the concern.”

NOTICE PERIOD: THE HONEST ARITHMETIC

As a fresher your “notice” is usually your **joining date**, and that is decided by your exam and result timeline. Be exact, not optimistic.

Write down and be ready to say, in one breath:

- Last exam date.
- Expected result date.
- Provisional certificate date, and degree certificate date.
- Whether you can join with a provisional certificate.
- Any period you are **not** available (a pre-booked family event).

If you are already working, the honest form is: “My notice period is 30 days as per my contract. I can ask about a shorter release, but I cannot promise it, and I will not leave without serving what I owe.” An employer hearing that knows exactly how you will treat **them** on your way out.

TRAP

Do not promise an impossible joining date to win the offer. A missed joining date is a cancelled offer at many companies, and your college’s placement cell finds out. Say the real date. If the real date is a problem for them, you want to know that now.

Question 12 — The rapid-fire set

These come fast, often at the end. Each needs a short answer, 20 to 40 seconds. Long answers here hurt you.

12a. “What do you know about our company?”

TRAP

Weak. “Your company is a leading IT services company founded in India with offices all over the world and lakhs of employees.” – a Wikipedia first line, and it proves nothing.

SOLUTION

Strong. “Three things, and one of them is a question. You are primarily a services business, so most engineers work inside a client’s system rather than on your own product. Your last annual report put the largest revenue share in banking and financial services. And the role I applied for sits in application support, which is the maintenance side rather than new build.

What I could not find out from outside is how a fresher gets assigned to a domain – is it allocated, or is there any choice?”

Three facts and one honest gap. The gap makes it a conversation. It also proves the facts were researched rather than recited, because a reciter would not know where the information stopped.

12b. “Rate yourself out of 10 in your best skill.”

TRAP

Weak. “9 out of 10 in Java.” Followed by one question you cannot answer, and now the number is the story instead of the skill. Equally weak: “5 or 6, sir” said about your own main skill, which invites “then why did you apply?”

SOLUTION

Strong. “In Node and Express, 6. I can build and debug a REST API with a database, handle async properly, and read someone else’s code in it. What a higher number would need, and I do not have yet, is production experience – running it under load, and knowing what breaks at scale. Where I have been weakest is testing, and that is what I have been fixing this semester.”

SHORTCUT

Give the number, then define the number. “6, and here is what 6 means and what 8 would need.” A defined 6 is far stronger than an undefended 9, because it proves you know the shape of the skill above you.

12c. “What if you get a better offer after joining us?”

TRAP

Weak. “No sir, I will never leave your company, I will stay for my whole career.” Nobody believes this, and an unbelievable promise makes your other statements cheaper.

SOLUTION

Strong. “I am not going to say I will never move – you would not believe me and it would not be true. What I can commit to is this: I will not accept this offer and then not turn up, and I will not leave inside my agreed period. Beyond that, I would want to be here as long as I am learning at the rate I want, and I would tell my manager before I started looking rather than after.”

12d. “How do you handle pressure or a tight deadline?”

SOLUTION

Strong. “By cutting scope early rather than working faster at the end. In our fest project we were two days from the event with the payment part unfinished. I proposed we ship with cash only and a simple order list, and keep payments for after. It was less impressive, but it worked for three days and 200 orders. I would rather deliver a smaller working thing on the date than a bigger broken one.”

This is a behavioural answer inside an HR question. Chapter 4 shows how to build these properly with STAR. The point here: even a rapid-fire answer is better with one concrete example than with three adjectives.

12e. “Are you comfortable working in any technology we assign?”

SOLUTION

Strong. “Yes, and I will tell you honestly what that costs. I am strongest in Node and SQL. If you put me on a Java or a mainframe project I would be slow for the first two months and I would need to be told that is expected. What I can show you is that I have picked up an unfamiliar codebase before – the attendance tool I inherited was in code I had not written, and it took me two weeks of reading before I changed anything. So: yes to any technology, with the caveat that I will ask a lot of questions in month one.”

12f. “Do you have any other offers?”

THE RULE: TRUE, BRIEF, NO LEVERAGE GAMES

If you have none: “No, this is my first process this season.” That is fine. Freshers with no offers are hired every day.

If you have one: say so simply, without naming a number and without threatening. “Yes, I have one offer from another company in this same process. I have not accepted it. I am here because this role is closer to backend work than that one is.”

Never invent an offer to create pressure. Recruiters in the same city talk to each other, and campus processes share information. If the bluff is checked, the offer is withdrawn.

12g. “Can you join immediately?”

SOLUTION

Strong. “My last exam is on 28 May and results usually come in early July. I can join any time after 1 June if a provisional certificate is acceptable for onboarding, and from mid-July if you need the final degree certificate. If you need someone before June, I should say now that I cannot do that.”

12h. “Why have you applied for a non-technical role with a technical degree?” (or the reverse)

SOLUTION

Strong. Same structure as the branch question: TRUE START, TURNING POINT, PROOF OF COMMITMENT. “I took the degree because I was good at maths and it was the sensible option at 17. The turn was in my third year when I ran registrations for our tech fest – 400 people, and the part I enjoyed was the coordination, not the website. Since then I have done two more organising roles and a business communication course. My technical background is not wasted in this role because I can talk to engineers without a translator.”

TRAP

The rapid-fire trap: length. Every answer in this section is 20 to 40 seconds. If you take 90 seconds on “rate yourself out of 10”, the interviewer learns that you cannot judge what a question is worth – and that is a real mark against you, separate from the content.

Delivery — the part nobody practises

Content is most of the score, but delivery is the part that is visible in the first ten seconds.

SIX FIXABLE DELIVERY HABITS

- 1. Filler words.** “basically”, “actually”, “like”, “you know”, “so yeah”. Fix: record 60 seconds, count them, then deliberately pause in silence instead. A silent pause sounds thoughtful. “Basically” does not.
- 2. Rising tone.** Ending every sentence as if it were a question makes true statements sound uncertain. Fix: drop your pitch on the last two words of a sentence.
- 3. The run-on.** Joining five sentences with “and”. Fix: full stop, breathe, next sentence.
- 4. Speed.** Nervous students speak at 180 words a minute. Fix: aim for 130. It will feel slow to you and normal to them.
- 5. Apologising.** “Sorry, I am not sure, maybe I am wrong, but...” said before a correct answer. Fix: say the answer, then state the uncertainty once if it is real.
- 6. Not stopping.** The worst habit. You finish a good answer, feel the silence, and start again – and the second half undoes the first. Fix: finish, then stop. Let them talk.

TRAP

The silence trap. Some interviewers stay silent for a few seconds after you answer. Many candidates panic and fill the gap, usually by weakening the answer they just gave. The silence is often just the interviewer typing notes. Let it sit. If it goes past about five seconds, ask “Would you like me to go deeper on any part of that?”

On video calls

Item	What to do
Camera	At eye level. Look at the lens when making your main point, not at your own face
Light	A window or lamp in front of you, never behind you
Sound	Wired earphones beat a laptop mic. Test with a recording, not with a friend
Network	Have your phone hotspot ready. Say so at the start: “If I drop, I will rejoin in a minute”
Notes	One page, on the wall behind the camera, big letters. Not a scrolling document

Background	Plain. Tell your family the time in advance
Name	Set your display name to your real full name, not a nickname

A full HR round, annotated

Here is a short round end to end, with the interviewer's private note after each answer. The student is the fresher from Example 4 – no internship, 7.1 CGPA, one small project.

TRANSCRIPT

I: "Hi Arun, tell me about yourself."

A: Gives the NOW-PATH-PROOF-POINT answer from Example 4. 70 seconds. Mentions the two bugs he had to fix after release. **Note:** speaks in paragraphs, not lists. Has shipped something. Says his CGPA before I ask. Flag: no internship – probe how he spends time.

I: "You said a date shifted by a day. What exactly happened?"

A: "I stored the submission time as a local-time string. When a teacher marked something at 11:40 pm, the report grouped it into the next day. I changed the column to store UTC and converted only when displaying. I also added one test with a 11:59 pm time, because that was the case I kept getting wrong by hand." **Note:** the project is real. He can describe a bug at the level of the column. The test he added is the mature part.

I: "Your CGPA is 7.1. What happened in second year?"

A: "Second year I did badly – 6.4 overall, and I failed one paper in the fourth semester, Discrete Mathematics, which I cleared on the next attempt. The honest reason is that I was not attending and I had no study method. What changed was small: I started doing the previous year papers four weeks before the exam instead of two days. Fifth semester was 7.6, sixth was 8.0." **Note:** does not blame the college or a teacher. Names the failed paper without being asked twice. Upward trend is real. Comfortable.

I: "Why our company?"

A: Gives the FACT-FIT-FUTURE answer, quoting the JD's mention of application support. **Note:** read the JD. Rare.

I: "Any location preference?"

A: "No preference – anywhere in India. I have been in a hostel for three years." **Note:** direct answer, no negotiation. Good.

I: "Questions for me?"

A: "Two. What does the first three months look like here for a fresher? And when someone breaks something in their first month, what actually happens?" **Note:** asked about the work and about failure, not about leave. Recommend proceed.

Count what won that round. One shipped project. Two bugs he could describe precisely. An honest CGPA story with numbers going up. Twenty minutes of reading their job description. None of that required an internship, a top college, or perfect English.

Practice

Do these with a recording. Writing them is 30 percent of the benefit; saying them out loud is the other 70.

PRACTICE · TIER 1 target: 90 minutes, spread over a week

1. Write your NOW sentence. 25 words maximum. Rewrite it three times, cutting one word each time.

2. List every project, course, job, volunteer role and college activity you have done. For each, write one sentence containing a number you can actually defend. Mark the ones where you had to guess – those need the word “about”.
3. Write your full NOW-PATH-PROOF-POINT introduction. Run the word count. Target 130–190 words.
4. Record it. Play it back. Count your filler words. Write the number down.
5. Record it again, with silence instead of fillers. Compare.
6. Write a strength using NAME-PROOF-USE. Then apply the test: “could someone bad at this job also claim it?” If yes, choose again.
7. Write a weakness using REAL-COST-FIX-EVIDENCE. The COST must be a real event with a real cost. If you cannot name one, the weakness you picked is not real.
8. Take one company you will actually apply to. Spend exactly 20 minutes on research. Write down three checkable facts. Then write FACT-FIT-FUTURE.
9. Apply the swap test: replace the company name with a competitor’s. Does the answer still work? Fix it until it does not.
10. Write your five-year answer using DIRECTION-NEXT STEP-ANCHOR, with no job title in it.
11. Write your three questions to ask them. For each, write what a good answer and a bad answer from them would sound like.
12. Write the truthful answer to relocation and shift. If it is a yes, write the evidence. If it is a no, write the single-sentence version.
13. Build your salary range. Write the two sources you used and the bottom number. Then ask yourself in writing: “would I sign for the bottom number?” If no, the bottom is wrong.
14. Write the four bond questions as one sentence you can say calmly.
15. Write your joining-date paragraph with real dates: last exam, result, provisional certificate, degree certificate, earliest join.
16. Answer all eight rapid-fire questions in writing, each in 40 words or fewer.

ANSWER KEY

1. Cut adjectives first, then any word that does not change meaning. “Highly motivated final-year student“ loses two words and no information.
2. Any project touched by a real user beats any project that was only demoed. Say how many users, even if it is four.
3. Under 130 words means you have no PROOF yet. Over 190 means you have two PROOFS – keep the stronger one.
4. Most students find between 6 and 15 fillers per minute on the first recording. That is normal. Three recordings usually halve it.
5. The second recording almost always sounds slower to you and better to a listener. Trust the listener.
6. Good strengths are behaviours. Bad ones are adjectives. If it fits on a certificate, it is probably an adjective.
7. If no cost comes to mind, you have chosen a fake weakness. Real ones hurt somewhere, and you remember where.
8. Checkable facts come from: the job description, the engineering blog, recent news, the stack in other open roles.
9. If the answer survives the swap test, it is about a company, not about **the** company. Add something only they have.
10. Titles are promises the interviewer cannot make. Capabilities are things you can start earning in week one.
11. A question whose answer you cannot evaluate is a question you do not really want answered.

12. The single-sentence “no” is stronger than a three-sentence one. State it once, then stop.
13. Whatever range you say, expect the bottom of it. If the bottom makes you unhappy, you have written a wish, not a range.
14. “Could you tell me about the service agreement – the duration, the amount, when the clock starts, and whether any original documents are held?”
15. Use real dates, not “around June”. “Around” is how joining dates get missed.
16. If any rapid-fire answer runs past 40 words, cut the second sentence first – it is usually the one repeating the first.

PRACTICE · TIER 2 target: 60 minutes

1. Write a location reason for one regional market that does not mention salary, “abroad exposure”, or settling. It must survive the question “why not the same work in India?”
2. Find the actual work-pass or visa page for that country, from the government site, not a blog. Write down three facts: who applies, roughly how long it takes, and what the company must do.
3. Write your three-fact availability answer: status, earliest start date, passport validity.
4. Rewrite your PROOF paragraph so that it names one cross-border or multi-market complication – a currency, a time zone, a language, or two people doing the same thing at once. Only if it is true for your project. If it is not, write instead one sentence on what you would expect to be hard, and label it clearly as an expectation.
5. Write two questions specific to working in a mixed-nationality team that are about the work, not about relocation benefits.

ANSWER KEY

1. Durable reasons: the problem is here, the scale is here, the decisions are made here, I want a team that does not share my assumptions. Fragile reasons: money, weather, permanent residence.
2. If a blog and the government page disagree, the government page is the fact. Quote the fact, not the blog.
3. Three facts, said flatly, in under 30 seconds. No apology, no enthusiasm, no guessing.
4. If your project had none of these complications, saying so and describing what you **expect** is honest and still earns credit. Inventing a multi-currency feature does not.
5. Examples: “When your team disagrees across three offices, how is it settled – the call or the document?” and “Which parts of the product change per country and which stay the same?”

PRACTICE · TIER 3 target: 2 hours

1. Take your PROOF paragraph. Write the three “why?” questions an interviewer could ask about it. Answer all three in writing. If any answer is “I do not know”, that is your study list.
2. Add to your PROOF one place where your first guess was wrong. It must be true. Write what changed your mind.
3. Write your “why us” for one product company using only facts you can cite – a job description line, a blog post, a launch. Name your source for each.
4. Write your calibrated gap: one thing normal-for-your-stage that you have not done, in one sentence, with what you think transfers anyway.
5. Write a five-year answer that would still be true if this company’s plans changed.
6. Do a 20-minute mock with a friend where their only job is to ask “why?” after every answer, five times. Note where you run out.

ANSWER KEY

1. Three levels is the normal probing depth. Level one: what did you do. Level two: why that and not the alternative. Level three: how did you know it worked.
2. Wrong first guesses make a story credible. A story where everything went right sounds rehearsed, and it usually is.
3. If you cannot cite a source, it is not a fact, it is a vibe. Vibes fail the swap test.
4. Good: “I have never worked on a system with real users beyond a few hundred, so I expect my instincts about performance to be wrong at scale – what should transfer is the habit of measuring first.” Bad: admitting you cannot do the core job.
5. If your answer depends on their current roadmap, it breaks when the roadmap does. Anchor to the kind of work instead.
6. Most people run out at level three. That is the level worth preparing.

ONE-PAGE REVISION CARD

The five private scores. Can you speak clearly · is the resume true · will you join · will you stay · are you easy to work with. Nothing else.

The substitution rule. Keep the structure. Replace every fact with your own real experience. If a slot cannot be filled truthfully, the answer is not ready.

Structures.

Question	Structure
Tell me about yourself	NOW – PATH – PROOF – POINT
Strength	NAME – PROOF – USE
Weakness	REAL – COST – FIX – EVIDENCE
Why us	FACT – FIT – FUTURE
Five years	DIRECTION – NEXT STEP – ANCHOR
Why hire you	MATCH – EDGE – RISK
Branch change	TRUE START – TURNING POINT – PROOF OF COMMITMENT
Relocation / shift	Direct answer first, then one reason, then one constraint at most

Time budget. 130 words per minute. Intro 60–90s. Strength or weakness 45–60s. Why us 45–60s. Five years 40–50s. Behavioural story 90–120s.

Three tests.

- **Swap test** – replace the company name. If it still works, it is not an answer.
- **Adjective test** – could someone bad at this job claim your strength? Then choose again.
- **Cost test** – does your weakness have a real event with a real cost? If not, it is fake.

Money, bond, notice – the safe defaults.

- **Salary.** Deflect once and ask for their band. If pushed, give a range with two named sources. The bottom of your range is the number you will get.
- **Bond.** Ask four things: duration, when the clock starts, the amount, and whether originals are held. Then read the actual clause in the offer letter before signing.
- **Notice / joining.** Real dates only: last exam, result, provisional, degree, earliest join. Never promise a date you cannot keep.

Always. Hedge numbers you did not measure (“about 200”). Name your CGPA trend before they ask. Read the job description before you write “why us”. Ask two questions and follow up on one answer.

Never. Claim work you did not do. Say yes to a location or shift you will renegotiate. Guess about a visa status. Fill a silence by weakening your own answer. Say “myself Meera”.

STAR & Behavioural Depth

Twelve stories from your own life, told in one shape, that survive three levels of "why?"

Pattern in one page

A behavioural question starts with “Tell me about a time when...”. It is not a test of your opinion. It is a test of **evidence**. The interviewer believes that what you did before is the best available guess at what you will do next.

So an answer that says what you **would** do is worth almost nothing. An answer that says what you **did** is worth everything.

WHAT A BEHAVIOURAL ROUND IS SCORING

1. **Is there a real event?** With a date, a place, other people, and a consequence.
2. **What did YOU do?** Not the team. You. The interviewer is hiring you, not your group.
3. **Did you decide anything?** A story with no decision in it is a diary entry.
4. **How do you know it worked?** A result, measured or observed.
5. **Does it hold up under “why?”** Asked three times, in a row, about the same thing.
6. **Would you do it again, and what would you change?** Reflection is scored separately from the event itself, and many candidates never offer it.

TRAP

The hypothetical trap. Question: “Tell me about a time you disagreed with a teammate.” Answer: “If I disagreed with a teammate, I would first listen to their point of view, then I would explain mine calmly, and we would find a middle ground.”

That is a **policy**, not a **story**. It scores zero on all six lines above. The fix is one word at the start: “In February, in our database project, ...” – a specific time and place forces a real event out of your memory.

STAR — the four parts, labelled

S - T - A - R

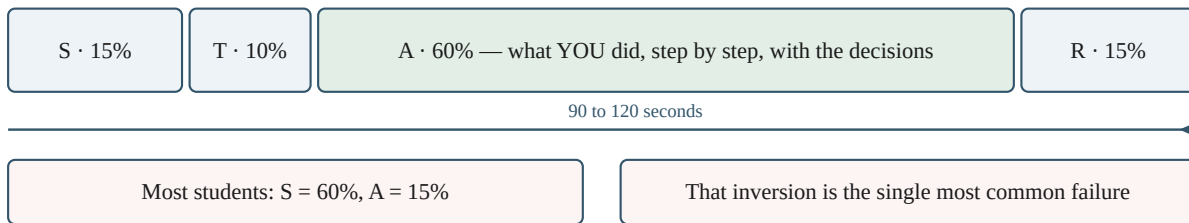
S — Situation. Where and when, in one or two sentences. Enough context that the rest makes sense. Nothing more. This is the part every student overgrows.

T — Task. What **you** were responsible for, and what “done” meant. One or two sentences. This is the part every student skips, and skipping it is why the story feels aimless.

A — Action. What **you** did, step by step, in order, with the decisions visible. This is 60 percent of the answer. If you are not saying “I”, you are not in the Action.

R — Result. What changed, with a number or an observed outcome, plus one sentence of what you learned or would do differently.

Say the labels out loud if it helps you at first. “The situation was...” “My task was...” “So what I did was...” “The result was...” Interviewers do not mind. It makes you easy to score, and easy to score is good for you.



Where your words should go. Action is the story; the rest is framing.

Check your own split with a script

You can measure this instead of guessing. Split your story into four strings and run it.

```
star.py — is your story spending its words in the right place? PY

TARGET = {"S": 0.15, "T": 0.10, "A": 0.60, "R": 0.15}

def report(name, parts):
    counts = {k: len(v.split()) for k, v in parts.items()}
    total = sum(counts.values())
    print(f"--- {name} ({total} words, ~{round(total/130*60)}s spoken)")
    for k in "STAR":
        share = counts[k] / total
        flag = "ok"
        if share < TARGET[k] - 0.07: flag = "too thin"
        if share > TARGET[k] + 0.10: flag = "too fat"
        print(f"  {k}: {counts[k]:3d} words {share:5.0%} target {TARGET[k]:.0%} {flag}")
```

Run on a weak story and a strong one:

```
Measured output TEXT

--- weak (70 words, ~32s spoken)
S: 47 words    67% target 15% too fat
T:  5 words    7% target 10% ok
A:  9 words   13% target 60% too thin
R:  9 words   13% target 15% ok
--- strong (156 words, ~72s spoken)
S: 17 words   11% target 15% ok
T: 15 words  10% target 10% ok
A: 95 words  61% target 60% ok
R: 29 words  19% target 15% ok
```

The weak story is not weak because it is short. It is weak because two-thirds of it is scene-setting. Nine words of Action is nothing to score. Fix the **split** first, then the length.

WARM-UP

build the reflex

Example 1 [warm-up]

Here are the two stories the script measured. Read the weak one and name the four faults before reading the strong one.

TRAP

Weak.

(S) “In my sixth semester we had a group project for the database subject and there were four of us in the team and the college gave us six weeks and my friends were also busy with placement preparation so everybody had a lot going on that month.”

(T) “We had to build something.”

(A) “So I did my part and we finished it.”

(R) “It went fine and sir gave us good marks.”

SOLUTION

Strong. Same project. Same student. Nothing invented.

(S) “In my sixth semester our four-person team had six weeks to build a library database project.”

(T) “I owned the search feature, and the team needed a working demo by week five.”

(A) “I listed the three queries the demo needed and timed them on a copy of the data. Title search took about two seconds on nine thousand rows, because it scanned the whole table. I added an index on the title column and rewrote the query to filter before joining. I then asked two classmates to try breaking it, and they found that an empty search string returned every row, so I added a minimum of two characters. I wrote each timing into a shared sheet so the team could see what changed and why.”

(R) “Search dropped from about two seconds to under a tenth of a second, the demo ran on time, and I reused the same index check in my next project.”

Name exactly what changed

What	Weak	Strong
Situation	47 words of excuses (busy friends, placements)	17 words: who, how many, how long, what
Task	“build something”	A named feature she owned, and a deadline
Voice	“we”, “my friends”, “everybody”	“I” in every action sentence
Decisions	None visible	Timed first · indexed · rewrote join · asked for testers · wrote it down
Numbers	None	9,000 rows · 2s · under 0.1s · 2-character minimum
Result	“good marks” – the teacher’s opinion	A measured change plus a reused method
Follow-ups invited	Nothing to ask	Why an index? Which join? What did the testers break?

SHORTCUT

The excuse detector. If your Situation contains why things were hard for you – busy, no time, bad teammates, weak internet – delete those words. They feel like context to you. They read as excuse-building to the interviewer, before anything has even gone wrong.

The “I” rule

WE BUILT IT. I DID WHAT?

Interviewers count pronouns. It is the fastest way to tell a participant from a contributor.

Use “we” for the Situation and for genuinely shared context. “We had six weeks.”

Use **“I”** for every sentence in the Action. “I timed the queries.” “I added the index.” “I asked two classmates to test it.”

If a sentence in your Action cannot honestly start with **“I”**, it does not belong in your Action. Move it to the Situation, or cut it.

And do not steal. If a teammate wrote the code, say so: “Rohit wrote the parser; my part was the schema and the queries.” Naming a teammate’s contribution makes your own claims more believable, not less.

Example 2 [Infosys · pattern]

Rewrite this Action so every sentence is honest and starts with **“I”**.

“We decided to change the database, so we moved everything to Postgres, and then we tested it and it worked.”

SOLUTION

The honest rewrite depends on what you actually did. Two possible true versions:

If you led it: “I compared our query times on SQLite and Postgres with a copy of the data and found the concurrent writes were the problem, not the read speed. I proposed the move in our Tuesday meeting with those two numbers. I wrote the migration script and ran it on a copy first. Rohit updated the application code to the new client.”

If you did not lead it: “Rohit proposed moving to Postgres. My job was to check it would not break the reports, so I ran our six report queries against both and found one that used a SQLite date function with no direct equivalent. I rewrote that one query and wrote down the difference so nobody else would hit it.”

Both are strong. The second one is a smaller role, honestly told, and it still shows a decision, a method and a result. **A small true role beats a large false one every single time.**

Build your story bank before you prepare any answer

You cannot invent a story in an interview. You can only retrieve one. So the work is done in advance: collect 8 to 10 real events from your own life, then map each one to several themes.

THE EIGHT THEMES THAT COVER MOST BEHAVIOURAL QUESTIONS

Theme	Question it answers
Conflict	Disagreement with a teammate; a difficult person
Failure	A mistake you made; something that went wrong
Deadline	Working under pressure; missing or saving a date
Leadership	Leading without a title; taking charge
Learning fast	Unfamiliar technology; steep learning curve
Helping someone	Mentoring; supporting a struggling teammate
Data decision	Using evidence; changing your mind with numbers
Pushed back	Disagreeing with someone senior; saying no

One event can serve three or four themes. The library project above serves deadline, data decision, and (with a different emphasis) learning fast. You need **events**, not answers.

bank.js — which themes are still empty?

JS

```
const THEMES = [  
  "conflict", "failure", "deadline", "leadership",
```

```

    "learning-fast", "helped-someone", "data-decision", "pushed-back",
  ];

  // Replace every line below with your own real story.
  const myStories = [
    { title: "Canteen site: menu cache",      themes: ["deadline", "data-decision"] },
    { title: "Group project: teammate quit",  themes: ["conflict", "leadership"] },
    { title: "Attendance tool: wrong schema", themes: ["failure", "learning-fast"] },
    { title: "Taught juniors Git",            themes: ["helped-someone",
"leadership"] },
  ];

  const covered = new Map(THEMES.map((t) => [t, []]));
  for (const s of myStories) {
    for (const t of s.themes) {
      if (!covered.has(t)) { console.log(`unknown theme: ${t}`); continue; }
      covered.get(t).push(s.title);
    }
  }

  let gaps = 0;
  for (const t of THEMES) {
    const list = covered.get(t);
    if (list.length === 0) { gaps++; console.log(`${t.padEnd(16)} EMPTY <- write one`); }
    else console.log(`${t.padEnd(16)} ${list.length} (${list.join("; ")} )`);
  }
  console.log(`\nstories: ${myStories.length}    themes covered: ${THEMES.length - gaps}/${THEMES.length}`);

```

Output

TEXT

```

conflict      1 (Group project: teammate quit)
failure       1 (Attendance tool: wrong schema)
deadline      1 (Canteen site: menu cache)
leadership    2 (Group project: teammate quit; Taught juniors Git)
learning-fast 1 (Attendance tool: wrong schema)
helped-someone 1 (Taught juniors Git)
data-decision 1 (Canteen site: menu cache)
pushed-back   EMPTY <- write one

```

stories: 4 themes covered: 7/8

Four real events covered seven of eight themes. That is the point of the bank: you are not writing 20 answers, you are collecting 8 events and learning to angle each one. The empty theme is your homework, and the honest way to fill it is to go and **do** something that fits.

“But nothing has ever happened to me”

It has. You are looking in the wrong place. Behavioural stories do not require an internship or a job. They require a moment where you decided something and it mattered to somebody.

Look here	What it can prove
A group project where someone did not deliver	Conflict, leadership, deadline
A bug that took you days	Failure, persistence, learning

A college fest, sports team, or club role	Leadership, priorities, working with strangers
Teaching a junior or a classmate	Helping someone, communication
A family responsibility you carried	Priorities, reliability under pressure
A part-time job or tuition you took	Customer handling, deadlines, ownership
An open-source issue or a bug report you filed	Initiative, working with strangers' code
A competition you entered and lost	Failure, reflection, second attempt
Something you organised for 20 people	Planning, dealing with people who change their minds

TRAP

Do not upgrade the story. A four-person college project does not become “I led a team of ten”. A class assignment does not become “a client deliverable”. The interviewer does not expect a fresher to have run a company. They **do** expect the story to be true, and they will find the seam – usually by asking a boring logistical question like “how often did the team meet?”

The twelve stories

Each one below is a **theme**, a **structure**, and a **filled example**. Say it once more to yourself: the filled example is a made-up student's life, shown so you can see the shape. **You must substitute your own real experience into every slot.** An answer copied from this page will break on the first follow-up, because none of it happened to you.

TIER 1

TCS NQT · Accenture · Infosys · Wipro · Capgemini

Story 1 — Conflict with a teammate

Example 3 [TCS · pattern]

“Tell me about a time you had a disagreement with someone on your team. How did you handle it?”

TRAP

Weak.

(S) “In our final year project one of my team members was very lazy and never did his work.” (T) “I had to manage everything.” (A) “I told him many times but he did not listen, so finally I did his part also and completed the project.” (R) “We submitted on time. Some people are just like that.”

Faults: the other person is a villain, there is no attempt to understand why, the “solution” is doing their work silently, and the closing line says this student will quietly resent a colleague for months rather than raise anything. Every one of those is a hiring risk.

SOLUTION

Strong.

(S) “In our final-year project, four of us, I owned the backend and a teammate owned the report screens. In week three his screens had not started.”

(T) “My job was the backend, but the demo needed both halves, and I was the one who had noticed first.”

(A) “I did not raise it in the group chat, because I did not want him to defend himself in public. I asked him to sit for ten minutes and I asked one question: what is blocking you? It turned out he had never used the chart library and had been stuck on it for a week without saying so, and he thought asking would look bad. So we split it: I spent one evening getting one chart working end to end as an example, and he did the remaining four from that

pattern. I also proposed to the group that we each post a two-line update every Monday, so this would surface in a week rather than in three.”

(R) “He finished the four screens in five days, and the Monday updates caught a second problem later that month – our data export was pointing at the old schema. What I learned is that ‘not doing the work’ usually has a reason underneath, and the reason is often that asking felt expensive.”

What	Weak	Strong
The other person	A villain with a fixed character	A person with a specific, findable blocker
First move	Public complaining, then silence	A private question: “what is blocking you?”
Solution	Did his work for him	Unblocked him with one worked example
System change	None	A weekly two-line update – and it caught a second issue
Result	“we submitted”	Five days, plus a second problem found by the new process
What it says about you	Will resent colleagues quietly	Will surface problems early and fix causes

SHORTCUT

The best conflict stories end with a mechanism, not a truce. “We talked and understood each other” is fine. “We talked, and then we changed how we work so it does not recur” is a level above, and it is the answer a team lead wants to hear.

Story 2 — A failure you owned

Example 4 [Wipro · pattern]

“Tell me about a time you failed, or made a mistake.”

THE FAILURE STRUCTURE: OWN - COST - FIX - PREVENT

OWN: say what you did wrong, in the first sentence, without softening it. **COST:** who was affected and how much. Be exact. **FIX:** what you did immediately. **PREVENT:** the change that stops it recurring. This is the part that is actually scored.

Choose a real failure with a real cost. “I once worked too hard and burned out” is the fake weakness in story clothing. Interviewers can spot it instantly.

TRAP

Weak. “Honestly I cannot think of a big failure. In one project the requirements were changed by our guide at the last minute and we could not finish, but that was not really our mistake.”

Two faults: refusing the question, and then blaming someone else inside the refusal. If you say you have never failed, the interviewer concludes either that you have never tried anything hard, or that you do not notice when you break things. Both are worse than any failure you could describe.

SOLUTION

Strong.

(S / OWN) “I deleted a week of real data from our college attendance tool. It was my mistake – I ran a cleanup query on the live SQLite file instead of the copy, because both terminals looked the same and I did not check which directory I was in.”

(T / COST) “Three faculty members had entered about 200 attendance records that week. There was no backup, because I had never set one up.”

(A / FIX) “I told the faculty member the same afternoon, before anyone noticed, and I said plainly that it was my error and that the records were gone. Then I found that the tool wrote a plain-text log line for every entry,

which I had added months earlier for debugging. I wrote a script to parse the log and rebuild about 180 of the 200 rows. For the remaining 20 the log lines were incomplete, so I listed exactly which sessions were missing and asked the two teachers to re-enter those from their own registers.”

(R / PREVENT) “About 90 percent recovered, 20 re-entered by hand, roughly two hours of other people’s time lost. I then did three things: a nightly copy of the database file, a confirmation prompt on any delete in that script, and I changed my terminal prompt to show the directory in colour. Nothing like it has happened since, but the real change was the backup – the mistake will happen again to someone, and the backup is what makes it survivable.”

What	Weak	Strong
Ownership	Deflected to the guide	First sentence names her own error
Cost	Unstated	200 records, 3 people, 2 hours of others’ time
Honesty	Hid behind “requirements changed”	Told the affected person the same day
Recovery	None	Rebuilt 90 percent from a debug log
Prevention	None	Backups, a confirm prompt, a visible directory
Maturity	–	“The mistake will happen again – the backup is the fix”

Notice the strongest single line in that answer: she designed for the **next** person’s mistake, not just her own. That is the sentence a senior engineer remembers.

Story 3 — A hard deadline

Example 5 [Capgemini · pattern]

“Tell me about a time you had to work under a tight deadline.”

TRAP

Weak. “Our project deadline was very near and we had a lot of work left, so we all sat the whole night and finished it somehow. We submitted at 9 am and it was accepted.”

“Somehow” is the problem. The story proves stamina, not judgement. Every fresher can stay up. The scored skill is **what you decided to drop**.

SOLUTION

Strong.

(S) “Two days before our college fest, the canteen pre-order site I was building still had no payment integration.”

(T) “My task was a working ordering system for three days of the fest, not a complete product. I had about 14 usable hours left.”

(A) “I wrote down the three things left – online payment, an admin dashboard, and order history – and asked which of them would actually stop the fest working. Only ordering itself would. So I proposed to the canteen manager that we go cash-on-pickup for the fest: the site takes the order and prints a token, and money is handled at the counter as it always was. He agreed, because that was already how his counter worked. I spent the 14 hours on the token screen, the counter’s order list, and testing two people ordering the last item at the same time. I left payment and the dashboard undone and I told the team in writing that they were undone, so nobody assumed otherwise on the day.”

(R) “The site handled about 200 orders over three days with no manual intervention. The missing payment feature was never even noticed by students, because they were paying at the counter anyway. What I took from it is that cutting scope early is a decision, and cutting it at 4 am is just damage.”

SHORTCUT

Deadline stories are scope stories. The interviewer is not asking “can you suffer?”. They are asking “when the date cannot move, do you know what to drop, and do you tell people?” Make the cut visible, make it early, and make it communicated.

Story 4 — Leading without a title

Example 6 [Accenture · pattern]

“Give me an example of when you showed leadership.”

LEADERSHIP WITHOUT AUTHORITY: NOTICED - PROPOSED - MADE IT EASY - FOLLOWED UP

A student has no authority. Nobody has to listen. So the story cannot be “I told them”. It must be **noticed** a problem, **proposed** something small, **made it easy** for others to say yes, and **followed up** so it did not die in week two.

SOLUTION

Strong.

(S) “In our fourth semester, six of us were preparing for the same subject and everyone was re-deriving the same past-paper solutions separately.”

(T) “Nobody asked me to fix this. I noticed we were duplicating about four hours a week each. **(NOTICED)**”

(A) “I proposed something deliberately small, because a big plan would have been ignored: one shared document, one person per unit, solutions written out fully rather than just answers. **(PROPOSED)** To make saying yes easy, I did the first unit myself, completely, before asking anyone – so they could see the format and the effort rather than imagining it. **(MADE IT EASY)** Two people did not deliver in the first week. I did not push it in the group; I asked each one privately whether they wanted a different unit, and one swapped to a shorter one and then finished. **(FOLLOWED UP)** I also set the rule that if a solution was wrong, whoever spotted it fixed it rather than complaining, and I fixed two of my own that way.”

(R) “All six units were done in nine days, and eleven people outside our group used the document by exam week. Four of us scored above our own previous semester average in that subject. What I learned is that the way to lead people who do not report to you is to do the first unit yourself.”

The weak version of this story says	Why it fails
“I was the team leader so I assigned work”	A title, not a behaviour
“I motivated everyone to work hard”	Unverifiable and unmeasurable
“I did most of the work myself”	That is not leadership, that is absorption
“I told them to finish on time”	No mechanism, no follow-up, no easy yes

Story 5 — Learning something fast

Example 7 [Cognizant · pattern]

“Tell me about a time you had to learn something new quickly.”

SOLUTION

Strong.

(S) “I inherited a half-finished attendance tool written by a senior who had graduated. It used a query builder library I had never seen, and there was one comment in 1,800 lines.”

(T) “I had to add a report screen in two weeks without breaking what already worked.”

(A) “I gave myself two days of not writing any code. I drew the six screens on paper and traced which tables each one touched, by reading the code and writing it down. That produced a one-page map, which I later put in

the README. Then instead of reading the whole library documentation, I took the three queries the tool already used and rewrote each one as plain SQL to check I understood what the library was generating. Two of them I got wrong on the first try, which told me exactly which part of the library I had misunderstood – it was doing an implicit join I had not noticed. Only then did I start on the report screen, and I built it by copying the shape of the closest existing screen rather than inventing one.”

(R) “The report shipped in nine days, and the one-page map is still the first thing in that README. The method I keep using is: translate the unfamiliar thing into something I already know, and be happy when I get it wrong, because that is where the gap is.”

SHORTCUT

Learning stories should contain a moment of being wrong. “I read the documentation and then I built it” is not a learning story, it is a reading story. The interviewer wants to see your **method for discovering you were wrong**, because that is what you will be doing on their codebase in month one.

Story 6 — Helping someone else

Example 8 [Infosys · pattern]

“Tell me about a time you helped a teammate who was struggling.”

SOLUTION

Strong.

(S) “Two juniors in my department kept losing work because they were emailing zip files to each other instead of using version control, and one of them lost two days of work that way.”

(T) “Nobody asked me to teach them. I had used Git for a year and I could see this repeating.”

(A) “I did not run a workshop, because I had sat through those and nobody remembers them. I sat with them for 40 minutes on their **own** project, not a sample one, and we used exactly four commands: status, add, commit, push. I refused to explain branching that day, even when they asked, because I wanted the four commands to become automatic first. Then I wrote a half-page card with those four commands and the two error messages they had already hit, with the fix for each. Two weeks later I checked their repository and saw they had 30 commits but all on one branch, so we did a second 30-minute session on branches, which now had a reason to exist.”

(R) “Both of them used Git for their final year project, and one of them taught it to his own group. The thing I would repeat is teaching four commands properly rather than twenty badly, on their real work rather than a toy.”

This story has no technology difficulty in it at all. It is scored on judgement: teaching less on purpose, using their real project, and going back two weeks later to check. Helping stories are about **whether the help stuck**, not about what you knew.

TIER 2

Singapore & Thailand · Grab · Shopee · GIC · DBS · Agoda · SCB

Story 7 — Deciding with data

Example 9 [Agoda · pattern]

“Tell me about a decision you made based on data. What did the data say that you did not expect?”

TRAP

The weak version: “We analysed the data and decided to optimise the database, and performance improved a lot.” No number, no alternative considered, no surprise – and the last clause, “improved a lot”, is an opinion wearing a number’s clothes.

SOLUTION

Strong.

(S) “Our canteen order page took about four seconds to load, and users were complaining during a trial run before the fest.”

(T) “I had one evening to make it usable, and I had to choose where to spend it.”

(A) “My assumption was that the server was slow, and I was about to add caching on the server. Before doing that I put timers around three sections: the network call, the JSON parse, and the render. On my phone the numbers were roughly 80 milliseconds for the server call, 15 for the parse, and about three seconds in the render. So my assumption was wrong by a wide margin – the server was not the problem at all. The render was slow because I redrew all 120 menu items whenever any quantity changed. I changed it to update only the row that changed, and I cached the menu in memory so it was fetched once per session instead of per click. I re-measured on the same phone, in the same place, because our campus wifi varies by building and I did not want to fool myself.”

(R) “It went from about four seconds to about 0.7. The lesson I actually use now is not about rendering – it is that I should measure before I optimise, because my instinct was confidently wrong and would have cost me the whole evening.”

What	Weak	Strong
Data	“we analysed”	Three timings, named, with the device
Surprise	None	Her assumption was wrong – the server was fine
Alternative	Not considered	Server caching, explicitly rejected by evidence
Rigour	–	Re-measured in the same place to avoid fooling herself
Result	“improved a lot”	4s to about 0.7s
Lesson	About the fix	About the method – which transfers to any job

Story 8 — Disagreeing with someone senior

Example 10 [Grab · pattern]

“Tell me about a time you disagreed with your manager, your professor, or a senior. What did you do?”

THE PUSH-BACK STRUCTURE: UNDERSTAND - EVIDENCE - PROPOSE - DEFER

UNDERSTAND: state their position fairly, in their words, first. **EVIDENCE:** bring something, not just an opinion. A measurement, an example, a cost. **PROPOSE:** offer a concrete alternative, ideally a cheap test rather than a full reversal. **DEFER:** say what you did after the decision – including when it went against you. That last part is what is actually being scored. Anyone can disagree. The question is whether you commit afterwards, or sulk.

SOLUTION

Strong.

(S) “Our project guide wanted us to build a mobile app for the attendance tool. I thought a mobile-friendly web page would serve the same users for a fraction of the work.”

(T) “I had one meeting to make the case, and it was five weeks before submission.”

(A) “First I made sure I understood his reason rather than assuming – he wanted offline entry, because the labs have poor signal. That was a real requirement I had not known about. So my disagreement got smaller: it was not app versus web, it was how much of the app we needed. I went away and measured two things: how long it took me to get a basic offline-capable web page storing entries locally, which was about six hours, and how many of the three teachers actually had phones we could install a test app on – it turned out one of them had a phone the toolchain we knew did not support well. I brought both facts to him and proposed a two-week test: web version first, and if offline entry failed in the lab we would still have three weeks for the app.”

(R) “He agreed to the test. The web version worked offline in the lab, so we kept it and never built the app. But the important part for me is what I did when he pushed back a second time on the report format – I disagreed

once, he explained, and I dropped it and did it his way that same day, because that one was his call and not mine. I try to separate ‘I have evidence’ from ‘I have a preference’.”

SHORTCUT

Shrink the disagreement before you argue it. Most disagreements are smaller than they look – you are often arguing about a requirement you did not know existed. Asking “what is this for?” first either dissolves the disagreement or makes it precise. Either is better than arguing.

Story 9 — Ambiguity: nobody told you what to build

Example 11 [Sea · pattern]

“Tell me about a time you had to work with unclear requirements.”

SOLUTION

Strong.

(S) “The canteen manager’s entire brief was ‘students should be able to order in advance’. No screens, no rules, no idea what happened when the kitchen ran out.”

(T) “I had to turn that into something buildable in three weeks, on my own.”

(A) “I did not start with code and I did not ask him to write a specification, because he would not have. I spent 30 minutes behind his counter at lunch watching what actually happens. Three things showed up that nobody would have told me: he refuses orders after 1:30 pm because the kitchen stops, some items sell out and he shouts it across the counter, and students pay after collecting, not before. Then I wrote eight sentences describing the flow in his words, and read them back to him. He corrected two – he wanted to close ordering at 1:00, not 1:30, to give the kitchen slack. I built the smallest version of those eight sentences and showed it to him after four days, on his phone, at the counter, rather than waiting until it was finished.”

(R) “The eight sentences became my requirements document, and two of the eight never got built because he told me after seeing the demo that they did not matter. Watching for 30 minutes was worth more than any meeting – people describe what they think they do, but they show you what they actually do.”

Ambiguity stories are about **how you reduce uncertainty**, not about coping with it. Named techniques: observe the real work, write it back in their words, demo early on their device, build the smallest version first. Use whichever ones you truly did.

TIER 3

Google · Amazon · Microsoft · Goldman Sachs · D. E. Shaw · Adobe

Story 10 — Going past what was asked

Example 12 [Amazon · pattern]

“Tell me about a time you did more than your role required. Why did you do it?”

TRAP

Weak. “I always take extra responsibility and I never say no to extra work.” A claim about character, with no event, and it accidentally says “I have poor judgement about priorities”.

SOLUTION

Strong.

(S) “My part of the library project was the search feature. During testing I noticed that the issue-return screen, which was a teammate’s, allowed a book to be returned twice, which double-counted the stock.”

(T) “Strictly it was not my code and we were four days from the demo.”

(A) “I did not silently rewrite his code, because that would have wasted the four days and insulted him. I wrote the smallest possible reproduction – return the same book twice, watch the count go from 3 to 5 – and sent him those four lines with the two screenshots. I also checked whether my own search code had the same class of bug, since it was really a missing idempotency check, and found that my ‘recently searched’ list had a similar double-count. He fixed his in an hour. I offered to write the test for the return screen since I had already written the reproduction, and he took that.

The reason I did it: our demo would have been graded as one system, and a wrong stock count on stage would have been the only thing anyone remembered.”

(R) “Both bugs were fixed before the demo. Afterwards we added one rule – before any demo, each of us tests somebody else’s screen for 15 minutes – and that caught two more issues in the next project. What I would not do is fix a teammate’s code without telling him. Finding it is helpful; taking it over is not.”

What	Weak	Strong
Event	None – a character claim	A specific bug, on someone else’s screen, 4 days out
Method	Implied heroics	Minimal reproduction, handed over, not taken over
Self-check	None	Looked for the same bug class in her own code
Motive	“I like responsibility”	A concrete consequence: the demo is graded as one system
Boundary	None	Explicitly says what she would NOT do
Durability	–	A 15-minute cross-test rule that caught two more

Story 11 — A difficult user or customer

Example 13 [Microsoft · pattern]

“Tell me about a time a user was unhappy with something you built.”

SOLUTION

Strong.

(S) “A senior faculty member told me, in front of two colleagues, that my attendance tool was ‘useless’ and that she was going back to paper.”

(T) “She was one of only three users. If she left, the tool was effectively dead.”

(A) “My first instinct was to defend it, and I did start to – I said the feature she wanted was there. That was a mistake and it made it worse. So I stopped and asked if I could watch her do one class’s entry. It took her four and a half minutes. The reason was that she taught two sections of the same subject and my screen made her re-select the subject and the date for each one, because I had built it for a teacher entering one class a day, which is how the other two worked. She was not wrong at all; my mental model of the user was wrong. I added a ‘same subject, next section’ button, which was about 20 lines, and I sat with her again while she used it. Entry went to about 50 seconds.”

(R) “She kept using it, and she reported two more issues over the next month, which is the actual win – an unhappy user who keeps talking to you is worth more than a silent one. What I changed in myself is that I now watch someone use the thing before I explain the thing.”

SHORTCUT

Include the moment you got it wrong in the room. Here it is “I started to defend it, and that made it worse.” Interviewers trust a story more when it contains a small, unflattering, specific detail – because invented stories almost never have one.

Story 12 — Competing priorities

Example 14 [Goldman Sachs · pattern]

“Tell me about a time you had more work than you could finish. How did you choose?”

SOLUTION

Strong.

(S) “In the same fortnight I had final-year project submission, two course assignments, and I had committed to running registrations for our department’s tech fest.”

(T) “All four had the same week. I could not do all four properly and I knew it by the Wednesday before.”

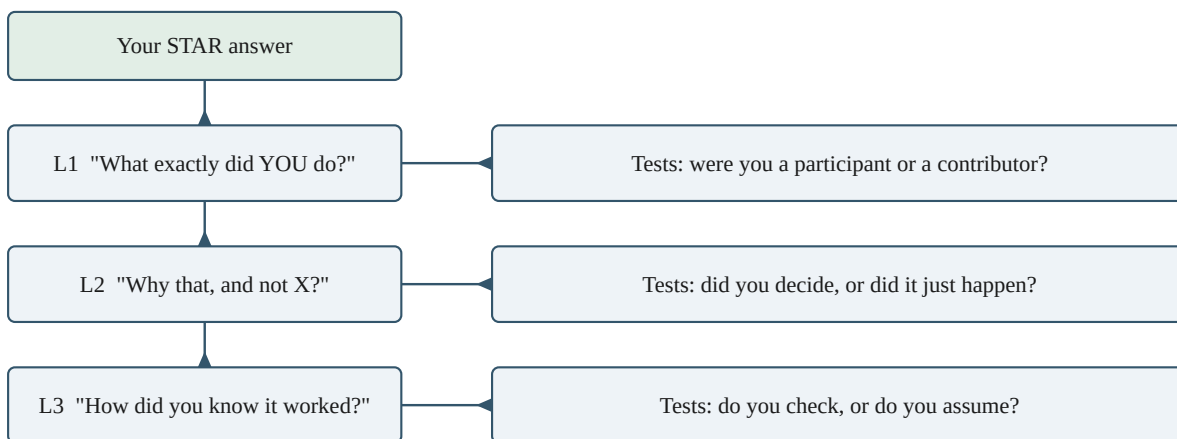
(A) “I wrote down what actually happened if each one slipped, which took about ten minutes and changed everything. The project could not slip – it was graded and there was no resit. One assignment allowed a three-day late submission at a 10 percent penalty, which I checked in the handbook rather than assuming. The fest registrations could not slip, but they could be **handed over**, which is different. So: I told the fest coordinator on the Wednesday, not the Saturday, that I could do the setup but not the desk on the day, and I trained two juniors on the registration sheet in one evening. I took the 10 percent penalty on the assignment deliberately, and I told that professor before the deadline rather than after. I put the rest into the project.”

(R) “Project submitted on time, one assignment three days late with the penalty I had chosen, the fest desk ran without me, and the two juniors ran registrations again the following year. The part I would repeat is deciding on Wednesday instead of Saturday – the same decision on Saturday would have been an apology instead of a plan.”

The scored skill is not “worked hard”. It is: **decided early, checked the real cost of each option, communicated before the deadline rather than after, and delegated one thing properly**. Any student with a busy week has a version of this story. Yours will have different items.

Follow-up probing — where most answers die

Tier 3 interviewers do not accept the first answer. They dig. Usually three levels, sometimes five, and always on the part you were vaguest about.



The probe tree. Each level tests a different thing.

PREPARE EVERY STORY TO LEVEL THREE

For each story, write the answers to:

1. **L1.** What exactly did you do, personally, in that step?
2. **L2.** Why that choice and not the obvious alternative? What did you give up?
3. **L3.** How did you know it worked? What did you measure or observe?
4. **L4 (bonus).** What would you do differently now?
5. **L5 (bonus).** What is the one thing about that story you are least proud of?

If any answer is “I do not remember”, say exactly that. **“I do not remember the number, but it was roughly a quarter of the page”** is a completely acceptable answer. **Inventing** the number is not, and a follow-up will usually expose it, because invented numbers do not stay consistent.

Example 15 [Amazon · pattern]

Here is Story 7 (the four-second page) taken down five levels. Read the probes, not just the answers – you are going to run this drill on your own stories.

SOLUTION

I: “You said you put timers in. Where exactly, and how?” (**L1 – is this real?**)

A: “Three performance.now() calls in the click handler: before the fetch, after the response was parsed, and after the DOM update, and I logged the three differences to the console. It was about six lines and I deleted them afterwards, which in hindsight I should not have done.”

I: “Why not just add server caching, which is what you had planned?” (**L2 – was it a decision?**)

A: “Because the plan was based on a guess and it would have cost me the evening. Measuring cost me about 15 minutes. The general rule I was using is that when a fix is expensive and the diagnosis is cheap, do the diagnosis first.”

I: “How did you know the fix actually worked, and not that the wifi was just better?” (**L3 – do you check?**)

A: “That was a real risk on our campus. I measured on the same phone, in the same room, within the same half hour, five clicks before and five after, and I used the median rather than the best. The before numbers were about 3.9 to 4.3 seconds and the after were 0.6 to 0.9.”

I: “What would you do differently?” (**L4**)

A: “Two things. I would not have deleted the timing code – I would have put it behind a flag so the next person can turn it on. And I would have checked on a slower phone, because mine was fairly new and the render cost is exactly what varies by device.”

I: “What part of it are you least proud of?” (**L5**)

A: “That I was one command away from spending an evening optimising something that took 80 milliseconds. The measuring habit was not mine yet at that point – I only did it because a classmate asked me how I knew the server was slow, and I did not have an answer.”

SHORTCUT

The L5 answer is the one that is remembered. Notice it gives credit to the classmate. Giving credit costs you nothing and it is one of the strongest honesty signals available, because people who invent stories always star in them alone.

What to say when you genuinely do not know

Say this	Not this
“I do not remember the exact number. It was roughly a third.”	A precise invented number
“I did not measure that. What I observed was that nobody complained again.”	“It improved by 40 percent”
“That part was my teammate’s – I can tell you what I saw from outside.”	Claiming it
“I did not consider that alternative at the time. Thinking about it now...”	“Yes we considered it” (you did not)
“I got that wrong. What I should have done is...”	Defending a decision you now think was bad

Every line on the left makes you **more** hireable, not less. Interviewers are testing calibration – whether you know the difference between what you measured, what you observed, and what you assume. Most candidates do not, and that is a much bigger problem than a missing number.

Principle-driven behavioural rounds

Some large product companies publish a written list of leadership principles or values, and their interviewers are trained to score your stories against specific items on that list. Amazon is the best-known example; several others run a similar system under different names.

HOW TO PREPARE FOR ONE, IN FOUR STEPS

- 1. Read their actual published list.** It is on their careers site, in their own words. Do not rely on a summary, and do not rely on this book – lists get revised.
- 2. Group the items into the themes you already have.** Most published lists reduce to a familiar set: ownership, customer focus, high standards, acting without perfect information, digging into detail, disagreeing and then committing, frugality or doing more with less, and developing others.
- 3. Map two stories to each group.** Two, not one. Interviewers in these loops compare notes, and telling the same story in three rounds looks thin.
- 4. Prepare each story to L5.** These loops probe harder than anything else you will face as a fresher. One story, five levels, is worth more than five stories, one level.

The vocabulary shift

You do not need to quote their principle names. In fact, quoting them mechanically (“this shows my customer obsession”) sounds rehearsed. What you **should** do is make sure the behaviour is visible in the story.

If the principle is about...	Your story must contain...
Ownership	You acted on something that was not assigned to you, and you stayed with it to the end
Customer focus	A real user, their actual words, and a change you made because of them
High standards	A time you were not satisfied with something that had already passed
Acting fast with incomplete data	A decision made without full information, and what risk you accepted
Digging into detail	A number, a log line, a query plan – something at the level of the actual thing
Disagree then commit	A disagreement you lost, and what you did the next day
Doing more with less	A constraint – no budget, no server, no time – and a smaller solution
Developing others	Someone who can now do something they could not, and how you know

Example 16 [Amazon · pattern]

“Tell me about a time you had to make a decision without all the information you wanted.”

SOLUTION

(S) “Two days before the fest, I had to decide whether the pre-order site could handle the lunch rush. I had never load-tested anything and I had no way to get 200 real students to click at once.”

(T) “The decision was binary: tell the canteen manager to use it for lunch, or tell him to keep paper for the busiest hour. Getting it wrong in the direction of over-confidence meant a queue and a bad first impression the site would never recover from.”

(A) “I could not get certainty, so I tried to bound the risk instead. I wrote a 20-line script that fired 50 concurrent orders against my laptop server and I ran it three times; nothing broke and the slowest response was about 400 milliseconds. That is not proof for 200 real users on campus wifi, and I said so explicitly rather than pretending the test was more than it was. So I proposed a fallback rather than a decision: use the site for lunch, but keep the paper token pad at the counter, and agree in advance on the trigger for switching – if the counter sees any order not appearing within a minute, he stops using the site and we go back to paper for that day. I also made sure the order list page worked without JavaScript, so a slow phone would still show him the queue.”

(R) “We used it, it held, and the paper pad was never opened. What I would keep is the agreed trigger. The decision I actually made was not ‘the system will work’ – it was ‘I know how we find out cheaply and what we do if I am wrong’.”

SHORTCUT

Acting-without-information stories should contain a reversal plan. The scored behaviour is not recklessness, it is moving forward while keeping the cost of being wrong small. “Here is how we would have found out within a minute, and here is what we would have done” is the whole answer.

One event, four questions

You do not need 12 events. You need 8, and the skill of **angling** one event at whatever was asked. Here is a single event – the library search feature from Example 1 – answering four different questions. The facts never change. The emphasis does.

HOW TO ANGLE A STORY IN TEN SECONDS

1. Hear the question. Name the **theme** it is really asking about.
2. Pick the event that has the strongest evidence for **that** theme.
3. Move the matching moment to the front of your Action, and shrink everything else.
4. End the Result with a lesson that answers **their** question, not the story’s most obvious one.

The thing you are changing is **which two sentences get the detail**.

Example 17 [TCS · pattern]

Question A: “Tell me about a time you improved something.”

SOLUTION

(A, emphasis on the improvement) “...I timed the three queries the demo needed. Title search took about two seconds on 9,000 rows because it scanned the whole table. I added an index on the title column and rewrote the query to filter before joining. It went to under a tenth of a second.”

(R) “Search went from about 2 seconds to under 0.1, and I have used the same ‘time it first’ check on every project since.”

Example 18 [Shopee · pattern]

Question B: “Tell me about a time you found a problem nobody else had noticed.”

SOLUTION

(A, emphasis on the discovery) “...Nobody had reported the search being slow, because with our test data of 50 books everything felt instant. I loaded a copy of the real catalogue – about 9,000 rows – before the demo, mostly out of caution, and that is when the two seconds appeared. So the problem only existed at real size, and our test data was hiding it. I then checked the other two queries at real size as well, and one of them was fine, which told me it was not a general database problem.”

(R) “We found it a week before the demo instead of during it. The habit I kept is loading realistic data early – toy data hides exactly the class of bug that matters.”

Example 19 [Wipro · pattern]

Question C: “Tell me about a time you worked with feedback from others.”

SOLUTION

(A, emphasis on the feedback loop) “...Once it was fast I asked two classmates to try to break it, before I told anyone it was done. They found something I would never have tried: an empty search returned all 9,000 rows and froze the page. I had only ever typed real words into it. I added a two-character minimum and a clear message instead of an empty list, and I asked them to try again rather than assuming my fix matched what they meant.”

(R) “The empty-search bug would have appeared in the demo in front of the class. What I do now is hand the thing to someone else before I call it finished, because I only test the inputs I already imagined.”

Example 20 [Adobe · pattern]

Question D: “Tell me about a time you had to explain something technical to someone who did not have your background.”

SOLUTION

(A, **emphasis on the communication**) “...My teammates could see the search was faster but not why, and one of them wanted to add the same index to every column ‘to be safe’. So instead of talking about B-trees I used the back of the book: I asked him how he would find a title in a 9,000-page book with no index, then with one, then asked what it would cost to keep an index for every word on every page updated. He worked out the trade-off himself in about a minute. I then wrote the three timings into our shared sheet so the reasoning was visible without me.”

(R) “He dropped the every-column idea and indexed one more column that genuinely needed it. The lesson I keep is to explain with something the person already has in their hands, and to write the numbers down so the explanation outlives the conversation.”

Four questions, one event, zero invention. Every fact appears in the original story. This is why the story **bank** matters more than a set of prepared answers: prepared answers only fit the questions you predicted.

Depth control: the 30-second and the 2-minute version

Not every behavioural question deserves two minutes. Some are asked in a 15-minute HR round as filler; some are the whole point of a 45-minute loop. Prepare both lengths for your top stories.

TWO LENGTHS, ONE STORY

The 30-second version. One sentence of Situation and Task combined, two sentences of Action with only the decisive step, one sentence of Result with the number. Then **offer the depth**: “I can go into how I found it, if that is useful.”

The 2-minute version. Full STAR, with the decisions visible and one moment where you were wrong.

How to choose. Short round, rapid questions, or a question asked in passing – go short and offer. Dedicated behavioural round, or the interviewer leans in and says “walk me through it” – go long.

Example 21 [Capgemini · pattern]

Compress Story 3 (the fest deadline) to 30 seconds without losing the decision.

SOLUTION

30-second version.

“Two days before our college fest, the ordering site still had no payment feature and I had about 14 hours. I listed what was actually blocking the fest, and only ordering itself was – so I proposed cash at the counter, which is how they already worked, and spent the time on the order flow instead. It handled about 200 orders over three days, and nobody missed the payment feature. I can go into how I chose what to cut, if that is useful.”

That is 78 words, about 36 seconds. The **decision** survives the compression, which is the test. If your short version loses the decision, you have compressed the wrong sentences.

TRAP

Do not compress by speaking faster. A two-minute answer delivered in 75 seconds is not a short answer, it is an unintelligible one. Compress by **removing sentences**, and keep your speed at about 130 words per minute.

Behavioural questions in disguise

Some questions do not begin with “tell me about a time”, but they are still scored on evidence. The fix is the same: answer with a specific event.

a. “How do you handle criticism?”

TRAP

Weak. “I take criticism positively and I see it as an opportunity to improve myself.” A policy statement. Everyone says it. Nothing is scored.

SOLUTION

Strong. “I will give you an example rather than a claim. A faculty member told me in front of her colleagues that my attendance tool was useless. My first reaction was to defend it, and I did start to, which made it worse. What worked was asking to watch her use it – she was doing an entry that took four and a half minutes because she taught two sections and my screen made her re-select everything. The criticism was accurate and my model of the user was wrong. I added one button, about 20 lines, and entry dropped to about 50 seconds.

What I changed in myself: I now ask to see the problem before I explain my design. The defending instinct has not gone away; I just do not act on it in the first minute.”

SHORTCUT

“The instinct has not gone away, I just do not act on it” is a strong sentence, because it is honest about a habit rather than claiming a cure. Claims of complete self-transformation are not believed by anyone who has managed people.

b. “How would you describe your working style?”

SOLUTION

Strong. “Three things, each with an example.

I write things down – the one-page map of which screen touches which table is still in that project’s README, and I wrote it for myself before anyone asked.

I measure before I change – I have been wrong about the cause often enough that I do not trust my first guess any more.

And I am slower than most to ask for help, which is the part I am actively working on with a 45-minute rule. That is a genuine cost, not a hidden strength.”

Ending a “working style” answer on a real limitation makes the first two claims credible. An answer with only good parts is a sales pitch, and it is heard as one.

c. “What motivates you?”

TRAP

Weak. “I am motivated by challenges and by learning new things.” True of everyone, evidence for nothing.

SOLUTION

Strong. “Somebody using the thing. The clearest signal I have is that I worked much harder on the attendance tool, which three teachers actually used, than on a bigger project that was only graded. With the tool I got messages when it broke, and that was oddly motivating – it meant it mattered to someone.

The honest opposite is also true: I lose energy on work that nobody will use, and I have to manage that deliberately rather than pretend it is not the case.”

d. “Tell me about a time you received feedback you disagreed with.”

SOLUTION

Strong. Structure: HEARD - CHECKED - ACTED.

“My project guide told me my report was too long and that I should cut the background section by half. I disagreed – I thought the background was what made the design choices make sense.

(HEARD) Rather than argue from my opinion, I asked what specifically was not working, and he said he could not find the actual contribution. That was a different complaint from ‘too long’.

(CHECKED) I gave the report to two classmates with one question: after reading it, what did I build? Both of them described the background rather than my part. So he was right about the problem, even though I still think ‘cut it in half’ was the wrong prescription.

(ACTED) I kept the background at about the same length but moved it after my contribution and added a one-paragraph summary at the top. He accepted it. What I take from that is to separate the observation from the prescription – people are usually accurate about what is wrong and less reliable about the fix.”

TIER 2

Singapore & Thailand · Grab · Shopee · GIC · DBS · Agoda · SCB

Cross-cultural behavioural questions

Regional rounds add a specific worry: this person has only worked with people exactly like themselves, in one language, in one office. The questions test whether you can adapt.

Example 22 [Grab · pattern]

“You would be working with people in three countries who do not share your first language or your working hours. Tell me about a time you had to adjust how you communicate.”

TRAP

Weak. “I am very adaptable and I can adjust with any kind of people. I have good communication skills and I believe respect is the most important thing.” Three claims, no event, and it would be said identically by someone who has never worked with anyone.

SOLUTION

Strong.

(S) “Our fest team included students from two states and a group whose classes ran in the evening, so we were rarely in the same room, and our shared language was English, which was the second or third language for most of us.”

(T) “I was coordinating registrations and I kept getting things done wrong – not through unwillingness, but because my instructions in a group chat were being read differently by different people.”

(A) “I changed three specific things. First, I stopped giving instructions verbally in the corridor and started writing them, because a written message can be re-read by someone who did not catch it the first time. Second, I stopped asking ‘is that clear?’, which always gets a yes, and started asking the person to tell me back what they were going to do – twice this revealed a completely different plan. Third, for the evening group I wrote a short end-of-day summary with what was decided and what was open, so they did not have to reconstruct it from 60 chat messages. I also started writing dates as ‘12 March’ rather than in numbers, after two people read the same date in different orders.”

(R) “Registration errors dropped to almost none in the last two weeks, and the end-of-day summary was kept for the next year’s team. What transfers to a multi-country team is the repeat-back habit – it is the only cheap way I know to find out that ‘yes’ meant ‘I did not follow’.”

Behaviour	Why it travels across cultures and time zones
Write it, do not say it	Re-readable by a non-native speaker, and by someone who was asleep

Ask for a repeat-back, not “is it clear?”	In many cultures “yes” is politeness, not agreement
Written end-of-day summary	The only handover a different time zone actually gets
Unambiguous dates and numbers	Date order and decimal marks differ by country
Name the decision and the owner	Prevents “everyone thought someone else had it”
Ask, do not assume, about how disagreement is shown	Directness is normal in some places, rude in others

Example 23 [DBS · pattern]

“Tell me about a time you worked with someone whose way of working was very different from yours.”

SOLUTION

(S) “In my final-year team, one teammate planned everything on paper before writing a line, and I prefer to build a rough version and fix it. We annoyed each other for about two weeks.”

(T) “We shared the backend, so our two styles were colliding in the same files.”

(A) “I stopped treating it as a personality clash and tried to find where each style was actually better. His planning was clearly better for the database schema, because changing a schema after data exists is expensive – I had already caused one painful migration by guessing. My way was better for the screens, where the fastest way to find out what was wrong was to show somebody something. So we split it that way explicitly, and I said out loud that the schema decision was his to make. I also asked him to keep his plan to one page, because his first version was six and nobody would have read it.”

(R) “We stopped colliding, the schema only changed once after that, and the screens went through four quick rounds with our guide. What I took from it is that a working-style difference is usually a disagreement about where the expensive mistakes are, and those are different in different parts of a system.”

SHORTCUT

Never resolve a style difference by declaring one style correct. The answer that scores is the one that finds where each is cheaper. It shows you can work with people you disagree with, which is the actual thing being tested.

A behavioural round, annotated

Twenty minutes, three questions, with the interviewer’s notes.

TRANSCRIPT

I: “Tell me about a time you disagreed with someone and lost the argument.”

A: Gives Story 8 – the app-versus-web disagreement – and ends on the **second** disagreement, the report format, which he lost and then implemented the same day. Note: answered the question as asked, including the losing part, which half of candidates skip. Committed the same day. Distinguishes evidence from preference.

I: “You said you measured six hours to build the offline web version. How did you know it was six and not three days?” L3 probe on the one number in the story.

A: “I did not estimate it – I built it. I spent one evening, about six hours, getting a page that stored entries in local storage and pushed them when the signal returned. It was ugly and it only handled one field. But I brought a working thing to the meeting rather than an estimate, because my estimates at that point were not worth much.” Strong. The number was measured, not guessed, and he knows the difference. Prototyped instead of arguing.

I: “What is a mistake you made in that project?”

A: Gives the deleted-data story. Owns it in the first sentence. 200 records, 3 people, recovered about 180 from a debug log, then backups, a confirm prompt and a coloured prompt. Told the faculty member the same

afternoon before it was discovered. The prevention is systemic, not a promise to be careful. Would hire on this answer alone.

I: “Last one – what would you have done if the log file had not existed?” Testing whether the recovery was luck or method.

A: “Then about 200 records were simply gone and my only honest option was the same one I took for the last 20 – go to each teacher, tell them exactly which sessions were affected, and ask them to re-enter from their registers. It would have been about ten times the apology and roughly a day of their time. The log made the recovery possible, but the log was an accident – I had added it months earlier for a different reason. That is exactly why I set up the nightly copy: I do not want the next recovery to depend on luck.” Does not pretend the log was foresight. Separates luck from method. Recommend hire.

The anti-pattern list

TRAP

Twelve ways a behavioural answer fails.

1. **Hypothetical.** “I would...” instead of “I did...”.
2. **All Situation.** Two-thirds scene-setting, nine words of action.
3. **The “we” fog.** No sentence in the Action starts with “I”.
4. **No decision.** Things happened; nobody chose anything.
5. **No result.** The story just stops.
6. **Fake numbers.** Precise figures for things you never measured.
7. **Villains.** A lazy teammate, an unreasonable professor, a stupid user.
8. **The refused failure.** “I cannot think of a failure.”
9. **Upgrading.** A four-person project becomes “a team of ten”.
10. **Same story three times.** A round has three interviewers and they compare notes.
11. **No reflection.** Nothing learned, nothing you would change.
12. **Endless.** Four minutes with no structure. If you cannot see the end, stop and ask: “Is that the level of detail you wanted, or should I go deeper on one part?”

SHORTCUT

The rescue line. If you are two minutes in and lost, do not push on. Say: “Let me get to the point – what I actually did was...” and give your Action and Result. Interviewers respect a self-correction far more than a slow collapse, and every one of them has seen candidates never recover from a rambling start.

The one-page story sheet

Write one of these for each of your 8 to 10 events. Handwriting is fine. Keep them in one file and reread them the night before an interview – not to memorise the words, but to reload the facts.

STORY SHEET TEMPLATE — COPY THIS EIGHT TIMES

Event name (for you only):

When and where:

Themes it can serve:

S (max 2 sentences, no excuses):

T (what I owned, what “done” meant):

A (every sentence starts with “I”; mark each decision with a star):

R (number, plus what I would change):

Numbers, marked M / O / G:

What teammates did (credit them by name):

L1 — exactly what I did:

L2 — why that, not the alternative:

L3 — how I knew it worked:

L4 — what I would do differently:

L5 — least proud of:

30-second version (about 70 words):

The sheet is a fact store, not a script. On the day you will say it differently, and that is correct. What must not change between tellings are the facts and the numbers – inconsistency across two interviewers in the same loop is one of the few things that reliably fails a candidate.

Practice

PRACTICE · TIER 1 target: 3 hours, spread over a week

1. List 10 real events from your life: projects, group work, a fest, teaching, a job, a family responsibility, a competition. One line each. No judging yet.
2. Fill in the `myStories` array of `bank.js` with your own events and themes. Run it. Write down which themes come back EMPTY.
3. For the empty themes, either find a forgotten event, or accept the gap and plan something real in the next month to fill it. Do not invent one.
4. Take your strongest event and write it out in full STAR, with the four labels written in.
5. Split it into four strings and run `star.py`. Fix the split until every part says “ok”.
6. Rewrite your Action so every sentence starts with “I”, and cut any sentence that cannot honestly do so.
7. Find every number in your story. For each, mark it M (measured), O (observed) or G (guessed). Rewrite every G with “about”, or remove it.
8. Delete every excuse from your Situation: busy, no time, bad teammates, poor internet.
9. Record yourself telling it. Target 90 to 120 seconds.

10. Repeat for a failure story, using OWN-COST-FIX-PREVENT. Choose a failure with a real cost.
11. Fill one complete story sheet, including the 30-second version, for your strongest event.
12. Take that one event and angle it at four different questions: improved something, found a problem, used feedback, explained something. Change only which sentences get the detail.

ANSWER KEY

1. Ten events is plenty. Most students stop at three because they only look at “projects”.
2. Two or three empty themes is normal on the first run. Empty is information, not failure.
3. The honest fill is: take a club role, teach a junior, file a real bug report, help on an open-source issue. A month is enough for a true small story.
4. If you cannot write the T line, you did not own anything in that event – pick a different one.
5. Almost every first draft comes back “S too fat, A too thin”. Cut the Situation to two sentences first; that alone usually fixes both.
6. Sentences that resist “I” are usually background. They belong in S, or nowhere.
7. Interviewers do not punish “about”. They punish precision you cannot defend.
8. Excuses in the Situation predict excuses on the job. That is exactly how they are read.
9. Under 60 seconds means no Action detail. Over 150 means you are telling two stories.
10. If your failure has no cost, it is not a failure, it is a humblebrag in disguise.
11. The 30-second version is right when the decision survives. If only the outcome survives, you cut the wrong sentences.
12. If angling requires you to add a fact that is not in the original event, stop – you have started inventing. Pick a different event for that question.

PRACTICE · TIER 2 target: 2 hours

1. Write a story where you decided something using evidence, and the evidence **surprised** you. The surprise is the point.
2. Write a disagreement story using UNDERSTAND-EVIDENCE-PROPOSE-DEFER where **you lost**. Say what you did the next day.
3. Write an ambiguity story. Name the specific technique you used to reduce the uncertainty – watching the real work, writing it back in their words, demoing early, building the smallest version.
4. Take one story and rewrite the Result so it contains a measured number, an observed outcome, and a lesson that is about your **method**, not about the specific fix.
5. Write a story involving someone whose assumptions were different from yours – a different department, a different year, a non-technical user, a different language. What did you change in how you communicated?

ANSWER KEY

1. No surprise means you probably did not use the data, you just decorated a decision you had already made.
2. The “I lost and committed anyway” story is rarer and more valuable than the one where you were right. Most people only prepare the second.
3. Vague coping (“I just figured it out”) scores nothing. A named technique scores, because it can be repeated on their team.
4. Measured: “0.6 to 0.9 seconds, median of five.” Observed: “she stopped asking for the paper form.” Method lesson: “measure before optimising.”
5. Good answers change something concrete: shorter sentences, a written summary after the call, a diagram instead of a paragraph, confirming understanding by repeating it back.

PRACTICE · TIER 3 target: 4 hours, and one mock

1. Take your three strongest stories. For each, write the L1, L2, L3, L4 and L5 answers in full.
2. Mark every number in those answers M, O or G again. Any G that survives to L3 will break you.
3. For each story write the sentence you would say if you did not remember a detail. Practise saying it without apologising.
4. Find the published values or principles list of one company you are applying to, from their own site. Group the items into themes. Map two of your stories to each group. Note where you have nothing – that is your gap, and it is better to know it now.
5. Write one story that contains a decision made without enough information, including the reversal plan and the trigger for using it.
6. Do a 30-minute mock where a friend asks one behavioural question and then only asks “why?” or “how did you know?” until you run out. Write down the level at which you ran out.
7. For your weakest story, write the answer to “what are you least proud of in that?”

ANSWER KEY

1. Most candidates are solid at L1, thin at L2, and empty at L3. Preparing L3 is the single highest-value hour in this chapter.
2. Guessed numbers drift between tellings. A second interviewer in the same loop will notice.
3. “I do not remember the exact figure – it was roughly a third” said calmly, once, with no apology, reads as confidence, not weakness.
4. Two stories per group. If three groups have nothing, you do not have a story problem, you have an experience problem – and the fix is real work in the world, not better writing.
5. The reversal plan is the answer. Speed without a way back is recklessness, and it is scored as such.
6. Running out at L2 means you are describing events, not decisions. Running out at L4 is normal and fine.
7. If nothing comes to mind, you are probably still defending the story. Every real story has a part you would rather not mention; that part is the most credible thing you own.

ONE-PAGE REVISION CARD

STAR, and the word budget. S 15% · T 10% · A **60%** · R 15%. Ninety to 120 seconds. Say the labels out loud if it helps. Most students invert S and A – that is the top failure.

The “I” rule. “We” in the Situation. “I” in every Action sentence. If a sentence cannot honestly start with “I”, it is background – move it or cut it. Name what teammates did; it makes your own claims believable.

The four sub-structures.

Story type	Structure
Failure	OWN – COST – FIX – PREVENT
Leadership without a title	NOTICED – PROPOSED – MADE IT EASY – FOLLOWED UP
Disagreement	UNDERSTAND – EVIDENCE – PROPOSE – DEFER
Deadline	It is a scope story: what did you drop, how early, and who did you tell

The eight themes. conflict · failure · deadline · leadership · learning fast · helping someone · data decision · pushed back. Eight to ten real events cover all of them. Collect **events**, not answers.

The probe ladder. L1 what did YOU do · L2 why that and not X · L3 how did you know it worked · L4 what would you change · L5 what are you least proud of. Prepare your top three stories to L5.

Calibration beats precision. M (measured) / O (observed) / G (guessed). Say “about” for anything guessed. “I do not remember the exact number” is a strong answer. An invented number is a failed interview.

The substitution rule, one last time. Every filled example in this chapter belongs to an invented student. Keep the structures. Replace every fact with your own real experience. If a slot cannot be filled truthfully, the answer is not ready – go and do something real, however small, and then fill it.

Never. Hypotheticals. Villains. Upgraded team sizes. Invented numbers. “I have never failed.” The same story in three rounds. A four-minute answer with no end in sight.

Defending Your Project

They are not testing the project. They are testing whether you built it.

Pattern in one page

Your resume says you built something. The next twenty minutes decide whether the interviewer believes you. That is the whole round. Not “is this a good project”. Not “is this a big project”. Only: **did this person actually build this, and do they understand what they built?**

THE FOUR THINGS A PROJECT ROUND SCORES

- 1. Ownership.** Can you say what YOU did, in a group of four, without vague words like “we worked on” and “I was involved in”?
- 2. Depth.** When they push one level deeper than your prepared line, is there anything there? Most candidates have one layer. Good candidates have four.
- 3. Trade-off thinking.** Every choice you made had an alternative. Do you know what it was, and why you did not pick it?
- 4. Honesty.** Do you say “I do not know” cleanly, or do you bluff? Bluffing is the fastest way to fail this round. An interviewer who catches one bluff stops believing everything else you said.

Everything in this chapter is a **structure**, not a script. Every filled example uses a made-up project so you can see the shape. **You must replace every fact with your own real project.** If you memorise my example and say it in a room, the first follow-up question will destroy you, because you cannot answer follow-ups about a project you did not build.

Why candidates fail this round

What the interviewer hears	What they write down
“We used MongoDB because it is fast.”	Repeating a tutorial. No real reason.
“My friend did the backend, I did the frontend, but I know all of it.”	Cannot separate own work from team work.
“There were no major problems.”	Either the project is trivial or he is hiding.
“It can handle any number of users.”	Never measured anything. Never load tested.
A confident wrong answer about their own code.	Bluffing. Stop trusting this candidate.
“I would add more features.”	No engineering reflection, only product wishes.

The Project Defence Sheet

Before any interview, fill this sheet for **each** project on your resume. One page per project. Written by hand is fine. You prepare this once and it serves every company.

THE 12 LINES YOU MUST BE ABLE TO SAY WITHOUT THINKING

- 1. Problem.** Who had a pain, and what was the pain? One sentence, no technology words.
- 2. Users.** Real or imagined? How many? Be honest: “a demo with 30 test users from my class” is a perfectly good answer.
- 3. What it does.** Three features, named. Not ten.
- 4. Stack.** Every technology, and **one honest sentence of why** for each.

5. **My part.** What you personally wrote, by file or module or feature.
6. **Their part.** What your teammates wrote. Naming this makes your own claim believable.
7. **Data model.** The 4–6 main tables or collections and how they link.
8. **One request end-to-end.** A user taps a button. Trace it to the database and back.
9. **One number.** Anything you measured: rows, requests, milliseconds, file size, users, test count. One real number beats ten adjectives.
10. **What broke.** One real bug or failure, with the cause and the fix.
11. **A trade-off.** One choice with a named alternative you rejected, and why.
12. **What you would change.** Two things, engineering, not features.

SHORTCUT

Line 9, the **one number**, is the highest-value line on the sheet. Candidates who have a number sound like engineers. Candidates who have none sound like students describing a poster. If you never measured anything, go measure something tonight — even `console.time()` around your slowest page gives you a number you own.

A filled sheet — the example project used in this chapter

I will use one made-up project all chapter so you can watch it survive hard questions. **Your project is different. Substitute yours everywhere.**

Line	Filled
Problem	Our hostel mess had a paper sign-up sheet for meals. Cooks over-ordered food and threw it away.
Users	78 students in one hostel block, 6 weeks of real use. Not a company, a real but small user base.
What it does	(1) A student marks meals in or out for tomorrow. (2) The cook sees a count per meal by 9 pm. (3) A weekly waste report.
Stack	Node + Express API, PostgreSQL, plain HTML/JS front end, deployed on one small cloud VM.
My part	The API, the database schema, the 9 pm cut-off job, and the report query.
Their part	Ravi wrote the front end. Meera did the deployment and the login screen.
Data model	<code>students</code> , <code>meals</code> , <code>opt_ins</code> , <code>daily_counts</code> .
One request	Student taps “skip dinner” → <code>POST /optin</code> → check cut-off time → upsert row in <code>opt_ins</code> → return new count.
One number	The report page took 4.1 s with 6 weeks of data; after one index it took 0.2 s.
What broke	Two taps on a slow phone created two rows and the count went wrong twice.
Trade-off	Chose Postgres over MongoDB because the weekly report is a join and a group-by.
Would change	(1) Make the opt-in endpoint idempotent properly. (2) Store counts in one place, not two.

The 90-second project pitch

Almost every project round opens the same way: “**Tell me about this project.**” You get roughly 90 seconds before they interrupt with a real question. Those 90 seconds decide which questions they ask.

PITCH STRUCTURE — SIX BEATS, ABOUT 15 SECONDS EACH

1. **Pain.** Who suffered, and how.
2. **Fix.** What the thing does, in one line.
3. **Scope.** How big and how real. Honest size.

4. **Stack.** Named, with one reason.
5. **My part.** Your own words: “I wrote...”
6. **Result.** The one number, plus one honest limit.

Then **stop talking**. Silence invites them to choose the next question, and you have just given them six safe places to choose from.

Example 1 [Infosys · pattern]

“Tell me about the project at the top of your resume.”

TRAP

WEAK answer

“So basically this is a full-stack web application which I have made using the MERN stack. It has login, signup, user dashboard, admin dashboard, and CRUD operations. I have used React for the frontend because React is very popular and fast, Node and Express for backend, and MongoDB as database because it is a NoSQL database and it is scalable. We worked on it as a team of three and it was a great learning experience. It is fully responsive and can handle any number of users.”

SOLUTION

STRONG answer

“Our hostel mess used a paper sheet for meal sign-ups, so the cooks guessed the count and threw food away most nights.

We built a small web app where a student marks tomorrow’s meals in or out, and the cook sees the counts at 9 pm.

It ran for six weeks in one block, about 78 students, so it is small but the users were real.

It is a Node and Express API with Postgres. We chose Postgres because the weekly waste report is a join and a group-by, and I wanted SQL for that.

I wrote the API, the schema and the 9 pm cut-off job. Ravi wrote the front end and Meera did deployment.

The report page originally took 4.1 seconds; adding one index on `opt_ins(meal_date)` brought it to 0.2 seconds. The main weakness is that the opt-in endpoint is not properly idempotent — I can talk about that if it is useful.”

Name exactly what changed

Beat	Weak version	Strong version
Pain	Missing. Starts from technology.	One concrete human problem: food thrown away.
Fix	“CRUD operations, dashboards.”	Two named features a person can picture.
Scope	“Can handle any number of users.”	78 students, 6 weeks. Small and honest.
Stack reason	“React is popular”, “Mongo is scalable”.	A reason tied to this app’s query shape.
My part	“We worked on it as a team.”	Named modules for self and for teammates.
Result	“Great learning experience.”	4.1 s → 0.2 s, plus a named weakness.
Ending	Trails off into buzzwords.	Stops, and offers one door to walk through.

TRAP

The buzzword trap. “Scalable”, “robust”, “user-friendly”, “optimised”, “industry standard” are worth zero marks each. They are adjectives with no evidence. Every one of them invites the question “how do you know?” — and if you cannot answer that, the adjective has cost you, not helped you.

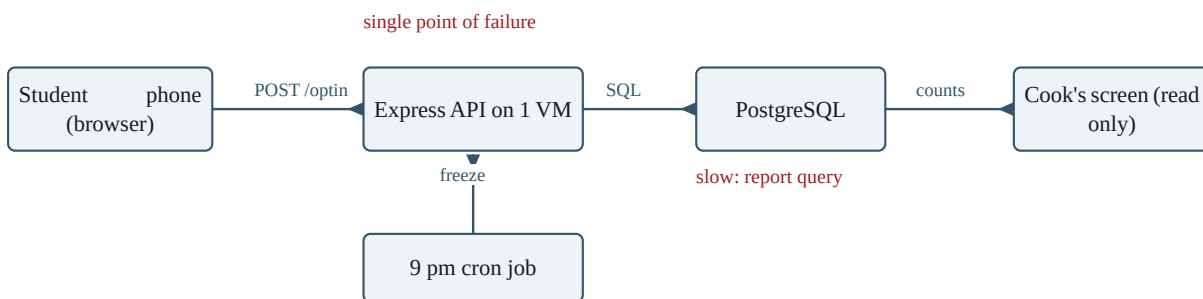
Say the pitch out loud with a timer. If it runs past 100 seconds you will be interrupted mid-sentence and lose the ending, which is where your number lives. Cut, do not speed up.

Draw the architecture in 60 seconds

Many interviewers will say “draw it”. On paper, on a whiteboard, or by sharing a screen. Practise this until it is automatic. Boxes and arrows only.

THE DRAWING RULES

- **Left to right:** user → client → server → data store. Nothing else on the first pass.
- **Label every arrow** with what travels on it, not just an arrowhead.
- **Draw only what exists.** Do not draw a load balancer you never deployed.
- **Mark the slow part** and the **failure point** yourself, before they ask. That single act moves you from “student who followed a tutorial” to “engineer who knows his system”.



The whole system on one line, plus the two things you point at yourself.

SHORTCUT

Point at your own weak spot while drawing: “this VM is a single point of failure — if it dies, the cook has no counts, and our fallback was literally the old paper sheet.” Interviewers love this. You have taken their best question and answered it first.

The hardest small question: “What did YOU do?”

Group projects are normal. Pretending you did everything is not. The interviewer has interviewed your classmates. Claiming all four parts is how you get caught.

THE OWNERSHIP STRUCTURE — FOUR MOVES

1. **Claim narrowly.** Name your modules, not the project.
2. **Credit clearly.** Name what teammates owned.
3. **Show the seam.** Explain one interface between your part and theirs — this proves you understood the whole, without claiming the whole.
4. **Offer depth.** “I can go deepest on the API and the schema.”

Example 2 [Wipro · pattern]

“There were three of you. What exactly was your contribution?”

TRAP

WEAK. “Basically I worked on all parts. Mainly backend but I also know the frontend code, I helped in deployment also. Whenever anyone was stuck I used to help them, so I can say I contributed in every module.”

SOLUTION

STRONG. “I owned the API and the database. Concretely: the schema for the four tables, the four endpoints under /optin and /counts, the 9 pm cut-off job, and the weekly report query.

Ravi owned the front end, Meera owned login and deployment.

The seam between Ravi and me was the response shape from POST /optin. We argued about it — he wanted the whole updated list back so his page could re-render, I wanted just the new count so the response stayed small. We agreed on returning the count plus a version number, and he refetched only when the version jumped.

I can go deepest on the API and the schema. I know the front end well enough to read it, but I did not write it.”

What changed	Why it matters
“All parts” → four named pieces	Checkable. A checkable claim reads as true.
Teammates named	Someone who credits others is usually telling the truth.
Added the interface argument	Proves system understanding without over-claiming.
States the limit of own knowledge	Sets the depth of follow-ups where you are strong.

TRAP

Never say “I know all of it” about code you did not write. The next question will be “good — how does the front end store the auth token?” and the silence that follows costs you more than admitting the boundary would have.

WARM-UP

build the reflex

Warm-ups. One idea each. Answer in two sentences, out loud, from your own project.

Example 3 [warm-up]

“In one sentence, what problem does your project solve?”

SOLUTION

Structure: [who] could not [do what], so [thing] lets them [do it].

Filled: “The mess cook could not know tomorrow’s head count, so the app lets students mark meals in or out before 9 pm.”

Check: no technology word appears. If you cannot describe the problem without naming a framework, you do not yet understand the problem.

Example 4 [warm-up]

“How many users did it have?”

SOLUTION

Give the true number and the true nature of the users. Small and true beats big and vague.

Filled: “78 students in one hostel block, for six weeks. Real users, but one block only — I have never run it above about 80 people.”

Never say “it can handle any number of users”. You have not tested that, and the next question will be “what makes you say that?”

Example 5 [warm-up]

“Why did you choose this database?”

SOLUTION

Structure: [shape of my data / queries] → [the store that fits that shape] → [what I gave up].

Filled: “My main read is a weekly report that joins opt-ins to meals and groups by day, so I wanted SQL and a real GROUP BY. I gave up schema flexibility — every new field needs a migration, and I wrote three migrations in six weeks.”

The “what I gave up” clause is the part that scores. Every choice costs something. Saying the cost proves you chose rather than copied.

Example 6 [warm-up]

“How long did it take, and how did you split the work?”

SOLUTION

Filled: “Five weeks of evenings. Week 1 was schema and endpoints, weeks 2–3 were the front end and the cut-off job in parallel, week 4 was the report, week 5 was fixing the double-tap bug and the report speed. We split by layer: I had the API, Ravi the UI, Meera deployment.”

Dates and a split make the story checkable. Checkable stories are believed.

TIER 1

TCS NQT · Accenture · Infosys · Wipro · Capgemini

Standard service-company project questions. Direct, factual, one level deep. Answer these perfectly and you clear the round at this tier.

Example 7 [TCS NQT · pattern]

“Explain the flow when a user does the main action in your app.”

SOLUTION

This is the **end-to-end trace**. Walk it like a packet, not like a feature list. Name each hop and what it checks.

Structure: client event → request → auth → validation → business rule → data write → response → what the UI does next.

Filled:

1. Student taps **Skip dinner** on the phone.
2. The page sends POST /optin with the meal id and the choice, plus the session cookie.
3. Express checks the cookie and loads the student id. No cookie, 401.
4. It validates that the meal id exists and belongs to tomorrow.
5. The business rule: if the server clock is past 21:00, reject with 409 and the message “counts are frozen for tomorrow”.
6. It writes one row into opt_ins for (student_id, meal_id).
7. It reads back the fresh count for that meal and returns it as JSON.
8. The page swaps the button label and shows the new count.

Then volunteer the weak point: “Step 6 is where our bug lived — I will come back to it if you want.”

SHORTCUT

Numbered steps are far easier to follow out loud than a paragraph. Practise counting on your fingers as you speak. It stops you from skipping a hop, and skipping a hop is what makes an interviewer suspect you never wrote it.

Example 8 [Capgemini · pattern]

“Show me your database design. Why these tables?”

SOLUTION

Structure: name the tables, name the keys, name the one relationship that carries the product, then name the one thing you would normalise differently.

Filled:

Table	Main columns	Why it exists
students	id, roll_no, block	One row per person. roll_no is unique.
meals	id, meal_date, slot	One row per meal. (meal_date, slot) unique.
opt_ins	student_id, meal_id, choice, created_at	The join table. One row per decision.
daily_counts	meal_id, count_in, frozen_at	The frozen 9 pm snapshot the cook reads.

“opt_ins is the heart: it is many-to-many between students and meals, with the choice on the link. The unique key is (student_id, meal_id) — that is what should have stopped our double-row bug, and adding it is exactly how we fixed it.

What I would change: daily_counts stores a number I can already compute from opt_ins. That is duplicated state, and duplicated state drifts. I would keep it only as a deliberate frozen snapshot and rename it meal_snapshots so nobody treats it as live truth.”

TRAP

The invented normal form. Do not say “it is in 3NF” unless you can state what 3NF is and point at the column that would violate it. A one-word claim you cannot unpack is worse than no claim.

Example 9 [Accenture · pattern]

“What was the most difficult part, technically?”

SOLUTION

Pick a **real** difficulty with a **cause you understood**. Difficulty means “I did not know why it was happening”, not “there was a lot of work”.

Structure: symptom → what I first believed → how I found the real cause → fix → proof.

Filled:

- **Symptom:** twice in the first week, a meal count was one higher than the number of students who had actually opted in.
- **First belief:** I assumed the cook’s screen was caching an old value, so I added a no-cache header. It happened again.
- **Finding the cause:** I printed every row in opt_ins for that meal and saw two rows for the same student, created_at 380 ms apart. That is a double tap on a slow phone, not a cache.
- **Fix:** a unique constraint on (student_id, meal_id), and the insert became an upsert so the second tap updates instead of inserting.
- **Proof:** I wrote a small script that fires the same request twice in parallel; before the fix it produced two rows, after the fix one.

Why this scores: it contains a wrong first guess. Real debugging has wrong guesses. Stories with no wrong guess sound rehearsed.

Example 10 [Cognizant · pattern]

“How did you test it?”

SOLUTION

Be honest about the level you actually reached. Every level below is respectable if you say what it covers and what it misses.

Level	How to say it	What it does not cover
Manual only	“I tested by hand with a checklist of 11 cases that I kept in the repo.”	Regressions. Nothing re-runs when you change code.
A few unit tests	“9 tests on the cut-off rule and the count query, run with <code>node --test</code> .”	The HTTP layer and the database.
API tests	“6 tests that start the app and hit the real endpoints against a test database.”	The browser and real concurrency.
Load / concurrency	“A script that fires 50 parallel opt-ins to check the unique constraint holds.”	Sustained load, real network.

Filled: “Honestly, mostly manual with a written checklist. After the double-row bug I added 9 unit tests around the cut-off rule and a small parallel-request script, because that was the bug that actually hurt us. I did not have browser tests.”

Never say “we tested everything”. Say the level, say the hole.

Example 11 [Infosys · pattern]

“How is the application secured?”

SOLUTION

Answer only what you truly did, in layers, and name the gap.

Filled:

1. **Authentication:** session cookie, `HttpOnly` and `SameSite=Lax`, signed by the server. Passwords hashed with `bcrypt`, not stored in plain text.
2. **Authorisation:** a student can only write their own opt-in. The student id comes from the session, never from the request body — that was a deliberate decision, because if I had trusted the body, anyone could opt out for anyone.
3. **Input validation:** meal id must be an integer and must exist. All queries are parameterised, so no string concatenation into SQL.
4. **Transport:** HTTPS through the host’s certificate.
5. **The gap:** no rate limiting. Someone could hammer the login endpoint. On a hostel app with 78 users nobody did, but on a public app that is a real hole and it is the first thing I would add.

SHORTCUT

Point 2 — “**the id comes from the session, never from the body**” — is a genuine security idea that most freshers have never thought about. If it is true of your project, say it. If it is not true of your project, go fix your project tonight; it is usually a two-line change.

Example 12 [TCS Digital · pattern]

“Why Node and Express? Why not Java Spring or Django?”

SOLUTION

There is no “correct” answer. There is only a **reasoned** answer plus a **fair** description of the alternative. Never insult the alternative.

Structure: my constraint → why my pick fits it → what the alternative is genuinely better at → honest admission if the real reason was familiarity.

Filled: “The honest first reason is team familiarity: all three of us already wrote JavaScript for the front end, and with five weeks of evenings, one language across the stack saved real time.

The technical reason that held up: our workload is many small, short requests that mostly wait on the database, and Node’s single event loop handles waiting work well without me managing threads.

Spring would have been better if we needed strong typing, mature transaction management, and a big team touching the same code — those are real advantages and on a bank-style project I would expect to lose that argument. Django would have given us an admin panel almost free, which we hand-built and which took Meera two evenings.“

What changed vs the usual answer	Effect
Admits familiarity as a real reason	Sounds true, because it is the true reason for most student projects.
Gives one workload-shaped technical reason	Shows you know what the runtime actually does.
Praises the alternative specifically	Shows you can lose an argument gracefully — a teamwork signal.

PRACTICE · TIER 1 target: say each out loud, 90 s each

Use **your own** project for every one of these. Write the answer in six lines, then say it without reading.

1. Describe your project to a non-technical person in 60 seconds. No technology words.
2. Trace your main user action from tap to database and back, in numbered steps.
3. Draw your schema. State the primary key and one foreign key out loud.
4. Name the single slowest thing in your app and say how you know it is slowest.
5. What exactly did you write, and what did each teammate write?
6. How did you test it, and what does your testing **not** cover?
7. Name one thing in your app that is not secure, and how you would fix it.
8. Why that language and framework? Name one thing the alternative does better.

ANSWER KEY

There is no fixed answer key here — the answers are yours. Grade yourself against this checklist, one mark each, out of 8: (1) no buzzword without evidence; (2) at least one real number; (3) at least one named alternative you rejected; (4) at least one honest limitation stated by you before being asked; (5) modules claimed narrowly; (6) no sentence longer than about 20 words; (7) you stopped talking within 100 seconds; (8) nothing you said was untrue. Below 6 out of 8, redo it before moving to Tier 2.

TIER 2

Singapore & Thailand · Grab · Shopee · GIC · DBS · Agoda · SCB

Regional and business-flavoured questions. The interviewer is a working engineer or a manager in Singapore, Bangkok or Jakarta. They care about scale, money and users, not about definitions.

Example 13 [Grab · pattern]

“Suppose this app went from 78 users to 78,000. What breaks first?”

SOLUTION

They are not asking you to design a distributed system. They are asking: **do you know where your own system’s limits are?** Answer in order of what breaks first, with a reason.

Structure: find the biggest cost per user → multiply → name the first thing to hit a wall → say the cheapest fix → then the next wall.

Filled:

1. **First wall: the report query.** It scans `opt_ins` for a date range. At 78 users and six weeks that is about 6{,}500 rows. At 78{,}000 users it is about 6.5 million rows per six weeks. Even indexed, that is a heavy scan on a 1 GB VM, and the cook’s page would time out. **Cheapest fix:** keep a rolled-up daily count table written once at 9 pm, and have the report read the roll-up, not the raw rows.
2. **Second wall: the 9 pm spike.** Everyone opts in just before the cut-off, so traffic is not flat — it is a spike in the last ten minutes. With 78 users that is nothing; with 78{,}000 it is maybe 300–800 writes per second for

ten minutes, on one VM with one Postgres. **Fix:** more than one app process behind a load balancer, and move the cut-off to a per-hostel time so the spikes do not overlap.

3. **Third wall: the single VM.** One machine means one reboot equals a full outage. **Fix:** two app instances, and the database on its own managed instance with backups.

What I would measure first: I would not guess. I would load test the opt-in endpoint and the report separately and find which one falls over at the lower number.

SHORTCUT

The phrase “**what I would measure first**” is worth a lot at Tier 2 and Tier 3. Guessing the bottleneck confidently is a junior habit. Naming the measurement is a senior habit, and it costs you one sentence.

Example 14 [Shopee · pattern]

“What was the business value? How would you know if it worked?”

SOLUTION

Engineers at this tier are asked to justify work in outcome terms. Practise the translation from feature to money or time.

Structure: who benefits → the metric that would move → the baseline → what we actually saw → what I could not prove.

Filled: “The person who benefits is the cook, and the metric is food thrown away per meal.

Baseline: before the app, the mess recorded leftovers roughly, about 12–15 plates a night by the cook’s own estimate.

What we saw in six weeks: the cook’s estimate dropped to about 5 plates on the nights when more than half the block used the app.

What I cannot prove: we had no proper measurement before and after, the cook’s numbers were eyeballed, and there was a holiday week in the middle. So I would call it a promising signal, not a proven result. If I ran it again I would weigh the leftovers for two weeks before launching anything.”

Move	Why it scores at Tier 2
Names one metric, not five	Focus. Managers distrust five-metric answers.
Gives a baseline	Without a baseline a number means nothing.
States what the evidence cannot prove	Intellectual honesty is a hiring signal here.
Says what you would measure next time	Shows you learned the method, not just the result.

Example 15 [Agoda · pattern]

“What would this cost to run per month, and how would you reduce it?”

SOLUTION

Most freshers have never thought about cost. Thinking about it at all puts you ahead.

Structure: list what you pay for → put a rough number on each → find the biggest → name one lever.

Filled: “Today it is one small VM and a domain — roughly 8–10 US dollars a month, and the database runs on the same VM, which is why it is cheap and also why it is fragile.

If I moved the database to a managed instance with backups, that alone would be the biggest line, likely more than the app server.

The cheapest real lever is the report: it is the only heavy query, it is read by one person once a day, and it is the reason I would otherwise need a bigger machine. Pre-computing it at 9 pm turns a heavy query into a tiny read, and that keeps me on the small instance. So the cost lever and the performance fix are the same change.”

You are not expected to quote exact cloud prices. You are expected to know **what** costs money — compute time, storage, data transfer, managed services — and which of your components is the expensive one. Ranges are fine. Say “roughly” and mean it.

Example 16 [DBS · pattern]

“Your report showed a wrong number to the cook for two days. Walk me through how you handled it.”

SOLUTION

This is a **behavioural** question wearing a project costume. Use STAR, and label the parts in your head. Here they are labelled on the page so you can see the shape.

S — Situation. “In week two, two meal counts on the cook’s screen were one higher than the true number of opt-ins. The cook had already ordered food based on them.”

T — Task. “I owned the API and the counts, so it was mine to find and fix. Two things mattered: tell the cook before he trusted a third wrong number, and find the real cause rather than patching the display.”

A — Action. “First I told the cook the same evening and asked him to use the paper sheet for one more night — that cost us credibility but it was the honest move. Then I dumped the raw `opt_ins` rows for those meals and found duplicate rows for one student, 380 ms apart, which pointed at a double tap and not a caching problem as I had first assumed. I added a unique constraint on (`student_id`, `meal_id`), changed the insert to an upsert, and wrote a script that fires two parallel requests to prove the second one no longer creates a row. Finally I recomputed the two bad days from the raw rows and showed the cook the corrected figures.”

R — Result. “No count mismatch in the remaining four weeks, and the parallel script is still in the repo as a regression check. The lesson I actually carry: I spent an evening on my first theory because it was convenient, not because I had evidence. Now I look at the raw data before I form a theory.”

This is my example, from a project I invented for this book. Yours must be a real thing that really went wrong in your project. If nothing ever went wrong in your project, you either did not use it with real people or you are not remembering hard enough.

TRAP

The blameless-to-the-point-of-blame-shifting trap. “The cook misread the screen” or “the front end sent it twice” both move the fault away from you. Even when partly true, they read as excuses. Own the part that was yours: the server accepted two rows it should have rejected. Servers must not trust clients. That is your fault line, and saying so is stronger than being right about the phone.

PRACTICE · TIER 2 target: 5 min each, written

For **your own** project:

1. At 1000× your current users, what breaks first? Give the number you used to decide.
2. What is the business or human value, what is the baseline, and what can you **not** prove?
3. What does your project cost to run, and what is the single biggest lever?
4. Tell, in STAR form with the four parts labelled, one time your project produced a wrong result for a real user.
5. Which one component, if it died at 3 a.m., takes the whole app down?

ANSWER KEY

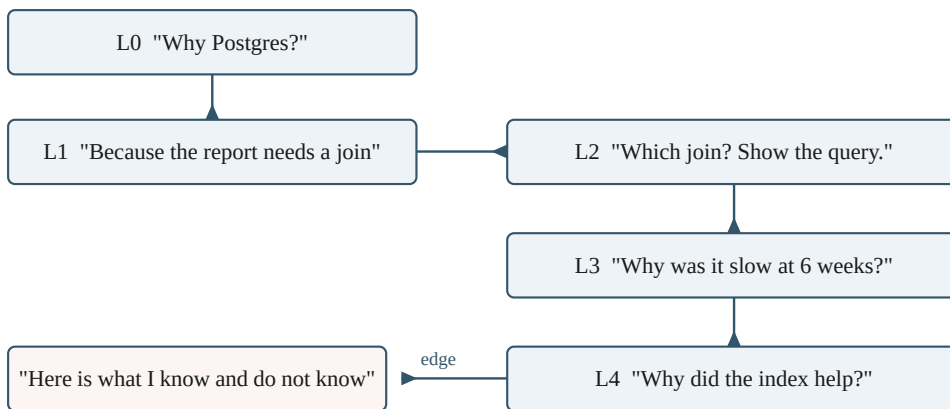
Self-grade. Full marks need: (1) an actual arithmetic step, not just “it would be slow”; (2) one named metric with a baseline; (3) a cost driver named, not a price quoted; (4) all four STAR parts present, with the R containing a lesson stated as a changed behaviour; (5) a single named component, plus what the fallback was.

TIER 3

Google · Amazon · Microsoft · Goldman Sachs · D. E. Shaw · Adobe

Deep probing. At this tier the first answer is never the end. Expect three to five follow-ups on one thread until you reach the edge of what you know. **Reaching that edge is normal and expected.** How you behave at the edge is what is being scored.

The follow-up ladder



One answer, four levels of probe. Most candidates have material for level 1 only.

Example 17 [Amazon · pattern]

The ladder, played out. Four probes on one choice.

SOLUTION

Probe 1 — “Why Postgres and not MongoDB?”

“My heaviest read is the weekly report: for each day, count opt-ins joined to meals and grouped by slot. That is a relational shape. In Mongo I would either denormalise and keep counts on the meal document, or use an aggregation pipeline — both are possible, but I would be re-implementing a join by hand and I preferred to let the database do it. I also wanted a unique constraint across two columns, which is exactly what stopped my duplicate bug.”

Probe 2 — “Show me that query.”

“Roughly:

```
SELECT m.meal_date, m.slot, COUNT(*) FROM opt_ins o JOIN meals m ON m.id = o.meal_id
WHERE m.meal_date BETWEEN $1 AND $2 AND o.choice = 'in' GROUP BY m.meal_date, m.slot;
```

I am writing it from memory so the exact syntax may be off by a comma, but that is the shape: filter by date range, join, group by day and slot.”

Probe 3 — “Why did it take 4.1 seconds at only six weeks of data?”

“Because there was no index that matched the filter. The filter is on `meals.meal_date`, so Postgres scanned `meals` fully and then, for each matching meal, scanned `opt_ins` to find its rows. `opt_ins` had no index on `meal_id` at all — only the primary key. So the work was roughly `rows_in_opt_ins × matching_meals`, not `matching_rows`. Small tables, bad plan.”

Probe 4 — “Why did adding the index fix it? What does the index actually do?”

“A B-tree index on `opt_ins(meal_id)` keeps the meal ids in sorted order with pointers to the rows. So instead of reading every row to find the ones for meal 412, the database walks down the sorted structure — a few steps instead of a full pass — and then follows the pointers. That turns the per-meal lookup from work proportional to the whole table into work proportional to a logarithm of it, plus the rows actually returned.

Here is the edge of what I verified: I confirmed the improvement with `EXPLAIN ANALYZE` before and after and saw the plan change from a sequential scan to an index scan. I have not studied Postgres’s planner cost model, so I could not tell you why it chose one join strategy over another. That is on my list.”

SHORTCUT

Notice the two protective phrases used above, and steal both:

- **“I am writing it from memory so the syntax may be off”** — buys accuracy forgiveness without weakening the idea.
- **“Here is the edge of what I verified”** — turns “I don’t know” into a boundary you drew deliberately, which reads as self-awareness rather than a gap.

Example 18 [Google · pattern]

“What would you do differently if you rebuilt it today?”

SOLUTION

The common failure is answering with **features**. They asked an engineering question.

Weak kind of answer	Strong kind of answer
“Add a mobile app.”	“Make writes idempotent with a client-supplied key.”
“Add payment integration.”	“Remove the duplicated count and keep one source of truth.”
“Improve the UI.”	“Add structured logs so I can debug without SSH-ing into the VM.”
“Use microservices.”	“Write the load test first so I know the limit before users find it.”

Filled, three items, in priority order:

1. **Idempotency done properly.** The unique constraint fixed duplicate rows, but the client still has no way to retry safely after a network timeout — it cannot tell “my request never arrived” from “it arrived and the reply was lost”. I would have the client send a request key, store it server-side, and return the stored reply on a repeat. That is the general fix; the constraint was the specific patch.
2. **One source of truth for counts.** `daily_counts` duplicates what `opt_ins` already knows. Two places holding the same fact will eventually disagree. I would keep it only as an explicitly frozen snapshot with a `frozen_at` timestamp, and make every live read go to `opt_ins`.
3. **Observability before features.** When the count was wrong I had no logs, so I debugged by printing rows by hand. One structured log line per write, with the student id and the request key, would have shown me the duplicate in a minute instead of an evening.

What I would not change: the stack. It was right for the problem and the team.

TRAP

The microservices trap. “I would split it into microservices” on a project with 78 users is an instant negative signal at Tier 3. It says you are repeating something you read. If you do want to talk about splitting a system, you must first name the scaling pain that justifies it — independent deployment, separate scaling needs, or team boundaries — and your hostel app has none of those.

Example 19 [Microsoft · pattern]

“You said the fix was a unique constraint. What happens to the user experience when that constraint fires?”

SOLUTION

This probe checks whether your fix was reasoned or copied. A constraint that fires throws a database error — what did you **do** with that error?

Filled: “Two cases, and they need different behaviour.

Case A — the same choice twice (the double tap). The second request should look successful to the user, because their intent is already recorded. So I made the write an upsert: `INSERT ... ON CONFLICT (student_id, meal_id) DO UPDATE SET choice = EXCLUDED.choice`. The user sees success, and there is one row.

Case B — a changed choice (opted in at 6 pm, opts out at 7 pm). The same upsert handles it: the row updates. That is correct and it is why I chose upsert over ‘ignore the conflict’.

The case I got wrong at first: I initially used `DO NOTHING`, which made Case A correct and Case B silently broken — the student saw ‘saved’ and the choice had not changed. I found it because Ravi tested exactly that path on the UI. So the honest version is that the constraint alone was not the fix; the constraint plus the right conflict action was.“

SHORTCUT

An answer that contains “**the case I got wrong at first**” is nearly impossible to fake and very strong to hear. Hunt through your project for these moments. Every real project has them; students forget them because they only remember the final working state.

Example 20 [Amazon · pattern]

“Tell me about a technical disagreement on this project and how it ended.”

SOLUTION

STAR, four parts labelled. The scoring is on **how you handled disagreeing**, not on whether you were right. Answers where you were right and everyone agreed score badly.

S — Situation. “Ravi, who owned the front end, wanted `POST /optin` to return the whole updated list of meals so his page could re-render in one go. I wanted it to return only the new count for the meal that changed.”

T — Task. “We were two days from the demo and the endpoint was blocking his screen, so this had to be settled that evening, and it had to be settled in a way he would actually build on.”

A — Action. “I asked him what problem the full list solved for him, and the real answer was that his page drifted out of date when two devices belonged to the same student. That is a genuine problem, and my small response did not solve it. So I stopped arguing about payload size and proposed the thing that solved his problem cheaply: return the count plus a version number for the day, and let him refetch the full list only when the version jumped. I wrote the version column that night; he wired the refetch.”

R — Result. “It shipped in time, his stale-screen bug disappeared, and the response stayed around 40 bytes instead of a few kilobytes. What I took from it: I had been arguing against his **solution** for two days without asking what his **problem** was. Now I ask that first — it also saved an argument about the report format later.”

Substitute your own disagreement. If you genuinely never disagreed with anyone, that is itself worth examining: it usually means one person made all the decisions, and you should be ready to say which one of you that was.

Example 21 [Goldman Sachs · pattern]

“How do you know your data is correct? Convince me.”

SOLUTION

A correctness question. The expected move is to name your **invariants** — statements that must always be true — and say how each one is enforced.

Filled:

Invariant	Enforced by	Honest status
One opt-in row per student per meal	Unique key (<code>student_id</code> , <code>meal_id</code>)	Enforced by the database, not by code. Solid.
No writes after 21:00 for tomorrow	A time check in the handler	Weak. Code-level only. A direct SQL write bypasses it.
A frozen count never changes	<code>frozen_at</code> set once by the cron job	Weak — nothing stops an update. Should be a trigger or a read-only view.

Every opt-in points to a real meal	Foreign key <code>opt_ins.meal_id</code> → <code>meals.id</code>	Enforced by the database. Solid.
------------------------------------	--	----------------------------------

“So two of my four invariants are enforced where they cannot be bypassed, and two live in application code, which means a second writer — a script, a future service, me at 2 a.m. with `psql` open — could break them. If this handled money instead of dinner, all four would need to be in the database.”

SHORTCUT

The sentence “**if this handled money instead of dinner**” shows you can scale your standard of rigour to the stakes. That is exactly the judgement a bank or a payments team is listening for.

Example 22 [Uber · pattern]

“Your cron job runs at 9 pm. What happens if it does not run?”

SOLUTION

Failure-mode probing. Answer as a list of failure modes, each with detection and recovery. This is the single most common Tier-3 project probe and most candidates have never thought about it for one second.

Filled:

Failure	Effect	Detected by	Recovery
VM was down at 21:00	No snapshot. Cook’s page empty.	Nothing, in my version. That is the real answer.	Run it by hand the next morning — the raw rows are still there.
Job ran twice	Snapshot written twice	Nothing	Harmless only because the write is an upsert keyed on <code>meal_id</code> . That was luck at first, then deliberate.
Job ran but the query failed	Stale snapshot from yesterday shown as today	Nothing — worst case, because it looks correct	Add <code>frozen_at</code> to the cook’s screen so a stale date is visible.
Clock/timezone wrong on the VM	Cut-off fires at the wrong hour	Students complain	Store times in UTC, convert at the edge. We did not, and it bit us during a server move.

“The honest summary: my design has no detection at all. The cheapest real improvement is a **heartbeat** — the job writes a row with a timestamp when it completes, and the cook’s page shows a warning if the newest heartbeat is older than today. That is about ten lines and it converts a silent failure into a visible one. Silent failures are the ones that cost you.”

Example 23 [Adobe · pattern]

“If I gave you two weeks and one engineer, what would you build next — and why not the other things?”

SOLUTION

A prioritisation probe. The structure is: **pick one, justify by cost and risk, and explicitly reject the alternatives**. Rejecting alternatives out loud is the part that scores.

Filled: “I would spend the two weeks on the idempotent write plus the heartbeat, in that order.

Why those: both are silent-failure classes. A wrong count damages trust in the whole system, and once the cook stops trusting it he goes back to paper and the project is dead regardless of features. They are also small — I would estimate three days for idempotency with tests, two for the heartbeat, leaving room for the load test.

What I would not build, and why:

- **A mobile app.** The web page already works on phones. It would consume the whole two weeks and change nothing about trust.
- **Multi-hostel support.** Real demand, but it needs the report roll-up first or it makes the slow query slower. Wrong order.
- **Notifications.** Cheap and tempting, but they would remind students to use a system that sometimes shows the wrong count. Fix truth before you increase traffic to it.

What would change my mind: if the cook told me the count was fine and the real problem was that students forget to fill it in, then notifications become first and idempotency waits. I would ask him before starting.“

SHORTCUT

End a prioritisation answer with “**what would change my mind**”. It proves your priority is an argument from evidence, not a preference — and it is the single line that most often turns a good Tier-3 answer into a strong one.

Code you must be able to explain

You do not need to recite your whole repository. You need to be able to reproduce, on paper, the three or four **ideas** in your code. Here are the three that carry most student projects. Run them, change them, break them.

Idea 1 — the N+1 query

This is the most common real performance bug in a student project, and one of the most common “what would you change” answers.

N+1 versus one batched fetch — counting database calls

JS

```
const orders = [
  { id: 1, userId: 10 }, { id: 2, userId: 11 },
  { id: 3, userId: 10 }, { id: 4, userId: 12 },
];
const userTable = new Map([[10, "Asha"], [11, "Bilal"], [12, "Chen"]]);

let dbCalls = 0;
function findUser(id) { dbCalls++; return userTable.get(id); }
function findUsersIn(ids) { dbCalls++; return new Map(ids.map(i => [i,
userTable.get(i)])); }

// SLOW: one query for the list + one query per row
dbCalls = 1; // the query that fetched orders
const slow = orders.map(o => ({ ...o, name: findUser(o.userId) }));
console.log("N+1 version -> db calls:", dbCalls);

// FAST: one query for the list + ONE query for all users
dbCalls = 1;
const ids = [...new Set(orders.map(o => o.userId))];
const users = findUsersIn(ids);
const fast = orders.map(o => ({ ...o, name: users.get(o.userId) }));
console.log("batched version -> db calls:", dbCalls);
console.log("same result? ", JSON.stringify(slow) === JSON.stringify(fast));
```

Output of `node s1.js`

TEXT

```
N+1 version -> db calls: 5
batched version -> db calls: 2
same result? true
```

How to say it in the room: “For a page showing 4 orders I made 5 database calls — one for the list and one per row. With 200 orders it is 201 calls. I collected the ids first and fetched all the users in one query, so it became 2 calls regardless of list length. Same output, constant number of round trips.”

TRAP

[...new Set(ids)] matters. Without the Set, user 10 is fetched twice. Interviewers do notice the deduplication, and it is a free point.

Idea 2 — why offset pagination is wrong

If your project has a “load more” button or a paged list, expect this question.

Offset pagination repeats a row when data arrives between requests

JS

```
let rows = ["r1","r2","r3","r4","r5","r6"]; // newest first
const page = (n, size) => rows.slice(n*size, n*size+size);

const shown0 = page(0, 2);
console.log("offset page 0:", shown0);
rows = ["NEW", ...rows]; // a new row arrives
console.log("offset page 1:", page(1, 2), "<- r2 is shown TWICE");

// Cursor pagination: remember the last id you saw, not a page number.
let data = ["r1","r2","r3","r4","r5","r6"];
function after(cursor, size) {
  const i = cursor === null ? 0 : data.indexOf(cursor) + 1;
  return data.slice(i, i + size);
}
const p0 = after(null, 2);
console.log("cursor page 0:", p0);
data = ["NEW", ...data];
console.log("cursor page 1:", after(p0[p0.length-1], 2), "<- no repeat, no skip");
```

Output of `node s2.js`

TEXT

```
offset page 0: [ 'r1', 'r2' ]
offset page 1: [ 'r2', 'r3' ] <- r2 is shown TWICE
cursor page 0: [ 'r1', 'r2' ]
cursor page 1: [ 'r3', 'r4' ] <- no repeat, no skip
```

How to say it: “OFFSET 2 means ‘skip two rows as the table looks right now’. If a row was inserted at the top between page 0 and page 1, everything shifted down by one, so the user sees r2 twice. If a row was deleted, they miss one. A cursor says ‘give me what comes after the row I actually last saw’, which does not move. The cost is that you can no longer jump to page 7.”

Also name the cost, every time. Cursor pagination cannot jump to an arbitrary page number, and it needs a stable sort key. An answer that names only the benefit sounds like a memorised line.

Idea 3 — idempotency

The same request twice must not charge twice

JS

```
const charges = [];
const seen = new Map(); // key -> the reply we already sent
```

```
function chargeNaive(userId, amount) {
  charges.push({ userId, amount });
  return { ok: true, total: charges.length };
}

function chargeIdempotent(key, userId, amount) {
  if (seen.has(key)) return seen.get(key); // replay the old reply
  charges.push({ userId, amount });
  const reply = { ok: true, total: charges.length };
  seen.set(key, reply);
  return reply;
}

chargeNaive(7, 500); chargeNaive(7, 500); // user tapped Pay twice
console.log("naive -> charges stored:", charges.length);

charges.length = 0;
chargeIdempotent("req-abc", 7, 500);
chargeIdempotent("req-abc", 7, 500);
console.log("idempotent -> charges stored:", charges.length);
```

Output of `node s3.js`

TEXT

```
naive -> charges stored: 2
idempotent -> charges stored: 1
```

How to say it: “The client generates a key once per user intent and sends it with the request. If the network times out, the client retries with the **same** key. The server stores the key with the reply it sent, so a repeat returns the stored reply instead of doing the work again. The key point is that the key comes from the client — the server cannot tell a retry from a genuine second purchase on its own.”

TRAP

In this teaching version `seen` is a Map in memory, so it is lost on restart and not shared between two servers. Say that yourself before you are asked. In a real system the key goes in the database or a shared cache with an expiry, and the check plus the write must be in one transaction or one atomic operation — otherwise two parallel retries can both pass the `has` check.

The “I don’t know” protocol

You **will** be pushed past what you know. That is the design of the round. What matters is the shape of your reply.

FOUR MOVES WHEN YOU HIT THE EDGE

1. **Say it plainly.** “I do not know that.” Two seconds. No apology tour.
2. **Give the nearest true thing.** “What I do know is...” — the adjacent fact you are sure of.
3. **Reason out loud, labelled as reasoning.** “If I had to guess, and this is a guess...”
4. **Close the loop.** “How would you approach it?” or “I will read about it tonight” — and if you say that, mean it.

Example 24 [Google · pattern]

“How does Postgres decide between a hash join and a nested loop join here?”

TRAP

WEAK. “It depends on the optimiser. Basically the optimiser is very intelligent, it uses statistics and cost-based optimisation, and it automatically chooses the best plan for performance.” — This is empty. It reuses the words in the question and adds nothing. The interviewer now suspects every earlier answer.

SOLUTION

STRONG. “I do not know the planner’s cost model, so I cannot tell you how it decides.

What I do know from my own project: I ran EXPLAIN ANALYZE before and after adding my index, and the plan changed from a sequential scan to an index scan, and the measured time went from 4.1 seconds to 0.2.

If I had to guess — and this is a guess — a nested loop makes sense when one side is tiny and the other is indexed, because you look up a few rows; a hash join makes sense when both sides are large and unindexed, because you pay once to build a hash table instead of scanning repeatedly. I have not verified that.

Is the planner something you use day to day on your team?”

What changed	Effect
Admits the gap in the first sentence	Removes all suspicion of bluffing.
Supplies an adjacent verified fact	Proves the gap is a boundary, not emptiness.
Labels the guess as a guess	You may now reason freely without risk.
Ends with a real question	Turns a dead end into a conversation.

TRAP

Never answer a question you did not understand. Ask for it again: “Do you mean how the database chooses, or how I would choose?” Answering the wrong question confidently is scored the same as bluffing.

When your project feels too small

Many students think “my project is just a CRUD app / a clone / a college mini project, so I will lose”. You will not lose for that. You lose for not knowing your own small project.

HOW TO DEFEND A SMALL OR TUTORIAL-DERIVED PROJECT — HONESTLY

Say what it is. “I built this by following a tutorial series, and then I changed three things.” — This is fine, and it is true for most people’s first project. Hiding it is what hurts.

Then earn the round on the changes. The three changes are now your project. Know them at four levels of depth: what, why, what broke, what you would do differently.

Add one thing the tutorial did not have. One index with a measured before and after. One test file. One deliberate constraint. One rate limit. One log line. Any of these gives you a number and a decision that are genuinely yours.

Never inflate. Do not turn a to-do app into “an enterprise task management platform”. An interviewer who has seen a thousand resumes will ask one question and the inflation will collapse.

Example 25 [Accenture · pattern]

“This looks like a standard to-do application. What is special about it?”

TRAP

WEAK. “Sir, it is not a normal to-do app, it is a complete task management system with many advanced features like priority, categories, reminders, and it is fully scalable and production ready.”

SOLUTION

STRONG. “Honestly, the core is a standard to-do app — I built it from a tutorial to learn Express. Nothing special there.

Two things in it are mine. First, I added recurring tasks, and that turned out to be harder than I expected: I first stored 365 rows for a daily task, which made the list query slow and made editing the series impossible. I changed it to store one rule row plus an exceptions table and generate the occurrences on read.

Second, I added a soft delete, because a friend testing it deleted a list by accident. So a delete sets `deleted_at`, and every query filters on it — which then caused a bug where a deleted task still blocked a unique title, and I had to make the unique index partial.

So the app is ordinary. The recurrence model and the soft-delete interaction with the unique index are the two things I actually learned from, and I can go as deep as you like on either.“

What changed	Effect
Admits the tutorial origin first	Removes the interviewer’s main suspicion immediately.
Narrows the claim to two changes	Two defensible things beat six inflated ones.
Each change has a failure and a redesign	This is engineering, and it is not fakeable.
Ends by inviting depth	Signals confidence in the narrow claim.

Every resume line is a question

Before an interview, take your own resume and write the question each line invites. If a line invites a question you cannot answer, either learn the answer or delete the line.

Your resume line	The question it guarantees
“Built a REST API using Node.js and Express”	“What makes it REST? Which verb did you use for the update, and why?”
“Implemented JWT authentication”	“Where is the token stored, and what happens when it is stolen?”
“Optimised database queries”	“Which query, from what to what, measured how?”
“Deployed on AWS”	“Which service, and what happens when the instance restarts?”
“Used Redis for caching”	“What is the expiry, and what happens on a cache miss during a spike?”
“Worked with a team of 4 using Agile”	“What was your sprint length and what did you do in a stand-up?”
“Integrated payment gateway”	“What happens if the user’s payment succeeds and your server crashes before saving?”
“Improved performance by 40%”	“40% of what, measured with what tool, on what data size?”
“Machine learning model with 95% accuracy”	“95% on what split? What was the class balance? What is the baseline?”

TRAP

The percentage with no origin. “Improved performance by 40%” is the single most questioned line on fresher resumes. If you did not measure it, remove it. If you did, memorise the before number, the after number, the tool, and the input size. Three of those four are usually missing and the interviewer notices instantly.

Example 26 [Infosys · pattern]

“Your resume says you implemented JWT authentication. Walk me through it.”

SOLUTION

Structure: what the token contains → who signs it → where the client keeps it → how the server checks it → what breaks and what you did about it.

Filled: “On login the server checks the password hash, then creates a token whose payload has the user id, a role, and an expiry — 24 hours in my case. The server signs it with a secret from an environment variable, so a client cannot forge one.

The browser stores it — and this is the part I got wrong first. I originally kept it in `localStorage` because every tutorial does. That means any script on my page can read it, so one cross-site scripting hole gives away the token. I moved it to an `HttpOnly` cookie so JavaScript cannot read it, which meant I then had to think about cross-site request forgery and set `SameSite`.

On each request, middleware verifies the signature and the expiry, and puts the user id on the request object. The handlers never read the id from the body.

The honest limitation: my tokens cannot be revoked. If a token is stolen it is valid until it expires, because I have no server-side list of valid or blocked tokens. The usual fix is short-lived access tokens plus a refresh token you can revoke, and I have read about it but not built it.”

SHORTCUT

Notice the pattern that runs through this whole chapter: **state the thing, state the thing you got wrong, state the limitation that remains**. Three beats. It is the most convincing structure available to a student, because nobody who copied a tutorial has the middle beat.

Internship projects: the extra questions

If your project was an internship, two new questions appear that never come up for a college project.

WHAT IS DIFFERENT ABOUT DEFENDING INTERNSHIP WORK

You must separate your work from the team’s even more carefully. In college nobody checks. In industry the interviewer may know someone at that company.

You must respect confidentiality without hiding behind it. “I cannot discuss that” for every question is useless. Describe the **shape** of the problem and your **technical decisions** without naming customers, internal systems or numbers you were told to protect.

You will be asked about process, not just code: how work reached you, how it was reviewed, how it was released.

Example 27 [Accenture · pattern]

“What was your day-to-day work in the internship, and what did you actually ship?”

TRAP

WEAK. “I was in the backend team. I learned a lot about how industry projects work. I did bug fixes and small tasks given by my mentor and attended daily stand-ups. It was a great learning experience and I got exposure to real-world codebases.”

SOLUTION

STRONG. “I was on a four-person team that owned a reporting service. My work came as tickets from my mentor, and I shipped three things in five months.

The largest was a retry for a nightly export that failed when the downstream system was slow. I added a bounded retry with a growing wait between attempts, and a dead-letter table so a permanently failing job stopped being retried forever instead of looping.

The smallest was a one-line fix to a timezone bug that made one report a day late for one region — that one taught me the most, because finding it took two days and the fix took two minutes.

Process: everything went through a pull request with one required review. I could not merge my own code. Releases were twice a week, and I watched two of them.

I cannot share the customer names or the volume figures, but the technical shape I have just described is all mine.“

What changed	Effect
“Bug fixes and small tasks” → three named deliveries	Countable work. Countable work is believable work.
Adds the retry design in one sentence	A real engineering decision, not a task list.
Includes the trivial fix that took two days	Honest, memorable, and shows debugging.
Describes the review and release process	Proves you worked in a team, not beside one.
Confidentiality handled in one clause	Respectful without hiding.

The extension question

At Tier 2 and Tier 3 you will often be asked to **extend your own project live**. This is a small system-design question with a huge advantage: you already know the system.

HOW TO ANSWER ANY EXTENSION QUESTION — FIVE MOVES

1. **Clarify the requirement.** Two questions, maximum. Who uses it, and how often?
2. **State the data change first.** New table, new column, or new relationship.
3. **State the write path,** then the read path.
4. **Name the hard case.** Concurrency, failure, or an edge in the existing data.
5. **Give a smaller version** you could ship this week.

The fifth move is what most candidates skip, and it is the one that shows engineering judgement rather than ambition.

Example 28 [Grab · pattern]

“Add guest meals to your app: a student can bring up to two guests, and the cook needs the guest count separately.”

SOLUTION

Clarify: “Two questions. Does a guest need to be identified, or is a count enough? And can a guest be added after the 9 pm cut-off?” — Assume: a count is enough, and the cut-off applies equally.

Data change: “A column `guest_count` on `opt_ins`, default 0, with a check constraint `guest_count BETWEEN 0 AND 2`. I would not create a `guests` table, because there is no guest entity — there is no name, no identity, nothing to join to. A column on the existing row is the honest model.”

Write path: “`POST /optin` takes an optional `guests` field. The same cut-off check applies. The upsert sets both `choice` and `guest_count`, so a second tap changing the guest count updates rather than inserting — the existing unique key already handles this.”

Read path: “The count query changes from `COUNT(*)` to `COUNT(*) + COALESCE(SUM(guest_count), 0)` for the total plates, and I would return both numbers, not one. The cook needs to know the guest share because guests often skip, and a single merged number hides that.”

Hard case: “The existing rows. Every row already in `opt_ins` has no guest count, so the column needs a default of 0 and a backfill — otherwise `SUM` returns null for old dates and my report shows null instead of a number. That is exactly the kind of quiet break that would show up in the weekly report only.”

Smaller version I could ship this week: “The column with the default, the API field, and the two-number report. I would leave out any per-guest rules, and I would not add a `guests` table unless someone later needs guest names for billing.”

SHORTCUT

“I would not create a table, because there is no entity” is a sentence worth owning. The instinct to create a table for every noun in the requirement is the most common modelling mistake. Ask: **does this thing have an identity and a life of its own?** A guest count does not. A guest with a name, a phone number and a history does.

Example 29 [Amazon · pattern]

“Now make it work for 40 hostels across the campus, each with its own cook and its own cut-off time. What changes?”

SOLUTION

Clarify: “Do cooks ever need to see another hostel’s numbers, and does a student ever eat in a different hostel?” — Assume: cooks see only their own, and yes, students occasionally eat elsewhere, which turns out to be the interesting part.

Data change:

- A `hostels` table with `id`, `name`, `cutoff_time`, `timezone`.
- `students.hostel_id` — their home hostel.
- `meals.hostel_id` — a meal now belongs to a hostel, so (`hostel_id`, `meal_date`, `slot`) becomes the unique key.
- `opt_ins` needs no new column, because the meal already carries the hostel. **That is the test of a good model: the new dimension enters in one place.**

Write path: “The cut-off check reads the cut-off from the **meal’s** hostel, not the student’s, because the deadline belongs to the kitchen doing the cooking. That is the subtle one — if I read the cut-off from the student, a visiting student gets the wrong deadline.”

Read path: “Every cook’s query gains `WHERE hostel_id = :id`, and that column goes into the index: `opt_ins(meal_id)` stays, but the meals index becomes `meals(hostel_id, meal_date)`. Without the hostel in the index, every cook scans all 40 hostels’ meals.”

Authorisation: “New requirement. A cook must not see another hostel’s data, so the hostel id must come from the cook’s session, never from a query parameter. Right now my app has one cook and no such check, so this is genuinely new code, not a config change.”

Hard cases:

1. **Visiting students.** A student opting into another hostel’s meal is now allowed by the model. Should it be? I would ask, and if yes, the home cook’s count must go down while the visited cook’s goes up — which means the count is per-meal and correct automatically, but the **waste report per hostel** now needs care about who paid.
2. **Cut-off spread.** 40 cut-offs at different times is actually good news: the 9 pm spike I worried about earlier spreads out.
3. **Migration.** The existing 78 students and six weeks of rows need a hostel id. I would add the column as nullable, backfill it to hostel 1, then make it `NOT NULL` — three steps, because a single step locks the table and fails if any row is null.

Smaller version: “Ship hostels, the hostel column on meals, the per-hostel cut-off and the session-based authorisation. Do **not** ship visiting students in version one — it is the only part with a money question attached, and it needs someone to decide policy, not engineering.”

Move in the answer	What it demonstrates
New dimension enters in exactly one table	Data modelling judgement.
Cut-off read from the meal, not the student	You thought about whose rule it is.
Index changed along with the query	You know an index must match the filter.
Hostel id from the session, not the URL	Authorisation instinct.
Three-step migration for <code>NOT NULL</code>	Operational awareness — the thing freshers never mention.
Deliberately cuts a feature from v1	Prioritisation, and knowing which problems are not technical.

SHORTCUT

The migration point — “**nullable, backfill, then NOT NULL**” — is a small, true, practical detail that almost no fresher says. Details like this are worth more than a fluent paragraph, because they can only come from having actually changed a table that had data in it.

If your project is not really yours yet

Some students reach the interview knowing, privately, that they cannot defend the project on their resume. Perhaps a teammate wrote most of it. Perhaps it was bought or copied. Here is the honest path, and it is not a comfortable one.

THE THREE LEGITIMATE MOVES

Move 1 — Rebuild the part you will be asked about. You usually have days or weeks, not hours. Take the single main feature and rewrite it yourself from an empty file. Not copied, not pasted — typed. One feature you truly wrote beats four you cannot explain.

Move 2 — Narrow the claim on the resume. Change “Built a full-stack e-commerce platform” to “Built the cart and checkout API for a team e-commerce project”. A narrower true claim is stronger than a broad false one, and it steers the questions to where you are strong.

Move 3 — Remove it. A resume with one project you own beats a resume with four you cannot discuss. Removing a line costs you nothing at screening and saves you a collapse in the room.

The move that is not available: claiming work you did not do. It is not a strategy with a downside — it is a strategy that fails, because the follow-up questions in this chapter cannot be survived by someone who did not build the thing.

Example 30 [any company · pattern]

You are asked about a project module a teammate wrote, and you cannot explain it.

SOLUTION

The recovery, in the room: “I should be straight with you — that module was Priya’s, and I can describe what it does but I would be guessing about how it works inside.

What I can take you through properly is the part I wrote: the cart and the checkout endpoints, including the double-submit problem we hit and how the unique key on the order reference fixed it.

Would that be useful, or do you specifically need the recommendation module?”

Why this survives: it is one sentence of admission, then an immediate redirect to real material, and it treats the interviewer as someone with a goal. Compare it to two minutes of vague guessing, which ends the interview mentally even if it continues on the clock.

And then fix it afterwards. Narrow that line on your resume tonight.

Red flags that end the round

Red flag	Say this instead
“It is fully scalable / production ready.”	“It ran for six weeks with 78 users. I have not load tested past that.”
“There were no bugs.”	“The one that cost us most was the duplicate count. Here is what caused it.”
Cannot name a single file or function.	“The cut-off check lives in <code>opt.in.js</code> , in the handler, before the write.”
Describes a diagram from the internet, not the code that exists.	Draws only the boxes that were deployed, and says where the drawing ends.

Blames teammates for everything broken.	Names what was yours to own, first.
Answers a question about the front end in detail, having not written it.	"I did not write that part; here is what I know from reading it."
Uses "we" for every sentence.	Uses "I" for your modules and names teammates for theirs.

PRACTICE · TIER 3 target: 45 min, with a partner if possible

Have a friend ask these about **your** project and refuse to accept the first answer — they must ask "why?" or "how do you know?" three times per question.

1. Name every technology in your stack and, for each, one thing the alternative does better.
2. Which invariants in your data are enforced by the database and which only by your code?
3. What is the most likely silent failure in your system — a failure nobody would notice?
4. Your main write endpoint gets called twice with the same payload. Trace what happens now, and what should happen.
5. Give the arithmetic for what breaks first at 1000× users.
6. Your background job does not run tonight. Who notices, and how?
7. In STAR form, four parts labelled: a technical disagreement you had and how it ended.
8. You have two weeks and one engineer. What do you build, and what do you explicitly not build?
9. What is the edge of what you actually verified, as opposed to what you believe?

ANSWER KEY

Score each answer 0, 1 or 2. **2** = you survived three "why"s with true, specific content. **1** = you survived two and then went vague. **0** = you repeated the question back, used a buzzword, or said something you are not sure is true.

Target: at least 14 out of 18 before a Tier-3 interview. Any **0** on questions 2, 4 or 9 is the most urgent to fix — those three are where bluffing gets caught.

The 20-minute self-drill

You cannot practise this round by reading. You practise it by being asked. If you have no partner, be your own interviewer with a timer and a voice recorder on your phone.

RUN THIS THE NIGHT BEFORE, FOR EACH PROJECT

Minutes 0–2. Say the 90-second pitch. Record it. Do not stop for mistakes.

Minutes 2–5. Play it back. Count three things: buzzwords with no evidence, sentences over 20 words, and whether a number appeared. Write the counts down.

Minutes 5–10. Draw the architecture from memory on paper. Then open your repository and check it. Every box you drew that does not exist, and every piece that exists but you forgot, is a question you would have fumbled.

Minutes 10–15. Pick one line from your resume. Ask yourself "why?" four times in a row, out loud, and answer each one. Note the level at which you ran out. That level is your honest depth today.

Minutes 15–18. Answer these two, timed at 60 seconds each: "what broke?" and "what would you change?" These are the two questions candidates most often have nothing prepared for.

Minutes 18–20. Write one sentence: the thing you would most hate to be asked. Then prepare that one. It is almost always the one that gets asked.

SHORTCUT

The recording is the part people skip and the part that works. You cannot hear your own filler words, your own trailing sentences, or your own three-minute answers while you are speaking. You can hear all of them on playback within ten seconds.

Questions to ask them about the code

At the end of a project round you are usually asked “do you have any questions?”. Technical questions about **their** system are far stronger than questions about perks, and they continue the conversation on your strongest ground.

Ask	What it signals
“What does a typical pull request look like on your team — size, and how many reviewers?”	You have worked in a reviewed workflow and care about it.
“What is the oldest part of the system that people are still afraid to change?”	You know real systems have those, and you are not scared of legacy.
“How do you find out something is broken in production — alerts, or a user telling you?”	Observability awareness, which is rare in a fresher.
“What would I be shipping in my first month?”	Practical, and it gives you real information about the role.
“What is the test and release process — how often do you deploy?”	Process awareness, and it tells you a lot about the team’s health.

Ask two, not six. Then stop. The same discipline that ends your pitch at 90 seconds applies here — a candidate who keeps asking questions past the interviewer’s time is remembered for that, not for the questions.

ONE-PAGE REVISION CARD

The round scores four things: ownership, depth, trade-off thinking, honesty.

Prepare the 12-line Defence Sheet per project. Problem · users · what it does · stack with reasons · my part · their part · data model · one request end-to-end · one number · what broke · one trade-off · what you would change.

The 90-second pitch: pain → fix → honest scope → stack with one reason → “I wrote...” → one number plus one limit. Then stop.

Ownership: claim narrowly, credit teammates by name, explain one interface between your part and theirs, state where your knowledge ends.

Every choice needs an alternative. “I chose X over Y because Z, and I gave up W.”

One real number beats ten adjectives. Scalable, robust, optimised and user-friendly are worth zero without evidence.

Volunteer your weak spot. Point at the single point of failure and the slow query before they ask. It converts their best question into your best moment.

Failure modes: for any background job or external call — what happens if it does not run, runs twice, or runs and silently fails? Name detection and recovery. Silent failures cost the most.

Invariants: know which are enforced by the database and which only by code. Code-level ones can be bypassed by a second writer.

The “I don’t know” protocol: say it plainly → give the nearest true thing → label any guess as a guess → ask a real question back.

Small project? Say it came from a tutorial, then defend the two or three things you changed, at four levels of depth. Never inflate.

Rebuild answers are engineering, not features: idempotency, one source of truth, logging, load tests — not “add a mobile app”.

Substitute your own project everywhere. Every filled example in this chapter is invented to show the shape. A memorised story dies on the first follow-up.

Hard & Awkward Questions

Salary, bonds, backlogs, gaps, rejections — the questions nobody prepares for

Pattern in one page

Every question in this chapter is a **risk question**. The interviewer is not curious about your life. They are checking one thing: **if we hire this person, what could go wrong, and how badly?**

That reframing changes your whole answer. You are not defending yourself. You are removing a risk from their mind.

WHAT EACH HARD QUESTION IS REALLY ASKING

They ask	They mean
“What is your expected salary?”	Can we afford you, and will you leave in six months for 5% more?
“Are you willing to sign a two-year bond?”	Will you accept our training cost, or run?
“You have a backlog.”	Will you fail to graduate, or repeat this pattern under pressure?
“There is a one-year gap here.”	Is there something you are hiding?
“Your CGPA is 6.4.”	Can you learn, or did you stop trying?
“You were rejected here last year.”	Did anything change, or is this the same candidate?
“What is your notice period?”	When can the seat actually be filled?
“Do you have other offers?”	How fast must we move, and are you being honest with us?
“Why should we not hire you?”	Do you have self-knowledge, or only rehearsed lines?

Three rules that apply to every question in this chapter

THE THREE RULES

Rule 1 — Never lie. Not once, not slightly. A lie about a backlog, a CGPA, a current salary or a notice period is checkable at the offer stage. Offers are withdrawn for this, and in some companies the candidate is blacklisted. A weakness told honestly costs you far less than a lie discovered later.

Rule 2 — Do not over-explain. The most common failure is not the fact; it is the four extra minutes spent apologising for it. Say it once, briefly, then move to evidence. Length signals shame, and shame signals that the thing is worse than it is.

Rule 3 — Never apologise twice. One acknowledgement is maturity. Three is a request for reassurance, and the interviewer is not there to reassure you.

Nothing in this chapter is a script. Every filled answer uses an invented person so that you can see the shape of the reply. **You must put your own true facts into these structures.** A memorised answer about someone else’s backlog will not survive the question “which subject was it?”

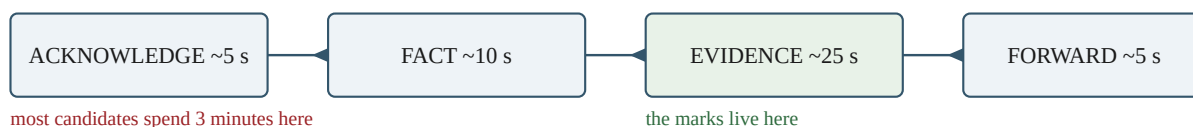
The four-move structure

Every awkward question in this chapter uses the same four moves. Learn the moves once and you can answer a question you have never seen.

A · F · E · F

1. **ACKNOWLEDGE** — one short sentence. No flinching, no drama. “Yes, I have one backlog.”
2. **FACT** — the plain, specific truth. Subject, year, number, reason. Ten to twenty words.
3. **EVIDENCE** — proof that it is handled or handled-in-progress. A date, a score, a certificate, a project, a habit you changed. **This is the part that scores.**
4. **FORWARD** — hand the conversation back. A question, or a bridge to what you want them to ask next.

Time budget: about 45 seconds total. If you are past 90 seconds you are pleading.



Most candidates spend all their time in the first two boxes. The marks are in the third.

WARM-UP

build the reflex

Example 1 [warm-up]

Practise the **ACKNOWLEDGE** move alone. Say each of these out loud, once, flatly, with no extra words. Then stop and count to three in silence.

SOLUTION

- “Yes, I have two backlogs; one is cleared.”
- “Yes, there is a fourteen-month gap after my degree.”
- “My CGPA is 6.4.”
- “My notice period is 60 days.”
- “Yes, I applied here last year and was not selected.”

The exercise is the **silence afterwards**. Most candidates cannot hold it and start explaining before being asked. The silence is where the interviewer decides you are calm. Practise it with a timer until three seconds feels normal.

Example 2 [warm-up]

Which of these is the **EVIDENCE** move, and why are the others not?

- (a) “I really regret that semester.”
- (b) “I cleared it in the December supplementary exam with 71%.”
- (c) “It was not fully my fault, the lab timing clashed.”
- (d) “I promise it will not happen again.”

SOLUTION

(b) is the evidence. It has a date and a number, and a third party could verify it.

(a) is emotion. Emotion is not evidence, and repeated emotion reads as instability. (c) is an excuse. Even true excuses shift responsibility, and shifting responsibility is the exact risk they are screening for. (d) is a promise about the future. Anyone can promise. Nobody can promise evidence.

Test for your own evidence: could a stranger check it? A mark sheet, a certificate date, a repository, a score, an attendance record, a manager’s name. If nobody can check it, it is not yet evidence — go and create some.

Example 3 [warm-up]

Fix this answer to “your CGPA is low”:

“Sir, actually my CGPA is low but I am very hardworking and I have strong practical knowledge and I will prove myself if given a chance.”

SOLUTION

Three problems: no fact, no evidence, and a plea at the end.

Rewritten with A·F·E·F:

A: “Yes, my CGPA is 6.4.”

F: “It is low because my second and third semesters were poor — 5.1 and 5.4 — while I was figuring out how to study at this level.”

E: “My last four semesters were 7.6, 7.9, 8.1 and 8.3, so the trend is the honest picture. Outside class I have shipped two projects and solved a bit over 300 practice problems, and my last internship review is in my folder if that is useful.”

F: “Would it help if I walked through the backend project, since that is the closest to this role?”

The plea is gone. Nothing in the strong version asks for a chance. It presents a trend and offers the next topic.

TIER 1

TCS NQT · Accenture · Infosys · Wipro · Capgemini

Salary — the real strategy

This is the question students handle worst, and it is the one with money attached. Read this section twice.

First: understand what you are being offered

Indian offers are quoted as **CTC** — cost to company. CTC is not your salary. Parts of it never reach your bank account.

THE FIVE LAYERS OF AN OFFER

1. **CTC** — everything the company spends on you, including things you never see.
2. **Variable / bonus** — paid only if targets are met. Assume **zero** when comparing offers until you know the historical payout.
3. **Fixed pay** — CTC minus variable. This is the real base.
4. **Gross cash** — fixed pay minus employer PF, gratuity, insurance premium. Money that actually flows.
5. **In-hand** — gross cash minus your own PF share, minus income tax. **This is what you live on.**

The one question that cuts through all of it: “What is the monthly in-hand for a new joiner at this level, after standard deductions?”

Two offers, and why the bigger CTC is the smaller pay cheque

JS

```
function inHand({ ctc, employerPF, gratuity, variablePct, insurance, taxRatePct }) {  
  const variable = ctc * variablePct / 100;  
  const fixed    = ctc - variable;  
  // money that never reaches your bank account each year:  
  const notCash  = employerPF + gratuity + insurance;  
  const grossCash = fixed - notCash;  
  const employeePF = employerPF;           // your own 12% share is also deducted  
  const taxable    = grossCash - employeePF;  
  const tax        = taxable * taxRatePct / 100;  
  const yearly     = grossCash - employeePF - tax;  
}
```

```

    return { fixed, variable, grossCash, yearly, monthly: Math.round(yearly / 12) };
}

const offerA = inHand({ ctc: 600000, employerPF: 21600, gratuity: 11538,
                        variablePct: 10, insurance: 8000, taxRatePct: 3 });
const offerB = inHand({ ctc: 550000, employerPF: 21600, gratuity: 10577,
                        variablePct: 0, insurance: 5000, taxRatePct: 2 });

for (const [name, o] of [ ["Offer A 6.0 LPA", offerA], ["Offer B 5.5 LPA", offerB] ])
{
    console.log(name, "-> fixed", o.fixed, "| variable", o.variable,
                "| take-home/month ~", o.monthly);
}
console.log("A minus B per month:", offerA.monthly - offerB.monthly);

```

Output of `node s4.js`

TEXT

```

Offer A 6.0 LPA -> fixed 540000 | variable 60000 | take-home/month ~ 38579
Offer B 5.5 LPA -> fixed 550000 | variable 0 | take-home/month ~ 40117
A minus B per month: -1538

```

SHORTCUT

Read that output again. The **6.0 LPA offer pays about 1,538 rupees a month less** than the 5.5 LPA offer, because 10% of it is variable and the deductions differ. Students accept the bigger headline number every year and are confused in month one.

The numbers above are illustrative, and tax slabs and PF rules change — run the script with **your** offer's real components. The lesson is the method, not my figures.

TRAP

Never compare two offers on CTC. Compare on **fixed pay** first, and on **in-hand** second. A company that will not tell you the fixed-versus-variable split before you sign is telling you something about how it will behave later — ask directly and in writing.

Who says a number first?

The old advice is “never say a number first”. That advice is written for experienced hires in countries with different norms. For a fresher in India it is often impractical: many companies have a fixed band and will simply ask.

DECIDE BY SITUATION

Campus or mass-hiring drive (TCS, Infosys, Wipro, Accenture, Capgemini): the package is usually fixed and public. Do not negotiate the base. Answer briefly and truthfully, and put your energy into the role and the location instead.

Off-campus fresher role at a product or mid-size company: give a **researched range**, not a point. Anchor at the level you can justify with evidence.

Experienced hire: try to get their band first. “What range has been budgeted for this role?” is a normal, professional question. If they insist, give a range with a reason.

Never give a number you have not researched. “Whatever you think is appropriate” is the weakest possible answer — it reads as no market awareness and no self-worth.

“What are your salary expectations?” — asked to a final-year student in a mass-hiring process where the package is standard.

TRAP

WEAK. “Sir, salary is not important for me at all. I just want to learn and grow. You can give me whatever is the company standard, I have no expectations.”

SOLUTION

STRONG. “I understand this role has a standard package, and I am comfortable with the band you publish for this profile.

What matters more to me at this stage is which unit I land in — I would like to be on a backend or data team, because that is where my project work is.

Is there any flexibility on which account or technology I am allocated to?”

Move	Weak	Strong
Honesty	“Salary is not important” — almost never true.	Accepts the band without pretending money is irrelevant.
Self-worth	“Whatever you give.”	Neutral acceptance, no self-lowering.
Use of the turn	Wasted.	Redirected to a thing that is actually negotiable — the team.
Ending	Passive.	A question that keeps you in the conversation.

SHORTCUT

In a fixed-package process, the negotiable items are almost never the money. They are: **role/technology allocation, base location, joining date, and shift**. Ask about those. That is where your leverage actually exists, and asking makes you look experienced.

Example 5 [off-campus · pattern]

“What is your expected CTC?” — off-campus, a 120-person product company, you are a fresher with one internship.

SOLUTION

Structure — five beats:

1. **Anchor on research, not on need.** Never mention rent, loans or family expenses. Those are your business and they weaken you.
2. **Give a range, not a point.** Roughly 15–20% wide.
3. **Justify with one piece of evidence.**
4. **State flexibility on the right thing.**
5. **Hand it back.**

Filled (substitute your own researched numbers for your city, year and role):

“Based on what I have seen for backend roles at this size of company in Pune for someone with an internship and shipped projects, I am looking at somewhere between 6.5 and 8 lakhs fixed.

I am saying fixed rather than CTC because I would rather compare on the base.

I put myself toward the middle of that because of the six-month internship where I owned the payments retry service, rather than at the top.

I am flexible on the structure — if the base is a little lower and there is a clear review at six or twelve months, that works for me.

Is that in line with the band you have for this role?”

TRAP

Never justify a number with your expenses. “I need at least 50,000 because of my education loan” invites the reply “that is unfortunate, but our band is 38,000”. Your price comes from the market and your evidence, never from your costs.

Example 6 [off-campus · pattern]

They reply: “That is above our band. We can offer 5.5 lakhs fixed.”

SOLUTION

Three honest paths. Pick one — do not stumble between them.

Path A — accept, and buy something else. “5.5 is below where I started, but I am genuinely interested in the work here. If the base is fixed, could we look at a written six-month review with a defined band, or a one-time joining amount? Either of those would make it work for me.”

Path B — a single, evidence-backed counter. “Could we look at 6.2? The gap between us is about 12%, and the reason I would ask for it is the internship — I came into that team on a service that was already in production and I owned the retry logic end to end. If 6.2 is impossible, tell me and I will give you a straight answer on 5.5.”

Path C — decline cleanly. “Thank you for being direct. 5.5 is below what I can accept right now, so I do not want to waste your time. If the band changes later I would be glad to talk again.”

Rules for all three:

- **One counter, not three.** A second counter after they move damages trust badly.
- **Never bluff a competing offer you do not have.** It is checkable more often than you think, and the cost is total.
- **Get the final number in writing** before you stop interviewing elsewhere.

Joining bonus, relocation allowance, review date, notice period, laptop, shift allowance. When base pay is frozen by a band, these are the levers that often are not. Ask for one, not five.

Example 7 [lateral · pattern]

“What is your current CTC?” — you are a candidate with two years of experience.

SOLUTION

Answer honestly. In India this is routinely verified against your salary slips and Form 16 at the offer stage, and inflating it is a common reason offers are cancelled.

Filled: “My current fixed is 7.2 lakhs, with a variable component of about 8% that paid out fully last year.

For a change I am looking at the 10 to 11 range, because the role you have described has on-call responsibility and direct ownership of a service, which my current role does not.

What band have you budgeted for this position?”

Why it works: the truth is stated without embarrassment, the ask is tied to a **difference in the work**, not to a percentage rule, and the turn ends with their number.

TRAP

Inflating current salary by “just a little” is the single most common self-inflicted wound in lateral hiring. Offer letters ask for documents. A 15% inflation discovered at document check does not become a 15% smaller offer — it usually becomes no offer.

Bonds and service agreements

Many Indian service companies and some product companies ask freshers to sign a **service agreement** — commonly one to two years, with a penalty if you leave early.

THE SEVEN QUESTIONS TO ASK BEFORE YOU SIGN

1. **Exact duration**, and the date it starts — joining date or training completion?
2. **Exact penalty amount**, and whether it reduces month by month.
3. **Is an original document being held?** Degree certificates should never be handed over. Ask this explicitly, and get the answer in writing.
4. **What counts as “leaving”** — does dismissal or a medical exit trigger the penalty?
5. **Is the penalty waived** if the company relocates you, changes your role, or does not confirm you?
6. **Can it be bought out**, and by a future employer?
7. **Is the amount mentioned in the offer letter itself**, or only in a separate document you will see on day one?

TRAP

The most important of these is question three. Handing over original educational certificates is a practice you should refuse. Companies that require it are relying on the difficulty of replacing those documents rather than on the agreement itself. A copy, notarised if they insist, is normal and reasonable; the original is not.

A note on enforceability, in plain words, not as legal advice. In India, an employment bond is generally treated as a contract, and a company can usually claim **reasonable compensation for actual losses** — chiefly the money genuinely spent training you. Courts have been reluctant to force a person to keep working for someone, and penalty amounts that look punitive rather than compensatory have been reduced or refused. In practice, though, companies commonly withhold the **experience letter and relieving letter**, and the absence of those causes real trouble with your next employer’s background check. Treat the bond as a serious commitment. **If a specific bond matters to your decision, get the actual document read by a qualified lawyer — a book cannot do that for you.**

Do the arithmetic before the emotion

Is the bonded offer actually better over the bond period?

JS

```
function compare({ months, bondedMonthly, freeMonthly, bondPenalty,
raisePctPerYear }) {
  let bonded = 0, free = 0;
  for (let m = 0; m < months; m++) {
    const year = Math.floor(m / 12);
    bonded += bondedMonthly; // bonded pay is
usually flat
    free += freeMonthly * Math.pow(1 + raisePctPerYear/100, year);
  }
  return {
    bondedTotal: Math.round(bonded),
    freeTotal: Math.round(free),
    breakEarly: Math.round(bonded) - bondPenalty,
  };
}
const r = compare({ months: 24, bondedMonthly: 33000, freeMonthly: 31000,
bondPenalty: 200000, raisePctPerYear: 15 });
console.log(r);
console.log("bonded job better over 24 months?", r.bondedTotal > r.freeTotal);
console.log("cost of walking out on day 1:", 200000);
```

Output of `node s5.js`

TEXT

```
{ bondedTotal: 792000, freeTotal: 799800, breakEarly: 592000 }
bonded job better over 24 months? false
cost of walking out on day 1: 200000
```

Read the result. The bonded job pays 2,000 rupees a month more **today**, and still loses over two years, because the unbonded job's 15% yearly raise compounds while the bonded pay stays flat. Run this with your own numbers before you decide anything. If you have no other offer, the bonded job is obviously better than no job — but know what you are choosing.

Example 8 [TCS · pattern]

“We require a two-year service agreement. Are you comfortable with that?”

TRAP

WEAK — version 1 (fake enthusiasm). “Yes sir, absolutely no problem, two years is nothing, I am ready to sign anything, I will stay for ten years.”

WEAK — version 2 (negotiating in the interview). “Actually is the bond really necessary? My friend said these are not legally valid anyway. Can it be reduced to one year?”

SOLUTION

STRONG. “Yes, I am comfortable with a two-year commitment — I am looking for a first role where I can go deep rather than move quickly, so two years matches what I want anyway.

I would like to read the agreement before I sign, and I have three specific questions: the penalty amount and whether it reduces over time, whether any original certificates are held, and whether the two years start from joining or from the end of training.

Those are the details, not objections. Who should I ask about the document?”

What changed	Why it matters
Commits first, asks second	Removes the flight-risk worry immediately.
Gives a genuine reason to stay two years	A reason is believable; enthusiasm alone is not.
Asks three precise questions	Reads as an adult who reads documents.
Says “those are details, not objections”	Stops the interviewer hearing it as resistance.
Does not argue legality	Arguing enforceability in an interview reads as planning to break it.

TRAP

Never say “bonds are not legally enforceable anyway”. Even where the legal position is genuinely complicated, saying it out loud tells the interviewer exactly one thing: this person is already thinking about how to leave. The interview is not the place for that argument. Read the document, ask your questions, and decide **before** you sign.

Notice period

Example 9 [lateral · pattern]

“What is your notice period? When can you join?”

SOLUTION

Structure: official period → what I will genuinely attempt → what I will not promise → their constraint?

Filled: “My notice period is 60 days as per my contract.

I will ask for early release and I think 45 days is realistic, because my current project has a natural handover point in mid-April. Some companies here also allow a buyout of the remaining days, and I am open to that if it matters to you.

What I do not want to do is promise you 30 days and then miss it — I would rather commit to a date I can hold. What is the date you are trying to fill this seat by?“

TRAP

The 15-day promise. Candidates panic and promise a joining date they cannot meet, believing the offer depends on it. Then they miss it. Missing your first committed date to a new employer is the worst possible first impression, and the previous employer’s relieving letter is not under your control. Quote the contract, state the realistic best case, and never promise a date that depends on someone else’s goodwill.

NOTICE-PERIOD FACTS WORTH KNOWING

- The **buyout** is usually the basic pay for the unserved days, and sometimes the full CTC equivalent — ask which. Some new employers reimburse it; ask, but ask after the offer, not during the first interview.
- **Leave balance** sometimes offsets notice days and sometimes does not. It is written in your contract. Read it before you resign, not after.
- The **relieving letter** is what background checks want. Leaving without serving notice can mean no relieving letter, which can block a future offer years later.
- If you are in a probation period the notice is usually much shorter. Check the clause.

Backlogs, gaps, and a low CGPA

Three different facts, one structure. A·F·E·F, with the weight on E.

Example 10 [Wipro · pattern]

“I see you have an active backlog.”

TRAP

WEAK. “Yes ma’am, actually that subject was very tough and the professor was very strict, most of my class also failed in it. But I have applied for the supplementary exam and I am confident I will clear it definitely. It will not affect my work at all, I assure you, I am very good in programming.”

SOLUTION

STRONG.

A: “Yes — one active backlog, in Digital Electronics.”

F: “I failed it in the fourth semester. The honest reason is that I put my time into a hackathon and a course project that semester and I under-prepared for that paper.”

E: “I sat the supplementary in December and missed by four marks, so I am re-attempting in the May window. I have been doing past papers twice a week since January with a classmate who cleared it. My other 34 papers are clear, and my last two semesters were 7.9 and 8.1, so the pattern is one bad decision rather than a general problem.”

F: “If it helps, I can tell you my exact exam date and send the result the day it is out.”

What changed	Effect
“The professor was strict” → “I under-prepared”	Ownership. This is the single biggest swing in the answer.
“Most of my class failed” removed	Comparing yourself to failing peers never helps you.
“I am confident I will clear” → a date and a method	Confidence is not evidence. A study routine is.

Adds “34 papers clear, last two semesters 7.9 and 8.1”	Puts the one failure in proportion with numbers.
Offers to send the result	Converts an open risk into a tracked item.

TRAP

If your company’s offer says “must have zero active backlogs at joining”, that is a hard rule and no interview answer changes it. Ask directly: “Is a cleared backlog before joining acceptable, or is the requirement zero backlogs throughout?” Knowing this early saves you months of false hope.

Example 11 [Capgemini · pattern]

“There is a gap of fourteen months after your graduation. What were you doing?”

SOLUTION

Gaps are common and are not, by themselves, a problem. **Unexplained** gaps are the problem. Name the reason plainly, then show the gap was not empty.

Three honest shapes, pick the true one:

Shape 1 — preparation. “I spent that year preparing for the banking exams. I sat two, cleared the first stage of one, and did not get through the final. At the end of it I decided software was the right direction for me, and since then I have completed a backend course and built two projects. I do not regret the year, but I am clear now.”

Shape 2 — family or health. “My father was ill and I was the person at home who could manage it. I did not work for about a year. That is settled now, and there is no ongoing constraint on my availability. During the second half of it I studied part-time and finished a database course.” — **You are not obliged to give medical details, yours or your family’s.** One sentence, the current status, and move to evidence.

Shape 3 — I did not find a job. “Honestly, I was job-hunting and not getting calls. My resume was weak and my practice was unstructured. What I did about it: I rebuilt my resume around two real projects, started solving problems daily — about 300 since June — and began contributing small fixes to an open-source tool, which is on my GitHub.”

The universal ending: “That is why my recent work matters more than the gap — would you like to go through the most recent project?”

SHORTCUT

For any gap, prepare **one artefact with a date in it**: a certificate, a repository with commit dates, a course completion, a freelance invoice, a volunteering letter. An artefact turns “I was studying” into “I was studying, here is the proof”. Prepare it now, not on the day.

Example 12 [Accenture · pattern]

“Your CGPA is 6.4. Our usual cut-off is 7.”

SOLUTION

When a hard cut-off is stated, there are only two useful moves: find out whether it is truly hard, and give the strongest honest counter-evidence.

Filled:

A: “Yes, 6.4 is below your cut-off.”

F: “The damage is in semesters two and three, 5.1 and 5.4. I moved cities for college and I handled it badly for a year.”

E: “From the fourth semester my scores are 7.6, 7.9, 8.1 and 8.3 — the last two years average about 8. If there is any weight given to a trend rather than an average, that is my case. I also placed second in the college hackathon, and my internship at a small analytics firm was extended from two months to four.”

F: “Is the 7.0 a hard filter in your process, or does the panel have discretion? If it is hard, I would rather know now.”

The last question is not weakness. It is the most valuable thing in the answer, because it either saves you three weeks or tells you the door is open.

TRAP

Never say “CGPA does not measure ability” or “marks are not everything”. It may be your honest belief, and it still reads as an attack on the person’s filter. Show the trend. Let the numbers argue for you.

PRACTICE · TIER 1

target: write each in 6 lines, then say it in 45 s

For each, write **your own true A·F·E·F** answer. Mark the four parts in the margin.

1. Your expected salary, in a fixed-package campus process.
2. Your expected salary, off-campus, with a researched range and one piece of evidence.
3. “We need a two-year service agreement.” — plus your three document questions.
4. Your notice period, or your earliest realistic joining date.
5. Your lowest semester, and the trend since.
6. Any backlog, gap or break you have — with one artefact you could show.
7. “Why should we not hire you?”

ANSWER KEY

Grade each answer on five points, one mark each: (1) the ACKNOWLEDGE is one sentence and contains the fact; (2) no excuse, no third party blamed; (3) the EVIDENCE contains at least one number or date a stranger could verify; (4) total length under 60 seconds spoken; (5) it ends by handing the turn back. Below 4 out of 5, rewrite before you move on. Anything untrue scores zero for the whole answer, regardless of the other points.

TIER 2

Singapore & Thailand · Grab · Shopee · GIC · DBS · Agoda · SCB

Regional questions. The interviewer is hiring into Singapore, Thailand, Indonesia or Vietnam. Two new risks appear: **will this person actually be able to work here**, and **will they leave after one year because they were homesick or underpaid for the city**.

Example 13

[Grab · pattern]

“What are your salary expectations for this role, in Singapore dollars?”

SOLUTION

The failure here is arithmetic, not confidence: candidates convert their Indian salary into the local currency and quote a number that is far below the local floor, or far above.

Structure:

1. **Quote in the local currency and the local unit** (monthly is the norm in Singapore, not annual lakhs).
2. **Anchor on the local market**, never on a conversion from home.
3. **Name the cost-of-living items** you have accounted for — this shows you did real research and will not be shocked in month two.
4. **Ask about the package’s local components**.

Filled (numbers are illustrative — research the current market for your role and level):

“I have been looking at monthly figures rather than converting from home, because a straight conversion is meaningless once rent is included.

For a backend engineer at this level I have seen a range I would place myself in, and I am comfortable discussing where in that range I sit based on your assessment.

I have budgeted for rent in a shared unit, CPF treatment for foreigners, and the difference in how the 13th-month component is handled, so I am not going to come back in three months and say I miscalculated.

Could you tell me what the package structure looks like — fixed, bonus, and whether relocation and the pass application are covered?”

TRAP

Never quote a low number hoping it makes you attractive. In regional hiring, a figure far below the local band signals that you do not understand the market, and it makes the company worry you will resign as soon as you learn what colleagues earn. Under-pricing loses offers here; it does not win them.

Example 14 [Shopee · pattern]

“Do you have the right to work here? Are you aware of the visa process?”

SOLUTION

Be plain and factual. Never guess at immigration rules, and never claim a status you do not have.

Filled: “No — I do not currently have work rights in Singapore. I would need the company to sponsor an Employment Pass.

What I do know: the application is made by the employer, there is a qualifying salary level that varies by age and sector, and the timeline is typically a number of weeks rather than days. I have my degree certificate, transcripts and passport ready to speed that up.

What I deliberately have not done is form an opinion about whether I would qualify — that is your team’s and the ministry’s call, not mine.

Two practical questions: does your process sponsor at this level, and is there a relocation allowance or a temporary accommodation period?”

SHORTCUT

The sentence “**I deliberately have not formed an opinion about whether I would qualify**” is strong. Immigration rules change, and a candidate who confidently quotes a qualifying salary figure that changed last quarter looks careless. Show preparation, not false expertise.

Example 15 [Agoda · pattern]

“You have never lived outside India. What makes you think you would stay?”

SOLUTION

This is a retention-risk question and it is asked seriously. Relocating a person is expensive, and a person who returns home in six months is a large loss.

Do not answer with enthusiasm alone. Answer with **evidence that you have thought concretely about the difficult parts.**

Filled:

A: “It is a fair question — I have not lived abroad, so I cannot give you experience as proof.”

F: “What I can give you is what I have actually checked. I have spoken to two seniors from my college who moved to Singapore and Bangkok, and I asked them specifically what they found hardest. Both said the first three months, and the cost of the first month before the first pay cheque.”

E: “So I have planned for those: I have savings for about three months, I have looked at shared accommodation near the office rather than assuming I would live alone, and I have been working with a distributed team on an open-source project for seven months, which is the closest thing I have to working across time zones and written-first communication. Also, I eat alone and cook — a small thing, but it is the thing my senior said broke people.”

F: “The real answer to your question is that I want to work on a product used across several countries, and that work is here rather than there.”

What changed vs the usual answer	Effect
----------------------------------	--------

Admits the lack of experience first	Stops the interviewer discounting everything after it.
Cites two named-source conversations	Research, not enthusiasm.
Concrete plan: savings, housing, food	Shows the hard parts were considered, not skipped.
Distributed open-source work as evidence	The nearest true proxy for the actual skill needed.
Ends with a work reason, not a lifestyle reason	“I want to travel” is the answer that loses this question.

Example 16 [DBS · pattern]

“Our team works in English but most of the office speaks Mandarin or Thai socially. Is that a concern for you?”

SOLUTION

Filled: “For the work itself, no — my written English is my strongest communication skill, and I have been in a code-review-based workflow where almost everything is written.

Socially, I expect to be outside some conversations at first, and I would rather say that honestly than claim it will not happen. My plan is simple and small: learn enough of the language for greetings, food and basic courtesy in the first two months, because that is what shows I am trying, and ask directly when a meeting slides into a language I do not follow rather than pretending I understood.

The one thing I would ask the team for is that decisions made in a corridor conversation get written down somewhere I can read. That is a reasonable ask in any language.”

SHORTCUT

That last request — “**decisions get written down**” — is a genuine remote-and-cross-culture working practice, and asking for it politely marks you as someone who has worked in a real team. Do not demand it. Ask for it.

Example 17 [SCB · pattern]

“Why are you applying to Southeast Asia rather than staying in India or trying the US?”

SOLUTION

The risk being screened: **are we your second choice, and will you leave the moment something else opens?**

Filled: “Three reasons, and I will give them in the honest order.

First, the work: the products here are built for several countries at once, with different payment methods and languages in one codebase. That is a harder and more interesting problem than the single-market products I have worked on.

Second, timing and reality: the route into this region is open to me now and clearly defined, and I would rather build a career on a real path than wait for a lottery.

Third, distance: it is a five-hour flight home, which matters to me and to my family, and I would rather tell you that than pretend it is irrelevant.

I am not going to claim I never considered anywhere else. I am telling you this is the choice I have actually made and why it holds.”

TRAP

Do not say “India has too much competition” or “salaries are low in India”. Complaints about what you are leaving make a weak case for what you are joining. Make the positive case for the work.

PRACTICE · TIER 2 target: 6 min each, written

1. Give a salary expectation in the local currency and monthly unit for a country you would actually move to. Show the three cost items you researched.
2. State your work-authorisation position in four sentences, with no guessing.

3. “What makes you think you would stay?” — with three concrete pieces of preparation.
4. “Why this region and not somewhere else?” — three reasons, honest order, no complaints about home.
5. You are asked to join in six weeks but your notice is 60 days. Write the reply.

ANSWER KEY

Full marks require: (1) local currency, local unit, and at least three named living-cost items; (2) no claimed knowledge of immigration thresholds; (3) at least two items of preparation that cost you something real (savings, a conversation, a habit); (4) zero sentences complaining about your home market; (5) a date you can actually hold, plus one alternative you can offer, such as a buyout.

TIER 3

Google · Amazon · Microsoft · Goldman Sachs · D. E. Shaw · Adobe

Structured behavioural probing on the uncomfortable material. Expect follow-ups, and expect the interviewer to stay on the topic longer than feels pleasant. That is deliberate. Staying calm is part of the answer.

Example 18 [Amazon · pattern]

“Tell me about a time you failed.”

SOLUTION

The trap is choosing a fake failure (“I worked too hard”), or a failure with no consequence. Choose a real one where **something was actually lost** and where **you caused part of it**.

STAR, four parts labelled. **This example is invented — substitute your own real failure.**

S — Situation. “In my internship I owned a small script that reconciled payment records against the vendor’s daily file. It ran every morning at 6 am.”

T — Task. “My task was to add a new vendor to it. I had two days, and I had done a similar change before.”

A — Action. “I added the new format and tested it on a sample file the vendor had sent in the first week. It passed, and I shipped it on Thursday evening. What I did not do was check that the sample matched what the vendor was **currently** sending — they had changed a date format three weeks earlier. On Friday morning the job failed, and because the failure was inside a try/catch that logged and continued — code I did not write but did not read either — nobody saw it. It was found on Monday when the finance team’s numbers did not match. Three days of reconciliation were missing.”

R — Result. “Nothing was lost permanently: we re-ran the job over the three days and the records matched, so it cost the team about half a day. I told my mentor on Monday morning before finance escalated, which he told me later was the part that mattered to him.

Two things changed in how I work. First, I never test against a sample that I did not pull from the real source that week. Second, before I touch code, I read the error path, not just the happy path — the try/catch that swallowed the error was the real reason a one-hour bug became a three-day bug. I raised that separately and we made it alert.”

Why this scores: a real cost, a clearly owned cause, an escalation done correctly, and two changed behaviours — one specific to the bug, one general.

TRAP

Failures that do not work as answers: “I am a perfectionist”, “I took on too much work”, a failure caused entirely by someone else, or a failure with no outcome at all. At Tier 3 the interviewer will keep asking until they get a real one. Prepare a real one and save everyone the time.

Example 19 [Amazon · pattern]

Follow-up probing on the same failure. Four probes, which is normal at this tier.

| SOLUTION

Probe 1 — “Why did you not check the current file format?” “Because I treated the sample as the specification. It was in the ticket, it was labelled with the vendor’s name, and it passed. The real reason underneath that is time pressure — I had two days and checking the live file meant asking someone for access. I chose the faster path, and that was the decision that failed.”

Probe 2 — “You said the try/catch was not your code. Are you blaming the previous author?” “No. The code was in the file I was editing, so it was mine to read. Whoever wrote it made a reasonable choice for a different situation — they wanted a single bad row not to kill a whole run. The mistake was that it did not distinguish a bad row from a broken format. I should have read it before I shipped on top of it.”

Probe 3 — “What would you have done if the records had not been recoverable?” “Then the honest answer is that my half-day mistake becomes a real financial problem and I would not be telling this story so calmly. What I would do is the same first step — tell my mentor immediately, with the exact window affected and what is known versus unknown — and then propose a manual reconciliation from the vendor’s copies. I would also expect to be part of a change that makes that data recoverable, because ‘we were lucky’ is not a control.”

Probe 4 — “Has this happened again since?” “Not this failure. I did have a related near-miss two months later — I almost deployed a change tested against a stale fixture. I caught it because the habit I described made me go and pull the current file first, and the format had indeed changed again. So the habit has been tested once and held once. I would not claim more than that.”

SHORTCUT

Probe 4’s answer is a model. **“The habit has been tested once and held once. I would not claim more than that.”** Precise, honest, unbluffable. Interviewers at this tier are trained to look for over-claiming, and calibrated language is scored positively.

Example 20 [Microsoft · pattern]

“You applied to us last year and were rejected. Why should this time be different?”

TRAP

WEAK. “Last time I was very nervous and I could not perform. I actually knew the answers but I blanked out. I think the interviewer was also in a hurry. This time I am much more confident.”

SOLUTION

STRONG.

A: “Yes — I interviewed for a similar role in August last year and did not clear the second round.”

F: “I know roughly why. I could not get past a brute-force solution on a string problem, and when the interviewer hinted at preprocessing, I did not take the hint because I did not understand it. The gap was that I had memorised patterns rather than understood why they worked.”

E: “What I did after that: I stopped doing new problems for six weeks and re-did forty old ones, writing out the invariant for each one before coding. Since then I have solved around 400 problems, but the measurable change is different — I now finish the hint-driven part. I also asked two seniors to run mock interviews with me and specifically to give me a hint mid-problem and see whether I used it. In the first three I did not. In the last six I did.

I also shipped a project with a real cache in it, which is exactly the preprocessing idea I could not see last year.”

F: “So the honest claim is not that I am a different person; it is that the specific gap that failed me has been worked on and tested. You will find out in the next round whether that is true.”

What changed	Effect
“I blanked out” → the exact technical gap	Shows you can diagnose yourself.
Blame on the interviewer removed	Blaming the panel ends the interview mentally.
“More confident” → a measured change in behaviour	Confidence is unverifiable; “6 of my last 9 mocks” is not.

Modest final claim

“You will find out in the next round” is calm and true.

Example 21 [Goldman Sachs · pattern]

“Do you have other offers? Where else are you interviewing?”

SOLUTION

The honest, useful answer depends on your actual situation, and all three versions below are true statements of some real situation. **Use the one that is true for you.**

If you have an offer: “Yes. I have one written offer from a mid-size product company, and I need to respond by the 28th. I would rather be straight with you about the date than create pressure later. Where is your process likely to be by then?”

If you are interviewing but have nothing: “I am in process with two other companies, one at final round. No offers yet. I am not going to name them, if that is alright — I would not want them to hear about yours from me either.”

If you have nothing at all: “No, nothing right now. I started applying recently and this is one of the first processes I am in.”

Three rules:

1. **Never invent an offer.** It is the most common lie in interviewing, and experienced interviewers ask one follow-up — “who is it with, and what did they offer?” — and hesitating there costs you the round.
2. **Never name the amount** of a competing offer unless you have decided to negotiate with it and can produce it in writing.
3. **Do give a real deadline** if you have one. Deadlines move processes; bluffs do not.

TRAP

Having no offers is not a weakness to hide. “No, nothing yet” said calmly is completely normal and costs you nothing. The damage comes from a nervous, over-long answer that makes the interviewer wonder what you are covering.

Example 22 [Uber · pattern]

“Your last role lasted seven months. Why did you leave so quickly?”

SOLUTION

Short stints are a genuine risk signal. Your job is to show it was a **specific** situation rather than **your pattern**.

Structure: state the fact → give the real reason in one sentence, without attacking anyone → show what you did before leaving → show what you now check before joining.

Filled: “Seven months, yes. I joined for a backend role, and after the first month I was moved onto a support rotation for a legacy product because the team that owned it lost two people. That was a real business need and I do not think anyone acted in bad faith.

Before I left I asked twice about returning to the original team — once informally at three months, once in my written review at five. My manager was honest with me that the rotation would last at least another year.

What I do differently now: I ask in the interview what the team’s on-call and support load is, what I would be doing in month one versus month six, and whether the role I am hired for is the role that exists. So — what would I be doing in my first month here?”

SHORTCUT

Ending a job-hopping answer with **the question you now ask** is the strongest available move. It proves the lesson is real, and it hands them the question as evidence rather than as a claim.

Example 23 [lateral · pattern]

“You were laid off. Tell me about that.”

SOLUTION

Being laid off is not a performance failure and should not be presented as one. Be brief, factual and unembarrassed.

Filled: “Yes. In March the company cut about 18% of engineering, and my whole team — eight people including the manager — was part of it. It was a cost decision about a product line, not individual performance; my last review was ‘exceeds’ and I can share it.

I had six weeks of notice, and I used them to hand over my two services properly, which mattered to me. Since then I have been interviewing and I have kept a small open-source contribution going so I am not writing nothing for three months.

My former manager is happy to be a reference and I can give you his contact.“

Three things that carry the answer: the **scale** of the cut (it happened to a group, not to you alone), a **verifiable** performance record, and a **reference from the same manager**. If any of those are true for you, say them.

If you were **dismissed for performance or conduct**, that is a different and harder conversation, and the rules do not change: do not lie, and do not volunteer more detail than the question asks. State it plainly, state what you understood the reason to be, state what you changed, and do not criticise the employer. If there is an ongoing legal matter, say only that you are not able to discuss the details. A short, calm, non-defensive answer is the best available outcome, and it is a genuinely achievable one.

Example 24 [panel · pattern]

“Are you planning to get married soon? Do you plan to have children in the next two years?”

SOLUTION

Questions about marriage, family plans, pregnancy, caste, religion, or a partner’s job are **not** job-relevant, and in many places asking them is improper or unlawful. You still have to survive the moment. You have three options, and all three are legitimate.

Option 1 — Answer the underlying business concern, not the personal question. This is usually the best move in the room. “I think what you are checking is whether I will be available and committed through the project cycle. On that: I am not planning any break, I am fully able to travel, and my plan for this role is a multi-year one. Is there a specific commitment period you need me to confirm?”

Option 2 — Decline politely and move on. “I would rather keep my personal plans out of it, if that is alright — but on availability and commitment, ask me anything.”

Option 3 — Answer it, if you genuinely do not mind. Some candidates do not mind, and that is a valid choice. It is **your** choice and nobody else’s.

What not to do: do not argue about the legality of the question in the room. You will not win the job by winning that argument. Note it, handle it, and afterwards decide whether you want to work there — that information about the culture is worth having.

TRAP

If this happens, **write down the question and the date afterwards**. If you later want to raise it with the company’s HR, a recruiter, or a campus placement cell, the specific wording matters far more than a general complaint. And weigh it honestly in your own decision: how a company screens is a preview of how it manages.

Example 25 [Google · pattern]

“Why should we not hire you?”

SOLUTION

A self-awareness probe. The failure modes are the humble-brag (“I care too much”) and genuine self-destruction (“I am bad at coding”).

Structure: one real limitation that is true and survivable → the concrete cost it has had → what you do about it → the condition under which it would genuinely be a problem here. That last clause is what separates a Tier-3 answer.

Filled: “If your team needs someone productive in a large existing codebase in week one, I am the wrong hire. My experience is on small codebases I helped start, and the one time I joined a big existing service I was slow for about three weeks — I kept reading whole files instead of following one request path through.

What I do about it now: I start by tracing a single request end to end and writing the path down, rather than reading broadly. That cut my ramp-up on the last project to about four days.

So the honest condition is this: if this role needs independent output in a mature codebase inside two weeks, weigh that. If the first month has any onboarding at all, it is not a real risk.”

What changed vs a normal “weakness” answer	Effect
A limitation that could actually cost them	Answers the question honestly instead of dodging it.
A measured past cost: “slow for three weeks”	Specific, so it reads as true.
A method, plus a measured improvement	Shows the weakness is being worked, not narrated.
Names the exact condition where it matters	Treats the interviewer as someone making a decision.

The other “will you leave us” questions

Several questions look different but are the same worry underneath: **this person’s real plan is somewhere else**. The answer shape is always the same — do not deny the other interest if it is real; show why **this** is the choice, and give it a horizon.

TIER 1

TCS NQT · Accenture · Infosys · Wipro · Capgemini

Example 26 [TCS · pattern]

“You studied Mechanical Engineering. Why are you applying for a software role?”

TRAP

WEAK. “Sir, actually there are no jobs in the core mechanical field nowadays and the salary is also very less, so that is why I am shifting to IT. Anyway, coding can be learned by anyone.”

SOLUTION

STRONG.

A: “Yes, my degree is Mechanical and I am applying for software. It is a real switch and I will not pretend otherwise.”

F: “It started in my second year with a MATLAB assignment for a heat-transfer problem. I found that I enjoyed the part where I was writing the solver far more than the part where I was analysing the result, and that did not go away.”

E: “So I did something about it over two years rather than at the last minute: a data-structures course I finished in my third year, about 300 practice problems, a final-year project where I wrote the entire sensor-logging backend while my teammates did the mechanical design, and one internship at a small firm doing Python automation. My mechanical background is not wasted either — the logging project needed me to understand what the sensor data actually meant, and that is why the validation rules in it are right.”

F: “The reason I am comfortable saying this is a real switch and not a fallback is that it cost me two years of evenings. Would it help if I went through the logging project?”

What changed	Effect
“No jobs in core” → a specific moment of interest	Turns a push into a pull. Pulls are believable.
Adds a two-year trail with dates	The switch is demonstrated, not announced.

Connects the old degree to the new work	Makes the background an asset, not a scar.
Removes “coding can be learned by anyone”	That sentence insults the job you are asking for.

Example 27 [Wipro · pattern]

“Are you planning to do an MS, or prepare for GATE or a government exam?”

SOLUTION

There are only two honest answers, and both are survivable. The unsurvivable one is a lie, because you will resign in eight months and everyone will remember.

If you are not planning it: “No. I considered GATE in my third year and decided against it — I want to work on real systems now, and if I ever do a master’s it would be much later and part-time, because I would want a work problem to take into it. My plan for the next few years is this kind of role.”

If you are genuinely considering it: “I will be straight with you. I have thought about a master’s in two or three years, but I have not applied anywhere, I am not preparing for any exam right now, and I am not going to make a decision about it before I have a couple of years of real work. If I ever do decide, you will hear it from me with proper notice rather than as a surprise.”

Why the second one does not sink you: it is what a large share of engineers actually think, the interviewer knows it, and the promise of notice is the thing they actually want. What sinks candidates is a confident “never” followed by a resignation letter next spring.

TRAP

Some companies will genuinely reject a candidate who mentions future higher studies. That is their right, and you cannot control it. What you can control is that you did not lie to get a job you would leave in a year — which, apart from being the right thing, is also the version where nobody’s reference is burned.

Example 28 [Cognizant · pattern]

“You have eleven certificates on your resume but no work experience. What does that tell me?”

SOLUTION

This is a sharp question and it is fair. A long certificate list can read as collecting rather than building.

Filled: “Fairly, it tells you I have spent more time consuming courses than shipping things. That is a real criticism and it is why I stopped.

Of those eleven, three produced something: the backend course, where the final project is the meal-counting app on my resume; the SQL one, which is where I learned the indexing that took my report from 4.1 seconds to 0.2; and the Linux one, which is why I can debug on the server instead of only on my laptop. The other eight are, honestly, watched-and-forgotten, and I should probably cut them from the page.

Since January I have changed the ratio: no new courses, two projects and daily practice instead. Would you like the three that produced something, or should I just walk through the project?”

The move that scores: agreeing with the criticism, then separating the small real part from the noise, and offering to trim your own resume in front of them. That is the opposite of defensiveness and it is very hard to argue with.

Location, shift and on-call

Example 29 [Infosys · pattern]

“Are you willing to relocate anywhere in India, and to work in night shifts?”

SOLUTION

Answer truthfully. A false yes gets discovered at allocation, and refusing an allocation after joining is far more damaging than a qualified yes now.

Structure: a clear yes to what you can truly do → the single genuine constraint, if any → the reason, stated once → what you can offer instead.

Filled, unconstrained: “Yes to relocation — I have no constraint on the city, and I would prefer Hyderabad or Pune if there is a choice, but I am not making it a condition. Night shifts are fine; I have done a rotating schedule during my internship and I know what it costs me, so I am not agreeing to it blind.”

Filled, with a real constraint: “Relocation anywhere in India, yes. On night shifts I want to be accurate: I can do a rotation, and I have done one. What I cannot commit to is a permanent night shift for more than a few months, because I am the only person at home for my mother’s medical appointments. I would rather tell you now than accept and ask for a change in month three. If the role needs permanent nights, tell me and I will step back from it.”

Why the second version is strong: the constraint is named once, the reason is given once without a plea, and the candidate offers to withdraw. That last offer is what makes the constraint read as honesty rather than as negotiation.

On-call is worth asking about, in every role. Ask: how often is the rotation, what is the expected response time, how many pages came in last month, and is it compensated? These are normal questions, the answers vary enormously, and a team with a healthy on-call will answer them happily.

TIER 2

Singapore & Thailand · Grab · Shopee · GIC · DBS · Agoda · SCB

Example 30 [Sea / Shopee · pattern]

“We would hire you on a local contract, not on expatriate terms. There is no housing allowance and no annual flight home. Is that acceptable?”

SOLUTION

This is a real and common situation for junior regional hires, and candidates often either accept without understanding it or react with offence. Do neither.

Structure: confirm you understand what a local contract means → state the two or three items you need to check → ask about the items that often do exist → give a clear conditional answer.

Filled: “I understand — a local contract means local salary, local benefits and local tax treatment, with no relocation package attached to it.

Two things I would want to check before answering properly: the one-time costs of the move, which for me are the flight, the deposit on a room, and the first month before the first pay cheque; and whether the pass application fee and any medical check are covered by the company, since I believe those are usually the employer’s cost.

I am not asking for housing or annual flights. I am asking whether there is any one-time joining support, because that is the part I cannot spread over the year.

If the answer is no on all of it, the offer can still work for me — I have savings for about three months — but I would want the base to reflect the fact that I am carrying the move myself.”

Move	Why it works
Restates the terms accurately	Shows you know what you are being offered.
Asks about one-time costs, not ongoing perks	A modest, specific, hard-to-refuse ask.
Explicitly drops housing and flights	Removes the impression that you expected expat terms.
Gives a conditional yes with a reason	Keeps the offer alive and still argues for the base.

SHORTCUT

“One-time costs” is the right frame for any relocation conversation at junior level. Ongoing allowances belong to senior transfers and are rarely available to you. A joining amount, a flight, or two weeks of temporary accommodation are small numbers for the company and large ones for you — which is exactly what makes them the right things to ask for.

TIER 3

Google · Amazon · Microsoft · Goldman Sachs · D. E. Shaw · Adobe

Stress tactics, and how to stay in the chair

Some interviewers apply pressure deliberately. Some are simply blunt, or tired, or running late. You cannot tell the difference from inside the room, and the correct response is the same either way.

THE FOUR PRESSURE MOVES AND THE REPLY TO EACH

- 1. The interruption.** You are cut off mid-sentence, repeatedly. **Reply:** stop immediately, answer what was asked, and shorten every subsequent answer. Being interrupted usually means you are too long, not that they are rude.
- 2. “That is wrong.”** Said flatly, with no explanation. **Reply:** “Let me check my reasoning” — then re-derive out loud from the start. If you find the error, say what it was. If you do not, say: “I still get the same thing. Where do you see it going wrong?” Never simply flip your answer because they frowned.
- 3. The long silence.** You finish, and they say nothing for eight seconds. **Reply:** say nothing. This is the most common trap and the most common failure. Candidates fill the silence by weakening a good answer or inventing an extra claim. Let it be quiet.
- 4. The escalating “why”.** Four or five “why”s on one thread. **Reply:** keep going honestly until you reach the edge, then name the edge: “That is as far as I have actually verified.” The escalation is **supposed** to reach your limit. Finding it is the point of the exercise, not a failure of it.

Example 31 [stress · pattern]

Mid-answer, the interviewer says: “Honestly, that sounds like a very basic project. Do you have anything harder?”

TRAP

WEAK — version 1 (collapse). “Sorry sir, yes it is basic, I know it is not a big project, maybe I should have built something bigger, I am still learning...”

WEAK — version 2 (defensive). “It is not basic at all. It has authentication, a database, deployment, cron jobs, a full REST API and a report module.”

SOLUTION

STRONG. “It is a small project, yes — 78 users and four tables. I would not call it hard in scope.

The part I found hard was narrower: two counts were wrong in week two, and the cause turned out to be duplicate rows from a double tap rather than the caching problem I spent an evening chasing. Fixing it properly meant understanding the difference between a unique constraint and an idempotent write, and I got that wrong once before I got it right.

If you want something harder in scope, the honest answer is that I have not built anything larger — my other work is practice problems, around 400 of them. If it is useful I can take you through how I would extend this one to 40 hostels, which is where the modelling starts to get interesting.”

What changed	Effect
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Agrees with the true part of the criticism	Removes the argument. You cannot be pushed off a position you already conceded.
Redefines “hard” from scope to a specific problem	Legitimate, and puts you back on your strong ground.
Admits having nothing larger	Truthful, and far safer than inflating.
Offers the extension question	Turns an attack into your best available topic.
No apology anywhere	Conceding a fact is not apologising for existing.

Example 32 [Amazon · pattern]

“Tell me about a time you disagreed with your manager — and lost.”

SOLUTION

The “and lost” is the whole question. Stories where you were right and eventually won are common and score little. STAR, four parts labelled. **Substitute your own real disagreement.**

S — Situation. “In my internship I wanted to add a database constraint to stop a duplicate row class of bug that had already caused two incidents. My mentor wanted to fix it in the application code instead, in that sprint, and add the constraint later.”

T — Task. “I thought the constraint was the real fix and the code check was the weak one, and I had to decide how hard to push and when to stop.”

A — Action. “I made the case once, properly: I wrote a short note with the two incident dates, what the code-level check would not catch — a second writer or a script — and how long the migration would take. I asked for fifteen minutes to talk it through rather than arguing in the pull request comments. He listened and then explained the thing I did not know: the table was being written to by an older service outside our team, and adding a constraint without coordinating with that team risked breaking their writes in production. I had been thinking about correctness only, and he was thinking about a dependency I could not see.

So I dropped it, and I implemented his version properly rather than half-heartedly — that part mattered to me. I also wrote the constraint work up as a ticket with the coordination step in it.”

R — Result. “The code fix held for the rest of my internship. The ticket was picked up about two months after I left and the constraint went in with the other team involved.

What I took from it: I had treated my disagreement as a technical argument when the missing piece was context I did not have. Now, before I push on a decision a second time, I ask ‘what do you know about this that I do not?’ — which is a faster way to find the missing piece than being right about the data model.”

SHORTCUT

The strongest element above is **“I implemented his version properly rather than half-heartedly.”** Interviewers at this tier are explicitly listening for whether you can commit to a decision you argued against. It is a named expectation in several companies’ behavioural frameworks, and most candidates never say it.

Background verification: what actually gets checked

Every honest answer in this chapter is also the practical one, because most of it is verified after the offer.

WHAT A TYPICAL BACKGROUND CHECK LOOKS AT

1. **Education:** degree, dates, and often the final result, checked with the university or a verification agency. Backlogs and year gaps appear here.
2. **Employment:** joining and leaving dates, designation, and sometimes salary, checked with the previous employer’s HR. This is where an inflated current CTC is found.
3. **Relieving letter:** whether you left properly. Leaving without serving notice can show up as “not eligible for rehire”.

4. **Identity and address**, and in some sectors a criminal record check.
5. **References**: usually a former manager, sometimes only confirming the basic facts.

The two facts worth remembering:

- Verification usually happens **after** you have resigned from your current job. A discrepancy found then is the worst possible timing, which is why people who inflate get hurt so badly.
- A disclosed problem is almost always survivable. An **undisclosed** one that surfaces in verification is usually not, because it is then a question of honesty rather than of a backlog or a date.

TRAP

Common small lies with large consequences: rounding a 6.4 CGPA to “about 7”, moving a leaving date by two months to hide a gap, describing a three-month internship as six, and listing a certification you started but did not finish. Each of these is checkable, each is found, and each converts a minor weakness into a disqualifying integrity finding. Fix the resume instead.

The offer-stage questions nobody teaches

ASK THESE BEFORE YOU ACCEPT ANYTHING — POLITELY, AND BY EMAIL

1. “Could I have the offer with the **fixed** and **variable** split written out?”
2. “What was the **actual** variable payout for this level last year?”
3. “Is there a service agreement, and can I see the document before I sign the offer?”
4. “Are any **original documents** required at joining?”
5. “What is the base **location**, and is it subject to change?”
6. “What is the shift, and is there **on-call**? Is it compensated?”
7. “Is the joining date firm? What has the recent gap between offer and joining been?”
8. “Who would I report to, and can I speak to someone on the team for fifteen minutes?”

Send this as one short, polite email, not as eight separate messages. Nobody minds a candidate who reads carefully. Everybody minds a candidate who drips questions for a week.

The uncomfortable one: renege on an accepted offer

You may end up with a better offer after accepting one. Here is the honest position, not a comfortable one.

WHAT IS ACTUALLY TRUE ABOUT BACKING OUT

It is not usually illegal to decline a job you accepted before joining, but it is not free either. Real costs: the company may blacklist you and, in campus hiring, your college may bar you from further drives. Recruiters and hiring managers move between companies and remember names.

What reduces the damage:

- **Decide fast.** Withdrawing in week one is a small problem. Withdrawing the night before joining is a large one.
- **Tell them by phone, then in writing.** Not by disappearing. Disappearing is what people remember.
- **Do not explain at length or invent a family emergency.** “I have accepted another role that is a better fit for my career, and I am sorry for the disruption” is enough.
- **If a service agreement was already signed and training was already given, expect a real claim** and read what you signed.

The genuine way to avoid the situation: do not accept an offer as insurance while you keep interviewing. Ask instead for more time — “could I have until the 28th?” — which is a normal request that is often granted. An honest deadline request costs you far less than a retracted acceptance.

PRACTICE · TIER 3 target: with a partner, 40 min, they must ask 3 follow-ups each

1. A real failure of yours, in STAR with the four parts labelled, where something was actually lost. Then survive three probes on **your** share of the cause.
2. A rejection or a setback you had, and the **specific** gap, plus how you measured the fix.
3. “Do you have other offers?” — answer truthfully for your real situation right now.
4. A short stint, a gap or a break in your own history, ending with the question you now ask employers because of it.
5. “Why should we not hire you?” — one real limitation, its measured cost, your method, and the condition where it would genuinely matter.
6. An improper personal question, handled with Option 1 — redirect to the business concern.
7. You accepted an offer on Monday and a better one arrives on Friday. Say out loud exactly what you will do.
8. “That sounds like a very basic project. Do you have anything harder?” — concede the true part, redefine hard, and offer your strongest topic. No apology.
9. A disagreement with a senior where **you lost**, in STAR, ending with what you did after the decision went against you.
10. “Are you planning higher studies?” — answer with whichever of the two honest versions is true for you, and include the notice promise.
11. Have your partner interrupt you mid-answer twice and then stay silent for ten seconds after you finish. Your job: shorten, and do not fill the silence.

ANSWER KEY

Score 0/1/2 per question as in Chapter 5. Additional failing conditions, which score zero regardless of delivery: any statement you are not certain is true; blaming a named person for a failure; an invented competing offer; a promised joining date that depends on someone else’s goodwill; arguing with the interviewer about a question’s fairness in the room.

Target before a Tier-3 interview: 15 out of 18, with no zeros on questions 1, 5 and 9 — those three are the ones that are always asked.

Question 11 has no marks, only a pass or fail: you pass if you shortened your answers after the interruptions and said nothing during the silence. Most people fail this the first three times they try it. Keep doing it until you pass twice in a row.

ONE-PAGE REVISION CARD

Every hard question is a risk question. Answer by removing the risk, not by defending yourself.

The three rules: never lie, never over-explain, never apologise twice.

A · F · E · F — Acknowledge (5 s) · Fact (10 s) · Evidence (25 s) · Forward (5 s). Under 60 seconds total. The marks are in the evidence.

Evidence test: could a stranger verify it? A date, a score, a certificate, a repository, a manager's name. Emotion, excuses and promises are not evidence.

Salary: CTC is not pay. Compare **fixed** first, **in-hand** second. Ask “what is the monthly in-hand for a new joiner at this level?” A bigger CTC with a variable component can pay less every month — run the numbers.

Never justify a number with your expenses. Justify it with the market and your evidence. One counter, not three. Nothing is real until it is in writing.

Fixed-band process? The negotiable items are team, location, joining date and shift, not money.

Bonds: ask duration, penalty and whether it reduces, original documents, what counts as leaving, buyout, and whether it is in the offer letter. Never hand over original certificates. Never argue enforceability in the room. Have a lawyer read anything that matters.

Notice period: quote the contract, offer a realistic best case and a buyout, never promise a date that depends on someone else.

Backlog / gap / low CGPA: own the cause in your own words, then show the trend with numbers and one artefact that has a date in it. Never blame a professor, a paper, or your class.

Regional roles: quote local currency in local units, research three living costs, never guess immigration thresholds, never complain about your home market, and prove you have thought about the hard first three months.

Failure stories need a real cost, your own share of the cause, correct escalation, and two changed behaviours. Calibrated language — “tested once, held once” — beats confident claims.

Other offers: tell the truth, give a real deadline if you have one, never invent one.

Improper personal questions: redirect to the business concern, or decline politely. Do not argue in the room. Write it down afterwards and weigh it in your decision.

Every filled answer in this chapter is invented to show the shape. Put your own true facts in. A borrowed story dies on the first follow-up.

Interview Puzzles & Guesstimates

They are not testing your answer. They are testing how you think out loud.

WHAT THIS CHAPTER GIVES YOU

- A 5-step protocol (PAUSE) you say out loud for **every** puzzle.
- 24 worked puzzles, warm-up to Tier 3, with every step shown.
- 7 worked guesstimates using one repeatable funnel.
- A short list of anchor numbers to memorise. Nothing else is memorised.
- The exact words for “I am stuck” and “I do not know” that keep you in the game.
- Weak answer vs strong answer, side by side, with the change named.

Why this round exists

A puzzle round is cheap for the company and hard to fake. In 10 minutes the interviewer learns four things that a coding test does not show.

What is scored	What a weak candidate does	What a strong candidate does
Structure	Jumps straight to guessing numbers	Names the method before computing
Assumptions	Hides them, or invents them silently	States each one out loud, with a reason
Arithmetic under pressure	Freezes, or produces 7 digits	Rounds to easy numbers, keeps going
Coachability	Defends a wrong answer	Takes the hint, folds it in, restarts cleanly

The score sheet in most companies has a line called **communication** that is worth as much as **correctness**. A candidate who reaches the wrong number while thinking clearly out loud often scores above a silent candidate who reaches the right one.

TRAP

The single biggest mistake. Silence. You think for 90 seconds, then say “42”. The interviewer has no data. From their side, a long silence and a lucky guess look identical. Say every step.

The PAUSE protocol — say these five things every time

PAUSE

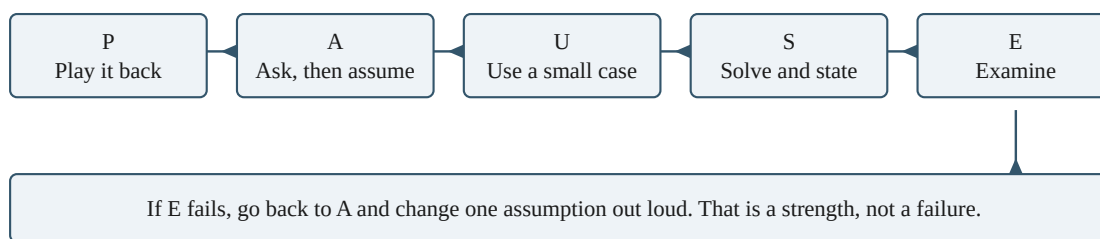
P — Play it back. Repeat the question in your own words. “So we have 12 chips, exactly one is fake, and I do not know if the fake is heavy or light. Correct?”

A — Ask, then assume. Ask at most two questions. If the interviewer says “you decide”, **state** the assumption: “I will assume the balance is exact and gives three outcomes.”

U — Use a smaller case. Solve it for 3 chips or 2 people first. Always. This is the step that separates a pass from a fail.

S — Solve and state. Do the general case. Say the numbers as you write them.

E — Examine. Sanity-check. “Is this in the right range? Does the extreme case still work?”



The same five steps, every puzzle, every time.

Weak answer vs strong answer, same puzzle

The puzzle: **a water tank has a leak that doubles in size every day. The tank is empty on day 30. On which day was it half empty?**

TRAP

WEAK. “Day 15.”

(pause)

“Because 30 divided by 2 is 15.”

SOLUTION

STRONG. “Let me play it back: the leak doubles daily and the tank is fully empty on day 30. I want the day it was half empty.

I will assume ‘doubles’ means the water lost per day doubles, so the amount remaining halves as I step backwards one day.

Small case first. Suppose the tank emptied on day 3 instead. Day 3 is empty, so day 2 is half. So the answer is always one day before the end, not half the days.

General case: day 30 is empty, so day 29 is half empty.

Examine: does 15 make sense? On day 15 the leak is 2^{15} times smaller than on day 30, so almost nothing had drained yet. So 15 is clearly wrong and 29 is right.”

WHAT EXACTLY CHANGED

1. The strong answer **restated the problem** — this buys you 5 seconds of thinking time for free.
2. It **named the assumption** about what “doubles” means.
3. It **ran a 3-day version first**. That is where the insight came from.
4. It **checked the tempting wrong answer** (15) and explained why it fails.
5. Same length. Roughly 40 seconds either way. The difference is structure, not speed.

Read that strong answer again. It is not a script. It is the PAUSE skeleton with **this** puzzle’s words poured into it. When you practise, do not memorise the sentences — memorise the five slots and fill them with whatever puzzle you are given.

The six puzzle families

Almost every interview puzzle is one of six shapes. Recognising the shape in the first 20 seconds is most of the work.

Family	Tell-tale sign	Opening move
Information / weighing	Balance, tests, questions, “minimum number of”	Count the outcomes vs the answers
Scheduling / crossing	Limited resource, must return, timings	Send the two slowest together

Invariant / parity	Toggling, flipping, colouring, “can you ever”	Find something that never changes
Counting / pigeonhole	“At least”, “guarantee”, drawers, socks	Assume the worst case, then add one
Probability / expectation	Random, chance, expected number of	Enumerate the sample space first
Game / backward induction	Two players, optimal play, voting	Solve the last move first

SHORTCUT

When you cannot tell the family, ask: “**what would happen with 2 instead of 100?**” Two is small enough to draw on paper and large enough to show the pattern.

WARM-UP

build the reflex

These are reflex builders. Aim for under 60 seconds each, spoken out loud.

Example 1 [warm-up · rates]

Five packing machines fill five cartons in five minutes. How long do 100 machines take to fill 100 cartons?

SOLUTION

U — small case. Five machines, five cartons, five minutes. So **one** machine fills **one** carton in five minutes. The five machines are working in parallel, not sharing one carton.

S — general. One machine takes 5 minutes per carton. Give 100 machines one carton each. They all work at the same time. Total time is still 5 minutes.

E — examine. Machines per carton is 1 in both cases, so nothing changed. Good.

Answer: 5 minutes

TRAP

The trap answer is 100 minutes. It comes from reading “5, 5, 5” as a ratio to scale up. Always ask **how many machines per carton**, not how many machines in total.

Example 2 [warm-up · algebra in disguise]

An ID badge and its lanyard cost Rs 110 together. The badge costs Rs 100 more than the lanyard. What does the lanyard cost?

SOLUTION

P. Two items, total 110, difference 100.

S. Let the lanyard be x . Then the badge is $x + 100$.

$$x + (x + 100) = 110$$

$$2x = 10$$

$$x = 5$$

So the lanyard is Rs 5 and the badge is Rs 105.

E. Check both conditions. Sum: $5 + 105 = 110$. Difference: $105 - 5 = 100$. Both hold.

Answer: Rs 5

TRAP

Almost everyone says Rs 10 first. Then the badge is Rs 110 and the total is Rs 120, not Rs 110. The check step catches it in three seconds. Never skip the check.

Example 3 [warm-up · counting]

Nine interns are in a room. Every intern shakes hands with every other intern exactly once. How many handshakes happen?

SOLUTION

U — small case. With 2 people: 1 handshake. With 3 people: 3. With 4 people: 6.

S. Each of the 9 people shakes 8 hands, giving $9 \times 8 = 72$. But that counts each handshake from both sides, so divide by 2.

$$\frac{9 \times 8}{2} = 36$$

E. Test the formula on the small case:

$$\frac{4 \times 3}{2} = 6$$

. Matches.

Answer: 36 handshakes

Example 4 [warm-up · relative speed]

Two delivery riders start 60 km apart and ride towards each other, one at 25 km/h and one at 35 km/h. A drone flies back and forth between them at 50 km/h until they meet. How far does the drone fly?

SOLUTION

A — assume. The drone turns instantly and never stops.

The insight. Do not track the zig-zag. Track **time**.

S — step 1: when do the riders meet? They close the gap at $25 + 35 = 60$ km/h.

$$\text{time} = \frac{60}{60} = 1 \text{ hour}$$

Step 2: how far does the drone fly in that hour? It never stops, so

$$50 \text{ km/h} \times 1 \text{ h} = 50 \text{ km}$$

E. The drone is slower than the two riders combined, so it should cover less than 60 km. $50 < 60$. Good.

Answer: 50 km

SHORTCUT

Any “back and forth until they meet” puzzle is solved by finding the **meeting time** and multiplying. Never sum the individual legs.

Example 5 [warm-up · pigeonhole]

A drawer holds 12 black socks, 12 blue socks and 12 grey socks, all mixed, in the dark. How many socks must you take out to be **sure** of a matching pair?

SOLUTION

A — assume. “Sure” means in the worst case, not on average.

S. Worst case: you pull one of each colour — black, blue, grey. Three socks, no pair. The fourth sock must repeat one of those three colours.

$$\text{colours} + 1 = 3 + 1 = 4$$

| **E.** Could 3 ever be enough? Yes, by luck, but not **guaranteed**. The question says “sure”, so 4.

Answer: 4 socks

Example 6 [warm-up · clocks]

In 12 hours, how many times do the hour hand and the minute hand overlap?

SOLUTION

S. The minute hand does 12 laps in 12 hours; the hour hand does 1. The minute hand gains $12 - 1 = 11$ laps on the hour hand. Each full lap gained is exactly one overlap.

So there are 11 overlaps in 12 hours, spaced $\frac{12}{11} \approx 1.09$ hours apart.

E. Naively you might say 12, once per hour. But between 11:00 and 12:00 there is no separate overlap — the 11th overlap **is** 12:00. That is why the count is 11, not 12.

Answer: 11 times

TIER 1

TCS NQT · Accenture · Infosys · Wipro · Capgemini

Tier 1 puzzles are the standard set. They are template-able: learn the move, and the whole family falls.

Example 7 [Infosys · pattern]

Four lab partners must cross a narrow catwalk at night. They have one working headlamp. At most two may cross at a time, and the headlamp must be carried each way. Alone they take 1, 2, 7 and 10 minutes. A pair moves at the **slower** person’s speed. What is the fastest everyone is across?

SOLUTION

Call them A(1), B(2), C(7), D(10).

U — small case, 2 people. A and B cross together: 2 minutes. Done. No return needed.

The greedy idea that people try first. Always send the fastest person as the escort:

Move	Who	Cost
cross	A + D	10
return	A	1
cross	A + C	7
return	A	1
cross	A + B	2
Total		21

The better idea. The two slow people, C and D, each cost a full crossing. Pair them so that **one** crossing pays for both:

Move	Who	Cost
cross	A + B	2
return	A	1
cross	C + D	10
return	B	2
cross	A + B	2
Total		17

E — why is 17 the floor? D must cross, costing at least 10. C must cross, and if C crosses alone or with a fast person that is 7 more, giving 17 at best; pairing C with D makes C free but then you need two fast people already on the far side to ferry the lamp, which is the $2 + 1 + 2 + 2$ above. Either way you land on 17.

Answer: 17 minutes

Here is a small search that proves it, by trying every legal sequence.

catwalk.js — brute force over all crossing plans

JS

```
const T = { A: 1, B: 2, C: 7, D: 10 };
const names = Object.keys(T);
let best = Infinity, bestPlan = null;

function subsets(arr, k) {
  return k === 0 ? [[]]
    : arr.flatMap((v, i) => subsets(arr.slice(i + 1), k - 1).map(r => [v, ...r]));
}

function go(left, right, time, lampLeft, plan) {
  if (time >= best) return; // prune
  if (left.length === 0) { best = time; bestPlan = plan; return; }
  if (lampLeft) {
    for (const pair of [...subsets(left, 2), ...subsets(left, 1)]) {
      const cost = Math.max(...pair.map(p => T[p]));
      go(left.filter(x => !pair.includes(x)), [...right, ...pair],
        time + cost, false, [...plan, `-> ${pair.join('+')} (${cost}`]);
    }
  } else {
    for (const p of right) {
      go([...left, p], right.filter(x => x !== p),
        time + T[p], true, [...plan, `<- ${p} (${T[p]}`]);
    }
  }
}

go(names, [], 0, true, []);
console.log('minimum crossing time =', best, 'minutes');
console.log('plan:', bestPlan.join(' '));
```

output

TEXT

```
minimum crossing time = 17 minutes
plan: -> A+B (2) <- A (1) -> C+D (10) <- B (2) -> A+B (2)
```

SHORTCUT

The rule for the whole family: **ship the two slowest together, and pre-position a fast pair to ferry the lamp back.** It generalises to 5, 6 or 20 people.

Example 8 [TCS NQT · pattern]

You have a 5-litre can and a 3-litre can, no markings, and a tap. Measure exactly 4 litres.

SOLUTION

A — assume. You may fill, empty, and pour from one can into the other until it is full.

S — the steps.

Step	Action	(5L, 3L)
1	Fill the 5 L can	(5, 0)
2	Pour 5 L into 3 L until full	(2, 3)
3	Empty the 3 L can	(2, 0)
4	Pour the 2 L across	(0, 2)
5	Fill the 5 L can	(5, 2)
6	Pour into the 3 L can — it takes only 1 more litre	(4, 3)

Four litres are now in the 5 L can.

E. Every step conserves water or uses the tap. Step 6 works because the 3 L can already held 2, so it accepted exactly 1.

Answer: Six steps; 4 L sits in the 5-litre can

jugs.js — BFS finds the shortest sequence

JS

```
function jugs(cap1, cap2, target) {
  const seen = new Set(['0,0']);
  let frontier = [[[0, 0], []]];
  while(frontier.length) {
    const next = [];
    for (const [[a, b], path] of frontier) {
      if (a === target || b === target) return [...path, `(${a},${b})`];
      const moves = [
        [[cap1, b], `fill ${cap1}L`], [[a, cap2], `fill ${cap2}L`],
        [[0, b], `empty ${cap1}L`], [[a, 0], `empty ${cap2}L`],
        [[a - Math.min(a, cap2 - b), b + Math.min(a, cap2 - b)], `pour ${cap1}L ->
${cap2}L`],
        [[a + Math.min(b, cap1 - a), b - Math.min(b, cap1 - a)], `pour ${cap2}L ->
${cap1}L`],
      ];
      for (const [st, label] of moves) {
        const k = st.join(',');
        if (!seen.has(k)) { seen.add(k); next.push([st, [...path, `${label} ->
(${st})`]]); }
      }
    }
    frontier = next;
  }
  return null;
}
for (const step of jugs(5, 3, 4)) console.log(' ' + step);
```

output

TEXT

```
fill 5L -> (5,0)
pour 5L -> 3L -> (2,3)
empty 3L -> (2,0)
pour 5L -> 3L -> (0,2)
fill 5L -> (5,2)
pour 5L -> 3L -> (4,3)
(4,3)
```

The general rule: with cans of size a and b you can measure any multiple of $\gcd(a, b)$ up to $\max(a, b)$. Here $\gcd(5, 3) = 1$, so every whole number from 1 to 5 is reachable. Say this line in the interview — it shows you saw the family, not just the instance.

Example 9 [Wipro · pattern]

Nine identical-looking power banks. Exactly one is heavier than the rest. You have a balance scale. Find the heavy one in two weighings.

SOLUTION

The counting bound first. Each weighing has 3 outcomes: left down, balanced, right down. Two weighings give $3 \times 3 = 9$ outcomes. There are 9 possible answers. So 2 weighings is exactly enough — and this tells you to split into **three** groups, not two.

S — weighing 1. Split into groups of 3: $X = \{1,2,3\}$, $Y = \{4,5,6\}$, $Z = \{7,8,9\}$. Weigh X vs Y .

- X goes down $\Rightarrow$ the heavy one is in X .
- Balanced $\Rightarrow$ it is in Z .
- Y goes down $\Rightarrow$ it is in Y .

Either way you are left with 3 candidates.

S — weighing 2. Take those 3 candidates, weigh one against another.

- One side down $\Rightarrow$ that is the heavy one.
- Balanced $\Rightarrow$ it is the third one, the one you left out.

E. Two weighings, always. Never three.

Answer: 2 weighings

SHORTCUT

A balance gives $\log_3$ information, not $\log_2$. For n items with one known-heavy fake you need $\lceil \log_3 n \rceil$ weighings. For $n = 27$ that is 3. Always split into thirds.

Example 10 [Capgemini · pattern]

You have two lengths of fuse cord. Each burns away completely in exactly 60 minutes, but the cord is uneven, so half the cord does **not** mean 30 minutes. Using only these two cords and a lighter, measure exactly 45 minutes.

SOLUTION

The insight. You cannot trust length, but you **can** trust that lighting **both ends** halves the time, whatever the unevenness — the two flames together consume the whole cord, and the whole cord is 60 minutes of burn.

S — the steps.

1. At time 0, light cord A at **both** ends and cord B at **one** end.
2. Cord A is gone after 30 minutes. At that moment, cord B has 30 minutes of burn left in it.
3. Immediately light cord B's **other** end. Now B is burning from both ends, so its remaining 30 minutes of burn takes 15 minutes.
4. $30 + 15 = 45$ minutes when cord B is gone.

E. Does unevenness break anything? No. Step 2 relies only on “total burn time”, not on position.

Answer: 45 minutes

TRAP

The wrong move is to try to fold a cord in half. Uneven burning means the midpoint of the **length** is not the midpoint of the **time**. Say this out loud — interviewers wait for it.

Example 11 [Accenture · pattern]

Three switches outside a sealed server room control three bulbs inside. You may flip switches as much as you like, but you may open the door and enter **once**. Match each switch to its bulb.

SOLUTION

A — ask first. “Are these filament bulbs that get warm?” If the interviewer says yes, the heat channel is open. If they say LED, you must ask for a different extra channel (a dimmer, a helper, or a second entry). **Asking this is the point of the puzzle.**

S — with warm bulbs.

1. Turn switch 1 ON. Wait 10 minutes.
2. Turn switch 1 OFF. Turn switch 2 ON. Enter immediately.
3. Inside:
 - the bulb that is **lit** → switch 2
 - the bulb that is **off but warm** → switch 1
 - the bulb that is **off and cold** → switch 3

E — why does this work? One visit gives you only one bit per bulb if you look at light alone, and you need to tell three states apart. Heat adds a second channel, so each bulb carries 2 bits worth of state: (lit or not) times (warm or not).

Answer: Use light AND heat as two separate signals

The strong candidate says the sentence “I need more than one bit per bulb, so I need a second physical channel.” That sentence is what gets scored, not the switch order.

Example 12 [Cognizant · pattern]

A corridor has 100 lockers, all closed. Student 1 opens every locker. Student 2 changes every 2nd locker (2, 4, 6, ...). Student 3 changes every 3rd locker. This continues to student 100. Which lockers are open at the end?

SOLUTION

U — small case, 10 lockers. Track locker 6. It is touched by students 1, 2, 3 and 6 — that is 4 touches, an even number, so it ends **closed**. Track locker 9: students 1, 3 and 9 — 3 touches, odd, so it ends **open**.

The pattern. Locker n is touched once per **divisor** of n . A locker ends open exactly when n has an odd number of divisors.

Why would divisors be odd? Divisors normally come in pairs: 1×12 , 2×6 , 3×4 — an even count. The pairing only breaks when a number is its own partner, that is when $n = k \times k$.

S. So the open lockers are the perfect squares:

1, 4, 9, 16, 25, 36, 49, 64, 81, 100

That is 10 lockers.

E. $\sqrt{100} = 10$, so exactly 10 squares are at or below 100. Consistent.

Answer: The 10 perfect squares: 1, 4, 9, 16, 25, 36, 49, 64, 81, 100

Example 13 [TCS Digital · pattern]

A digital clock shows hours and minutes in 24-hour form, from 00:00 to 23:59. On how many minutes of the day do all four digits read the same forwards and backwards (a palindrome), like 12:21?

SOLUTION

A — assume. Format is HH:MM with leading zeros, so 01:10 counts.

S. A palindrome means digit 1 = digit 4 and digit 2 = digit 3. Write the time as $H_1 H_2 : M_1 M_2$. So $M_2 = H_1$ and $M_1 = H_2$. The minutes are fully decided by the hour. So **count valid hours**.

Go hour by hour and check that the forced minutes are legal (minutes must be 00 to 59, so $M_1 \leq 5$, which means $H_2 \leq 5$).

- Hours 00 to 09: $H_1 = 0$, H_2 runs 0 to 9, but we need $H_2 \leq 5$. That is 00, 01, 02, 03, 04, 05 → **6 hours**.

- Hours 10 to 19: $H_1 = 1$, need $H_2 \leq 5$: 10, 11, 12, 13, 14, 15 → **6 hours**.
- Hours 20 to 23: $H_1 = 2$, H_2 runs 0 to 3, all are ≤ 5 → **4 hours**.

$$6 + 6 + 4 = 16$$

E — spot check. Hour 15 forces minutes $M_1 M_2 = 51$, so 15:51. Reversed that is 15:51. Correct. Hour 16 would force 16:61, which is not a real minute — and indeed we excluded it.

Answer: 16 minutes per day

TIER 2

Singapore & Thailand · Grab · Shopee · GIC · DBS · Agoda · SCB

Tier 2 puzzles wear business clothes. The maths is still small, but the setup is a product or operations story and there are usually two steps.

Example 14 [Grab · pattern]

A warehouse has three sealed racks. Exactly one holds a working GPU; the other two are empty. You point at rack 1. The technician, who knows what is inside every rack, opens rack 3 and shows it is empty. He offers to let you switch to rack 2. Should you switch?

SOLUTION

P. Three racks, one prize, you pick, a knowing host reveals an empty one that is not yours, then offers a swap.

A — the assumption that decides everything. The technician **knows** the contents and will **always** open an empty rack that you did not pick. State this out loud. If he opened a rack at random and it happened to be empty, the answer changes.

U — enumerate all three cases. Suppose the GPU is equally likely in each rack and you always pick rack 1.

GPU is in	Technician must open	Stay wins?	Switch wins?
Rack 1	rack 2 or 3	yes	no
Rack 2	rack 3 (forced)	no	yes
Rack 3	rack 2 (forced)	no	yes

S. Staying wins in 1 of 3 equally likely worlds. Switching wins in 2 of 3.

$$P(\text{stay}) = \frac{1}{3}, \quad P(\text{switch}) = \frac{2}{3}$$

E — why it feels wrong. Your first pick was made with no information and stays a $\frac{1}{3}$ shot. The technician's action moves **all** the remaining probability onto the one rack he did not open. Switching doubles your odds.

Answer: Switch. It wins 2/3 of the time.

racks.js — 200,000 simulated trials confirm the split

JS

```
function trial(switchDoor, rnd) {
  const prize = Math.floor(rnd() * 3);
  const pick = Math.floor(rnd() * 3);
  const open = [0, 1, 2].filter(r => r !== pick && r !== prize)[0];
  if (!switchDoor) return pick === prize;
  const final = [0, 1, 2].filter(r => r !== pick && r !== open)[0];
  return final === prize;
}

function mc(switchDoor, n = 200000) {
  let seed = 12345;
```

```
const rnd = () => (seed = (seed * 1103515245 + 12345) % 2147483648) / 2147483648;
let win = 0;
for (let i = 0; i < n; i++) if (trial(switchDoor, rnd)) win++;
return win / n;
}

console.log('stay wins about', mc(false).toFixed(3), '(exact 1/3 =',
(1/3).toFixed(3) + ')');
console.log('switch wins about', mc(true).toFixed(3), '(exact 2/3 =',
(2/3).toFixed(3) + ')');
```

output

TEXT

```
stay wins about 0.328 (exact 1/3 = 0.333)
switch wins about 0.672 (exact 2/3 = 0.667)
```

SHORTCUT

If the interviewer will not accept $\frac{2}{3}$, scale the puzzle up out loud: “**imagine 100 racks. You pick one. The technician opens 98 empty ones. Would you still keep your first pick?**” Almost nobody says yes. That is the same argument.

Example 15 [Shopee · pattern]

A QA batch has 1000 sachets. Exactly one is contaminated. A test strip shows a result after 24 hours, and you have only 24 hours before shipping. What is the smallest number of test strips that guarantees you find the bad sachet?

SOLUTION

P. One round of tests, all in parallel, 1000 candidates.

A — assume. A strip can be dipped in a **mixture** of many sachets, and it turns positive if any drop of the contaminated sachet is present.

The information count. Each strip returns 1 bit: positive or negative. With k strips you can tell apart 2^k situations.

$$2^9 = 512 < 1000, \quad 2^{10} = 1024 \geq 1000$$

So $k = 10$ strips.

S — the construction. Number the sachets 0 to 999 and write each number in 10-bit binary. Strip i receives a drop from every sachet whose bit i is a 1.

After 24 hours, read the strips: positive = 1, negative = 0. Those 10 bits, read as a binary number, **are** the sachet number.

Worked case. Suppose sachet 37 is bad. In binary, $37 = 0000100101$. So strips 0, 2 and 5 turn positive and the rest stay negative. Reading the pattern back gives 37.

E. Could 9 strips ever work? 9 bits name only 512 sachets, and $1000 > 512$, so two different sachets would have to share a pattern. Then you could not tell them apart. So 10 is the minimum.

Answer: 10 test strips

Say the words “each test gives me one bit, so k tests give me 2^k distinguishable outcomes”. That sentence is the whole puzzle. The binary labelling is just bookkeeping after it.

Example 16 [Agoda · pattern]

A training batch has 30 people. What is the chance that at least two of them share a birthday? Estimate it, then say whether you would bet on it.

| SOLUTION

A — assume. 365 days, all equally likely, no twins, ignore leap years. Say all four aloud.

The move. “At least two share” is messy. Its opposite, “all 30 are different”, is a clean product. Compute the opposite and subtract.

S. Line the 30 people up. The first may have any birthday. The second must dodge 1 day, the third must dodge 2, and so on.

$$P(\text{all different}) = \frac{365}{365} \times \frac{364}{365} \times \frac{363}{365} \times \dots \times \frac{336}{365}$$

That is 30 factors. Each is slightly under 1, and they multiply down fast.

Fast mental estimate. The number of **pairs** is

$$\frac{30 \times 29}{2} = 435$$

. Each pair collides with chance $\frac{1}{365}$. So the expected number of collisions is about $\frac{435}{365} \approx 1.19$. When the expected count of a rare event is above 1, the chance of “at least one” is already well past a half.

S — the actual number. The exact product gives about **0.706**, roughly 71%.

E. Yes, I would bet on it. And note the crossover: 23 people is where it passes 50%.

Answer: About 71%

birthday.js — exact chance for several batch sizes

JS

```
function sharedBirthday(n) {
  let allDifferent = 1;
  for (let i = 0; i < n; i++) allDifferent *= (365 - i) / 365;
  return 1 - allDifferent;
}
for (const n of [10, 23, 30, 50, 60]) {
  console.log('n =', String(n).padStart(2), '->', (sharedBirthday(n) *
100).toFixed(1) + '%');
}
```

output

TEXT

```
n = 10 -> 11.7%
n = 23 -> 50.7%
n = 30 -> 70.6%
n = 50 -> 97.0%
n = 60 -> 99.4%
```

SHORTCUT

Memorise one number: **23 people, 50%**. From there, 30 is “about 70”, 50 is “about 97”. That is enough to answer any version of this question without computing.

Example 17 [Sea / Shopee · pattern]

You must drop-test a new phone from a 100-floor test tower to find the highest floor it survives. You have exactly **two** test units. A broken unit cannot be reused. Minimise the number of drops in the **worst case**.

SOLUTION

P. Find the highest safe floor f (0 to 100) using at most 2 phones, minimising worst-case drops.

U — with one phone only. You cannot skip: if it breaks you learn nothing about the floors below. So you climb 1, 2, 3, ... and the worst case is 100 drops.

The naive two-phone plan. Drop phone 1 every 10 floors. Worst case: it survives 10, 20, ..., 90 and breaks at 100 (10 drops), then phone 2 climbs 91 to 99 (9 drops). Worst case = 19.

The fix — balance the work. The problem with fixed steps of 10 is that a late break costs the most. Make the steps **shrink**. If the first jump is k floors, then after one drop you have $k - 1$ drops of budget left for phone 2, so the next jump should be $k - 1$, then $k - 2$, and so on.

S. Total floors covered must reach 100:

$$k + (k - 1) + (k - 2) + \dots + 1 = \frac{k(k + 1)}{2} \geq 100$$

Try $k = 13$:

$$\frac{13 \times 14}{2} = 91$$

. Not enough. Try $k = 14$:

$$\frac{14 \times 15}{2} = 105 \geq 100$$

. Enough.

So the first drop is from floor 14, then 27, 39, 50, 60, 69, 77, 84, 90, 95, 99, 100.

E — check the worst case. If it breaks at floor 14, phone 2 tests 1 to 13, which is 13 more drops: $1 + 13 = 14$. If it survives everything to 99 and breaks at 100, the count of drops is also 14. The plan is balanced, which is the sign of an optimum.

Answer: 14 drops in the worst case, starting at floor 14

TRAP

Do not answer “binary search, so 7”. Binary search needs a phone you can keep breaking. With only two units the **first** break must leave a range you can walk linearly.

Example 18 [LINE MAN · pattern]

A snack pack contains one of 5 different stickers, equally likely. On average, how many packs must a customer buy to collect all 5?

SOLUTION

The move. Split the journey into 5 stages: the wait for the 1st new sticker, then the 2nd, and so on. Expectations add, even when the stages are not independent.

S — stage by stage. When you already hold i stickers, the chance a fresh pack is new is $\frac{5-i}{5}$. The average number of packs to get one success at probability p is $\frac{1}{p}$.

Stickers held	Chance next pack is new	Average packs to wait
0	$\frac{5}{5}$	1.00
1	$\frac{4}{5}$	1.25
2	$\frac{3}{5}$	1.67
3	$\frac{2}{5}$	2.50
4	$\frac{1}{5}$	5.00

$$E = \frac{5}{5} + \frac{5}{4} + \frac{5}{3} + \frac{5}{2} + \frac{5}{1} = 5 \left(1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} \right) \approx 11.42$$

E. It must be more than 5 (you cannot do better than perfect luck) and the last sticker alone costs 5 packs on average. 11.4 sits sensibly between.

Answer: About 11.4 packs

stickers.js — formula against 100,000 simulated customers

JS

```
function expectedPackets(k) {
  let e = 0;
  for (let i = 1; i <= k; i++) e += k / i;
  return e;
}
console.log('exact expectation for 5 stickers =', expectedPackets(5).toFixed(3));

let seed = 99;
const rnd = () => (seed = (seed * 1103515245 + 12345) % 2147483648) / 2147483648;
let total = 0;
const runs = 100000;
for (let r = 0; r < runs; r++) {
  const seen = new Set();
  let c = 0;
  while (seen.size < 5) { seen.add(Math.floor(rnd() * 5)); c++; }
  total += c;
}
console.log('simulation average          =', (total / runs).toFixed(3));
```

output

TEXT

```
exact expectation for 5 stickers = 11.417
simulation average          = 11.439
```

Example 19 [DBS · pattern]

You need to pick one of two team members fairly, but the only randomiser you have is a bent coin that lands heads with some unknown probability p , where $0 < p < 1$. How do you get a fair 50-50 decision?

SOLUTION

The insight. One toss is biased. But a **pair** of tosses has two outcomes that are equally likely no matter what p is.

S. Toss the coin twice.

$$P(\text{HT}) = p(1 - p), \quad P(\text{TH}) = (1 - p)p$$

These are exactly equal for every p . The other two outcomes, HH and TT, are useless.

The procedure.

1. Toss twice.
2. If you see HT, pick member A. If you see TH, pick member B.
3. If you see HH or TT, throw the pair away and start again.

E — does it always end? The chance of a useful pair is $2p(1 - p) > 0$, so the chance of never finishing shrinks to zero. If $p = 0.9$, a pair is useful with chance $2(0.9)(0.1) = 0.18$, so you need about $\frac{1}{0.18} \approx 5.6$ pairs, that is roughly 11 tosses. Slow, but exactly fair.

Answer: Toss in pairs; HT means A, TH means B, repeat on HH or TT

TIER 3

Google · Amazon · Microsoft · Goldman Sachs · D. E. Shaw · Adobe

Tier 3 puzzles need one non-obvious idea. You will not find it by grinding. You find it by asking “what stays the same?” or “what does the last move look like?”

Example 20 [Goldman Sachs · pattern]

Twelve identical chips. Exactly one is a fake. The fake may be **heavier or lighter** than a genuine chip, and you do not know which. Using a balance scale three times, identify the fake **and** say whether it is heavy or light.

SOLUTION

The counting bound first — always say this. There are $12 \times 2 = 24$ possible truths (each chip, heavy or light). Three weighings give $3^3 = 27$ outcomes. $27 \geq 24$, so three is **just** enough. That tells you the weighings must be near-perfectly balanced: each of the three outcomes should split the remaining candidates into nearly equal thirds.

A — assume. The balance is exact, and each weighing has exactly three outcomes.

S — weighing 1. Weigh $\{1, 2, 3, 4\}$ against $\{5, 6, 7, 8\}$.

Case A: balanced. The fake is one of 9, 10, 11, 12, and chips 1 to 8 are known genuine. That is 8 truths left, and $3^2 = 9 \geq 8$, so two weighings suffice.

• **Weighing 2:** $\{9, 10, 11\}$ vs $\{1, 2, 3\}$ (known genuine).

- ▶ **Balanced** $\Rightarrow$ the fake is 12. **Weighing 3:** 12 vs 1 tells you heavy or light.
- ▶ **Left down** $\Rightarrow$ one of 9, 10, 11 is heavy. **Weighing 3:** 9 vs 10. Down side is the fake; balanced means 11.
- ▶ **Right down** $\Rightarrow$ one of 9, 10, 11 is light. **Weighing 3:** 9 vs 10. The **up** side is the fake; balanced means 11.

Case B: the left group went down. Now one of $\{1, 2, 3, 4\}$ is heavy, or one of $\{5, 6, 7, 8\}$ is light. Call the down-group $A = \{1, 2, 3, 4\}$ and the up-group $B = \{5, 6, 7, 8\}$. Chips 9 to 12 are known genuine. That is 8 truths and 2 weighings left, so again it fits — but only if the second weighing splits 8 into roughly 3, 2, 3.

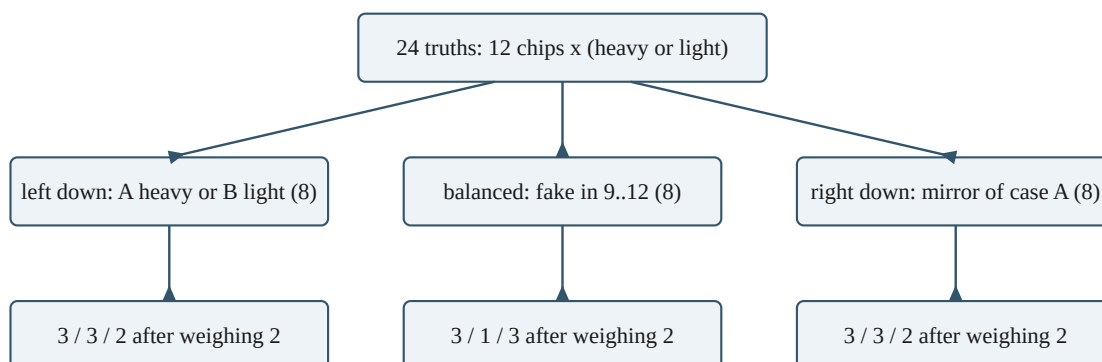
• **Weighing 2:** $\{A_1, A_2, B_1\}$ vs $\{A_3, B_2, 9\}$.

- ▶ **Balanced** $\Rightarrow$ none of those is the fake, so the fake is A_4 heavy, B_3 light, or B_4 light. **Weighing 3:** B_3 vs B_4 . If balanced, the fake is A_4 and it is heavy. Otherwise the pan that **rose** holds the light fake.
- ▶ **Left down** $\Rightarrow$ either A_1 or A_2 is heavy, or B_2 is light. **Weighing 3:** A_1 vs A_2 . Down side is a heavy fake; balanced means B_2 is light.
- ▶ **Right down** $\Rightarrow$ either A_3 is heavy or B_1 is light. **Weighing 3:** B_1 vs a genuine chip. Balanced means A_3 is heavy; otherwise B_1 is light.

Case C: the right group went down. Swap the names of the two groups and reuse Case B exactly.

E. Every branch ends with exactly one chip and a direction. 24 truths, all covered.

Answer: 3 weighings, using the split 4-vs-4, then a mixed 3-vs-3, then 1-vs-1



Weighing 1 splits 24 truths into 8 + 8 + 8. Every later split must stay near-equal.

```

function weigh(left, right, fake, heavy) {
  let d = 0; // + means the LEFT pan goes down
  if (left.includes(fake)) d += heavy ? 1 : -1;
  if (right.includes(fake)) d -= heavy ? 1 : -1;
  return d > 0 ? 'L' : d < 0 ? 'R' : '=';
}

function strategy(fake, heavy) {
  const W = (l, r) => weigh(l, r, fake, heavy);
  const r1 = W([1, 2, 3, 4], [5, 6, 7, 8]);

  if (r1 === '=') { // fake is in {9,10,11,12}
    const r2 = W([9, 10, 11], [1, 2, 3]);
    if (r2 === '=') return [12, W([12], [1]) === 'L'];
    const hi = (r2 === 'L');
    const r3 = W([9], [10]);
    if (r3 === '=') return [11, hi];
    if (hi) return [r3 === 'L' ? 9 : 10, true];
    return [r3 === 'L' ? 10 : 9, false];
  }

  const [A, B] = r1 === 'L' ? [[1,2,3,4], [5,6,7,8]] : [[5,6,7,8], [1,2,3,4]];
  const g = 9; // known genuine
  const r2 = W([A[0], A[1], B[0]], [A[2], B[1], g]);

  if (r2 === '=') {
    const r3 = W([B[2]], [B[3]]);
    if (r3 === '=') return [A[3], true];
    return [r3 === 'L' ? B[3] : B[2], false];
  }
  if (r2 === 'L') {
    const r3 = W([A[0]], [A[1]]);
    if (r3 === '=') return [B[1], false];
    return [r3 === 'L' ? A[0] : A[1], true];
  }
  const r3 = W([B[0]], [g]);
  if (r3 === '=') return [A[2], true];
  return [B[0], false];
}

let ok = 0;
const bad = [];
for (let fake = 1; fake <= 12; fake++) {
  for (const heavy of [true, false]) {
    const [id, h] = strategy(fake, heavy);
    if (id === fake && h === heavy) ok++; else bad.push([fake, heavy, id, h]);
  }
}
console.log('counting bound: 3^3 =', 3 ** 3, 'outcomes for 24 answers');
console.log('cases solved correctly:', ok, 'of 24');
console.log('failures:', bad.length === 0 ? 'none' : bad);

```

output

TEXT

counting bound: $3^3 = 27$ outcomes for 24 answers
cases solved correctly: 24 of 24
failures: none

Example 21 [Amazon · pattern]

A hundred ants are placed at random points on a 1-metre pole, each facing left or right at random. They all walk at 1 metre per minute. When two ants meet head-on, both instantly reverse. An ant that reaches an end of the pole falls off. What is the longest possible time before the pole is empty?

SOLUTION

The insight — the one idea in this puzzle. The ants are identical. When two ants collide and both reverse, the **picture** is exactly the same as if they had walked **through** each other and swapped identities. Nothing observable changes.

S. So ignore collisions completely. Each ant now walks in a straight line at 1 m/min until it falls off. The longest any single ant can walk is the full length of the pole, 1 metre, which takes 1 minute.

Therefore the pole is empty after at most **1 minute**, no matter how the 100 ants are arranged.

E — sanity check with 2 ants. Two ants at the very ends, walking towards each other. They meet in the middle at 0.5 min, reverse, and each walks 0.5 m back to its own end, arriving at 1 min. Matches.

E — why “at most” and not “exactly”? If every ant happens to be facing the nearer end, they all leave sooner. The worst case is one ant starting at one end facing the far end.

Answer: 1 minute

SHORTCUT

The “pass through instead of bouncing” swap works whenever the objects are **indistinguishable** and the collision rule is symmetric. Look for it in any bouncing-particle or swapping-token puzzle.

Example 22 [D. E. Shaw · pattern]

Five engineers, ranked 1 (most senior) to 5, must split a bonus of 100 indivisible units. The most senior **surviving** engineer proposes a split. Everyone still in the room votes. If at least half vote yes, the split stands. If not, the proposer is removed from the process and the next most senior proposes. Every engineer wants, in order: to stay in the process, then to get the most units, then to see a senior removed. What should engineer 1 propose?

SOLUTION

The method. Never start at 5 engineers. Start at the **end state** and walk backwards. This is backward induction.

Case: only 5 is left. Engineer 5 proposes “100 to me” and votes yes. 1 of 1 is at least half. So 5 gets 100. **This means engineer 4 must never let it reach this stage.**

Case: 4 and 5 are left. Engineer 4 proposes. With 2 voters, “at least half” is 1 vote — and 4’s own yes is enough. So 4 proposes “100 to me” and it passes. Result: (4 : 100, 5 : 0). So **engineer 5 gets nothing** if it reaches the pair stage, and would accept even 1 unit to avoid it.

Case: 3, 4, 5 are left. Engineer 3 needs 2 of 3 votes: his own plus one. Who is cheapest? Engineer 5 gets 0 in the next stage, so 1 unit buys his yes. Engineer 4 gets 100 next stage, so he is unbuyable. Result: (3 : 99, 4 : 0, 5 : 1).

Case: 2, 3, 4, 5 are left. Engineer 2 needs 2 of 4 votes: his own plus one. In the next stage, 4 gets 0. So 1 unit buys engineer 4. Result: (2 : 99, 3 : 0, 4 : 1, 5 : 0).

Case: all five. Engineer 1 needs 3 of 5 votes: his own plus two. In the next stage (the 2-3-4-5 case) engineers 3 and 5 both get 0. So 1 unit each buys them both.

(E1 : 98, E2 : 0, E3 : 1, E4 : 0, E5 : 1)

E — verify engineer 3's choice. If he votes no, engineer 1 is removed and 3 lands in the 2-3-4-5 case where he gets 0. Voting yes gets him 1. He votes yes. Same logic for 5. So the proposal passes 3 votes to 2.

Answer: 98 / 0 / 1 / 0 / 1

The scoring point here is not the numbers. It is that you **said out loud** “I will solve the last stage first.” Interviewers stop listening to the arithmetic once they hear that sentence.

Example 23 [Microsoft · pattern]

You toss a fair coin repeatedly. On average, how many tosses until you first see the pattern **HH**? And until you first see **HT**? Are they the same?

SOLUTION

Guess first, then check. Most people say both are 4, since each 2-letter pattern has probability $\frac{1}{4}$. That is wrong for HH, and explaining **why** is the whole answer.

S — HT first. Wait for the first head; that takes 2 tosses on average. After a head, keep tossing until a tail; that also takes 2 on average. Nothing can “undo” the head you already have — a second head simply replaces it as the newest head.

$$E[\text{HT}] = 2 + 2 = 4$$

S — HH. Set up states by how much of the pattern you already hold.

- E_0 = expected tosses from “nothing yet”
- E_1 = expected tosses from “one head so far”

From state 0, one toss lands you in state 1 (heads) or back in state 0 (tails):

$$E_0 = 1 + \frac{1}{2}E_1 + \frac{1}{2}E_0$$

From state 1, one toss finishes (heads) or **resets you all the way to state 0** (tails):

$$E_1 = 1 + \frac{1}{2}(0) + \frac{1}{2}E_0$$

Solve. From the first equation,

$$\frac{1}{2}E_0 = 1 + \frac{1}{2}E_1$$

, so

$$E_0 = 2 + E_1$$

. Substitute into the second:

$$E_1 = 1 + \frac{1}{2}(2 + E_1) = 2 + \frac{1}{2}E_1$$

, giving

$$E_1 = 4$$

and

$$E_0 = 6$$

.

E — why the difference? After a head, a tail **completes** HT but **destroys** HH. That asymmetry costs HH two extra tosses on average. The patterns have the same probability per window but different **overlap** structure.

Answer: HT takes 4 tosses on average; HH takes 6

Example 24 [Google · pattern]

A hundred new joiners stand in a line, each wearing a red or blue lanyard chosen at random. Each person can see every lanyard in front of them but not their own and not those behind. Starting from the back, each must say one

word — “red” or “blue” — loud enough for all to hear, and it must be a guess at their own lanyard. They may agree on a strategy beforehand but not communicate afterwards. How many can be guaranteed correct?

SOLUTION

P. 100 people, each hears all earlier guesses and sees all lanyards ahead. One word each, back to front.

U — small case, 2 people. Person 2 (at the back) sees person 1. If person 2 simply says what he sees, person 1 repeats it and is certain to be right. Person 2 is a coin flip. So 1 guaranteed out of 2.

The generalisation. The back person is the sacrifice. He must broadcast **one bit** that helps all 99 people ahead of him. One bit cannot name 99 colours — but it can name a **parity**.

S — the strategy.

1. Agree beforehand: “red” means **the number of red lanyards I can see is even**; “blue” means it is odd.
2. Person 100 (at the back) counts the reds in front of him and says the matching word. He is right only by luck.
3. Person 99 counts the reds **he** can see. If his count has the same parity as announced, his own lanyard is blue. If the parity differs by one, his own is red. He says it, and he is certain.
4. Person 98 heard both the original parity **and** person 99’s true colour. He updates the running parity and repeats the same reasoning. And so on down the line.

Worked micro-example with 4 people. Lanyards front-to-back are R, B, R, ? . Person 4 sees R, B, R — that is 2 reds, even — so he says “red”. Person 3 sees R, B: 1 red, odd. Announced parity was even, so person 3’s own must be red. He says “red”, correctly. Person 2 now knows the total parity was even and that person 3 is red, so the parity among persons 1 and 2 must be odd. He sees R in front, which is 1 red, already odd, so his own must be blue. Correct.

E. Everyone from 99 down to 1 is certain. Person 100 is a 50-50 shot.

Answer: 99 guaranteed correct; the 100th is a coin flip

SHORTCUT

Whenever one person must help many with a single message, the answer is almost always **parity**. It is the only thing that compresses a whole line into one bit.

Part 2 — Guesstimates

A guesstimate asks for a number nobody knows: cups of tea sold in a city, routers replaced per year, the weight of all the luggage on a flight. There is no correct answer. There is a correct **method**.

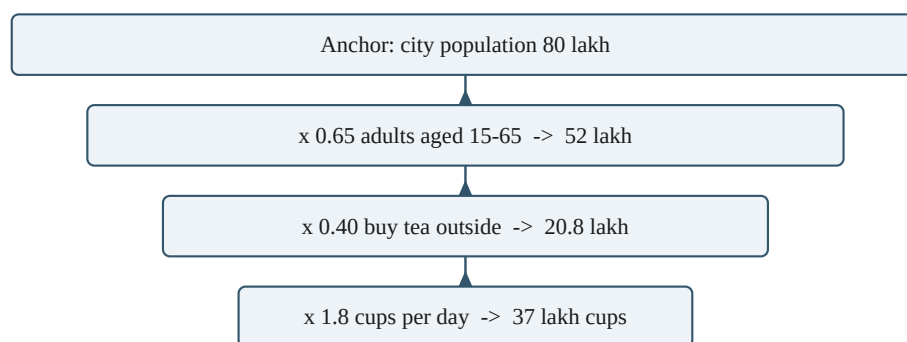
THE RULE THAT DECIDES YOUR SCORE

You are judged on the **chain**, not the number. An answer within a factor of 3 of the interviewer’s own estimate is a pass. An answer with no visible chain is a fail even if it is exactly right.

The funnel — five slots, always the same

THE GUESSTIMATE FUNNEL

1. **Define.** Say exactly what you are counting, in what unit, over what time. “Cups sold per day by roadside stalls, in one city, on a normal weekday.”
2. **Anchor.** Start from one number you actually know — usually a population.
3. **Filter.** Narrow with 3 to 5 multiplications, each one a stated assumption.
4. **Compute.** Round hard. Say the running total after every step.
5. **Sanity-check.** Cross-check with a second, different route, or a per-person feel test.



Each step is one stated assumption. Say the running total out loud after each one.

Anchor numbers — the only things you memorise

Round these. Precision is worthless here; a wrong order of magnitude is fatal.

Anchor	Use
India population	1.4 billion = 140 crore
Indian metro city	80 lakh to 2 crore
Indian tier-2 city	10 to 30 lakh
Average Indian household size	about 4 people
Share of India aged 15-65	about 65%
Singapore population	about 6 million
Thailand population	about 70 million; Bangkok about 10 million
Indonesia population	about 280 million; Jakarta metro about 30 million
Days in a year	365, but use 300 working days / 50 weeks
Smartphone life	about 3 years
Laptop / router life	about 5 years
Car life	about 12 years

SHORTCUT

Never divide by 365 in your head. Use 400 and say “I am rounding 365 up to 400 to keep the arithmetic clean; that makes my answer about 10% low.” Interviewers **like** this sentence.

Weak answer vs strong answer, same guesstimate

The question: **how many cups of tea are sold by roadside stalls in one Indian metro city in a day?**

TRAP

WEAK. “Hmm. India has a lot of tea drinkers. I’d say... maybe 10 lakh cups? Actually maybe more. 50 lakh. Let’s say 50 lakh.”

SOLUTION

STRONG.

“**Define:** cups sold by roadside stalls in one metro, on a normal weekday. Not packaged tea, not home tea, not cafes.

Anchor: take a metro of 80 lakh people.

Filter 1: roughly 65% are aged 15 to 65 — that is the band that drinks tea outside. That gives 52 lakh.

Filter 2: of those, maybe 40% actually buy tea from a stall rather than making it at home or drinking office tea. I am using 40% because tea is cheap and commuting is common, but in a wealthier city I would drop it to 25%. That gives about 21 lakh buyers.

Filter 3: a regular buyer has about 2 cups a day — one on the way to work, one in the afternoon. I will use 1.8 to allow for the light drinkers. That gives about 37 lakh cups.

Answer: roughly 35 to 40 lakh cups a day.

Sanity-check, second route: a busy stall sells maybe 300 cups a day. 37 lakh divided by 300 gives about 12,000 stalls in a city of 80 lakh, or one stall per 650 people. Walking through any Indian metro, that feels about right — maybe slightly low. So I am comfortable with 35 to 50 lakh.

If you want, I can tell you which assumption my answer is most sensitive to.“

WHAT EXACTLY CHANGED

1. The strong answer **defined the unit** before touching any number. The weak one never said whether it meant a city, a state, or the country.
2. Every multiplier came with a **reason**, and one came with a **range** (“40%, but 25% in a wealthier city”).
3. It **restated the running total** after each step, so the interviewer could interrupt with a correction — which is exactly what they want to do.
4. It gave a **band**, not a single digit.
5. It **cross-checked from a completely different direction** (stalls, not people).
6. It ended by **offering the next step**. That converts a monologue into a conversation.

Do not memorise those sentences. Memorise the five slots — Define, Anchor, Filter, Compute, Sanity-check — and fill them with the numbers **you** believe. If you recite the tea answer for a question about helmets, the interviewer will hear it.

WARM-UP

build the reflex

Example 25 [warm-up · guesstimate]

How many chairs are in your engineering college?

SOLUTION

Define. All seating in one campus: classrooms, labs, library, canteen, offices, auditorium.

Anchor. 4 years times 3 branches times 60 students = 720 students. Round to 800.

Filter by place.

Place	Reasoning	Chairs
Classrooms	One seat per student, plus 30% spare rooms	1000
Labs	Half the students seated at a time, 2 shifts of rooms	500
Library	About 10% of students at once	80
Canteen	About 15% of students at once	120
Staff offices	1 teacher per 20 students = 40 staff, 2 chairs each	80
Auditorium	Seats the whole college at once	800
Total		about 2,600

Sanity-check. That is about 3.2 chairs per student. For a campus where every student has a classroom seat **and** a lab seat **and** a share of the auditorium, 3 sounds right. If I had got 0.5 or 30, I would go back.

Answer: About 2,500 to 3,000 chairs

TIER 1

TCS NQT · Accenture · Infosys · Wipro · Capgemini

Example 26 [Infosys · pattern]

How many Wi-Fi routers are sold each year in one Indian metro city of 80 lakh people?

SOLUTION

Define. Home broadband routers, retail plus those supplied by internet providers, in one city, per year.

Anchor. 80 lakh people.

Filter.

1. Households: $80 \text{ lakh} \div 4 = 20 \text{ lakh households}$.
2. Households with home broadband: I will say 35%. Mobile data is dominant in India, so this is well under 100%. That gives 7 lakh connected households.
3. Routers per connected household: 1.1, allowing for a few homes with a second unit. That gives 7.7 lakh routers **in use**.
4. Replacement rate: a router lasts about 5 years, so $\frac{1}{5}$ of the installed base is replaced each year. That gives 1.54 lakh.
5. Growth: new connections are still being added, maybe 10% on top. Call it 1.7 lakh.

Answer: roughly 1.5 to 2 lakh routers a year.

Sanity-check. That is about 1 router per 45 people per year. A router lasting 5 years, in a city where roughly 1 person in 11 lives in a connected household, works out to about 1 per 50 per year. Close enough.

Answer: About 1.5 to 2 lakh routers per year

funnel.js — the same chain, printed step by step

JS

```
function funnel(name, steps) {
  let v = 1;
  console.log('\n' + name);
  for (const [label, factor] of steps) {
    v *= factor;
    console.log('  x ' + String(factor).padEnd(12) + label.padEnd(38) + '=> ' +
fmt(v));
  }
  return v;
}

function fmt(n) {
  if (n >= 1e7) return (n / 1e7).toFixed(2) + ' crore';
  if (n >= 1e5) return (n / 1e5).toFixed(2) + ' lakh';
  if (n >= 1e3) return (n / 1e3).toFixed(2) + ' thousand';
  return n.toFixed(2);
}

const routers = funnel('Routers replaced per year, city of 80 lakh', [
  ['city population', 8_000_000],
  ['people per household', 1 / 4],
  ['households with home broadband', 0.35],
  ['routers per connected household', 1.1],
  ['fraction replaced each year (5-year life)', 1 / 5],
```

```
});  
console.log(' ANSWER ~', fmt(routers), 'routers/year');
```

output

TEXT

```
Routers replaced per year, city of 80 lakh  
x 8000000    city population          => 80.00 lakh  
x 0.25       people per household     => 20.00 lakh  
x 0.35       households with home broadband => 7.00 lakh  
x 1.1        routers per connected household => 7.70 lakh  
x 0.2        fraction replaced each year (5-year life) => 1.54 lakh  
ANSWER ~ 1.54 lakh routers/year
```

Example 27 [TCS NQT · pattern]

How many windows are there on a 20-storey office tower?

SOLUTION

Define. Outward-facing glass panes on one rectangular tower.

Anchor — geometry, not population. A typical floor plate is about 40 m by 25 m. The perimeter is $2(40 + 25) = 130$ m.

Filter.

1. A window module is about 1.5 m wide, so one floor has $130 \div 1.5 \approx 87$ panes. Round to 85.
2. Twenty floors gives $85 \times 20 = 1700$.
3. The ground floor is mostly entrance and service, so subtract roughly 40. Call it 1,660.

Answer: about 1,600 to 1,800 windows.

Sanity-check. 85 windows per floor for a floor of about 1,000 square metres is one window per 12 square metres of floor. In an open-plan office with continuous glazing, that is reasonable.

Answer: About 1,700 windows

SHORTCUT

Not every guesstimate anchors on population. The three anchors are **people**, **geometry** and **money**. Pick the one closest to the thing being counted.

TIER 2

Singapore & Thailand · Grab · Shopee · GIC · DBS · Agoda · SCB

Example 28 [Grab · pattern]

Estimate the number of food-delivery orders placed in Singapore on a normal weekday.

SOLUTION

Define. Completed restaurant-to-home orders through delivery apps, in Singapore, on a weekday.

Anchor. Singapore population about 6 million (including residents and pass holders).

Filter.

1. Delivery-app users: Singapore has very high smartphone use and small homes, so I will say 55% of the population uses a delivery app at least sometimes. That gives 3.3 million users.
2. Order frequency: a typical user orders maybe twice a week, so on any given weekday about $\frac{2}{7} \approx 0.29$ of users order. Round to 0.3. That gives about 1.0 million orders.
3. But an order often feeds more than one person. Divide by about 1.5 people per order, giving roughly **650,000 to 700,000 orders a day**.

Answer: roughly 0.6 to 0.8 million orders on a weekday.

Sanity-check from the supply side. Suppose there are about 12,000 active riders, each completing about 20 deliveries in a shift. That is 240,000 deliveries — which is a **lot** lower than my demand number. That gap is useful: either there are more riders than I assumed, or my order frequency is too high. I would revise down towards 300,000 to 500,000 and say so.

What the revision shows. Noticing and **naming** the gap between the two routes scores higher than either number alone.

Answer: Roughly 0.3 to 0.7 million orders per weekday, with the supply check pulling the estimate down

This example deliberately ends with a disagreement between two routes. That is realistic. Do not hide it. Say: “My two routes disagree by a factor of two. Here is which one I trust more and why.”

Example 29 [Agoda · pattern]

A hotel booking site wants to size its market. Estimate the number of hotel room-nights booked in Bangkok in a year.

SOLUTION

Define. Paid room-nights in hotels and serviced apartments in Bangkok, per year, both tourist and domestic business travel.

Two routes, both from the supply side and the demand side.

Route A — supply side.

1. Bangkok has roughly 2,000 to 3,000 registered hotels. Use 2,500.
2. Average size about 120 rooms. That gives 300,000 rooms.
3. Rooms available per year: $300,000 \times 365 \approx 110$ million room-nights of **capacity**.
4. Average occupancy about 65%. That gives about **71 million room-nights sold**.

Route B — demand side.

1. Bangkok receives roughly 20 million international visitors a year.
2. Average stay about 4 nights, and about 1.7 guests share a room. So room-nights from international visitors = $\frac{20 \text{ million} \times 4}{1.7} \approx 47$ million.
3. Domestic and business travel adds maybe 40% more: about 66 million.

Compare. 71 million and 66 million. The two routes agree within 10%, which is unusually good.

Answer: about 65 to 75 million room-nights a year.

Sanity-check. 70 million room-nights divided by 365 is about 190,000 rooms occupied on an average night, out of 300,000 rooms. That is 63% occupancy, which matches the assumption I started from. Consistent.

Answer: Roughly 65 to 75 million room-nights per year

TIER 3

Google · Amazon · Microsoft · Goldman Sachs · D. E. Shaw · Adobe

Example 30 [Amazon · pattern]

A photo-sharing app has 5 million monthly active users. Estimate its yearly cloud storage bill, and then say what you would change to cut it in half.

SOLUTION

Define. Cost of storing user photos for one year. Ignore bandwidth and compute; say so.

Step 1 — how many photos arrive per year?

- 5 million monthly actives; suppose 40% upload in a given month, so 2 million uploaders.
- Each uploader posts about 8 photos a month.
- $2 \text{ million} \times 8 \times 12 = 192$ million photos a year. Round to **200 million**.

Step 2 — how big is a photo?

- An original phone photo is about 3 MB.
- But a real app stores several sizes: original, a display copy, a thumbnail. Say $3 + 0.4 + 0.05 \approx 3.5$ MB per photo.
- $200 \text{ million} \times 3.5 \text{ MB} = 700 \text{ million MB} = 700 \text{ TB}$ per year of **new** data.

Step 3 — replication.

- Object storage keeps about 3 copies for durability, but that is usually already inside the quoted price. I will assume the price already includes it and say so.

Step 4 — cost.

- Object storage runs roughly USD 0.02 per GB per month. 700 TB is 700,000 GB.
- $700,000 \times 0.02 = \text{USD } 14,000$ per month for one year's worth of photos.
- But storage **accumulates**. In year 1 the average stored volume is only half the final amount, so year 1 costs about $14,000 \times 12 \times 0.5 \approx \text{USD } 84,000$.
- By year 3, with 3 years of photos on disk, the bill is roughly $14,000 \times 12 \times 2.5 \approx \text{USD } 420,000$ a year.

Answer: about USD 80,000 in year one, rising to roughly USD 400,000 a year by year three.

Step 5 — how to halve it. Rank by leverage, because in a pure product every factor counts equally, so attack the ones you can actually control:

1. **Tiered storage.** Photos older than 90 days are almost never viewed. Moving them to cold storage at about one-fifth the price cuts the bill by roughly 60% on its own. This is the single biggest lever.
 2. **Stop storing the original.** Re-encode to a modern format at about 1 MB. That is a 3x cut on new data, but it is irreversible, so it needs a product decision.
 3. **De-duplicate.** Identical re-uploads (memes, forwards) can be 5 to 15% of volume.
 4. **Delete on account closure.** Often forgotten, and often several percent.
- E. Levers 1 and 2 together are more than 2x, so halving the bill is clearly achievable.

Answer: About USD 80k in year 1, rising to roughly USD 400k by year 3; cold tiering is the main lever

Tier 3 guesstimates almost always end with “**and what would you do about it?**”. Practise the second half. A number with no recommendation is only half an answer.

Which assumption should you argue about?

In a pure chain of multiplications, every factor moves the answer by the same percentage. So do not spend your time polishing the factor you know best. Spend it on the one you are **least sure of**.

doubt.js — rank your assumptions by your own uncertainty

JS

```
const doubt = [
  { factor: 'city population',    spreadPct: 10 },
  { factor: 'share aged 15-65',  spreadPct: 8 },
  { factor: 'share who buy out',  spreadPct: 60 },
  { factor: 'cups per buyer/day', spreadPct: 100 },
];

console.log('WRONG - default sort compares text:');
console.log([100, 60, 10, 8].sort());

console.log('\nRIGHT - numbers need a comparator:');
console.log([100, 60, 10, 8].sort((a, b) => b - a));

console.log('\nattack this assumption first:');
for (const d of [...doubt].sort((a, b) => b.spreadPct - a.spreadPct)) {
  console.log(' +/- ' + String(d.spreadPct).padStart(4) + '% ' + d.factor);
}
```

output

TEXT

WRONG - default sort compares text:
[10, 100, 60, 8]

RIGHT - numbers need a comparator:
[100, 60, 10, 8]

attack this assumption first:
+/- 100% cups per buyer/day
+/- 60% share who buy out
+/- 10% city population
+/- 8% share aged 15-65

TRAP

JS trap you must know. `[100, 60, 10, 8].sort()` returns `[10, 100, 60, 8]`. The default sort converts every element to a string and compares text, so "10" comes before "100" which comes before "60". Numbers always need $(a, b) \Rightarrow a - b$ for ascending or $(a, b) \Rightarrow b - a$ for descending. This exact bug has cost people offers in coding rounds.

How close is close enough?

band.py — judging a guesstimate by order of magnitude

PYTHON

```
import math

def magnitude_gap(mine, theirs):
    return abs(math.log10(mine / theirs))

truth = 3_700_000
for guess in [400_000, 1_200_000, 3_000_000, 9_000_000, 40_000_000]:
    gap = magnitude_gap(guess, truth)
    verdict = "same ballpark" if gap < 0.5 else ("one zero off" if gap < 1.5 else
"way off")
    print(f"guess {guess:>12,} log10 gap {gap:.2f} -> {verdict}")

base = {"people": 8_000_000, "adult_share": 0.65, "buyer_share": 0.40, "cups": 1.8}

def answer(d):
    return d["people"] * d["adult_share"] * d["buyer_share"] * d["cups"]

print(f"\nbase answer = {answer(base):.0f}")
for k in base:
    hi = dict(base); hi[k] = base[k] * 1.5
    print(f"    raising {k:<12} by 50% changes the answer by {(answer(hi)/
answer(base)-1)*100:.0f}%")
```

output

TEXT

guess	400,000	log10 gap 0.97	-> one zero off
guess	1,200,000	log10 gap 0.49	-> same ballpark
guess	3,000,000	log10 gap 0.09	-> same ballpark
guess	9,000,000	log10 gap 0.39	-> same ballpark
guess	40,000,000	log10 gap 1.03	-> one zero off

```
base answer = 3,744,000
  raising people      by 50% changes the answer by 50%
  raising adult_share by 50% changes the answer by 50%
  raising buyer_share by 50% changes the answer by 50%
  raising cups        by 50% changes the answer by 50%
```

Look at the bottom block. Every factor moves the answer by exactly 50%. That is the mathematical proof of the rule above: in a product model there is no “unimportant” assumption, only assumptions you are more or less sure about.

When you are stuck — what to actually say

You **will** get stuck. Everyone does. The score depends entirely on what comes out of your mouth in the next 15 seconds.

THE RECOVERY STRUCTURE — FOUR MOVES, IN ORDER

1. **Name where you are.** “I have the setup but I cannot see the move yet.”
2. **Say what you have ruled out, and why.** “Brute force is out because the search space is 2^{100} .”
3. **Shrink the problem out loud.** “Let me drop to 3 people and see what happens.”
4. **Ask for a nudge, specifically.** “Am I right that the fake could be lighter, not just heavier? That is the branch I am stuck on.”

TRAP

WEAK when stuck. Long silence. Then: “Sorry... I’ve seen this one but I don’t remember the answer.”

Two separate failures. The silence gave no data, and admitting you were trying to **recall** rather than **derive** tells the interviewer the round is worthless.

SOLUTION

STRONG when stuck. “Let me say where I am. I know three weighings give 27 outcomes and I have 24 truths to separate, so it is tight but possible. My problem is the second weighing in the unbalanced branch — a plain 2-vs-2 leaves 4 candidates, and one weighing cannot split 4 into

1. So I need to **mix** a suspect-heavy chip with a suspect-light one on the same pan. Let me try that.”

WHAT EXACTLY CHANGED

1. It **named the exact sub-step** that failed, not a general “I’m stuck”.
2. It **showed the counting argument still standing**, proving the earlier work was real.
3. It **diagnosed why** the obvious move fails (“4 cannot split into 1”).
4. It **proposed the next experiment**. The interviewer can now nudge with one word.
5. It never said “I have seen this before”. Derive, do not recall.

Saying “I do not know” without losing the round

Sometimes you genuinely have nothing. There is still a right way.

TRAP

WEAK. “I don’t know.” (silence)

SOLUTION

STRONG. “I do not have a method for this one. Let me tell you how I would attack it if I had ten more minutes: I would first check whether some quantity stays fixed as the tokens move, because the question asks whether a state is **reachable**, and those are usually invariant problems. If you would rather move on, I am happy to, but I would like to come back to it at the end.”

WHAT EXACTLY CHANGED

The strong version admits ignorance **once**, in one short sentence, then spends the rest of the answer showing method. It also offers the interviewer control. You cannot fake an answer, but you can always show a plan of attack.

TRAP

Never bluff. If you half-remember a puzzle and state the answer with confidence but cannot reconstruct it, the very next question will be “why?” and you will have nothing. A derived wrong answer scores above a recalled right one you cannot defend.

Linking puzzles to the behavioural round

Interviewers often follow a puzzle with: “**tell me about a time you had to make a decision with incomplete information.**” The puzzle round and the behavioural round are testing the same muscle.

Use STAR. The four parts are labelled below so you can see the shape.

TRAP

WEAK. “In my final year project we didn’t have all the data but I managed it and it worked out fine. I’m good under pressure.”

SOLUTION

STRONG — with the four STAR parts labelled.

S — Situation. “In my sixth-semester project my team of four was building a bus-tracking app for our campus. Two weeks before the demo the college would not give us the real GPS feed from the bus contractor, and we had no idea if we would ever get it.”

T — Task. “I was responsible for the tracking module. I had to decide whether to keep waiting for the real feed or to build without it, knowing a wrong choice would cost us the demo.”

A — Action. “I listed what I actually knew: the feed, if it came, would be latitude, longitude and a timestamp every 30 seconds, because that is what the contractor’s public documentation said. I could not confirm anything else. So I wrote the app against that one assumption and put a small adapter layer in front of it. Then I built a replay service that pushed recorded GPS points from my own phone into the same adapter, so the app could run fully without the real feed. I told my guide in writing what assumption I had made, so that if it was wrong, it was a known risk and not a surprise.”

R — Result. “The real feed never arrived. We demoed on the replay service and scored the highest in our section. Two months later the feed was released, and swapping it in took one afternoon because only the adapter changed. My guide asked us to document the adapter pattern for the next batch.”

WHAT EXACTLY CHANGED

1. The weak version has no **S**, no **T**, no measurable **R**. “Worked out fine” is not a result.
2. The strong version names **one decision** under uncertainty and shows the **reasoning**, which is exactly what the puzzle round scores.
3. It states the assumption **in writing to the guide** — the interview equivalent of saying your assumption out loud.
4. It gives a **result you can verify**: highest in section, one afternoon to swap, reused by the next batch.

This is not your story. Do not use it. It is here so you can see the shape. Take one real project of your own, find the moment you had to decide without full information, and pour it into the same four slots. If you cannot find such a moment in your own work, you have not looked hard enough — every project has one.

Practice

PRACTICE · WARM-UP target: 10 minutes for all six

1. A tank of algae doubles every hour and is full at 12:00. At what time was it one quarter full?
2. Three printers print three pages in three seconds. How many pages do nine printers print in nine seconds?
3. A rope ladder hangs from a boat; its rungs are 30 cm apart and 4 rungs are above the water. The tide rises 1 metre. How many rungs are above the water now?
4. Twelve people at a reunion each hug every other person exactly once. How many hugs?
5. A box has 6 red, 6 green and 6 blue pens. How many must you draw blind to be sure of 3 of the same colour?
6. A book's pages are numbered from 1. The printer used 1,392 digits in total. How many pages?

ANSWER KEY

1. Step backwards one hour at a time, halving each step. Full at 12:00, so half full at 11:00, so one quarter full at **10:00**. 2. One printer does one page in three seconds, so in nine seconds each does 3 pages. Nine printers give **27 pages**. 3. **4 rungs**. The boat floats, so the ladder rises with the tide. This is a “does the situation actually change?” puzzle. 4. $\frac{12 \times 11}{2} = 66$ **hugs**. 5. Worst case is 2 of each colour, that is 6 pens, so the 7th must make a triple. **7 pens**. 6. Pages 1-9 use 9 digits; 10-99 use $90 \times 2 = 180$; running total 189. Remaining $1392 - 189 = 1203$ digits at 3 each gives 401 pages, starting at 100. So the last page is $99 + 401 = 500$ **pages**.

PRACTICE · TIER 1 target: 20 minutes

1. You have a 7-minute timer and an 11-minute timer, both sand glasses. Measure exactly 15 minutes.
2. Eight batteries look identical; one is flat and slightly lighter. Find it with a balance in two weighings.
3. A snail climbs a 10-metre well, rising 3 m each day and slipping 2 m each night. On which day does it get out?
4. There are 25 horses and a track that races 5 at a time, with no timer — only finish order. How many races to find the fastest 3?
5. A ferry takes exactly 1 minute to cross a river. Two ferries start from opposite banks at the same time. Where do they meet, and does the answer change if one is twice as fast?
6. Two candles burn for 6 hours and 4 hours respectively at constant rates. After how long is one exactly twice the height of the other, if they start at the same height and are lit together?

ANSWER KEY

1. Start both. At 7 min the 7-glass ends; flip it. At 11 min the 11-glass ends; the 7-glass has 4 min of sand in its bottom bulb. Flip the 7-glass so those 4 min run back. At 15 min it ends. **15 minutes**. (Do not restart — the running count from 11 plus 4 is the trick.) 2. Weigh 3 vs 3. If one side rises, the flat one is in that group of 3; if balanced, it is in the remaining 2. Second weighing: 1 vs 1 from the suspect group. In the group-of-3 case, the pan that rises holds it, or if balanced it is the third. In the group-of-2 case, the lighter pan holds it. 3. Net gain is 1 m per full day, but on the day it reaches the top it does not slip back. After day 7 it is at 7 m; on day 8 it climbs 3 m to 10 m and is out. **Day 8**. 4. **7 races**. Race the 5 groups (5 races). Race the 5 winners (race 6) — this ranks the groups. Only 6 horses can still be in the top 3: the 2nd and 3rd of the winning group, the 1st and 2nd of the second group, and the 1st of the third group — that is 5 horses. Race those 5 (race 7); the overall winner is already known, and this race gives places 2 and 3. 5. They meet when their combined distance equals one crossing, so at $t = \frac{1}{2}$ minute if equal speed, at the midpoint. If one is twice as fast, they still meet at t where combined distance is one width, and the fast one has covered $\frac{2}{3}$ of the river. **Meeting time is always the width divided by the sum of speeds; only the position moves**. 6. Let the starting height be 1. Heights are $1 - \frac{t}{6}$ and $1 - \frac{t}{4}$. Set $1 - \frac{t}{6} = 2(1 - \frac{t}{4})$, giving $1 - \frac{t}{6} = 2 - \frac{t}{2}$, so $\frac{t}{2} - \frac{t}{6} = 1$, so $\frac{t}{3} = 1$, so $t = 3$ hours.

PRACTICE · TIER 2 target: 30 minutes

1. A logistics hub has 5 loading bays. Trucks arrive at random and each uses one bay for 1 hour. On average 3 trucks arrive per hour. Estimate how often a truck has to wait, and state your assumptions.
2. Three boxes are labelled “cables”, “chargers” and “mixed”. Every label is wrong. You may pull one item out of one box, without looking inside. Can you relabel all three correctly?
3. Estimate the number of lifts (elevators) installed each year in Singapore.
4. You are given a coin that may or may not be fair. You toss it 10 times and see 8 heads. How suspicious should you be? Give a number, not a feeling.
5. A subscription app has 200,000 users and loses 4% of them each month. If it adds 12,000 new users a month, what does the user count settle at?
6. Estimate how many cups of coffee a 500-person office building consumes in a working year.

ANSWER KEY

1. State the assumption: arrivals are independent and spread out. With 3 arriving per hour and each holding a bay for 1 hour, the average number in service is 3, against 5 bays. A wait happens only when 5 or more overlap. Using the Poisson shape with mean 3, $P(5 \text{ or more}) \approx 18\%$. So roughly **1 truck in 5 waits**. Saying “I am assuming independent arrivals” is worth more than the exact percentage. 2. **Yes**. Pull one item from the box labelled “mixed”. Since all labels are wrong, that box is **not** mixed, so whatever you pull names it correctly. Say you pull a cable — that box is “cables”. Now the box labelled “cables” cannot be cables (its label is wrong) and cannot be the one you just named, so it is “mixed”. The last box is “chargers”. 3. Anchor: 6 million people, roughly 1.4 million households, mostly in high-rise blocks. Say one lift per 60 households in service, giving about 23,000 lifts, plus offices and malls, call it 35,000 total. A lift lasts about 25 years, so replacement is $35,000 \div 25 \approx 1,400$ per year, plus new construction, maybe 2,000 to 2,500. **Roughly 2,000 lifts a year**. 4. A fair coin gives 8 or more heads in 10 tosses with probability $(C(10, 8) + C(10, 9) + C(10, 10)) \div 2^{10} = \frac{45+10+1}{1024} \approx 5.5\%$. So it is unusual but not damning. **Mildly suspicious, about 1 in 18**. You would want more tosses. 5. Losses equal gains at the steady state: $0.04N = 12,000$, so $N = 300,000$. 6. 500 people, about 80% drink coffee = 400 drinkers, about 2 cups each per working day, 220 working days: $400 \times 2 \times 220 = \text{about } 176,000 \text{ cups a year}$.

PRACTICE · TIER 3 target: 45 minutes

1. Twenty-five engineers stand in a circle. Starting at engineer 1, every second engineer leaves the circle, going round and round, until one remains. Which position survives? Find the rule, not just the answer.
2. A jar holds 50 white and 50 black beads plus one spare white bead in your hand. You repeatedly draw two beads. If they are the same colour, you discard both and put a white bead in. If they differ, you put the black one back and discard the white. What colour is the last bead? Prove it.
3. Two numbers between 2 and 50 are chosen. One person is told their sum, another their product. Sketch how you would reason about who can deduce what, and say what type of puzzle this is.
4. A data centre’s monthly bill is USD 200,000. Break it into a chain of assumptions and identify the two levers that would cut it by 30%.
5. You must find the single defective chip among 100, but each test costs Rs 500 and can test any subset at once. What is the cheapest guaranteed plan, and what changes if a test can give a false negative 10% of the time?
6. Estimate the total weight of checked luggage on a full wide-body flight from Delhi to Singapore, and say which assumption you are least sure about.

ANSWER KEY

1. Write $25 = 16 + 9$, so $25 = 2^4 + 9$. The survivor in this “every second person leaves, starting the elimination at position 2” version is $2 \times 9 + 1 = \text{position } 19$. The general rule: write $n = 2^m + l$ with $0 \leq l < 2^m$; the survivor is $2l + 1$. Verify on $n = 5$: $5 = 4 + 1$, survivor = 3. Walk it by hand to confirm. 2. Track the **parity of black beads**. Two blacks discarded: black count drops by 2, parity unchanged. Two whites: black count unchanged. One of each: the black goes back, the white is discarded, so black count unchanged. Black parity **never** changes. It starts at 50, which is even, so it ends even — and the only even count with one bead left is 0. **The last bead is white**. 3. This is a **common knowledge** puzzle. Each statement of the

form “I cannot deduce it” removes a whole class of candidate pairs from both players’ worlds. You solve it by listing the pairs consistent with each announcement in turn and intersecting. Say the words “each ‘I don’t know’ is itself information” — that is the scored insight. **4.** Chain: bill = servers x cost per server + storage + network + staff. Typical split is roughly 60% compute, 20% storage, 15% network, 5% other. The two levers: **(a)** right-size over-provisioned instances and use committed-use pricing — 30 to 40% off compute alone, which is about 20% of the total; **(b)** tier cold storage and cut egress by caching at the edge. Together these comfortably reach 30%. **5.** With perfect tests, group testing needs $\lceil \log_2 100 \rceil = 7$ tests, costing Rs 3,500, using binary labelling. With a 10% false-negative rate you must **repeat** each test until the chance of a missed positive is acceptable: 3 repeats drop it to $0.1^3 = 0.1\%$. So about 21 tests, Rs 10,500. The scored point is that **reliability multiplies the cost, it does not add to it**. **6.** About 300 passengers, roughly 1.3 checked bags each, about 18 kg per bag: $300 \times 1.3 \times 18 \approx 7,000$ kg. Least certain assumption: **bags per passenger**, because it swings from 0.8 on a business-heavy route to 2.0 on a migrant-worker route. Say that.

ONE-PAGE REVISION CARD

The five spoken steps — PAUSE. Play it back. Ask, then assume. Use a small case. Solve and state. Examine.

The six families. Information/weighing · Scheduling/crossing · Invariant/parity · Counting/pigeonhole · Probability/expectation · Game/backward induction.

Openings that score.

- “Let me count the outcomes against the number of possible answers.”
- “Let me try this with 3 instead of 100.”
- “I am assuming X — tell me if that is wrong.”
- “What stays the same as the state changes?”
- “Let me solve the last move first.”

Numbers worth memorising. $3^3 = 27$ · balance gives $\log_3$ · 23 people = 50% birthday match · $\frac{n(n-1)}{2}$ handshakes · $\frac{k(k+1)}{2} \geq n$ for two-egg drops · coupon collector = $n(1 + \frac{1}{2} + \dots + \frac{1}{n})$.

Guesstimate funnel. Define → Anchor → Filter → Compute → Sanity-check from a second direction.

Anchors. India 140 crore · metro 80 lakh to 2 crore · household 4 people · 65% aged 15-65 · Singapore 6 m · Bangkok 10 m · 300 working days · phone 3 yr, router 5 yr, car 12 yr.

The three anchor types. People · Geometry · Money. Pick the closest one.

In a product model every factor moves the answer equally. So argue about the factor you trust least, not the one you know best.

When stuck, in this order. Name where you are → say what you ruled out and why → shrink the problem → ask one specific question.

When you truly do not know. Say it once, in one sentence. Then spend the rest of the answer on the method you **would** use. Never bluff. A derived wrong answer beats a recalled right one.

The three fatal habits. Silence · a bare number with no chain · defending a wrong answer after a hint.

And for the behavioural follow-up. S, T, A, R — all four, every time, with a result you can measure. Use **your own** project, never a sample one.

Group Discussion, Communication & Closing

The last three metres. Most candidates lose the offer after the technical round, not during it.

WHAT THIS CHAPTER GIVES YOU

- A 4-slot structure for speaking in a group discussion, with filled examples.
- The exact moves for entering a noisy GD, handling a dominator, and summarising.
- Email and message structures for every stage, with a worked example each.
- Questions to ask the interviewer, and the ones that quietly cost you the offer.
- Real strategy on salary, CTC versus in-hand, bond and service agreements, notice period and competing offers. With arithmetic, not platitudes.
- Weak answer versus strong answer, side by side, with the change named every time.

Everything in this chapter is a **structure**, not a script. Every filled example uses invented people, invented projects and invented numbers. If you recite one of them, the interviewer will hear it — these patterns are common enough that recruiters recognise borrowed lines instantly. Take the slots. Fill them with your own real work and your own real offer letter.

Part 1 — Group discussion

What a GD actually scores

A GD is usually 8 to 12 candidates, one topic, 10 to 20 minutes, and one or two silent evaluators with a grid. The grid does **not** have a row for “was right”.

Row on the grid	What earns the mark	Rough weight
Content	A point nobody else made, with a reason or an example	high
Structure	Signposting, sequencing, summarising	high
Initiative	Opening, or reviving a stalled discussion	medium
Team behaviour	Bringing in a quiet person, handling a clash	high
Clarity	Audible, unhurried, short sentences	medium
Listening	Referring to what someone else said, by name	medium
Aggression / rudeness	Negative marks. Interrupting, mocking, shouting	fatal

TRAP

The three GD myths.

Myth 1: speak the most and you win. False. Evaluators count **distinct contributions**, not seconds. Four good entries beat fourteen repetitions.

Myth 2: you must speak first. False. Speaking first with a weak point is worse than speaking third with a strong one. Speak first only if you have a real frame ready.

Myth 3: you must “win” the argument. False. A GD has no verdict. Candidates who try to defeat others usually collect the aggression mark.

THE REALISTIC TARGET

In a 15-minute GD with 10 people, aim for **4 to 6 entries of 25 to 40 seconds each**. That is about 2.5 minutes of speech. Any more and you are crowding; any less and the evaluator has no data.

The four topic types and the frame for each

Type	Example shape	Frame to open with
Factual	“Should public transport be free?”	Stakeholders: who pays, who benefits, who loses
Abstract	“A closed door”	Define it two ways, then pick one and build
Opinion	“Is remote work good for freshers?”	Split by context: for whom, in what role, at what stage
Case	“A retailer is losing customers. What now?”	Diagnose, then options, then a recommendation

SPEH — the four slots for every entry

SPEH

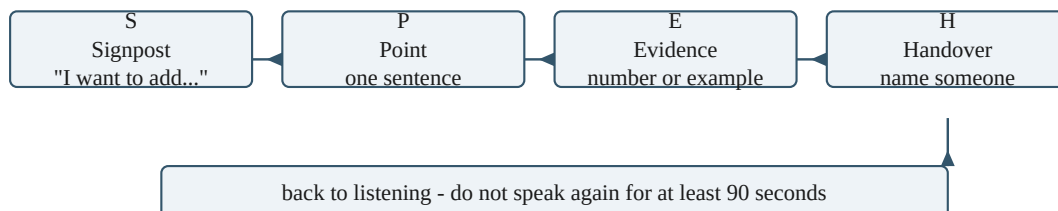
S — Signpost. Tell them what kind of thing you are about to say. “I want to add a second stakeholder.” / “I want to disagree with one part of that.”

P — Point. One sentence. One idea.

E — Evidence. A reason, a number, a mechanism, or a concrete example. Never just repeat the point louder.

H — Handover. Give the floor away deliberately. “Priya, you work part-time — does that match what you have seen?”

The handover is the slot almost everyone skips, and it is the one that scores on the **team behaviour** row.



One entry, 25 to 40 seconds. Four slots. The handover is what most candidates drop.

Weak entry vs strong entry, same topic

Topic: “Should companies require freshers to work from the office for the first year?”

TRAP

WEAK. “Yeah I agree with what he said, I mean office is better, because you learn more and there is more interaction and networking and all, and also work from home you get distracted, so yeah I think office is better.”

SOLUTION

STRONG — with the four slots labelled.

S — Signpost: “I want to sharpen Arjun’s point rather than repeat it.”

P — Point: “The benefit of the office for a fresher is not networking. It is **overheard correction**.”

E — Evidence: “In my internship I sat next to two senior developers. I learnt more from hearing them argue about a database index for ten minutes than from any document. On a video call that conversation would never have happened, because nobody schedules a meeting to think out loud. So the real question is not office versus home, it is whether the team has a place where juniors can overhear seniors reasoning.”

H — Handover: “Sneha, you did a fully remote internship — did you get that anywhere, or did you have to ask for everything explicitly?”

WHAT EXACTLY CHANGED

1. The weak entry **agrees** and adds nothing. Evaluators mark that as a repetition, not a contribution.
2. It uses filler (“I mean”, “and all”, “so yeah”) which costs the clarity mark.
3. The strong entry **reframes** the topic. “Not networking, but overheard correction” is a point nobody else in the room has.
4. The evidence is a **specific mechanism**, not a general belief.
5. It **ends by giving the floor away by name**. The next 30 seconds now belong to someone the evaluator saw you include.
6. Both are about 30 seconds. Length was never the difference.

How to enter when nobody will stop talking

THE THREE-STEP ENTRY

1. **Lean forward and raise a hand to chest height.** Not above your head. This is visible without being theatrical, and evaluators see it even if the room does not.
2. **Use a bridge phrase on the first pause — half a second is enough.** Bridges that work:
 - “Can I build on that?”
 - “One quick addition —”
 - “Two of us have said the same thing; may I offer the other side?”
3. **Say your point in the first 4 seconds.** Do not warm up. If you start with “I think that maybe in some cases...” you will be talked over before the verb arrives.

TRAP

Never say “excuse me, excuse me, EXCUSE ME”. Raising volume to enter is the fastest route to the aggression mark. Raise your **hand**, then speak at normal volume into the gap.

The four hard moments, and the words for each

Moment	What to do
Someone talks over you mid-sentence	Stop. Do not fight. When they finish: “I will finish the point I started — it was about X.” You now look composed and they look rude.
One person dominates the whole GD	Do not attack them. Attack the imbalance : “We have heard three of us in detail. Meera and Karthik have not spoken — can we hear them before we go deeper?” This scores heavily on team behaviour.
Someone states a wrong fact	Correct the claim , never the person. “I would put that number differently — my understanding is closer to X. Either way, the direction of the argument holds.” Softening lets them agree without losing face.
The discussion has stalled or gone in circles	Summarise and restart: “We have covered cost and convenience. We have not touched safety at all. Can we take that next?”

The summary — the highest-value 30 seconds in the room

If the evaluator asks for a summary, or if the GD is running out of time, take it. A summary is the single highest-scoring entry available.

THE SUMMARY STRUCTURE — FOUR SLOTS

1. Points of agreement. “We agreed on two things: ...” **2. Points of disagreement, stated fairly.** “We split on whether ... Some argued X; others argued Y.” **3. What we missed.** “We did not discuss ...” **4. A closing position.** “On balance the group leaned towards ...”

SOLUTION

Filled example (topic: office attendance for freshers).

“Let me pull the threads together. **We agreed** on two things: freshers need faster feedback than seniors do, and a blanket rule for the whole company is a blunt instrument.

We split on the reason. Arjun and Sneha argued the benefit is social — belonging and networking. Two others, including me, argued it is technical — overhearing how decisions get made.

We did not discuss cost at all, either the company’s office cost or the fresher’s commute and rent, which in a metro can be a third of a first salary.

On balance, the group leaned towards ‘office for the first six months, then the team decides’ — which is narrower than the question we were given, and I think that narrowing was the useful part of this discussion.”

WHY THAT SUMMARY SCORES

1. It names **other people’s** points accurately. That proves you were listening, which is its own grid row.
2. It states the disagreement **fairly**, including the side you were not on.
3. It adds **one new thing** — the cost angle — without starting a new argument.
4. It closes with a position, not a shrug.
5. Under 45 seconds.

TRAP

Do not summarise by saying “so basically everyone agreed that...” when they did not. A false consensus is the most common summary failure and evaluators catch it instantly.

Delivery: the small things that move the mark

Habit	Fix
Filler words: “basically”, “actually”, “like”, “I mean”	Replace every filler with silence . A half-second pause reads as thinking. “Basically” reads as padding. Record yourself once and count them; most people are shocked.
Speaking too fast, then trailing off	Aim for about 130 words a minute; say the first sentence deliberately slowly and the rest follows that pace. Land the last three words of each sentence — most candidates fade exactly where the point is.
Looking only at the evaluator	Speak to the group . Sweep across faces. Evaluators specifically mark candidates who perform at them instead of participating.
Notes	If paper is allowed, write only names and one keyword each. It makes the handover slot easy.

SHORTCUT

Write the other candidates’ names down in the first 30 seconds, in seating order. Using a person’s name is the cheapest way to score on listening and team behaviour, and almost nobody does it.

Part 2 — Written communication

Between rounds, almost all contact is by email or a messaging app. Recruiters forward these to hiring managers. Every message is part of your evaluation.

THE FOUR RULES FOR EVERY MESSAGE YOU SEND

1. **Subject line carries the fact.** “Interview reschedule request — Rahul Menon — 14 Oct slot”. Not “Regarding interview”.
2. **First sentence says what you want.** Recruiters read on a phone between meetings.
3. **Under 150 words.** Always.
4. **One ask per message.** Two asks means one gets forgotten.

Weak email vs strong email — the follow-up after silence

TRAP

WEAK.

Subject: Regarding my application

Respected Sir/Madam,

I am writing to you regarding the interview which I gave on the 3rd of this month for the role in your esteemed organisation. It has been quite some time and I have not received any update regarding the same. I am very much interested in this opportunity and I would be grateful if you could kindly do the needful and let me know about my status at the earliest. Awaiting your positive response. Kindly consider my request.

Thanking you, Rahul

SOLUTION

STRONG.

Subject: Following up - Backend Engineer interview, 3 Oct - Rahul Menon

Hi Anita,

I interviewed for the Backend Engineer role on 3 October with Deepak and Sowmya. You mentioned a decision by around the 14th, so I wanted to check in.

I am still very interested. One thing I did not get to mention: since the interview I finished migrating my side project to a queue-based design, which is close to the batching problem Deepak asked about. Happy to share the repository if useful.

I do have another process at a later stage, so any indication of timing would help me plan. No pressure if the decision is not ready.

Thanks, Rahul Menon · +91 98xxx xxxxx

WHAT EXACTLY CHANGED

1. The subject line **names the role, the date and the person**. It is searchable. “Regarding my application” is not.
2. It uses the recruiter’s **name**, and names the interviewers. That proves you are a real person in their pipeline, not a mass mailer.

3. It **quotes the timeline they gave you**. This is a reminder without being a complaint.
4. It adds **new information** — a reason to reply beyond “are we there yet”.
5. It mentions the other process **once, factually, without a deadline threat**. That is pressure done honestly.
6. It removes every hollow phrase: “esteemed organisation”, “do the needful”, “at the earliest”, “kindly”. These make you sound like a form letter.
7. 95 words instead of 110, and it says more.

TRAP

Phrases to delete from your professional English, permanently: “do the needful”, “revert back”, “esteemed organisation”, “kindly do the favour”, “at the earliest”, “Respected Sir/Madam”, “I would be highly obliged”. They are not wrong English, but in a hiring inbox they read as a template, and several sound like a demand dressed as politeness. Write “please”, “reply”, “let me know by Friday”.

Message structures for the other five moments

Each is a **structure**. The filled example is invented. Replace every detail.

A. THANK-YOU NOTE, SENT WITHIN 24 HOURS

Slot 1: thank them, name the role and date. **Slot 2:** one specific thing from the conversation that you found interesting — proof you were present. **Slot 3:** one sentence that closes a gap, if you fumbled something. **Slot 4:** a short, non-pushy close.

SOLUTION

Hi Deepak,

Thanks for the time this morning on the Backend Engineer role.

The part about moving retries out of the request path and into a queue stayed with me - I had not thought about retry storms as a load problem rather than a correctness problem.

One correction: you asked how I would prevent duplicate charges, and I gave you a database-lock answer. The cleaner answer is an idempotency key on the request, stored with the result, so a repeat of the same key returns the stored result instead of charging again. I wanted to give you the better answer rather than leave the weaker one standing.

Thanks again, Rahul

Slot 3 is powerful and almost nobody uses it. If you fumbled a question, a short written correction turns a weak moment into evidence that you keep thinking. Keep it to **one** topic. Do not re-answer the whole interview.

B. ASKING TO RESCHEDULE

Slot 1: the ask, in the first sentence. **Slot 2:** a one-line reason. Honest and brief. Do not over-explain. **Slot 3:** **three** concrete alternative slots with the time zone. **Slot 4:** an apology for the inconvenience, once.

SOLUTION

Hi Anita,

Could we move my interview from Thursday 14 Oct, 11:00 IST?

I have a university practical exam that was rescheduled onto the same morning.

I am free at any of these, all IST: Thu 14 Oct after 16:00, Fri 15 Oct 10:00-13:00, or Mon 18 Oct any time. I can also do an earlier or later hour if that suits the panel better. Sorry for the change, and thank you for accommodating it.

TRAP

Never ask to reschedule with “can we do it some other day?” and no alternatives. You have just handed the recruiter your work. Always give three slots, always with the time zone, especially for a company in Singapore or Thailand where the offset differs from India by 1.5 hours.

C. ACCEPTING AN OFFER

Slot 1: accept clearly, in the first line. No hedging. **Slot 2:** restate the three facts you are agreeing to: role, fixed pay, joining date. **Slot 3:** ask what documents are needed next. **Slot 4:** one line of genuine enthusiasm.

SOLUTION

Hi Anita,

I am glad to accept the Backend Engineer offer.

Confirming my understanding: fixed pay of Rs 7,20,000 per year plus a variable component of

Rs 80,000, joining on 1 December 2026 at the Pune office.

Please let me know what you need from me - I have my degree certificate, the last three months of

payslips, and I can start the background verification this week.

Looking forward to working with Deepak's team.

TRAP

Always restate the numbers in writing. Verbal offers get misremembered, and the number a recruiter says on a call is often the CTC, not the fixed pay. Writing it down is not rude; it is how every professional confirms an agreement. If the recruiter's reply corrects your numbers, you have just caught a misunderstanding for free.

D. DECLINING AN OFFER WITHOUT BURNING THE BRIDGE

Slot 1: decline clearly, early, and thank them. **Slot 2:** a short, true, non-comparative reason. You do not owe them the competitor's name or salary. **Slot 3:** something specific and positive about their process. **Slot 4:** leave the door open, sincerely.

SOLUTION

Hi Anita,

Thank you for the offer. After a lot of thought I am going to decline.

I have accepted a role that is closer to distributed systems work, which is the direction I want

to build in over the next few years. It was a genuinely hard decision.

I want to say that your process was the clearest I went through - Deepak's interview was the only

one where I learnt something, and the written feedback after round two was unusual and appreciated.

If there is a fit in the future I would be glad to talk again.

You will meet these people again. India and Singapore tech hiring circles are small, and recruiters move between companies constantly. A graceful decline is a real asset two years from now. A disappeared candidate — someone who simply stops replying — is remembered, and it is remembered by name.

E. ASKING FOR MORE TIME TO DECIDE

Slot 1: thank them and say you want to accept properly, not quickly. **Slot 2:** name a **specific** date, not “some more time”. **Slot 3:** a true reason at a high level. **Slot 4:** confirm your continued interest.

SOLUTION

Hi Anita,

Thank you for the offer - I am taking it seriously and want to give you a proper answer rather than a rushed one.

Could I confirm by Friday 24 October? That is eight days from today.

I am at the final stage of one other process and expect to hear by the 22nd. I would rather decide with complete information than accept and then reopen the conversation.

To be clear, I am genuinely interested in this role, and Pune works for me.

TRAP

Do not lie about a competing offer you do not have. It is the most common piece of bad advice in placement forums. Recruiters in the same city talk to each other, some ask for a written offer as proof, and if you are caught the offer is withdrawn. Asking for a week to decide needs no excuse at all — “I want to give you a proper answer” is a complete and honest reason.

Part 3 — Questions to ask the interviewer

At the end of almost every round you will hear “do you have any questions for us?” Saying “no, you have covered everything” is a wasted opportunity and is often recorded as low interest.

PREPARE SIX, ASK TWO OR THREE

Prepare six because three will get answered during the interview. Ask two or three. Then stop.

Round	What to ask
HR screen	Team size and where the role sits. Joining date and location. What the first six months look like. Whether there is a bond or training agreement — ask this early, in writing if possible.
Technical / hiring manager	What the team is actually building this quarter. How code review and deployment work. The last hard technical decision the team made, and what they chose. What a fresher is expected to do alone by month six.
Senior / leadership	What would make this hire a success in a year. What the team is worst at today. Why the last person in this role left or moved.

Weak questions vs strong questions

TRAP

WEAK.

- “What does your company do?” — you should already know.
- “How much leave do I get?” — ask HR after the offer, not the hiring manager before it.
- “Is there work-life balance here?” — nobody says no. You learn nothing.
- “What is the salary?” — not in round one; the process will reach it.
- “Do you provide training?” — reads as “I cannot learn by myself”.
- “How soon can I get promoted?” — reads as “I am leaving soon”.

SOLUTION

STRONG.

- “What is the team building this quarter, and where would a new engineer fit into that?”
- “What is the last technical decision the team argued about, and what did you choose?”
- “If someone joins in December, what would you want them doing on their own by June?”
- “What is the team worst at right now?”
- “Is the pay structure fixed plus variable, and how has the variable actually paid out in the last two years?”
- “Is there a service agreement or bond attached to this role? If so, what is the duration and the amount?”

WHAT EXACTLY CHANGED

1. Weak questions ask about **benefits to you**. Strong ones ask about **the work**.
2. Every strong question has an answer that is **specific and checkable**. “Work-life balance?” gets a brochure answer; “what is the team worst at?” cannot be answered with a brochure.
3. The strong list includes two hard-money questions — variable pay history and bond — asked plainly and without apology. That is not rude. It is how you avoid a bad surprise in the offer letter.
4. The strong questions give **you** information you will actually use to decide.

SHORTCUT

The single best question, and the one interviewers remember: “**what is the team worst at right now?**” A good interviewer will answer honestly, and their answer tells you more about the job than the entire job description. If they refuse to answer it at all, that is also information.

Part 4 — Money: CTC, in-hand, and negotiation

First: CTC is not salary

WHAT SITS INSIDE AN INDIAN CTC NUMBER

CTC (cost to company) = everything the company spends on you, including money you never see.

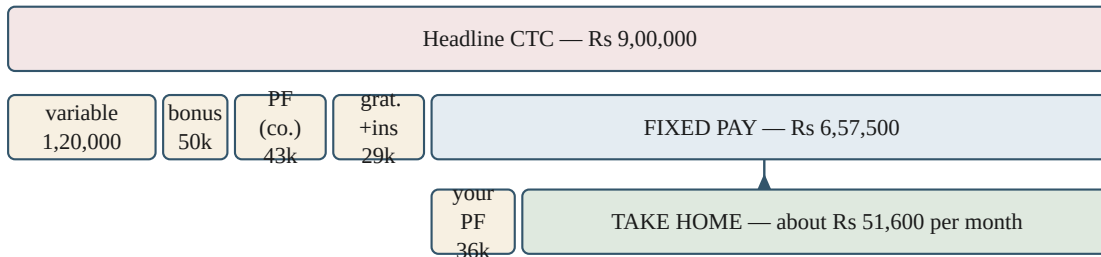
Typical components:

- **Fixed pay** — basic, HRA, allowances. This is the real salary.
- **Variable / performance pay** — paid only if targets are met. Often 10 to 20% of CTC.
- **Joining bonus** — paid once, often with a claw-back clause if you leave within a year.
- **Employer PF contribution** — goes to your provident fund, not your bank.
- **Gratuity provision** — you only receive it after (usually) five years of service.
- **Insurance premium** — the company pays an insurer, not you.

Only fixed pay, minus your own PF and tax, reaches your bank account.

TRAP

Two offers with the **same** CTC can differ by more than Rs 5,000 a month in take-home, depending on how much of the CTC is variable and how the basic is structured. And an offer with a **lower** CTC can pay you **more** every month. Never compare CTC. Compare monthly in-hand, and compare the fixed component.



A Rs 9 lakh CTC. The dark part is what reaches your bank.

ctc.js — turn an offer letter into a monthly number. Teaching figures only.

JS

```

function slabTax(x) {
    // simplified ladder, shape only
    const slabs = [[400000, 0], [800000, 0.05], [1200000, 0.10],
        [1600000, 0.15], [Infinity, 0.20]];
    let tax = 0, prev = 0;
    for (const [cap, rate] of slabs) {
        if (x > prev) tax += (Math.min(x, cap) - prev) * rate;
        prev = cap;
    }
    if (x <= 1200000) tax = 0; // rebate band
    return tax * 1.04; // + 4% cess
}

function inHand(o) {
    const fixed = o.ctc - o.variable - o.joiningBonus - o.employerPF - o.gratuity -
    o.insurance;
    const basic = fixed * o.basicShare;
    const employeePF = basic * 0.12; // your own PF cut, also out of fixed pay
    const tax = slabTax(fixed - employeePF - o.standardDeduction);
    const annual = fixed - employeePF - tax - o.profTax;
    return { fixed, employeePF, tax, monthly: annual / 12 };
}

const offerA = { name: 'Offer A', ctc: 900000, variable: 120000, joiningBonus: 50000,
    employerPF: 43200, gratuity: 17300, insurance: 12000, basicShare: 0.45,
    standardDeduction: 75000, profTax: 2400 };
const offerB = { name: 'Offer B', ctc: 780000, variable: 0, joiningBonus: 0,
    employerPF: 37440, gratuity: 15000, insurance: 10000, basicShare: 0.50,
    standardDeduction: 75000, profTax: 2400 };

for (const o of [offerA, offerB]) {
    const r = inHand(o);
    console.log('\n' + o.name + ' (headline CTC Rs ' + (o.ctc / 100000).toFixed(1) + '
    LPA)');
    console.log(' fixed pay Rs',
    Math.round(r.fixed).toLocaleString('en-IN'));
    console.log(' variable, NOT guaranteed Rs', o.variable.toLocaleString('en-IN'));
    console.log(' your PF cut Rs',
    Math.round(r.employeePF).toLocaleString('en-IN'));
    console.log(' income tax + cess Rs', Math.round(r.tax).toLocaleString('en-
    IN'));
    console.log(' TAKE HOME per month Rs',
  
```

```
Math.round(r.monthly).toLocaleString('en-IN'));
}
```

output

TEXT

Offer A (headline CTC Rs 9.0 LPA)

fixed pay	Rs 6,57,500
variable, NOT guaranteed	Rs 1,20,000
your PF cut	Rs 35,505
income tax + cess	Rs 0
TAKE HOME per month	Rs 51,633

Offer B (headline CTC Rs 7.8 LPA)

fixed pay	Rs 7,17,560
variable, NOT guaranteed	Rs 0
your PF cut	Rs 43,054
income tax + cess	Rs 0
TAKE HOME per month	Rs 56,009

READ THAT OUTPUT AGAIN

Offer B has a CTC that is **Rs 1.2 lakh lower**, and pays **Rs 4,376 more every month**. That is Rs 52,000 a year more in your bank.

The reason: Offer A packs Rs 1.2 lakh of its headline into variable pay that may or may not arrive, plus a one-time joining bonus that stops after year one.

This is why “what is your CTC expectation?” is the wrong conversation. The right conversation is “**what is the fixed component?**”

The tax ladder above is a simplified teaching shape, not a tax filing. Slabs, rebates and the standard deduction change. Use your own offer letter’s exact numbers and the current year’s rules. The **method** — strip out everything that is not fixed pay, then subtract PF and tax — does not change.

The salary question: what to actually say

You will be asked “what are your salary expectations?” — often in the very first HR call.

TRAP

WEAK answer 1. “Whatever the company policy is, sir. I just want the experience.”

You have told them you will accept the lowest number they have. You also sound like you have not researched the role.

TRAP

WEAK answer 2. “I want 12 lakhs.”

A single number with no basis. If it is too high you are screened out silently. If it is too low you have capped yourself before they ever said a figure.

SOLUTION

STRONG — the four-move structure.

Move 1 — deflect once, politely. “I would rather understand the role fully first — the responsibilities will shape what is fair. Do you have a band for this position?”

Move 2 — if they push, give a researched range, not a point. “Based on what I have seen for backend roles at this experience level in Pune, the range looks like Rs 7 to 9 lakh fixed. I am targeting that band.”

Move 3 — anchor on fixed, not CTC. “To be clear, I am talking about fixed pay. I would want to look at the variable component separately, since it is not guaranteed.”

Move 4 — close with flexibility that is real, not unlimited. “The number matters, but so does what I will be working on. If the role is strong I have some flexibility — though I do have a floor I need to respect.”

WHAT EXACTLY CHANGED

1. The strong answer **asks for their band first**. In many companies the recruiter will simply tell you, and then you never have to name a number at all.
2. It gives a **range**, and the range has a **stated basis** (“backend roles, this level, this city”). A number with a reason is negotiable; a number without one is a demand.
3. It says the word **fixed**. That one word reframes the whole negotiation, and most freshers never say it.
4. It signals flexibility **and a floor**. Unlimited flexibility is the same as having no number.
5. It never says “whatever you think is right”.

HOW TO BUILD YOUR RANGE IN 30 MINUTES

1. Ask three seniors from your own college, in the same role and city, what their **fixed** pay was. Ask for fixed, not CTC, or the data is useless.
2. Check two or three salary sites for the same role, level and city, and note that these skew high.
3. Take the **middle** of what you learn as your target, the bottom as your floor, and the middle plus 20% as your ask.
4. Write all three numbers down before the call. Under pressure you will otherwise say the first number that comes into your head.

TRAP

If you are asked for your current salary and you have one, you may answer it — but in several places asking is restricted, and you are always allowed to redirect: “My current fixed pay is X. That was set when I joined two years ago, so I am basing my expectation on the market for this role rather than on that number.” Never inflate a current salary. Payslips are checked during background verification, and a mismatch there ends the offer.

Can a fresher negotiate at all?

Often less than you hope, but more than zero. Know which door is open.

Situation	Realistic room
Mass campus hire, one common offer for 300 people	Almost none on salary. The number is fixed by policy. Negotiate the role , location or team instead — those often have room and matter more over three years.
Off-campus fresher hire, one-to-one process	5 to 15% is realistic , mostly if you have a second offer or a rare skill.
You hold a written competing offer	Real room. This is the only lever that reliably moves a number.
You have 1 to 3 years of experience	20 to 40% over current is a normal jump in Indian tech. Anchor on market, not on your current pay.
Offer already signed and accepted	None, and do not try. Reopening a signed offer damages you and sometimes gets it withdrawn.

THE COUNTER-OFFER STRUCTURE — FIVE SLOTS

Slot 1: enthusiasm, genuine and specific. You want the job. **Slot 2:** the gap, as a number, not a feeling. **Slot 3:** the basis for your number. **Slot 4:** an alternative if the base cannot move — joining bonus, earlier review, relocation support, a better title, a shorter bond. **Slot 5:** a clear commitment: if they meet it, you accept.

SOLUTION

Filled example — invented numbers, replace with yours.

“Anita, thank you — I want to say first that I want this role. Deepak’s team is doing the work I want to do.

I wanted to talk about the fixed component. The offer is Rs 6.5 lakh fixed. Based on what I have seen for backend roles at this level in Pune, and on the other written offer I am holding, I was targeting closer to Rs 7.5 lakh.

If the base genuinely cannot move, there are other ways to close the gap that would work for me: a joining bonus, a review at six months instead of twelve, or relocation support.

If we can get to Rs 7.2 lakh fixed, I will accept today and stop my other process.”

WHY THAT WORKS

1. It **opens and closes with commitment**. A negotiation that does not end in “then I accept” is just a complaint.
2. It names **one** number and **one** gap. Negotiating five things at once fails.
3. It offers **alternative currencies**. Base pay is often locked by band; a joining bonus usually is not, and an early review costs the company nothing today.
4. It gives a **precise close**: Rs 7.2 lakh, today, and the other process stops. The recruiter can now take a concrete proposal to their manager.
5. It is **honest**. It mentions a real competing offer. If you do not have one, delete that clause — the rest of the structure still works.

TRAP

Never negotiate over a chat message or in a rush. Ask for a 10-minute call. Text negotiations get forwarded, lose tone and get misread. And never give an ultimatum you are not willing to follow through on — recruiters call bluffs more often than students expect.

Part 5 — Bond and service agreements

Many Indian service companies, and some product companies, attach a **service agreement** (often called a bond) to a fresher offer: stay N months or pay a penalty.

THE SIX QUESTIONS TO ASK BEFORE YOU SIGN

1. **Duration.** How many months, and does it start from joining or from the end of training?
2. **Amount.** The exact rupee figure, and is it reduced pro-rata as you serve?
3. **Trigger.** Does it apply only if you resign, or also if the company releases you?
4. **Documents.** Do they hold your original degree certificate or marksheets?
5. **Relieving letter.** Is it withheld if you leave early?
6. **Transfer.** Does an internal move or a location change reset the clock?

TRAP

The one that actually hurts is not the money. It is question 5. A withheld **relieving letter** or a withheld original certificate can block your next job entirely, because background verification requires them. A candidate

who paid the penalty but cannot get a relieving letter is in a far worse position than one who paid nothing and left cleanly. Ask about the relieving letter in writing before you sign, and keep the reply.

How to price a bond

Spread the penalty over the lock-in. That converts a scary lump sum into a monthly number you can compare against the offer.

bond.js — the monthly price of your own flexibility

JS

```
function bondCost(penalty, months) { return penalty / months; }

for (const [p, m] of [[200000, 24], [200000, 12], [100000, 18], [50000, 12]]) {
  console.log('Rs ' + p.toLocaleString('en-IN') + ' over ' + m + ' months -> Rs ' +
    Math.round(bondCost(p, m)).toLocaleString('en-IN') + ' per month of lost
freedom');
}
```

output

TEXT

```
Rs 2,00,000 over 24 months -> Rs 8,333 per month of lost freedom
Rs 2,00,000 over 12 months -> Rs 16,667 per month of lost freedom
Rs 1,00,000 over 18 months -> Rs 5,556 per month of lost freedom
Rs 50,000 over 12 months -> Rs 4,167 per month of lost freedom
```

Read the first two lines together. The **same** Rs 2 lakh penalty is twice as expensive per month when the bond is short — because you are paying to escape a shorter remaining commitment. So when you are comparing two bonded offers, a longer bond with the same penalty is **less** costly per month, but it locks you for longer. Decide which of the two you care about, and say so out loud to yourself before signing.

Asking about the bond without sounding like you plan to leave

TRAP

WEAK. “What happens if I want to leave before two years?”

The interviewer now believes you intend to leave.

SOLUTION

STRONG. “I want to be sure I understand the full agreement before I commit to it, so a few straightforward questions: is there a service agreement attached, how long is it, and is it a fixed amount or reduced pro-rata as I serve? And are any original documents held during that period?”

WHAT EXACTLY CHANGED

1. The strong version frames the question as **diligence before committing**, which is exactly what it is, rather than as an exit plan.
2. It asks **all** the questions in one go, so you sound prepared rather than nervous.
3. It uses the correct term, **service agreement**, which signals you have read a contract before.
4. It slips in the document question last, quietly, and that is the answer that matters most.

Part 6 — Notice period and joining date

If you already have a job, the notice period drives everything. Work **backwards** from the date the new company wants you.

```
function addDays(d, n) { const x = new Date(d); x.setDate(x.getDate() + n); return x; }
function fmt(d) { return d.toISOString().slice(0, 10); }

function plan({ joinBy, noticeDays, handoverBuffer = 5, bgvDays = 10 }) {
  const join = new Date(joinBy);
  const lastWorkingDay = addDays(join, -1);
  const resignBy = addDays(lastWorkingDay, -noticeDays);
  return {
    join: fmt(join),
    lastWorkingDay: fmt(lastWorkingDay),
    resignBy: fmt(resignBy),
    safeResignBy: fmt(addDays(resignBy, -handoverBuffer)),
    tellNewCompanyEarliestJoin: fmt(addDays(new Date(), noticeDays +
handoverBuffer)),
    offerLetterExpected: fmt(addDays(new Date(), bgvDays)),
  };
}

console.log('today is treated as', fmt(new Date()));
for (const [k, v] of Object.entries(plan({ joinBy: '2026-12-01', noticeDays: 60 })))
{
  console.log('  ' + k.padEnd(28) + v);
}

console.log('\nIf notice is 90 days instead of 60:');
const q = plan({ joinBy: '2026-12-01', noticeDays: 90 });
console.log('  you must resign by          ' + q.resignBy);
console.log('  safe date with handover      ' + q.safeResignBy);

function buyout(monthlyBasic, daysToBuy) { return Math.round(monthlyBasic / 30 *
daysToBuy); }
console.log('\nBuying out 30 days on a Rs 30,000 monthly basic costs about Rs',
  buyout(30000, 30).toLocaleString('en-IN'));
```

output (run on 15 September 2026)

TEXT

```
today is treated as 2026-09-15
join                2026-12-01
lastWorkingDay      2026-11-30
resignBy            2026-10-01
safeResignBy        2026-09-26
tellNewCompanyEarliestJoin 2026-11-19
offerLetterExpected 2026-09-25

If notice is 90 days instead of 60:
  you must resign by          2026-09-01
  safe date with handover      2026-08-27

Buying out 30 days on a Rs 30,000 monthly basic costs about Rs 30,000
```

READ THE 90-DAY BLOCK CAREFULLY

It says you should have resigned on **1 September** — two weeks in the past. The script is telling you something true and uncomfortable: with a 90-day notice you **cannot** make a 1 December joining date.

So do not promise it. Your options are:

1. Ask the new company for a January joining date, now, before you accept.
2. Ask your current employer for a partial waiver of notice. Many will grant 15 to 30 days.
3. Buy out the remaining days, if your contract allows it. The last line prices that.

All three are honest. Promising a date you cannot meet is not, and it is the single most common way freshers and early-career candidates lose an offer **after** signing it.

The notice-period question in the interview

TRAP

WEAK. “Notice period is 90 days but I will try to manage it, maybe I can come in 30 days.”

“I will try to manage” is a promise with no mechanism. The recruiter writes down 30 days, plans around it, and when you cannot deliver, trust is gone.

SOLUTION

STRONG. “My notice period is 90 days from the date I resign. Realistically that puts my earliest joining date at around 20 December if I resign the day I sign.

There are two ways it could be shorter. My contract allows a buy-out of up to 30 days at basic pay, which would cost me about Rs 30,000 and bring it to late November. And my manager has released two people early in the past, though I cannot promise that.

So my honest answer is: 20 December is the date I can commit to. If either of those works out I will tell you immediately and join sooner.”

WHAT EXACTLY CHANGED

1. The strong version gives **one committable date** and treats everything else as upside.
2. It names the **mechanisms** — buy-out clause, manager discretion — with the cost of each.
3. It separates what it **can promise** from what it **cannot**, explicitly.
4. It commits to **telling them early** if the date improves. That is the sentence a recruiter remembers.
5. It never uses the words “I will try to manage”.

TRAP

Do not resign before you have the written offer letter in hand and background verification is clear. A verbal offer is not an offer. This is the most expensive mistake in this chapter: people resign on a phone call, the written offer is delayed or withdrawn, and they are left with neither job.

Part 7 — Working across borders

If you are interviewing with a company in Singapore, Thailand, Indonesia or Vietnam, three extra things get scored that never come up in a domestic interview: whether you have thought about the **work pass**, whether you have thought about the **real cost of living**, and whether your **communication style** will work in that office.

The work pass conversation — raise it first

TRAP

WEAK. Wait for them to bring it up, then say “I think the company will handle it.”

The company does handle the filing. But the **timeline** and the **eligibility** affect your joining date, and a candidate who has not thought about it looks like a candidate who will be a surprise in month two.

SOLUTION

STRONG. “Before we go further I want to check the pass situation, because it affects the date I can commit to. My understanding is that for this kind of role the company sponsors the application, and the processing takes a few weeks after the offer is signed, sometimes longer. Two questions: is there a minimum salary threshold for the pass type you would apply for, and does your offer process assume a joining date after approval or before it?”

WHAT EXACTLY CHANGED, AND WHAT TO ACTUALLY CHECK

The strong version raises it **early**, treats it as a **shared planning problem**, and asks the two questions that actually change your life: the salary threshold and whether the joining date assumes approval.

Check these yourself before any SE Asia interview, on the official government site, because the numbers change every year:

1. Which pass type applies to your role and experience level.
2. The **current** minimum qualifying salary for that pass, which usually rises with age and experience.
3. Whether the pass is tied to the employer (it almost always is), and what happens if you resign.
4. Typical processing time, and whether you may enter the country before approval.
5. Dependant rules, if that matters to you.

Never quote a threshold from a forum post or from this book. Quote the official figure you looked up this month.

Comparing an Indian offer with a Singapore offer

You cannot compare the headline numbers. The currencies, the tax systems, the retirement contributions and the rent are all different.

THE SIX-LINE COMPARISON — DO IT ON PAPER BEFORE YOU DECIDE

1. **Annualise both.** Is the SGD figure monthly? Is there a 13th month payment? Is the Indian figure CTC or fixed?
2. **Strip to fixed pay** on both sides. Remove variable, bonuses and employer retirement contributions.
3. **Subtract retirement contributions** — CPF in Singapore, PF in India. These are real savings, but they are not spendable income, so keep them in a separate line.
4. **Subtract income tax** using each country’s own rates. They are not comparable rates.
5. **Subtract rent and daily living.** This is where the comparison usually turns.
6. **Compare what is left**, converted at today’s rate, and separately compare the **savings** line.

Example 1 [Grab · pattern]

Offer A: Singapore, SGD 5,000 per month fixed, 12 months, no bonus. Offer B: Bengaluru, Rs 14,00,000 CTC of which Rs 2,00,000 is variable. Which one leaves you with more spendable money, and which leaves you with more savings? Use SGD 1 = Rs 65 and show every step.

SOLUTION

Step 1 — annualise.

- A: $5,000 \times 12 = \text{SGD } 60,000$ per year.
- B: CTC 14,00,000. Strip the variable: fixed is about Rs 12,00,000. Strip employer PF and gratuity, roughly Rs 80,000 together, leaving about **Rs 11,20,000** of real fixed pay.

Step 2 — retirement contributions. Assume the Singapore employee contribution is around 20% of the monthly wage up to a ceiling, and the Indian employee PF is 12% of basic.

- A: employee contribution roughly SGD 12,000 a year goes to CPF. Spendable base: SGD 48,000.
- B: employee PF roughly Rs 60,000 a year. Spendable base: Rs 10,60,000.

Step 3 — tax. Singapore’s resident rates on this income are low, a few percent effective. India’s effective rate on Rs 11 lakh is higher. Take rough effective rates of 3% and 7%.

- A: $48,000 \times 0.97 \approx \text{SGD } 46,600$.
- B: $10,60,000 \times 0.93 \approx \text{Rs } 9,86,000$.

Step 4 — rent and living. This is the step people skip and it decides the answer.

- A: a room in a shared flat, plus transport and food, realistically SGD 1,800 to 2,500 a month. Use SGD 2,200: that is SGD 26,400 a year.
- B: a 1BHK plus transport and food in Bengaluru, realistically Rs 35,000 a month. That is Rs 4,20,000 a year.

Step 5 — what is left.

- A: $46,600 - 26,400 = \text{SGD } 20,200 = \text{Rs } 13,13,000$ at Rs 65.
- B: $9,86,000 - 4,20,000 = \text{Rs } 5,66,000$.

Step 6 — the savings line, kept separate.

- A: CPF about SGD 12,000 plus an employer share on top = Rs 7.8 lakh or more, but **locked** and largely not withdrawable if you leave Singapore permanently before retirement age.
- B: PF about Rs 60,000 plus an employer share, withdrawable under Indian rules.

Answer. Offer A leaves roughly **2.3 times** more spendable money per year at these assumptions. But the savings in A are locked in a foreign retirement system and the pass is tied to the employer, so the risk profile is different. Both facts belong in the decision.

Answer: A, by a wide margin on spendable income — but say the CPF lock-in and the employer-tied pass out loud

TRAP

Every number in that example is an assumption, including the exchange rate, the tax rates, the CPF rates and the rent. Redo it with **your** offer letter, **this** month’s rates, and rent quotes from a real listings site for the district you would actually live in. The **method** is the transferable part; the numbers are not.

Why do you want to move to South East Asia?

This is asked in almost every Tier 2 interview, and it is a screen for whether you will leave after eight months.

TRAP

WEAK. “Singapore is a developed country with better opportunities and exposure. I want to grow my career and experience the international work culture.”

Nothing here is about the company, the product, or the work. It also applies equally to eleven other countries, which tells the interviewer you are optimising for “abroad”, not for this job.

SOLUTION

STRONG. “Two reasons, one about the work and one about me.

The work: the products in this region run across six or seven markets at once, with different languages, currencies, and payment rails in each. I have never built anything that had to handle that, and I think it is a genuinely different engineering problem from building for one market at scale. The last company I interviewed with in India was solving depth; you are solving breadth.

About me: I have an aunt’s family in Singapore and I have visited twice, so I am not romanticising it. I know the food is expensive, I know a room in a shared flat is what I could afford in year one, and I know the pass is tied to the employer. I would rather say that plainly than discover it in month three.”

WHAT EXACTLY CHANGED

1. It names a **technical** reason specific to the region — multi-market systems — not a lifestyle reason.
2. It contrasts with the alternative (“depth versus breadth”), which shows a real comparison was made.

3. It **removes the naivety risk**, which is the interviewer’s actual worry. Saying “I know rent is high and the pass is tied to you” is the single most reassuring thing a foreign candidate can say.
4. It is specific and personal without being a sob story.

As always, substitute your own reason. If you have never visited, say so honestly and replace that second half with what you **have** done: read about the cost of living, spoken to two seniors who moved, checked the pass rules. Honest preparation reads as well as personal experience. Inventing a relative who lives there does not.

Communication style — what actually differs

Situation	Common Indian-office habit	What travels better regionally
Disagreeing with a senior	Soften so much the disagreement disappears; say “yes sir” and then not do it	State the disagreement plainly, once, with a reason, then commit to whatever is decided
Saying you cannot meet a date	“I will try”, “we will manage”	“I cannot make the 14th. I can make the 21st. Here is why.”
Status updates	Report only when finished, or when it is already too late	Short, regular, boring updates, including “no progress this week, blocked on X”
Asking for help	Wait, in case asking looks weak	Ask after a stated time-box: “I gave this two hours, here is what I tried, here is where I am”
Email register	Very formal, long, deferential	Short, direct, first-name, no honorifics unless the office uses them
Meetings	Silence unless called on	One prepared point per meeting, said early

TRAP

The “I will try” problem. In many regional offices, “I will try” is heard as “yes”. So when the date is missed, the other person believes they were misled, even though you thought you had hedged. Replace **every** “I will try” with either a commitment or a refusal plus an alternative. This is the single highest-value communication change for an Indian engineer moving into a regional team, and it is also the reason the notice-period answer in Part 6 avoided that phrase.

TRAP

Do not confuse directness with rudeness. Direct means “I disagree with the caching approach, here is why”. It does not mean “that is a bad idea”. The point stays about the work, never about the person — which is the same rule as the GD in Part 1.

Part 8 — A ladder of worked interview moments

Same skills, three levels of pressure. Read them in order — the Tier 3 answers are only reachable because of the habits built in Tier 1.

TIER 1

TCS NQT · Accenture · Infosys · Wipro · Capgemini

Example 2 [TCS NQT · pattern]

“Are you willing to relocate anywhere in India?”

SOLUTION

The trap. Both extremes lose. “Anywhere, sir!” when it is not true means you will decline the posting later and burn the relationship. “Only my home city” at a company that posts freshers by allocation usually ends the process.

The structure — three slots. Slot 1 — a clear yes or a clear no. Slot 2 — one honest constraint, if you have one. Slot 3 — what you have already done to prepare, so it is not just words.

Filled example — substitute your own truth. “Yes, I am willing to relocate. I have lived away from home for my whole degree, so I know what it costs and I am comfortable with it.

One thing I should say honestly: my father has a medical condition and I visit every two months. So I would want to be somewhere with reasonable connectivity back to Nagpur. That rules out very few places.

I have already looked at accommodation in Pune and Chennai because those came up as likely locations, so I am not saying yes in the abstract.”

Answer: A clear yes, one real constraint, and evidence of preparation

If you truly cannot relocate, say so early and directly. Losing a process in week one costs you a week. Accepting a posting you cannot take and backing out in month three costs you the relationship and sometimes your campus’s relationship with that company.

Backlogs, study gaps, a low CGPA and rejection history are handled in full in Chapter 6. The pattern is the same three-sentence shape used above: the fact owned without blame, what changed and when, then the evidence that it stuck — and then stop talking. Never blame the university, a teacher or bad luck, and never invent a hardship that did not happen.

TIER 2

Singapore & Thailand · Grab · Shopee · GIC · DBS · Agoda · SCB

Example 3 [Shopee · pattern]

“Your English is good but your accent is quite different from the rest of the team, who are mostly based in Singapore and Jakarta. How will you handle that?”

SOLUTION

Why this is asked. It is not a test of your accent. It is a test of whether you get defensive when someone names a real friction.

The structure. Slot 1 — accept the observation without apologising for who you are. Slot 2 — name the actual mechanism that solves it. Slot 3 — a concrete habit you already use.

Filled example — substitute your own. “That is fair, and it is a real thing on calls, not an imagined one.

What I have found matters more than accent is **pace and structure**. When I slow down and say the conclusion first — ‘the job failed, here is why, here is what I need’ — people follow me easily. When I speak fast and build up to the point, they do not.

Two things I already do: I write the one-line summary in the chat before I say it out loud on a call, so there is a text version people can re-read, and I ask ‘does that make sense so far?’ once in the middle rather than only at the end. If someone needs me to repeat something I would rather they say so, and I will ask the team directly to tell me when that happens.”

Answer: Accept, then redirect to pace and structure, then give two concrete habits

TIER 3

Google · Amazon · Microsoft · Goldman Sachs · D. E. Shaw · Adobe

Example 4 [Amazon · pattern]

“Tell me about a time you disagreed with a decision, lost the argument, and then had to make the decision succeed anyway.” Then expect three follow-up probes.

SOLUTION

Why this is asked. It separates people who are merely opinionated from people who can disagree **and then commit**. The follow-ups are where most candidates fall apart, so prepare for them.

The answer, four STAR parts labelled — substitute your own experience.

S — Situation. “During my internship, my team was deciding how to store audit records for a payments service. I argued for a separate append-only table. The tech lead chose to reuse the existing orders table with a status column, because it was two days of work instead of two weeks and we had a compliance deadline.”

T — Task. “I was assigned to build it his way. I still believed it was the wrong long-term choice, and I was the one who would maintain it for the rest of my internship.”

A — Action. “I said my objection once more, in writing, in the design document, in two lines — so it was on record without being re-litigated in every stand-up. Then I stopped arguing and built it properly. I did three things to reduce the risk I was worried about: I put every audit write behind one function so a future migration would touch one file instead of forty; I added a test that fails if anyone writes an audit row outside that function; and I wrote a one-page note on what the migration would look like if the table ever became a problem, with an estimate.”

R — Result. “We shipped four days before the compliance deadline. Seven months later, after I had left, the team did split the table out — and the lead messaged me to say the migration took two days instead of the two weeks he had estimated, because of the single write path. He also said my objection note was what made them look at it at all.”

Follow-up probe 1: “Why did you stop arguing?” “Because the decision was made by the person accountable for it, with information I did not have — he knew the compliance deadline was immovable and I did not. Continuing to argue would have cost team time and changed nothing. Disagreeing once clearly is useful; disagreeing repeatedly after a decision is just noise.”

Follow-up probe 2: “What if you had been right and it had failed badly?” “Then the single write path and the migration note were exactly the insurance for that case. I did not just comply — I reduced the cost of being wrong. That is what I would do again.”

Follow-up probe 3: “What would you do differently?” “I would have asked **why** the deadline was immovable before arguing, not after. I spent twenty minutes arguing from a position that assumed we had time. If I had asked first, I would have argued for the same design but with a smaller scope, which might actually have won.”

Answer: Disagree once, in writing, then commit fully — and reduce the cost of being wrong

WHY THE FOLLOW-UPS MATTER MORE THAN THE STORY

Tier 3 interviewers almost never accept the first answer. They probe three times, and the probes follow a predictable shape:

1. **“Why?”** — they are testing whether your reasoning is real or rehearsed.
2. **“What if it had gone the other way?”** — they are testing whether you thought about risk.
3. **“What would you do differently?”** — they are testing self-awareness, and a candidate who says “nothing” fails here almost automatically.

Prepare all three for every story you plan to tell. If you cannot answer the third one honestly, the story is not ready.

Example 5 [Goldman Sachs · pattern]

The hiring manager says: “We can do Rs 16 lakh fixed. I know you mentioned 18. That is our band and I cannot go above it.” You do want the job. What do you say?

SOLUTION

The wrong instincts. Accept immediately (you leave value and look like you invented the 18), or repeat 18 (you have added nothing new, so you will get the same answer).

The structure — four moves. Move 1 — accept the constraint out loud. This is not surrender; it stops you arguing against a wall. **Move 2 — move to a different currency.** Base is banded. Other things are not. **Move 3 — make one specific, small, concrete ask. Move 4 — commit.**

Filled example — substitute your own numbers. “Understood, and thank you for being straight about the band rather than leaving me guessing.

If 16 is the ceiling on base, let me ask about the things outside it. Three options, and any one of them closes this for me:

First, a joining bonus of Rs 1.5 lakh. That is one-time and usually sits outside the band.

Second, a formal compensation review at six months instead of twelve, written into the offer. That costs nothing today.

Third, if neither is possible, I would ask for the title to be Engineer II rather than Engineer I, if the work justifies it — because that changes my band at the **next** review, which is worth more to me over three years than the Rs 2 lakh anyway.

If any one of those works, I will sign today and stop my other process.“

Answer: Accept the band, then negotiate joining bonus, review timing, or level — and close

WHY THIS WORKS WHEN REPEATING THE NUMBER DOES NOT

1. **Move 1 removes the fight.** The manager is now not defending the band, because you agreed with it.
2. **Move 2 is the actual insight:** base pay is set by a compensation band that a hiring manager genuinely cannot break. Joining bonus, review date and level are usually **different approval paths**, and often easier.
3. **Move 3 gives three options, not one demand.** The manager gets to choose, which makes saying yes easy.
4. **The third option is the smartest one** — level compounds over years, and Rs 2 lakh once does not.
5. **Move 4 makes it a deal, not a complaint.**
6. If all three are refused, you have lost nothing. You accept or decline on the same terms you already had, and you did it professionally.

TRAP

Everything above assumes the 18 figure was **real** — that you researched it and you meant it. If you invented it as an opening bluff, this conversation goes badly, because the follow-up is always “where did 18 come from?” and you will have no answer. Research first, then negotiate.

Part 9 — Closing behavioural questions, in STAR

The last round often circles back to behaviour, especially around teams and conflict — which is exactly what a GD is meant to predict.

Q: Tell me about a time you disagreed with your team.

TRAP

WEAK. “In my group project one guy wasn’t doing anything so I just did his part myself and we submitted on time. I don’t like conflict so I avoided it.”

SOLUTION

STRONG — four STAR parts labelled.

S — Situation. “In my seventh-semester project, four of us were building a campus complaint portal. Two weeks before submission, our team lead wanted to rewrite the whole front end in a framework none of us had used, because he thought it would look better to the panel.”

T — Task. “I thought the rewrite would cost us the deadline, and I was the one who would have to reconnect all the API calls. I had to either raise it or go along with it.”

A — Action. “I did not argue in the group chat. I asked for fifteen minutes with him in person. I said his goal was right — the front end did look dated — and then I put a number on the risk: I counted 23 API calls and 9 forms that would need rewiring, and estimated three full days minimum with nobody available to test. I proposed a smaller version: keep the structure, restyle the four screens the panel would actually see, and put the framework rewrite in our ‘future work’ slide, which the panel liked anyway. I offered to do the restyle myself over the weekend so he was not paying for my objection with his own time.”

R — Result. “He agreed the same evening. The restyle took me about eleven hours. We submitted two days early, and the ‘future work’ slide became a question in the viva, which we answered well. He and I worked together again on our final-year project, which I think matters more than the disagreement itself.”

WHAT EXACTLY CHANGED

1. The weak answer has **no disagreement in it at all**. It describes avoidance, then says “I don’t like conflict” — which answers the opposite of the question.
2. The strong **S** gives a real, specific technical disagreement with stakes.
3. The **T** says clearly what was at risk **for the speaker**, not just for the team.
4. The **A** shows four separate skills: choosing a private channel, agreeing with the other person’s goal first, quantifying the risk (23 calls, 9 forms, 3 days), and offering a compromise that costs the speaker something.
5. The **R** gives measurable outcomes **and** a relationship outcome. The last line is what separates a good answer from a great one: it proves the disagreement did not damage the team.
6. Length: about 50 seconds. Weak answer: 12 seconds. The gap is content, not padding.

Say it again: **this is not your story**. The project, the framework, the 23 API calls — all invented to show the shape. Open a notebook, list three real disagreements from your own projects, internships or part-time work, and write each one into the four slots. If a slot is empty, that story is not ready to use yet.

Q: Why should we hire you, as the last question of the day?

TRAP

WEAK. “Because I am a hard worker, a quick learner, and a good team player. I will give my 100% to this company.”

Every candidate that day said this. It contains no information.

SOLUTION

STRONG — three slots: match, proof, gap.

Match. “You said the team’s biggest problem this quarter is that the ingestion jobs fail silently and someone finds out the next morning.”

Proof. “That is close to what I spent my internship on. I added structured logging and a dead-letter queue to a batch job that was dropping about 2% of records without anyone noticing, and wrote the alert that caught it. I can show you the code.

Gap, stated honestly. “What I do not have is production experience at your scale. I have never run anything with more than a few thousand records an hour. I would need someone to review my work closely for the first couple of months, and I would rather say that now than discover it in month two.”

WHAT EXACTLY CHANGED

1. The strong answer **quotes the interviewer’s own words back**, which proves you listened for the whole hour.
2. It gives **one** piece of proof with a number and an offer to show the code.

3. It **names a real gap**. This is counter-intuitive and it works: admitting a limit makes everything else you claimed more believable. It also protects you — a candidate who oversells then struggles in month two is remembered badly.
4. It says nothing about being a hard worker, a quick learner, or giving 100%.

TRAP

Honest framing, never invention. If you have a backlog, a study gap, a low CGPA or a rejection history, say it plainly, in one sentence, and then spend the rest of the answer on **evidence of improvement** — the cleared backlog, the rising semester scores, the projects built during the gap. Never invent an experience you did not have. Background verification, a probing follow-up question, or a simple “show me the code” will expose it, and a discovered lie ends the process at any stage, including after joining.

Part 9b — A mock group discussion, scored line by line

Read this once for the flow, then read the score column. Six candidates, topic: “**Should companies be allowed to monitor employees’ work laptops?**” Twelve minutes. All names and lines are invented.

Time	Eliminator note what was said
0:10	Strong open. He did not just claim the first slot — he defined the term and proposed a structure . Initiative and structure both. start. I think we should first agree what 'mon- i- tor- ing' means, be- cause key- stroke log- ging and count- ing how many hours a lap- top is on are very

	different things. I'd suggest we take them separately. Let me start with the milder one."
0:50	Weak. Agreement plus a generic claim. No evidence, no new angle. This counts as participation but not contribution. agree with Ravi. Monitoring is important for security. Companies have data to protect."
1:10	Weak, and slightly worse. "That is basic" dismisses future disagreement. The repetitions are logged as repetitions, not as contributions. Small negative on team behaviour. exactly, and also productivity. If people

	ple are not work- ing the com- pany should know. That is ba- sic.” (He goes on to re- peat this same point three more times be- fore minute six.)
1:35	Vanhatrong. A signposted, narrowly scoped disagreement with a mechanism, not an opinion. This is the best entry “Can discussion so far. I take the other side on the pro- duc- tiv- ity point specif- i- cally? The prob- lem with mea- sur- ing ac- tiv- ity is

that
it
mea-
sures
the
wrong
thing.
If
I
am
stuck
on
a
bug
and
think-
ing,
my
lap-
top
looks
idle.
If
I
am
mind-
lessly
click-
ing
through
tick-
ets,
it
looks
pro-
duc-
tive.
So
ac-
tiv-
ity
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ing
does
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just
fail
to
mea-
sure
out-
put
—
it
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tively

	re-wards the wrong be-hav- iour.”
2:20	Negative for Arjun: interrupting, and a position with no support. Strong recovery for Sneha: she did not fight for the floor. Composure is visible and it is marked. (talks over Sneha’s last words) “But companies pay salaries so they have the right to check the time they paid for.” — Sneha: (stops, waits, then) “I’ll let Arjun finish and come back to the measurement point.”
3:05	Excellent. He separated two conflated arguments and proposed a sequence. This is the single highest-scoring move available in a GD. me connect

	Ar- jun and Sneha. Arjun's point is about rights — can they. Sneha's is about ef- fec- tive- ness — does it work. Those are dif- fer- ent ques- tions and I think we are mix- ing them. Can we fin- ish 'does it work' first, since that is where we ac- tu- ally dis- agree?"
4:00	Veigastrong. A concrete example, a named second-order effect, and a genuinely new angle. Speaking once at 4:00 (first time this is worth more than speaking five times with Meena's entries).

time
speak-
ing)
“On
whether
it
works
—
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is
a
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ond
ef-
fect
no-
body
has
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tioned.
Mon-
i-
tor-
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changes
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hav-
iour
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when
no-
body
is
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ing
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data.
In
my
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tern-
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we
had
screen-
time
track-
ing
and
peo-
ple
just
left
a
doc-
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open

	in an- other win- dow. So the com- pany got worse data than be- fore, and also lost trust. It was neg- a- tive on both axes.”
6:00	Supharak on team behaviour. She named the imbalance, not the person; she revived a dropped point; she “We invited Meena; and she invited the silent candidate. Four scores in one sentence. We have heard a lot from four of us. Meena said some- thing about se- cu- rity ear- lier that we dropped com- pletely, and we have not heard from

	Nikhil at all. Meena, can you take the se- cu- rity side prop- erly? That is the strongest case for mon- i- tor- ing and no- body has made it.”
6:20	Much stronger than her first entry. Being invited gave her a shape to fill. Note that Sneha’s move created this contribution. “Thank you — yes. The se- cu- rity case is nar- row but real: data loss pre- ven- tion. Not watch- ing what some- one types, but

	flag- ging when 4,000 cus- tomer records leave a lap- top. That is a spe- cific trig- ger, not sur- veil- lance, and I think that is the dis- tinc- tion we have been miss- ing.”
8:00	Nonbinding. A crisp test that resolves the whole debate. Late but decisive. Timing did not hurt him because the (first time) was new. “Build- ing on Meena — the dis- tinc- tion is tar- geted ver- sus con- tin- u- ous. A

trigger
on
a
specific
event
is
defensible.
Continuous
recording
is
not.
And
the
test
is
whether
the
employee
can
be
told
about
it
in
advance
without
the
control
becoming
useless.
A
data-loss
trigger
still
works
if
everyone
knows
it

	ex- ists. Key- stroke log- ging only 'works' if peo- ple for- get it is there — which tells you some- thing.”
11:00	Have the best 40 seconds in the room. Fair to both sides, names people accurately, adds the missing angle, closes with a position rather than a shrug. Position I sum- marise? We agreed mon- i- tor- ing is not one thing. We broadly agreed that tar- geted, an- nounced con- trols like data- loss trig- gers are de- fen- si- ble —

Meena
and
Nikhil
made
that
case.
We
split
on
con-
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ity
mon-
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tor-
ing:
Ar-
jun
and
Karthik
ar-
gued
from
the
employer's
right
to
the
time
paid
for;
Sneha
and
Priya
ar-
gued
it
mea-
sures
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wrong
thing
and
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grades
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data
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lects.
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test:
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you
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it."

THE RANKING THE EVALUATOR WOULD WRITE, AND WHY

1. Ravi — opened with a definition and a structure, separated two conflated arguments mid-way, and summarised fairly. Note he never “won” anything.

2. **Sneha** — one excellent mechanism-based disagreement, one composed recovery from being interrupted, and one move that pulled two silent people into the discussion.
 3. **Nikhil** — spoke **once**, at minute eight, and resolved the debate. Proof that frequency is not the metric.
 4. **Priya** — one new second-order effect with a concrete example.
 5. **Meena** — weak first entry, strong second. Would have ranked higher with Nikhil’s discipline of waiting for something worth saying.
 6. **Karthik** — spoke the most and ranked last. Repetition, plus “that is basic”.
- Arjun** is unranked in the top six because of the interruption. A single interruption is usually survivable; combined with an unsupported position, it was not.

SHORTCUT

Look at Nikhil again. He spoke **once**, in twelve minutes, at minute eight, and placed third. Copy that if you are nervous about volume. One genuinely new, well-formed point beats six mediocre ones, and it is a much easier target to hit when you are anxious.

TRAP

Do not copy Nikhil’s timing as a **strategy** while having nothing prepared. He placed third because his point was new and decisive. Staying silent while waiting for a brilliant idea that never arrives scores zero. If minute six arrives and you have not spoken, use SPEH on the best point you have and speak.

Part 10 — The 30 minutes before, and the 24 hours after

When	What to do
Night before	Re-read the job description and your resume. Write down the three stories you will use. Charge everything. Test the video link if it is remote.
30 minutes before	Water within reach, phone silenced but not off, one page of notes: six questions to ask, three story titles, your salary floor and target. Do not revise new topics — it raises anxiety and changes nothing.
First 60 seconds	Smile, use the interviewer’s name once, let them set the pace. If remote, confirm audio in one short sentence, then stop talking about the setup.
If the connection drops	Rejoin immediately; if it fails twice, call or message the recruiter within two minutes. Recruiters forgive bad networks; they do not forgive a ten-minute silence.
Immediately after	Write down every question you were asked and every one you fumbled. It is the most valuable revision document you will ever have, and it evaporates within an hour.
Within 24 hours / after the promised date	Send the thank-you note using the four slots from Part 2. If the decision date passes, wait two working days, send one follow-up, wait a week, send one more, then stop — and keep the relationship intact, because companies do come back.

TRAP

Do not send a follow-up every two days. Two messages over three weeks reads as interest. Six reads as desperation and gets you filtered.

Practice

PRACTICE · TIER 1 target: 30 minutes, written

1. Write a 30-second GD entry using SPEH on the topic “should engineering colleges make internships compulsory?” Label the four slots.
2. Rewrite this email in under 100 words with a searchable subject line: **“Respected Sir, I had given my interview last week and I have not heard anything so kindly do the needful and update me regarding my status at the earliest as I am very much interested.”**
3. You are asked “what is your salary expectation?” in the first HR call for a fresher backend role. Write your four-move answer with your own researched numbers.
4. Write the six bond questions as you would actually ask them in a call, in one paragraph.
5. Your notice is 60 days and the company wants you on 1 November. Today is 20 September. Can you make it? Show the arithmetic and write the sentence you would say.
6. Write five questions you would ask a hiring manager for a role you actually want. None may be answerable from the company website.

ANSWER KEY

1. Check your entry has all four: a signpost phrase, exactly one point, evidence that is a number or a concrete example, and a named handover. If your “evidence” is a restatement of your point, rewrite it. 2. Target shape — Subject: **Following up** - <Role> interview, <date> - <Your name>. Body: name the recruiter, name the role and date, state the timeline they gave you, add one new piece of information, one line about timing if you have another process, sign off with your phone number. Delete every instance of “respected”, “kindly”, “do the needful”, “at the earliest”. 3. Check: did you ask for their band **first**? Did you give a range with a stated basis? Did you say the word **fixed**? Did you signal both flexibility and a floor? If any is missing, the answer is not finished. 4. Should contain, in one breath: duration, start point (joining or post-training), exact amount, whether it reduces pro-rata, whether originals are held, and whether the relieving letter is withheld. If you did not ask about the relieving letter, you asked the wrong four questions. 5. From 20 September, 60 days of notice ends on 19 November, so the last working day is 19 November and you could join 20 November at the earliest. **You cannot make 1 November.** Say: “60 days from today puts my last working day at 19 November, so 20 November is the date I can commit to. If my manager waives part of the notice I will tell you immediately, but I do not want to promise 1 November and miss it.” 6. Each question must fail the website test: if the answer is on the careers page, delete it. Strong shapes: what the team is building this quarter; the last technical argument and its outcome; what the team is worst at; what a new joiner should do alone by month six; how the variable component has actually paid out.

PRACTICE · TIER 2 target: 30 minutes

1. You have an offer from a Singapore company at SGD 4,500 a month and an Indian offer at Rs 9 LPA CTC. List the six things you must know before you can compare them at all.
2. A recruiter in Bangkok asks “why do you want to move to South East Asia?” Write a 45-second answer that is not about money and not a platitude.
3. In a GD, one candidate has spoken for four of the first six minutes. Write the exact sentence you would use to redirect, without attacking them.
4. You are told the role requires relocating to a city where you know nobody. Write an honest answer that neither overpromises nor sounds reluctant.
5. Your visa or work pass timeline is uncertain. Write how you would raise it with the recruiter **before** accepting.
6. Write a decline email for an offer you would genuinely have liked, using the four slots.

ANSWER KEY

1. (a) Is the SGD figure monthly or annual, and is it 12 or 13 months? (b) What is the fixed versus variable split in each? (c) Employer CPF or PF contributions, which differ completely between the two systems. (d) Income tax in each place — the rates and brackets are not comparable. (e) Cost of living, especially rent, which dominates the Singapore side. (f) Visa or work pass

status, duration, and who pays for it. Only after all six can you compare. **2.** Strong shapes: a specific product or market you want to work on; regional scale that does not exist in your current market; a technical problem that is characteristic of that region such as multi-language or multi-currency systems at scale. Weak shapes: “better opportunities”, “exposure”, “growth”, “I like travelling”. **3.** Target: “**We have heard a lot of detail from two of us. Meera and Karthik have not come in yet — can we hear from them before we go further?**” Note that it names the imbalance, not the person, and it names the people you are inviting. **4.** Honest shape: say yes clearly if it is yes; name the one practical thing you would need (notice period, a few weeks to find accommodation); state one true reason you are comfortable with it. Do not say “I am ready to go anywhere” if you are not — you will be held to it. **5.** Raise it as a **shared planning problem**, not a confession: “Before I accept, I want us to plan the pass timeline together. My understanding is it takes X to Y weeks. Can we agree a joining date that assumes the slower end, so neither of us is surprised?” Never conceal a visa constraint. **6.** Check: did you decline in the first two lines? Is your reason true and non-comparative? Did you name something specific about **their** process? Did you leave the door open sincerely?

PRACTICE · TIER 3 target: 45 minutes

1. You hold two written offers. Offer 1: Rs 11 LPA CTC, of which Rs 2 lakh is variable, no bond, 90-day notice at the company you would join. Offer 2: Rs 9.5 LPA CTC, all fixed, a Rs 1.5 lakh two-year bond, 30-day notice. Compute the monthly in-hand for each using the method in Part 4, price the bond per month, and write the two-sentence justification for your choice.
2. Your preferred company has made an offer but is Rs 1.2 lakh below your target. You have one competing offer. Write the full counter-offer script using the five slots, then write the reply you would send if they say “the base cannot move at all”.
3. Write a STAR answer, all four parts labelled, for: “tell me about a time you had to deliver bad news to someone senior to you.” Use a real situation from your own life.
4. Write a STAR answer for: “tell me about a failure that was genuinely your fault.” State what you changed afterwards, with evidence.
5. You have a backlog on your transcript. Write the three-sentence honest framing: what happened, what you did about it, what the evidence of improvement is. No excuses, no invention.
6. A GD evaluator asks you to summarise a discussion in which your own view was clearly the minority one. Write the summary, keeping it fair to the majority.

ANSWER KEY

1. Method, not a single right answer. Offer 1’s fixed is about Rs 9 lakh before employer PF and gratuity are stripped out, so its real fixed is lower than Offer 2’s headline suggests; run both through the **inHand** breakdown. The bond on Offer 2 costs $\frac{150000}{24} =$ Rs 6,250 a month of lost freedom. The 90-day notice on Offer 1 costs you flexibility **later**, not now. A defensible justification names the in-hand gap in rupees per month, then the one non-money factor that decided it. An answer that compares only CTC is wrong regardless of which offer it picks. **2.** Check the five slots: enthusiasm, the gap as a number, the basis, alternative currencies, a precise close. For the “base cannot move” reply, the strong shape pivots immediately to the alternatives you already named — joining bonus, six-month review, relocation, title, a shorter bond — and still ends with a commitment: “if we can do X, I will accept today.” **3.** Check: does **T** say what was at risk **for you**? Does **A** show you told them **early** rather than hoping it would resolve? Does **A** include what you brought **besides** the problem — an option, a mitigation, a plan? Does **R** include the senior person’s reaction, not just the outcome? **4.** Check: is the failure genuinely yours, and not a disguised strength (“I worked too hard”)? Does **A** include what you did in the moment, not only afterwards? Does **R** include a **changed behaviour** with evidence that it stuck — a checklist you still use, a test you now always write, a habit someone else noticed? **5.** Target shape: “**I had a backlog in my fourth semester. I had taken on too much outside coursework and I misjudged how far behind I was until it was too late. I cleared it in the next attempt with , and my CGPA rose from X to Y over the following three semesters.**” Three sentences: fact, cause owned, evidence. No blame on the college, the teacher, or illness unless it is true and you are comfortable stating it. Then stop talking and let them move on. **6.** Check: did you state the majority view **first** and **accurately**? Did you avoid using the summary to re-argue your own position? A fair summary that clearly includes the view you disagreed with is one of the strongest signals available in a GD.

ONE-PAGE REVISION CARD

GD — the grid. Content · Structure · Initiative · Team behaviour · Clarity · Listening. Aggression is a **negative** mark. Four to six entries of 25 to 40 seconds is the target.

GD — every entry is SPEH. Signpost → Point → Evidence → Handover by name. The handover is the slot everyone drops and the one that scores.

GD — the four hard moments. Talked over: finish calmly afterwards. Dominator: name the **imbalance**, not the person. Wrong fact: correct the **claim**, not the person. Stalled: summarise and open a new angle.

GD — the summary. Agreements → disagreements, fairly → what we missed → a position. Never invent a consensus.

Email — four rules. Subject carries the fact · first sentence says the ask · under 150 words · one ask per message. Delete “do the needful”, “esteemed organisation”, “at the earliest”, “revert back”.

Questions to ask. Prepare six, ask two or three. Ask about **the work**, not the benefits. Best single question: “what is the team worst at right now?”

Money — the one rule. CTC is not salary. Strip out variable, joining bonus, employer PF, gratuity and insurance. What is left is fixed pay; subtract your PF and tax to get in-hand. A **lower** CTC can pay **more** every month.

Salary answer — four moves. Ask for their band → give a researched range, not a point → say the word **fixed** → signal flexibility **and** a floor.

Counter-offer — five slots. Enthusiasm → the gap as a number → the basis → alternative currencies (joining bonus, early review, relocation, shorter bond) → a precise close: “if we get to X, I accept today.”

Bond — six questions. Duration · amount · pro-rata or not · trigger · original documents · **relieving letter**. The last one is the one that can block your next job.

Notice period. Plan backwards from the joining date. Commit to **one** date you can actually meet. Buy-out and manager waivers are **upside**, never promises. Never resign before the written offer is in hand.

STAR, always all four parts. Situation, Task, Action, Result — and the Result must be measurable. Use **your own** experience. Never invent one.

Honest framing of a weakness. One sentence of fact, one of ownership, one of evidence that it improved. Then stop.

After the interview. Write down every question within the hour. Thank-you note within 24 hours. Follow up twice over three weeks, then stop.

Last-week quick reference

Every structure in this book on a few pages, plus the question bank and the checklist for the morning of.

This appendix teaches nothing new. It is the compressed form of Chapters 1–8, for the last week, when there is no time to re-read anything.

Use it in three passes. **Seven days out:** read section 1 and fill the gaps from the chapters. **The night before:** sections 2 and 3, out loud, on a timer. **The last hour:** section 5 only.

THE SIX STRUCTURES, IN SIX LINES

- **NOW · PATH · PROOF · POINT** — “tell me about yourself”. 5–7 sentences, 60–90 s. (Ch 3)
- **S · T · A · R** — every behavioural story. Situation, Task, Action, Result. 90–120 s. (Ch 4)
- **The 12-line Project Defence Sheet** — one per project on your resume. (Ch 5)
- **ACKNOWLEDGE · FACT · EVIDENCE · FORWARD** — every awkward question. 45 s. (Ch 6)
- **P · A · U · S · E** — every puzzle and guesstimate, said out loud. (Ch 7)
- **S · P · E · H** — every group-discussion entry. 25–40 s. (Ch 8)

1 · The question bank — what each answer must contain

Cover the right column. Say the answer. Uncover, and check that every listed part was actually in what you said.

A · The core HR questions (Chapter 3)

Question	Structure your answer must contain	Time
Tell me about yourself	NOW (who you are today) → PATH (one thread only) → PROOF (strongest concrete thing, with a number) → POINT (why that points at this job). Then stop.	60–90 s
What are your strengths?	Name one strength → one sentence defining what you mean by it → one story of 4–5 sentences where it produced a measurable result → the limit of it.	45–60 s
What is your weakness?	A real one that does not disqualify you → the cost it actually caused you once → the specific habit or tool you changed → the evidence it is working now. Never a disguised boast.	45 s
Why do you want to join us?	Something specific and checkable about them (product, stack, team, public engineering writing) → the link to something you have actually done → what you want to learn there. No adjectives about how great they are.	45–60 s
Where do you see yourself in five years?	A direction, not a job title → the skill you want to be good at → one sentence on what you would be contributing by then. Stay inside the company’s own reality.	30–45 s
Why should we hire you?	Three claims, each with one piece of evidence → one sentence on fit with this role. Evidence is a number, a shipped thing, or a named responsibility.	45 s
Why this branch / why this role?	The honest origin (even “I was assigned it”) → what you found in it that you now choose → one thing you built because of that choice.	30 s
Are you flexible on location / shift?	Answer the actual question in the first sentence → your real constraint stated plainly, if you have one → what you can commit to without anyone else’s permission. Never a fake yes.	20–30 s
Tell me about your family / hobbies	Two sentences, warm, ordinary, true → connect one detail to a working habit if it is genuine. Do not perform.	20–30 s

Do you have any questions for us?	Two questions minimum, about the work: what the first six months look like, how code gets reviewed, what the team is struggling with. Never “what is the salary” here.	–
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B · Behavioural stories — the twelve you must own (Chapter 4)

Every one in labelled STAR, with **I** not **we** in the Action, and a **measured** Result.

Story	The one thing this story must prove
1 Conflict with a teammate	You separated the person from the problem and reached a decision, without a villain.
2 A failure you owned	You state the cause in your own actions, and the fix you shipped after.
3 A hard deadline	What you cut , and who you told, and when. Not “I worked all night”.
4 Leading without a title	You created a mechanism (a list, a rota, a standup) that outlived the moment.
5 Learning something fast	The method you used to learn, and the thing you shipped with it, with dates.
6 Helping someone else	Their result improved, and you can say by how much.
7 Deciding with data	The number you looked at, the decision it changed, the outcome.
8 Disagreeing with someone senior	You disagreed in private, with evidence, then committed to the decision either way.
9 Ambiguity	How you narrowed an unclear problem to one testable first version.
10 Going past what was asked	Nobody asked; you measured the gain; it stayed in use.
11 A difficult user or customer	You found what they actually needed underneath what they said.
12 Competing priorities	The rule you used to choose, and what you explicitly dropped.

SHORTCUT

One real event usually feeds three or four of these twelve. Do not hunt for twelve separate dramas. Take your four biggest events and ask each of them all twelve questions.

C · The follow-up probes (Tier 3) — prepare these for every story

Probe	What it is really testing
“What did you do, specifically?”	Whether the story is yours or your team’s.
“Why did you choose that over the alternative?”	Whether you knew there was an alternative.
“What did the other person say?”	Whether the conversation happened at all.
“What would you do differently?”	Self-awareness; a real answer, not “nothing”.
“How did you measure that?”	Whether the number is real or decorative.
“What happened after?”	Whether the fix survived past the demo.

D · Defending your project (Chapter 5)

Question	What must be in the answer
90-second pitch	Problem → who used it and how many → three named features → stack in one line → your one number.
Draw the architecture	Boxes and arrows, 60 seconds, narrated: client → API → database, plus anything asynchronous.
What did YOU do?	Named files, modules or endpoints, plus an honest sentence naming what teammates wrote.
Trace one request end to end	Tap → route → validation → query → response → what the user sees. No gaps.
Why this technology?	One honest reason, the alternative you rejected, and the cost of your choice.
What broke?	One real bug: symptom, cause, fix, and how you would prevent the class of it.
What would you change?	Two engineering changes, not features.
Scale it 100x	Name the first thing that breaks, say how you would find out, then one fix.
Something you do not know	Say “I do not know” → the nearest true thing you do know → a guess labelled as a guess → a question back.

E · Hard and awkward questions (Chapter 6) — all in A · F · E · F

Question	Acknowledge → Fact → Evidence → Forward
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Salary expectation	Ask about the range first if you can. If pushed: give a researched range with the source named, say you are flexible on the whole package, and ask what band the role sits in. Never a single number, never “anything is fine”.
Bond / service agreement	Ask, politely and early: duration, start date, amount, who holds the certificates, and the exit triggers — in writing . Negotiate only after the written offer.
Notice period	State the real date. Promise only a date you can hit without anyone else’s permission.
Backlog	Number and subject, plainly → cleared or the exact attempt date → what changed in how you study → forward.
Gap in study or work	The length and the reason in one flat sentence → what you did in that time that you can show → forward. No apology tour.
Low CGPA	The number, said first → one sentence of honest cause → the upward trend or the evidence outside marks → forward.
Rejected here before	Yes, in which round → what you were missing then → the specific thing you built since → forward.
Other offers	Truth, without numbers or names → your honest ordering criterion → their timeline question.
Will you leave for higher studies?	Answer the plan you actually have. A vague answer reads as a yes you are hiding.
Stress or pressure tactics	Slow down, do not fill silence, ask for the question again if it was unclear, keep the same tone. The tactic is the test.

F • Puzzles and guesstimates (Chapter 7)

Say out loud	The words
P — play it back	“So we have N items, one is different, and I do not know if it is heavier. Correct?”
A — ask, then assume	At most two questions. Then: “I will assume ...” and continue.
U — use a smaller case	“Let me do it for 3 first.” This step is what separates a pass from a fail.
S — solve and state	Say every number as you write it. No silence longer than a few seconds.
E — examine	“Is that the right order of magnitude? Does the extreme case still hold?”

Guesstimate funnel	What you say
1 Define	What you are counting, in what unit, over what period.
2 Anchor	One number you genuinely know — usually a population.
3 Filter	Three to five multiplications, each one a stated assumption.
4 Compute	Round hard, say the running total after every step.
5 Sanity-check	A second route, or a per-person feel test.

The only things you memorise for this round are the anchor numbers in Chapter 7 (populations, household size, working days, device lifetimes). Everything else is derived out loud.

G • Group discussion and closing (Chapter 8)

Moment	What to do
Every GD entry	SPEH: Signpost → Point (one sentence) → Evidence (number, mechanism or example) → Handover (name someone).
Entering a loud GD	Signpost loudly and briefly at a breath-gap: “Can I add one thing on cost?” Then be short.
Being interrupted	Let them finish, then: “I will finish my point in one line, then come to yours.”
The summary	The highest-value 30 seconds: the two positions in the room, the point of agreement, the open question. Take it if nobody else does.
Questions to ask them	Two, about the work and the first six months. Never salary at this point.
CTC vs in-hand	Know the difference and do the arithmetic before you react to any number.
Follow-up email	Within 24 hours. Three short paragraphs: thanks, one specific thing from the conversation, one line of availability.
After silence	Follow up twice over three weeks, politely. Then stop.

2 • The thirty-second self-test

If you cannot fill this table from memory the night before, you are not ready — and every missing cell tells you exactly which chapter to reopen.

Can you say, right now ...	Chapter	Yes / No
Your NOW sentence, under 25 words	3	
One number from your best project, and how it was measured	2, 5	
Three STAR stories with a measured result in each	4	
What you personally wrote in your project, by module	5	
One request traced end to end	5	
One trade-off you made and the alternative you rejected	5	
Your salary range, with the source you got it from	6, 8	
Your real notice period or joining date	6, 8	
Your backlog / gap / CGPA sentence, flat, in under 15 words	6	
Two questions to ask the interviewer	3, 8	
The five PAUSE steps	7	
Four anchor numbers	7	

3 • Timing — the numbers that get people cut

Answer	Target	What going over sounds like
Tell me about yourself	60–90 s	A biography. The interviewer stops listening at 2 minutes.
A STAR story	90–120 s	You are re-living it instead of reporting it.
An awkward question	45 s	Past 90 seconds you are pleading, not answering.
Project pitch	90 s	You listed ten features instead of three.
A GD entry	25–40 s	You are dominating; the team-behaviour row drops.
Silence after a hard question	3 s	Filling it yourself is how candidates volunteer their own weaknesses.

4 • The last-week schedule

Day 7	Read this appendix, section 1. Mark every cell you cannot answer; reopen only those chapters.
Day 6	Record answers to the ten core HR questions. Play back at 1x. Cut every answer that ran long.
Day 5	Three STAR stories, probed three levels deep by a friend who has the probe table above.
Day 4	Project Defence Sheet, out loud, plus the architecture drawn in 60 seconds.
Day 3	The awkward questions in A–F–E–F, 45 seconds each, flat voice, three seconds of silence after.
Day 2	Four puzzles and two guesstimates out loud. Re-read the anchor numbers.
Day 1	Company research: 20 minutes on their product, their stack, their public engineering writing. Write the “why us” answer from it. Sleep.
The morning of	Section 5 below. Nothing else. Do not learn anything new today.

5 • Pre-interview checklist

THE NIGHT BEFORE

- Printed resume, 2 copies. Read your own resume top to bottom — **every line on it is a question you have invited.**
- Project Defence Sheet for each project on that resume, filled, in front of you.
- Your salary range and its source, written down. Your real notice period or joining date.
- Your two questions to ask them, written down.
- Company research done and reduced to three sentences.

- The route and the arrival time, or the meeting link tested end to end.
- Documents: ID, marksheets, offer letters, experience letters — in one folder.
- Clothes out. Phone charged. Sleep before midnight; being sharp beats one more revision.

THIRTY MINUTES BEFORE

- Read **only** section 1 of this appendix and your own story sheet. Nothing new.
- Say your NOW sentence and one full STAR story out loud, once, to warm up your voice.
- Water. Bathroom. Silence the phone — fully, not on vibrate.
- Online: camera at eye level, light in front of you not behind, headphones in, a second connection ready (a phone hotspot), background tidy, notifications off, name displayed correctly.
- In person: arrive 15 minutes early, not 45. Be polite to everyone in the building — reception often gets asked.

IN THE ROOM --- THE SIX BEHAVIOURS

1. **Answer the actual question first**, in the first sentence. Then support it.
2. **Say “I” for what you did** and “we” only for what the team did.
3. **Put one number in every answer that can hold one.**
4. **When you do not know:** say so, give the nearest true thing, label a guess as a guess, ask a question back.
5. **Let silence exist.** Three seconds is normal. Filling it is how people talk themselves down.
6. **Take the last question seriously** — “do you have questions for us” is scored.

WITHIN 24 HOURS AFTER

- Write down every question you were asked, while you still remember it. This is the most valuable study material you will ever have, and it evaporates within a day.
- Send the thank-you note: thanks, one specific thing from the conversation, one line of availability. Three short paragraphs.
- Note the two answers you were least happy with, and fix those two structures this week — whatever the outcome of this interview.

TRAP

Two things to **not** do on the morning of: learning a new structure, and reading someone else’s sample answers. Both crowd out the one thing that actually helps — your own material, recalled calmly.

ONE-PAGE REVISION CARD

The whole book in ten lines.

1. The rounds are a machine with stages; each stage scores a different thing. (Ch 1)
2. Every resume bullet: VERB + WHAT + HOW + NUMBER. A bullet with no number is a claim. (Ch 2)
3. “Tell me about yourself” is a pitch, not a biography: NOW → PATH → PROOF → POINT. (Ch 3)
4. Every behavioural answer is labelled STAR, and the Action is **I**, not **we**. (Ch 4)
5. Build the story bank from real events first; map them to questions second. (Ch 4)
6. Your project is defended from a 12-line sheet, and line 9 — the one number — carries the most weight. (Ch 5)
7. Awkward questions get 45 seconds of ACKNOWLEDGE → FACT → EVIDENCE → FORWARD, and the marks live in EVIDENCE. (Ch 6)
8. Puzzles are scored on thinking out loud: PAUSE, always, including the small case. (Ch 7)
9. In a group discussion, the handover and the summary score highest. (Ch 8)
10. Every structure in this book is a container. **You** supply the contents, and they must be true — an invented story dies at the third follow-up, and this book will not help you build one.